

ABSTRACTS COLLECTION

Selected Abstracts from The Royal College of Ophthalmologists
Annual Congress 2025 ACC, Liverpool, UK

Eye (2025) 39:191–318; <https://doi.org/10.1038/s41433-025-03831-0>

All content was reviewed and approved by the RCOphth Scientific Committee, which held full responsibility for the abstract selections.

AUDIT AND CLINICAL GOVERNANCE

P001

“A NOVEL AND EFFECTIVE OPHTHALMOLOGY MORBIDITY AND MORTALITY MEETING: EIGHT YEARS OF LEARNING AND IMPROVEMENT”

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Introduction: Morbidity and Mortality (M&M) meetings are commonplace in modern surgical practice, contributing to patient safety, education, and quality improvement. However, their adoption in ophthalmology has been limited, with inconsistent practices and no formal guidance from the Royal College of Ophthalmologists. In 2016, our ophthalmology department introduced M&M meetings as part of a monthly education and audit session, underpinned by a unique data collection tool used in our ophthalmic operating theatre: the “glitch book”. This poster highlights eight years of experience with this approach. As well as sharing our data and insights, we call for a broader adoption of structured M&M processes in ophthalmology to enhance patient care.

Method: The “glitch book” is a physical logbook placed in our operating theatre to capture intraoperative process deviations or “glitches”. These are defined as any unplanned or unintended event that could have, or did, result in patient harm, a definition similar to patient safety incidents (PSIs). Any member of the team, both clinical and non clinical, can enter anything into the book. Entries are reviewed and used to focus our M&M discussion on actionable change, patient safety and quality improvement within our department. Over eight years, the glitch book has captured approximately 40 entries annually, creating a substantial dataset. This gives us the unique ability to look back at a record of the “glitches” which have occurred in our department. In this poster we present a record of 8 years of this data and discuss how and why the introduction of M&Ms has changed the culture of our department.

Results: Since its implementation, the glitch book has captured 257 “glitches” which we split into 14 categories. Of these, 59% were related to cataract procedures, 13% of errors were equipment related, 6% drug related and 6% admin error. This data provides useful information on where to focus training, discussion and improvement within our department. Easy access to the physical copy of the glitch book within our M&Ms means patterns and similar errors can be identified and where needed to be acted upon, in addition to local hospital reporting procedures.

There were also limitations. The glitch book stays in the ophthalmic operating theatre and so misses many postoperative complications of ophthalmic surgery. For example, there were no retained fragments and no wrong refractive targets entered in the book despite these occurring during the 8 year period. One solution could be to run a parallel reporting process in eye casualty and other settings, such as another glitch book or reporting tool, to detect these complications.

Conclusion: The integration of a “glitch book” into the monthly Morbidity and Mortality (M&M) meetings has fostered a culture of safety, collaboration, and continuous learning in our ophthalmology department. Over eight years, this approach has provided a structured mechanism for capturing, categorising, and addressing intra-operative errors. The glitch book has empowered team members at all levels to contribute to the department’s improvement and culture of safety without fear of blame. As ophthalmology continues to develop its own standards for M&Ms, the lessons from our experience could serve as a valuable framework for other teams aiming to adopt similar practices.

Conflicts of interest: None

Funding: None

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P002

“SPARKLES, ANXIETY AND ENHANCED CATARACT CONSENT”: A PATIENT-CENTRED AUDIT TO ADDRESS EMOTIONAL WELLBEING AND IMPROVE INFORMED CONSENT PRIOR TO CATARACT SURGERY”

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Introduction: The current cataract surgery consent form primarily focuses on the surgeon’s perspective of potential complications, often neglecting the patient’s emotional and visual experiences during the procedure. This audit aimed to explore the importance of informing patients about the potential visual experiences they might encounter during cataract surgery. By incorporating this information into the consent process, the goal was to enhance patient understanding, alleviate anxiety, and reduce distress during the surgery.

This study was discussed with a patient focus group to understand their concerns about cataract surgery, particularly regarding anxiety and the unknown aspects of the procedure. Patients emphasised the lack of information about potential

visual phenomena, like flashes of light and shimmer, which contributed to their anxiety.

Method: To evaluate the impact of preoperative information about potential intraoperative visual phenomena on patient anxiety during cataract surgery and to compare outcomes between patients who were informed about these visual disturbances and those who were not

- **Study Design:** A prospective audit was conducted involving two groups of patients undergoing cataract surgery at a single surgical centre over a defined period (6months). One group received preoperative information about the possibility of visual phenomena during surgery, while the other group was not informed.

- **Group Classification:**

- **Group 1 (Informed group):** This information was included in the preoperative counselling session and added to the consent forms.
- **Group 2 (Non-informed group)**

Data Collection:

- **Patient Survey:** The survey included questions about their emotional state, whether they were surprised or distressed by any visual phenomena, and how well they felt prepared for the surgery.
- **Apprehension Scoring:** Likert scale (non continuous) along with open-ended questions to gather qualitative feedback on patient experiences and Visual Analogue Score (continuous) was also used.

Results: Preliminary data suggests that a significant number of patients experience distress due to the visual phenomena during cataract surgery. Incorporating this information into the consent forms is expected to better prepare patients for the procedure, thereby improving their overall experience and reducing unnecessary anxiety.

- **Informed Group:** The group that was informed about potential visual disturbances prior to surgery reported significantly lower levels of anxiety during the procedure. The mean apprehension score for this group was 2.1 ± 0.8 , with 72% of patients rating their apprehension as mild or minimal (scores 1–2). Statistical analysis revealed a significant difference in apprehension scores between the informed and non-informed groups ($p < 0.01$).
- **Non-informed Group:** The non-informed group showed a higher level of distress, with a mean apprehension score of 3.8 ± 1.2 . A larger proportion of patients in this group (43%) reported moderate to high distress (scores 4–5). The difference in distress between the two groups was statistically significant ($p < 0.001$).
- **Chi-Square Test Results:** The chi-square test indicated a significant difference in the frequency of patients reporting surprise or discomfort related to visual phenomena, with 55% of the non-informed group expressing surprise, compared to just 20% in the informed group ($\chi^2 = 14.68$, $p < 0.01$).
- **Confidence Interval:** The 95% confidence interval for the mean difference in apprehension scores between the informed and non-informed groups was (1.3, 2.4), suggesting a moderate-large effect size in the reduction of patient anxiety when preoperative information was provided.

Conclusion: Statistical analysis substantiated that reduction in anxiety was clinically and statistically significant ($p < 0.01$).

Cataract surgery remains a common procedure in ophthalmology in elderly patients, many with diverse co-morbidities.

A proposal is advanced to revise the informed consent documents and preoperative counselling protocols to include comprehensive details about the potential visual phenomena that patients may encounter postoperatively.

An enhanced cataract consent framework should be regarded as an integral aspect of evidence-based, patient-centred care, ensuring patients are thoroughly educated and psychologically prepared for surgery.

Conflicts of interest: None

Funding: None

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P003

AN AUDIT ASSESSING THE PROGRESSION OF PATIENTS FROM R2 TO R3A WITHIN THE DIABETIC SCREENING PATHWAY IN NORTH CENTRAL LONDON FROM 2018 TO 2024

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Introduction: Diabetic retinopathy has an increasing national disease burden and is a common cause of vision loss amongst people of working age in the UK. The NHS England Diabetic Eye Screening programme (DESP) recently underwent changes to the grading criteria including defining the R2 pre-proliferative level which will be implemented from October 2024. With these new criteria patients can be defined as being R2L (pre-proliferative retinopathy of lower risk) and be referred into DS (digital surveillance) pathway, which is part of DESP for screening within 13 weeks then reviewed on a 3,6-,9- or 12-month basis. Patients can also be categorised as R2H (pre-proliferative retinopathy of higher risk) and referred to HES (Hospital Eye Service) Medical Retina Clinic to be seen within 13 weeks.

Method: This audit looked retrospectively at patients within the DESP in North Central London with 3 main aims:

1. How many patients within the DESP progressed from R2 to R3a from 2018 to 2024?
2. Of these patients what was the frequency of follow up between progression?
3. Given the new changes to the guidance how many of these patients would have been classified as R2L or R2H prior to progression to R3a?

Patient data was collected using a code written to identify patients who progressed from R2 to R3 from 2018 to 2024. The code was run on the database of patients part of the North Central London DESP and used to generate a spreadsheet with patient details, where the last referral was from (DS or ophthalmology) and time taken to progress from R2 to R3 in months.

The clinical notes of patients identified to have progressed were then analysed to identify the frequency of follow up between progression, any DNAs from follow up, any referrals to HES, the appropriateness of referrals and evaluation of retinal imaging in the R2 stage to retrospectively determine whether these patients would have been identified as R2L or R2H.

Results: The data from 24976 patients within the digital surveillance, OCT surveillance, ophthalmology and routine digital surveillance from 2018- 2024 were reviewed to identify the number of patients who progressed from R2 to R3a. From this

dataset, 7 patients were identified in the DS pathway, 13 patients in the OCT pathway and 1 patient from RDS. Of these patients 1 patient in the DS pathway and 1 patient in the OCT pathway would have been categorised as R2H. The average time between encounters for patients progressing to R3a in months was 5 months in OCT, 8 months in RDS and 6 months in DS.

On analysis of retinal imaging and individual patient records key reasons identified for the delay in patient follow up prior to progression to R3a were that patients in the DS pathway who were identified as R2H retrospectively were downgraded by ophthalmology to R1. In OCT there was a lack of 6-field photography which may have limited the initial grading and the impact of DNAs (non-attendances), meaning patients progression was picked up in later screenings.

Conclusion: This audit demonstrates that whilst the diabetic retinopathy pathway prior to October 2024 had low numbers of patients progressing to R3a from R2, further implementation of R2 categorisation into R2L and R2H can ensure more timely follow up of higher risk patients. An important limitation of this audit is that some patients were discharged from HES to DESP (into DS or RDS) due to DNA's. The impact of DNA's played a large factor in patient progression and emphasises the importance of public health strategies to increase uptake of HES and screening appointments. Future lines of research include looking into factors effecting attendance to screening and re-auditing following the implementation of these new changes to the DESP in October 2024.

Conflicts of interest: None

Funding: None

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P004

AN AUDIT EVALUATING REFRACTIVE OUTCOMES IN PATIENTS UNDERGOING CATARACT SURGERY AT RUSSELLS HALL HOSPITAL

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Introduction: Cataracts are a clouding of the ocular lens that can reduce vision, confidence and quality of life. Successful treatment with cataract surgery and intraocular lens (IOL) can improve patients' lives and is cost-effective with approximately 516,000 of the UK population being admitted into hospital for this in 2023 alone.[1] With increasing population size and life expectancy, the estimated number of cataract procedures will increase by 50% between 2015 and 2035, demonstrating the need for optimal performance and precision due to the significant impact on a patient's health and satisfaction. [2] To standardise refractive outcome targets, the Royal College of Ophthalmologists established biometry guidelines in 2004 outlining that 85% of patients undergoing routine cataract surgery should achieve a spherical-equivalent postoperative refraction within 1 dioptre (D) of the predicted value, and 55% should fall within 0.5D.[3] This audit assesses whether the refractive outcomes of cataract surgeries performed at Russells Hall Hospital align with these benchmarks, ensuring high standards and reducing the likelihood of refractive surprise[4].

Method: A retrospective analysis was conducted using the Medisoft Ophthalmology Electronic Medical Records (EMR) database. Data included 965 patients aged 15–100 years who underwent phacoemulsification cataract surgery at Russells Hall Hospital between October 1, 2021, and October 1, 2024. All

surgeries that were included used Alcon AcrySof IQ SN60WF intraocular lenses. Inclusion criteria required routine NHS cataract cases with preoperative and postoperative spherical equivalent refraction data. Exclusion criteria included surgeries involving additional procedures or cases with complications. Patients without a recorded preoperative refractive aim or postoperative refractive outcome were also excluded. This audit analysed the deviation between the predicted and postoperative spherical equivalent refractions to calculate the percentage of eyes meeting the guideline targets, thereby serving as a direct measure of the quality and accuracy of biometry measurements and reflecting on surgical practice.

Results: The audit revealed that 87.67% of patients achieved a spherical equivalent refraction within $\pm 1.0D$ of the predicted value, while 62.59% achieved a refraction within $\pm 0.5D$. The mean postoperative refractive deviation was $+0.22D$, with deviations ranging from $-2.45D$ to $+4.50D$. These outcomes indicate that Russells Hall Hospital meets the published standards. The results obtained from this audit may only partially represent the patient population at Russells Hall as only 38% of operated eyes had a recorded postoperative refraction value. Additionally, the reliance on autorefractive data rather than subjective refraction could affect the accuracy of outcome measurements. Furthermore, data from the mediSIGHT platform does not use the locally preferred latest generation of recommended formulae as it is not licensed by the EMR so the PPOR is based on older generation formulae.

Conclusion: It is worth considering taking the mean deviation from the PPOR ($+0.22D$) into account when selecting the IOL power for the SN60WF Alcon IOL to improve visual outcomes and reduce the likelihood of refractive surprise and additional corrective surgeries. Alternatively, it may be equally beneficial to optimise the A-constant for the IOL, ensuring better results and minimising systematic refractive errors and re-auditing in six months to investigate possible differences. We would encourage the Medisoft team to acquire the license for Barrett Universal II Formula so that this can be used to guide IOL selections, as this is the gold-standard and is generally accepted to be the most accurate formula. Results have clearly demonstrated that the published standard for refractive aims has been met at Russells Hall Hospital, however, given that this is an increasingly performed procedure and we are aiming for excellent health care at the Trust, it is essential to continue improving performance and the refractive outcomes in these patients.

Conflicts of interest: None

Funding: None

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P005

ASSESSING UTILITY AND EFFICIENCY OF THE MINOR EYE CONDITIONS SERVICE IN PARTICIPANTS REGISTERED TO THE SPITALFIELDS PRACTICE, LONDON

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Introduction: The 'Minor Eye Condition Service (MECS) is a service funded by the NHS that helps to provide assessment and care of minor medical eye conditions, provided by local optometrists, to ease the burden on primary care and secondary ophthalmological services.

Method: To evaluate the utility and outcomes of MECS consultations in patients registered to The Spitalfields Practice (London) and whether the use of this service in primary care can be optimised.

Results: The service was accessed on 44 separate occasions during this period. Referrals primarily originated from GPs (52.3%), followed by undocumented sources (45.5%), with only 1 referral coming from the NHS 111 service. The most common presenting symptoms were eye discomfort (50%), discharge/watering (43.2%), and red eye (38.6%), with foreign body sensation reported in 4.5% of cases, while mild trauma was not observed. The majority of cases (54.5%) required no medication, while 40.9% involved over-the-counter (OTC) medications and 4.5% resulted in GP prescription requests. Many consultations (79.5%) required no follow-up, while 9.1% had follow-ups booked with the participating optometrist, 6.8% were referred to routine secondary care, and 4.5% required urgent secondary care referrals.

Conclusion: Although the sample size was small, the high percentage of MECS cases requiring no follow-up highlights the service's effectiveness in managing conditions within its inclusion criteria. Gauging the true proportions of the sources of referral proved difficult due to the inconsistent documentation from the MECS service providers. However, the relatively low number of referrals from the NHS 111 service recorded, compared to the higher proportion from GPs, suggests that the service may be underutilised, especially considering that patients using the NHS 111 service for minor medical eye conditions can often be redirected to primary care for further assessment.

Conflicts of interest: None

Funding: None

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P006

ASSESSMENT OF THE LEVEL OF HARM RESULTING FROM GLAUCOMA PROGRESSION

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Introduction: Reporting of harm and its disclosure to the patient is an ethical and statutory requirement in the UK. Attempts have been made to standardise harm reporting across the NHS although this can lead to inconsistencies when applied to different medical specialities. Glaucoma is a progressive condition which can lead to further inconsistencies when considering if an event caused harm. There is currently little guidance as to what levels of progression, either functional or structural, should be considered as harm. Disclosure of an event deemed to have caused moderate harm or above is a statutory requirement in the UK.

Our aim is to assess ophthalmologists' quantification of clinical harm from visual field (VF) loss or retinal nerve fibre layer (RNFL) loss in various levels of glaucoma severity.

Method: Web based questionnaire sent to UK and Eire Glaucoma Society members exploring scenario-based evaluation of harm severity for early, moderate and advanced glaucomatous visual field loss.

Results: Seventy-one respondents completed the questionnaire (85% were glaucoma consultants).

For early glaucoma (< -5 db loss) 85% considered > 2 dB VF loss to constitute functionally important loss. For moderate glaucoma (-5 to -15 db loss) 70% respondents considered > 2 dB VF loss to constitute functionally important loss. For advanced glaucoma (> -15 dB loss) 89% respondents considered

>1 dB VF loss to constitute functionally important loss. For all severity levels, paracentral VF loss was considered the most important in considering whether harm had occurred.

In the absence of VF loss 72% considered OCT changes alone could be considered to constitute harm.

A delay of beyond 25% from original planned review time was considered a delay contributing to harm by 93%.

Conclusion: For early and moderate glaucoma, 2 dB additional loss could be considered as significant functional loss whereas for advanced glaucoma 1 dB additional loss could be considered significant, in all cases paracentral loss was considered the most important area of loss. In the context of harm occurring, a delay of 25% from original review date was considered important. The scenarios depicted can help inform governance processes when moderate harm has occurred and there is a need to discharge duty of candour.

Conflicts of interest: None

Funding: None

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P007

AUDIT OF RETINAL SCREENING OF DIABETIC PATIENTS AT A NORTH LONDON PRACTICE

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Introduction: Diabetic retinopathy is the primary cause of blindness among working-age individuals in the UK. Each year, around 4200 people in England lose their sight due to this condition. However, with timely monitoring and intervention, blindness can be prevented. Our objective is to compare Arcadian Garden Surgery's retinal screening standards with those outlined by the NICE guidelines. This audit aims to identify any deficiencies in the practice regarding this aspect of patient care.

Method: The EMIS software was used to analyse the records of 505 patients within the practice to determine whether they had their annual diabetic screening. This data was collected from June to July 2024.

- Target criteria: All diabetic patients aged over 17 should have at least 1 annual retinal screening.
- Standard: 100% of patients eligible for this audit should have had retinal screening in the past 12 months.

Results: 85% of patients received screening in the past 12 months, while 15% did not.

Possible reasons for the 15% non-compliance include:

- Patient-related issues (Did Not Attend (DNA), requests for removal)
- Patients deemed unsuitable for screening by their optician
- Time constraints on GPs

The specific percentages of each patient group are as follows:

- Did not attend - 4.75%
- Screened > 12 months ago - 6.34%
- Not suitable - 0.20%
- Requested to be removed from national screening program - 0.59%
- Data unavailable - 3.17%

Conclusion: The lack of screening is not due to GP awareness, as they receive reminders for overdue patients. We suggest that providing additional training to all GPs on diabetic retinopathy screening and issuing more frequent reminders could improve referral rates. Furthermore, a large proportion of patients did not attend their annual screening. Educating patients on diabetic retinopathy and the importance of attending their yearly screening could increase future attendance. A re-audit is recommended in 2 years.

Conflicts of interest: None

Funding: None

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P008

BARRIERS TO PERFORMING FUNDOSCOPY IN GENERAL PRACTICE

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Introduction: Non-ophthalmologists report low confidence in performing direct ophthalmoscopy and difficulty in interpreting fundoscopic findings. The National Institute for Health and Care Excellence (NICE) guidance recommends fundoscopic examination for a number of conditions including for patients newly diagnosed with hypertension and presenting with headache.

We assessed the extent to which direct ophthalmoscopy was being performed in a general practice setting and to explore whether introducing a handheld fundus camera could improve adherence to guidelines.

Method: An initial audit of patients presenting with new hypertension or a new presentation of headache over a 6-month period at a single general practice was undertaken to determine the frequency with which direct ophthalmoscopy was performed. Following this, we administered a survey to general practitioners (GPs) and other clinicians within the practice to identify barriers to direct ophthalmoscopy and to understand barriers as well as potential solutions.

Subsequently, a digital handheld fundus camera was introduced. Training was provided to all relevant staff, and a standard operating procedure was developed to integrate fundus imaging into routine clinical workflows with minimal time impact. A follow-up audit was conducted over a subsequent 6-month period to determine the frequency of fundoscopic examination—either by direct ophthalmoscopy or by image capture using the handheld device, in patients in whom a new diagnosis of hypertension was made or those who presented with a new headache.

Results: During the initial 6-month audit period, direct ophthalmoscopy was performed in only one patient out of 158 with new hypertension and in none of the 359 patients presenting with headache. Survey responses prior to the introduction of the handheld fundus camera indicated that the principal barriers were a lack of confidence in performing direct ophthalmoscopy and uncertainty in interpreting fundoscopic findings. These findings were consistent with the established literature. There was broad agreement that a digital imaging device offering a wide field-of-view without dilation would improve utilisation.

Following the introduction of the handheld fundus camera, one patient out of 148 with new hypertension and seven patients out of 315 with new headache had a fundus photograph taken over the subsequent 6 months. No patients underwent examination with the direct ophthalmoscope during this period. Post-implementation survey data indicated that, although the handheld fundus camera produced easily interpretable, high-quality images and was viewed as a valuable diagnostic aid, the most significant barrier to its use was the additional time perceived to be required. Workflow integration challenges persisted despite efforts to streamline the process.

Conclusion: Our findings highlight considerable underuse of fundoscopy in primary care, even when clinically indicated. Initially, lack of confidence and diagnostic uncertainty were key barriers to direct ophthalmoscopy. After the introduction of a digital handheld fundus camera—intended to mitigate these knowledge and skill deficits—utilisation remained limited, this time primarily due to perceived workflow constraints and time pressures.

These results suggest that while advanced imaging tools can address gaps in clinical confidence and interpretation, their successful implementation requires careful consideration of practice logistics. There is perhaps an accepted culture in General Practice that fundoscopic examination offers limited clinical utility. Solutions must induce as little as friction as possible into clinical workflows to have a chance of success; these may include embedding fundus imaging into dedicated clinical pathways, ensuring rapid availability of equipment, and refining protocols to minimise any additional time burdens. With appropriate system-level support, handheld fundus cameras may become an integral part of routine examinations for patients, ultimately improving the quality of care and patient outcomes.

Conflicts of interest: None

Funding: None

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P009

BILATERAL OPTIC DISC SWELLING IN EYE CASUALTY – AUDITING THE UTILITY AND SAFETY OF ‘PAPILLOEDEMA PATHWAY’ TO IMPROVE ASSESSMENT ACCURACY

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Introduction: Suspected papilloedema with bilateral optic disc swelling frequently presents in the acute eye care setting. Accurate assessment is crucial because of the visual and life-threatening implications. A ‘papilloedema pathway’ was implemented to improve the timely investigation of children and adults presenting to the Eye Casualty. We report our audit findings with a view to sharing our experience and recommendations.

Method: Electronic patient records were used to collect data retrospectively on adults and children presenting with suspected papilloedema between March – June 2023 referred for ocular examination, neurological assessment, neuro-imaging within 48 hours, consideration of lumbar puncture (LP) and outpatient follow-up within 2–3 weeks as per ‘papilloedema pathway.’ Data included patient demographics, presenting symptoms, the results of examination, magnetic resonance neuroimaging (MRI), acute management and follow-up. We also evaluated assessment and diagnostic accuracies among these patients and the safety of

using the 'papilloedema pathway' by identifying if there is any patient safety incident reported.

Results: A total of 58 patients (9 males, 49 females) were managed using the 'papilloedema pathway'. The adult age range was 17–73, with a mean age of 33.4 years. There were 10 children (4 Male, 6 Female; age range 6–17; mean age of 11.2 years. The most common presenting complaints were headache (79.3%), nausea/vomiting (25.9%), visual symptoms including diplopia (46.6%) and tinnitus (20.7%). 48 patients (82.8%) recorded Snellen visual acuity $\geq 6/12$. The presence or absence of spontaneous venous pulsation was not documented in 78% of cases. Frisen grading was documented in 72.4% of patients to grade the severity of disc swelling, of which 10 cases (8 adults, 2 children, 17.2%) were deemed to be severe. 51 cases (87.9%) were referred to Neurology as per pathway guidelines. 38 patients (65.5%) underwent MRI, of which 23 cases (60.5%) were within 48 hours. 13 cases (22.4%) were admitted for further investigation. 28 patients had LP (25 adults, 3 children), of which 11 patients had normal LP opening pressure (8 adults, 3 children). 47 patients (81%) followed the 'Papilloedema Pathway', receiving the necessary investigation, diagnosis and management. The working diagnosis of papilloedema was mostly made by non-consultant grades (73.2%), of which only 58.6% were accurate. The final diagnosis were idiopathic intracranial hypertension (IIH) in 29 patients (50%), raised intracranial pressure due to medication (1, 1.7%), subdural haematoma (1, 1.7%) and space-occupying lesions (2, 3.4%), pseudo-papilloedema (21, 36.2%), optic neuritis (2, 3.4%), and ischaemic optic neuropathy (2, 3.4%). There was no patient safety incident reported from this audit. However, only 34 patients (58.6%) have outpatient clinic follow-up within 2–3 weeks as per pathway guidelines.

Conclusion: 'Papilloedema pathway' was safely utilised in the eye casualty setting in managing patients with suspected papilloedema. Continuous training of eye casualty staff is essential for accurate assessment and standard documentation of findings of optic disc swelling for appropriate diagnosis and treatment. Recommendations from this audit included trust-wide access to "Papilloedema pathway" guidelines via Sharepoint® and to identify reasons of delay in performing MRI scans and outpatient clinic follow-up in certain patients.

Conflicts of interest: No competing interest in all authors

Funding: None

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provided with a steroid alert card. This audit aimed to assess compliance to the NICE guidelines on corticosteroid prescribing, to identify areas of knowledge regarding systemic corticosteroid prescribing that clinicians perceive to be deficient, as well as to assess perceived access to relevant resources.

Method: A qualitative cross-sectional audit in the form of an anonymous Likert-scale questionnaire was undertaken in the Ophthalmology Department at the Royal Eye Infirmary in Plymouth. The questionnaire was distributed to all twenty-seven prescribing clinicians in the department. Sixteen responses were received, yielding a response rate of 59%. Data was collected and analysed using simple descriptive statistics.

Results: None of the clinicians that responded to the survey were providing patients with a steroid card when prescribing long-term (>3 weeks) systemic corticosteroids. Only 18.75% of respondents felt confident explaining the need for a steroid card to patients or knew where to access steroid cards, guidelines and patient information leaflets. Less than half of the clinicians surveyed (43.75%) knew the signs and symptoms of adrenal insufficiency, counselled patients on their recognition or knew how to identify patients at risk of HPA axis suppression. Only 18.75% of respondents reported counselling patients on 'sick day rules' for managing corticosteroid use during acute illness.

Conclusion: This audit identified that compliance to NICE guidelines on corticosteroid prescribing in our department is poor. None of the clinicians surveyed provided patients with steroid alert cards when prescribing long-term oral corticosteroids. Areas of perceived deficiency in knowledge were identified regarding potentially life-threatening adverse effects such as adrenal insufficiency, with fewer than half of clinicians aware of the signs and symptoms and able to counsel patients on appropriate 'sick day rules'. This represents a potential patient safety risk, leaving patients vulnerable to adrenal insufficiency during periods of acute illness. In addition, this audit identified a lack of access to steroid alert cards, guidance and patient information leaflets. We recommend a co-ordinated teaching program be implemented locally to improve knowledge amongst clinicians, in addition to making steroid cards, prescribing guidelines and patient information leaflets easily accessible.

Conflicts of Interest: None

Funding: None

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P010

COMPLIANCE TO NICE GUIDELINES ON CORTICOSTEROID PRESCRIBING IN AN OPHTHALMOLOGY DEPARTMENT

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Introduction: Systemic corticosteroids may be prescribed in a range of Ophthalmic conditions, such as giant cell arteritis, uveitis and inflammatory orbital disease. While highly efficacious, systemic corticosteroids carry significant risks associated with corticosteroid toxicity and hypothalamic-pituitary-adrenal (HPA) axis suppression. NICE Guidelines recommend that all patients on corticosteroids should be counselled on adverse effects and symptoms of adrenal insufficiency, and that patients on long-term steroids (>3 weeks duration) or high-dose steroids should be

P011

EVALUATING PATIENT COMMUNICATION IN THE RETINAL UNIT: A SURVEY-BASED STUDY ON INTRAVITREAL INJECTION THERAPY

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Introduction: Intravitreal injection (IVI) therapy used in the management of exudative retinal diseases has expanded significantly following the introduction of anti-vascular endothelial growth factor drugs. With an ageing population in the UK, the numbers of patients requiring IVI therapy are set to grow, with almost 750,000 procedures¹ performed in England in 2023. IVI procedures are a high-volume NHS activity² and a key indicator in the Department of Health & Social Care 2024 Vision Profile. The aim of this study was to evaluate communication with patients pre- and post-IVI therapy and to identify strategies to maximise

treatment adherence, increase patient knowledge and enhance patient wellbeing.

Method: A standardised 16-point questionnaire was developed as a survey tool with responses collected over a one-week period in the Retinal Unit (RU). Themes explored in the questionnaire included information satisfaction, symptom expectations, avenues of obtaining ophthalmic advice and patient confidence whilst undergoing IVI therapy. Demographic data including age, sex and ethnicity were collated.

Results: 198 patients participated in this study. The entire cohort were English speakers; 74% being native and the remainder speaking English as a second language. The most prevalent ocular condition for which IVIs were administered was neovascular age-related macular degeneration (53%) followed by retinal vein occlusion (16%). 48% of patients were 'very satisfied' with the information provided prior to IVI. 69% of patients received written information prior to the procedure. 83% of patients who received an information leaflet were aware of symptoms post-IVI requiring urgent ophthalmic review in comparison to 69% of patients who did not receive an information leaflet. 79% of patients received care instructions post-IVI. 83% of patients were confident contacting the RU if required, with the preferred mode of communication being telephone for the two-thirds of respondents. 12 patients developed an IVI related complication requiring emergency hospital attendance.

Conclusion: Providing patients with written communication on IVI therapy improves patient satisfaction and increases patient knowledge; empowering patients to make informed decisions on IVI treatment and encouraging patients to seek urgent ophthalmic advice when required.

Conflicts of interest: None

Funding: None

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P012

IMPLEMENTATION OF AN OPHTHALMOLOGY DATA MART FOR INTRAVITREAL INJECTIONS FROM THE EPIC ELECTRONIC HEALTH RECORD SYSTEM

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Introduction: The primary objective of this project was to develop an ophthalmology data mart specifically focused on patients undergoing intravitreal injections using the Epic electronic health record (EHR) system (Epic Systems Corporation, Verona, Wisconsin, USA) as the data source, which was implemented at the University Hospital Bern in March 2024. This data mart aims to improve clinical outcomes and support comprehensive research.

Method: Based on the Swiss Ophthalmological Society National Minimum Data Set (SOG-NMD), the data mart was designed as an SQL database to contain data on patient demographics, clinical history, treatment regimens, visual acuity scores, injection details (such as frequency and type of injected agents), adverse events and follow-up visits. The data mart extracts data directly from the data warehouse (DWH) in which it resides and provides its own extract-transform-load (ETL) process to meet specific analytical purposes. Archived data from the pre-Epic era was also fed into the data mart to minimise data loss during the transition to Epic. Clinician feedback was provided in iterative steps to confirm accurate data handling.

Results: The ophthalmology data mart was successfully implemented, providing a robust platform for advanced data visualisation and analysis. Data visualisation was implemented via the open source Apache Superset application (Apache Foundation, Wakefield, Massachusetts, USA) using interactive dashboards. The archived data required extensive data cleaning and standardisation, particularly in the handling of free-text diagnoses and adverse events. At the current stage of deployment, the data mart supports daily updates and queries. Initial analyses have demonstrated the utility of the data mart in supporting personalised treatment plans and providing a basis for benchmarking against the SOG-NMD.

Conclusion: The data mart facilitates real-time monitoring of treatment efficacy and adverse effects, contributing significantly to improved patient care and resource allocation. In addition, its integration with hospital-wide health databases via Epic opened up new opportunities for interdisciplinary research.

Conflicts of interest: None

Funding: This project was in part funded by an unrestricted grant from Bayer Switzerland

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P013

IMPROVING PATIENT ACCESS TO CVIS: A DATA-DRIVEN AND DIGITAL IMPROVEMENT PROJECT

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Introduction: The Certificate of Vision Impairment (CVI) can provide access to important financial and practical support services. Nationally, gaps in identifying eligible patients have been recognised as a significant issue. This project aimed to retrospectively identify potentially eligible patients who were not offered a CVI and to implement proactive measures to prevent future omissions.

Method: A multidisciplinary project team, including a Service Improvement Manager, three Eye Clinic Liaison Officers (ECLOs), a data analyst and a senior clinician met weekly to oversee the project. The team analysed data exported from the electronic patient record to identify patients with a visual acuity of 6/60 or less in the better seeing eye at last visit without documented evidence of CVI having been undertaken or offered. ECLOs scrutinised patient lists and updated CVI statuses, in some cases through communication with local authorities. The team reviewed existing processes, established accurate CVI status documentation in the electronic patient record, flagged potentially eligible patients, and generated automated weekly reports to highlight patients due for clinic review. These reports allowed the ECLOs and the ophthalmology medical team to take timely actions. Additionally, the team implemented wider education initiatives.

Results: An initial audit identified 1576 potentially eligible patients. Approximately 300 patients were found to be CVI registered elsewhere, allowing their records to be accurately updated. Of the remainder, to date, the ECLO team has reviewed 1,065 patients (83%). Of those reviewed, approximately 500 (45%) were confirmed as eligible, and 100% of these now have a CVI either initiated or issued.

We are already receiving positive feedback from patients about the improvement to the service and are continuing to monitor.

This was supported by a wider educational programme, including an all-staff event, targeted training sessions, "Lunch and Learn" initiatives, safety huddles and walkabouts, email reminders and information cascaded via service directors to raise awareness amongst clinical and non-clinical staff about the importance of CVIs and the ongoing project.

Further improvements to the electronic patient record system have also been identified to better support clinicians in identifying and managing eligible patients. These proposed improvements include adding clear flags linked to the CVI form, facilitating easy identification of eligible patients, and tracking their progress.

Conclusion: A cross-disciplinary, data-driven, and multi-pronged approach has significantly improved patient access to CVIs. These improvements, combined with planned future developments, are anticipated to positively influence the patient experience by streamlining the certification process, making it more comprehensive and accessible, and supporting patients in their daily lives. One limitation of the automated data extraction was inability to identify patients who meet eligibility criteria beyond visual acuity, namely visual field loss. However, wider education initiatives were implemented so that staff were more aware of eligibility criteria.

Improvements made to the process are being transitioned into business as usual for ongoing monitoring of failsafe reports by the ECLO team with the oversight of divisional Quality Governance Forums.

Conflicts of interest: None

Funding: None

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P014

OPHTHALMIC SPECIALTY TRAINING TEACHING VARIABILITY - COMPARING QUALITY AND QUANTITY OF TEACHING ACROSS THE DIFFERENT DEANERIES IN THE UK

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Introduction: There have been many discussions between trainees across the UK regarding the quality and quantity of teaching they receive in their deanery during the OST programme. We aimed to formally assess how teaching is delivered, comparing all the deaneries in the UK.

Method: A survey targeting ophthalmology trainees across all 22 UK deaneries was conducted via a secure online platform. A total of 200 responses from trainees of all grades were received, gathering both quantitative and qualitative data on the consistency, frequency, and quality of teaching. Responses were analysed to identify trends and main areas of disparities.

Results: The highest responses were from trainees in North London (12%), South London (11%) and Kent, Surrey, Sussex (8%). 11% of trainees did not get any formal weekly teaching and 32% of trainees had virtual teaching for majority of their sessions. 41% of respondents showed that trainees organised the weekly teaching rota; 38% reported that consultants organised it; the remainder were other doctors, nurses, allied health professionals, the medical education department, etc. 15% of trainees reported being pulled from teaching on a monthly basis to cover clinics / on calls / theatres, although this appeared to be very deanery dependent, where some deaneries had very strictly protected teaching time. 30% of trainees had teaching opportunities

combined with a different specialty e.g. neurology, endocrinology, oncology, radiology, etc. There were also significant disparities found in regional teaching across the different deaneries. Consultant-led teaching was frequently cited as the most valued, as were networking opportunities, competitions, being taught in a 'safe' environment, listening to consultant-led discussions and attitude towards teaching including openness of the deanery/trainee programme directors on changing the format if needed. Key areas of concern included insufficient coverage of subspecialties, lack of practical simulation-based sessions, difficulties around study leave/travel/expenses and inconsistencies in the delivery format, with a significant proportion of trainees reporting dissatisfaction with the overreliance of virtual teaching.

Conclusion: The findings highlight significant regional disparities in OST delivery, both positively and negatively as reported by current ophthalmology trainees. We found variability in the frequency of sessions, teaching resources, content, delivery formats and opportunities given in these settings. It would be beneficial for a more standardized teaching setup across the country, to help find a balance between clinical service demands with educational needs, and to maintain high training standards and support the future workforce in Ophthalmology.

Conflicts of interest: None

Funding: None

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P015

PATIENT SATISFACTION WITH AN ASYNCHRONOUS VIRTUAL KERATOCONUS CLINIC: A SERVICE EVALUATION

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Introduction: Keratoconus is a progressive corneal condition requiring ongoing monitoring to detect disease progression and guide management. Traditional in-person consultations can pose challenges for patients, including waiting and scheduling constraints. An asynchronous Virtual Keratoconus Clinic (VKC) offers an innovative solution, allowing clinicians to review patient data and provide feedback without requiring live interactions. This model leverages the wider healthcare team to enhance accessibility and convenience while maintaining high standards of care. This evaluation examines patient satisfaction with the asynchronous VKC at Aberdeen Royal Infirmary and identifies areas for improving service quality. Evaluating patient satisfaction with the asynchronous Virtual Keratoconus Clinic is essential for optimising this innovative model, which seeks to improve accessibility and efficiency in keratoconus care while overcoming challenges of traditional in-person consultations.

Method: Patients who participated in the VKC between 2020 and 2024 were invited to complete a patient satisfaction survey. The questionnaire, adapted from validated tools, assessed key dimensions of patient experience, including clinic efficiency, patient expectations, communication of clinical findings, convenience of the asynchronous format, and overall satisfaction. Quantitative data were analysed descriptively to identify satisfaction levels, while qualitative feedback was thematically analysed to uncover common themes and improvement opportunities.

Results: A total of 54 responses were received, yielding a response rate of 30.1%. Overall satisfaction with the asynchronous Virtual Keratoconus Clinic (VKC) was reported by 83.6% of

respondents. The model's key strengths included clinic efficiency (83.6%) and staff knowledge (70.9%). However, some challenges were identified: 14.5% of patients expressed dissatisfaction with receiving results at a different date by post, while 20% felt less reassured by the virtual visit. Additionally, 47.3% of patients expected to receive results during the clinic visit, and 16.7% expressed dissatisfaction with the timeliness and clarity of the communicated results. Key comments mentioned preference to see a clinician, desiring more detailing results, and difficulty attending the clinic from rural areas. These insights underscore both the strengths and areas for improvement in the asynchronous model.

Conclusion: The asynchronous Virtual Keratoconus Clinic (VKC) has demonstrated high patient satisfaction, particularly regarding clinic efficiency and staff knowledge. However, areas for improvement include the method of delivering results, as some patients expressed dissatisfaction with receiving results via post. Additionally, a significant portion of patients reported feeling less reassured by the virtual visit, with some expressing a preference for in-person consultations and desiring more detailed feedback. Patients from rural areas also indicated difficulties in attending the clinic. Addressing these challenges—particularly improving communication around results and increasing engagement with patients—will be key to optimising the service. Despite these areas for improvement, the asynchronous VKC model holds significant potential as an accessible and scalable solution for managing keratoconus, especially for patients in underserved areas.

Conflicts of interest: None

Funding: None.

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P016

PATIENT SATISFACTION WITH NURSE-LED CORNEAL COLLAGEN CROSS-LINKING: A TERTIARY CENTRE EXPERIENCE

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Introduction: Keratoconus is a progressive, bilateral disorder of corneal ectasia that causes thinning and cone-like bulging of the cornea, often leading to significant visual impairment if untreated. Corneal Collagen Cross-Linking (CXL) is the gold-standard treatment to halt disease progression and preserve vision. As the demand for CXL grows, the need for efficient models of care has become paramount. Nurse-led CXL represents an innovative approach to addressing this demand by utilising trained ophthalmic nurses to perform the procedure. This model not only alleviates resource constraints but also provides opportunities for nurses to expand their clinical roles. This study evaluates patient satisfaction with nurse-led CXL services at Aberdeen Royal Infirmary, identifies key areas for improvement, and ensures alignment with clinical governance standards, including safety, communication, and quality of care. Understanding patient satisfaction with nurse-led CXL services is crucial for evaluating the feasibility and effectiveness of this innovative care model, which has the potential to address increasing demand for keratoconus treatment while maintaining high standards of care.

Method: A service evaluation was conducted among patients who underwent nurse-led CXL between 2020 and 2023. Participants completed a satisfaction questionnaire adapted from previously validated tools, assessing multiple aspects of their experience. These included pre-procedure communication, procedural comfort and confidence, and post-procedure care.

Quantitative data were analysed using descriptive statistics to identify patterns and overall satisfaction levels, whilst qualitative feedback was thematically analysed to capture recurring issues and potential service improvements. National Institute for Health and Care Excellence (NICE) guidelines, emphasising effective communication, patient safety, and treatment outcomes, were adhered to.

Results: A total of 35 responses were received (36.1% response rate). Overall satisfaction with the nurse-led CXL service was reported by 97.2% of patients. Respondents expressed particular satisfaction with the conduction of the procedure (100%) and the management of anxiety throughout treatment (97.1%). These factors significantly contributed to patient confidence and comfort. However, 8.6% of respondents reported dissatisfaction with post-procedure pain management or follow-up care, indicating a need for enhanced recovery strategies. Additionally, 8.6% of patients highlighted dissatisfaction with waiting times. These findings suggest that while the service excels in critical areas, there are opportunities to refine patient care and improve the overall experience.

Conclusion: Nurse-led CXL has proven to be an effective, patient-centred model for managing keratoconus, demonstrating high levels of satisfaction and confidence among patients. Strengths of the service include clear communication, effective anxiety management, and seamless procedural execution. However, areas for improvement, particularly regarding post-procedure pain management and waiting times, are crucial for optimising service delivery. Addressing these challenges will ensure the service continues to meet clinical governance standards, improving both patient comfort and long-term treatment outcomes. These findings support the scalability of nurse-led CXL as a sustainable solution to the increasing demand for keratoconus care. Expanding this model to other centres can enhance accessibility for underserved populations, alleviate the burden on ophthalmologists, and improve healthcare efficiency.

Conflicts of interest: None

Funding: None

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P017 Abstract not included

P018

SURVEILLANCE OF DETERIORATION IN SIGHT DUE TO DELAY IN OPHTHALMIC REVIEW OR TREATMENT IN THE UK: CONTEMPORARY FINDINGS AND COMPARISON WITH THE 2017 BOSU STUDY

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Introduction: The NHS aspires to provide high-quality timely care that is safe and effective, free at the point of service and determined by clinical need. In 2017 a prospective BOSU surveillance study reported between 14 and 22 patients per month within the hospital eye service were losing significant vision because of delayed appointments. By replicating the methods of the initial study, this study establishes contemporary figures for the frequency of adult patients suffering harm due to delay in treatment or review in the UK and describes the similarities and changes.

Method: Adult patients with deterioration in vision in at least one eye of 3 lines of Snellen acuity or 15 letters on ETDRS chart or deterioration in visual field deviation of 3 decibels due to health

service initiated delay in review or care were ascertained between November 2023 and October 2024 through the British Ophthalmological Surveillance Unit using prospective active surveillance involving all UK ophthalmologists. Demographic details, diagnosis, cause and length of delay, and vision loss were then sought by questionnaire. To ensure comparability this study used the same case definition and questionnaires as the initial 2017 study.

Results: 309 cases were reported. 52 (17%) reports were out of the study period or did not meet the case definition. 84/257 questionnaires were returned (33%). Median age was 76 years (range: 51 to 95 years). Median delay was 31 weeks (range: 3 weeks to 4.9 years). 73% of cases experienced permanent reduction in visual acuity, and 28% experienced permanent deterioration in visual field. 11% experienced permanent reduction in visual field and visual acuity. 9% required an additional unplanned procedure. Main diagnoses were Glaucoma 41%, Age-related Macular Degeneration (AMD) 34% and retinal vein occlusions (RVO) 12%. 23% of patients were eligible for Severely Sight Impaired (SSI) or Sight Impaired (SI) registration. Main causes were delayed follow-up (85%), lost referral (7%) and delayed treatment (8%). Response to the BOSU reporting system for the study period was 47%.

In comparison to the 2017 surveillance study, median age remains at 76 years, and median delay has increased from 22 weeks to 31 weeks. The proportion of patients experiencing permanent visual loss and the high proportion of patients with chronic conditions are comparable to those previously reported. Whilst the demographic and diagnostic profile of the patient population remains unchanged, there is a strong likelihood that the frequency of harm has increased since the 2017 study when accounting for comparative participation and ascertainment rates.

Conclusion: These findings suggest patients continue to suffer preventable harm due to health service initiated delays, leading to permanently reduced vision. This occurs most consistently in patients with chronic conditions. Delayed follow up or review is the cause in the majority of cases, indicating that a lack of capacity persists within the hospital eye service and that changes and events since 2017 have not had an impact on reducing the frequency of harm.

Conflicts of interest: None

Funding: None

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P019

THYROID EYE DISEASE: CAN WE IMPROVE IDENTIFICATION OF DISEASE ACTIVITY AND DRIVERS?

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Introduction: Thyroid Eye Disease (TED) negatively impacts quality of life and can potentially have sight threatening complications. To determine appropriate management, clinical assessment should determine and grade the presence of active orbital inflammation. It also remains vital to identify and control drivers of disease, such as smoking and endocrine dysfunction. This two-cycle audit aimed to 1) identify if initial ophthalmology consultations at a tertiary centre documented assessment of disease activity and drives and 2) assess the impact of an informative poster intervention on improving this process.

Method: Retrospective chart review. Electronic medical notes were searched. For the first cycle of audit, patients who received a new diagnosis of TED from an ophthalmologist in a single tertiary centre between January 2020 and January 2024, were identified

and their clinic notes examined. Data retrieved included patient demographics, Clinical Activity Score (CAS), presence or absence of inflammatory features, smoking status, smoking pack-years, thyroid function, presence or absence of active sinusitis. Following this, an A4 poster intervention prompting ophthalmologists to assess and document CAS and disease drivers was displayed in each eye clinic room.

Subsequently, a re-audit was conducted on patients who received a new TED diagnosis at the same centre between September 2024 and December 2024.

Results: In the first audit cycle, 58 patients with a new diagnosis of TED were identified. CAS was documented in 19% of these patients and clinical findings of orbital inflammation in 52%. Smoking status and endocrine dysfunction were documented in 50% and 43% of patients, respectively. In the re-audit cycle, documentation rates improved, with clinical findings of orbital inflammation, smoking status and endocrine dysfunction recorded in 100% of patients.

Conclusion: In patients with newly diagnosed TED, there is a notable shortcoming in clear documentation of disease activity and identification of disease drivers, such as smoking. However, the introduction of informative poster prompts led to improvement during initial re-audit. This suggests that clear, visually accessible local guidelines help improve assessment of TED referrals. This will facilitate greater control of drives and allow more objective assessment of treatment response.

Conflicts of interest: None

Funding: None

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P020

UTILISING AI FOR MULTILINGUAL PATIENT EDUCATION: ENHANCING EYE DROP ADMINISTRATION AND BRIDGING HEALTHCARE INEQUALITIES

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Introduction: Effective patient education is essential for improving health outcomes, particularly for treatments requiring precise administration, such as eye drops. Misunderstandings in eye drop usage can compromise therapeutic efficacy, increase side effects, and frustrate patients. Language barriers and low health literacy exacerbate these challenges, particularly in diverse communities served by a London district general hospital with a large non-English speaking population.

This project leverages Artificial Intelligence (AI) to develop multilingual educational videos on eye drop administration, offering a scalable, accurate, and culturally sensitive solution. These efforts support equity, autonomy, and patient-centred care by making critical health information accessible to linguistically diverse populations.

Method: An AI-powered educational video was developed to address common challenges in eye drop administration. The video was translated into Arabic, Urdu, and Mandarin using AI-driven translation and voice-over tools, with subtitles added for enhanced comprehension. The content was made accessible via QR codes displayed in the eye department and online platforms.

Patient surveys were conducted at three stages:

1. Pre-video: Assessing baseline understanding and common challenges.
2. Immediate post-video: Evaluating clarity, confidence, and perceived usefulness.
3. One-month follow-up: Measuring retention, adherence, and satisfaction.

Refinements were made iteratively based on patient feedback. The project also mapped the patient journey to identify touchpoints for video integration, collaborated with IT for streamlined access, and ensured cultural sensitivity in translations.

Results: The intervention demonstrated significant improvements:

- Confidence Boost: Post-video surveys revealed a 28% increase in confidence for correct eye drop administration.
- Sustained Impact: One-month follow-ups showed improved adherence, reduced anxiety, and enhanced technique in 85% of patients.
- Accessibility: 92% of non-English speaking patients found the multilingual format “very helpful.”
- Patients valued the combination of voice-over and subtitles, which reinforced understanding and accommodated diverse literacy levels.
- Cultural Relevance: Feedback highlighted the effectiveness of culturally tailored content facilitated by AI.

Conclusion: This AI-driven approach improved patient education and demonstrated the potential for addressing health inequalities through scalable, cost-effective solutions. The project highlights the importance of combining linguistic accessibility with cultural sensitivity to bridge gaps in patient communication and education.

Moreover, the initiative raised concerns about inclusivity in AI. The predominance of English in AI development risks widening health disparities. Future efforts should focus on robust AI solutions in underrepresented languages to ensure equitable healthcare advancements.

This initiative highlights AI's transformative potential in healthcare education. By delivering accessible, accurate, and culturally sensitive materials, AI can enhance adherence and promote health equity. The model is adaptable and can be extended to other clinical areas, such as nasal spray education or pre-surgical consent using augmented and virtual reality tools.

Conflicts of interest: None

Funding: None

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CATARACT AND REFRACTIVE SURGERY

P021

IMPACT OF COMMISSIONED PRIMARY CARE ENHANCED CATARACT REFERRAL PATHWAYS ON CONVERSION RATE TO CATARACT SURGERY

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Introduction: Cataract extraction is the most common type of surgery performed under the NHS, with demand rising. GIRFT (Getting It Right First Time) reported the national mean Cataract Conversion Rate (CCR) as 71%, and made recommendations on managing increased demand include increasing CCR to 80–85%, supported by enhanced referral pathways utilising primary care

optometry. NICE (National Institute for Health and Care Excellence) guidelines for cataract surgery in adults include criteria for referral, namely that patients should be given appropriate information to make an informed decision about surgery, which should be based on a discussion with the patient.

While previous studies have shown improved CCR with enhanced referral services, these have focussed on single localities and a larger scale investigation across multiple localities is required.

We review the outcomes of cataract referrals through five commissioned integrated cataract pathways in England.

Method: An integrated cataract pathway comprises of a commissioned Enhanced Cataract Referral Service and Post-Cataract Service.

The Enhanced Cataract Referral Service aims to reduce unnecessary referrals by utilising primary care optometrists to assess the visual and lifestyle impact of cataract, relevant health information, and to carry out risk and benefit discussion ensuring those referred are suitable and willing to proceed. The Post-Cataract Service allows patients to have their postoperative assessment in primary care.

Enhanced Cataract Referrals made between March and September 2022 in five ICB areas across England were included; three in the North West, one in the South West and one in London.

Outcomes of referrals were determined by a) identifying patients transferred back for postoperative assessment in Primary Care following their surgery and b) for patients not transferred back, asking the surgical providers for the outcomes of the referrals.

Results: 6047 enhanced cataract referrals were included. 4582 (75.8%) received surgery and were transferred to an optometry practice for post-operative assessment, completing the integrated cataract pathway. Of these, 2643 received primary care post-operative assessment following first eye surgery, and 1939 continued to second eye surgery and second primary care postoperative assessment, meaning that 6521 primary care post-operative assessments were carried out in total.

The outcomes of the remaining 1465 referrals were requested from the surgical providers. Of these, 1111 attended their first hospital appointment and 636 listed for surgery. The total number of patients listed for surgery was therefore 5218 (4582+636). In total, 5693 (4582 + 1111) of referred patients attended their first hospital appointment meaning that the CCR (as defined by GIRFT) was 91.7% (5218/5693).

There was no significant difference in CCR across the five ICB areas, χ^2 (5,5693) = 0.19, p = 0.999, CCR 90.2%, 90.4%, 92.0%, 92.2% and 92.7%.

Based on a national mean CCR of 71%, an additional 1656 1st hospital appointments would be required to achieve the same number of cataract listings. Primary care post-operative assessments saved another 6521 appointments meaning the total number of secondary care appointments saved in these five areas over seven months is estimated at 8177.

Conclusion: Enhanced cataract pathways can support surgical providers in maximising their capacity and contribute to managing increasing demand for surgery. Patients benefit from avoiding unnecessary hospital visits, receiving care closer to home and being supported in making informed choice decisions about their cataract referral.

The results suggest that the GIRFT CCR target of 80–85% may be conservative, and 90% or more is achievable with well-established enhanced cataract pathways.

There is considerable scope to increase the impact of enhanced cataract referral pathways by consistent national commissioning and the deflection of all cataract referrals from other health

professionals and non-participating optical practices to an enhanced pathway.

Conflicts of interest: None

Funding: None

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P022. Abstract not included

P023

A REVIEW OF MINI-MONO VISION IN THREE DIFFERENT IOL CATEGORIES

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Introduction: Mini-mono vision is a vision correction technique that allows for a broader spectrum of spectacle independence while minimizing anisometropia. This narrative review aims to evaluate mini-monovision outcomes with three categories of intraocular lenses (IOLs): monofocal, enhanced monofocal, and extended depth of focus (EDOF).

Method: A comprehensive literature search was conducted to identify studies reporting mini-mono vision outcomes within the three categories of IOLs between April 2014 and August 2024. Inclusion criteria were studies with more than 20 patients, target refraction of emmetropia to -0.50D in one eye and up to -1.50D difference in the fellow eye, and minimum follow-up of 3 months. The primary outcome analysed is binocular uncorrected near visual acuity (BUNVA). Secondary outcomes include binocular uncorrected distance visual acuity (BUDVA), binocular uncorrected intermediate visual acuity (BUIVA), binocular corrected distance visual acuity (BCDVA), binocular corrected intermediate visual acuity (BCIVA), patient-reported outcome measures, complications, and rates of IOL exchange or excimer laser enhancement.

Results: 113 studies were screened, of which 19, with a total of 1530 patients, were eligible for inclusion in this review. Mean logMAR Binocular UCIVA was 0.16 ± 0.01 , 0.11 ± 0.06 , 0.08 ± 0.07 ($p = 0.41$), and mean logMAR BUDVA was 0.08 ± 0.05 , 0.04 ± 0.07 , 0.04 ± 0.04 ($p = 0.36$), in the monofocal, enhanced monofocal, and EDOF groups, respectively. The mean spectacle independence rate was $51\% \pm 22.1$, $55\% \pm 35.4$ and $63.4\% \pm 24.6$ ($p = 0.05$), respectively, in the monofocal, enhanced monofocal and EDOF groups. A comparable low incidence of halos and glare was observed when enhanced monofocal lenses were evaluated against traditional monofocal lenses. EDOF lenses have, however, demonstrated mixed results. The complications, IOL exchange, and excimer laser enhancement rates were generally low across all groups.

Conclusion: While enhanced monofocal and EDOF IOLs may provide slightly better intermediate visual outcomes and higher spectacle independence compared to monofocal lenses, with regards to mini-monovision and intermediate vision performance, the differences are not statistically significant. All three IOL types exhibit high patient satisfaction rates when choosing a mini-mono vision approach with decreased dependence on spectacles.

Conflicts of interest: None

Funding: None

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P024

A THREE-COUNTY AUDIT OF EXOGENOUS ENDOPTHALMITIS RATES IN THE PENINSULA REGION

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Introduction: Exogenous endophthalmitis is a rare but severe infection of the eye, most commonly occurring after cataract surgery and intravitreal injections. Postoperative endophthalmitis is associated with poor visual prognosis despite treatment. Current UK rates of endophthalmitis are 0.02% following cataract surgery and 0.02–0.06% following intravitreal injections.

Recently, the Peninsula region has seen significant growth in cataract surgery providers. While intraoperative complication rates are available from the National Ophthalmic Database, rates of serious post-operative complications are not published.

Additionally, while the population is more mobile than ever, the geography of the peninsular region, being bound on either side by the sea, is likely to capture some of these patients who are followed up after their procedure in another unit.

Method: Retrospective data was collected from 17 units, including seven NHS trusts, three ISTC, and seven private clinics in the Peninsula region for the calendar year of 2023. Data requested included the number of cataract surgeries, intravitreal injections, and cases of endophthalmitis. Vitreous biopsy details were also requested from regional microbiology laboratory databases to ensure comprehensive case capture.

Results: Data from all seven NHS trusts, two ISTC, and two private clinics has been obtained, with additional data pending. In 2022, 17,932 cataract surgeries and 40,496 intravitreal injections were performed in the region. There were four cases of endophthalmitis post-cataract surgery and eight cases following intravitreal injections. Additionally, three cases of endophthalmitis treated by NHS trusts were from procedures done in other units. The overall regional rate of endophthalmitis is 0.022% for cataract surgery and 0.020% for intravitreal injections. Including the three extra cases post-cataract surgery, the rate increases to 0.039%.

Conclusion: The rate of endophthalmitis in the Peninsula region aligns with the national average, but rises when including cases from other units.

Conflicts of interest: none

Funding: None

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P025

AN ARTIFICIAL-INTELLIGENCE-POWERED INTERVENTION TO AID THE INFORMED CONSENT PROCESS FOR CATARACT SURGERY

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Introduction: Informed consent is a critical component of patient care, particularly in elective procedures such as cataract surgery. Ensuring patients understand the risks, benefits, and alternatives of their treatment is fundamental to ethical and effective healthcare. However, traditional written consent forms often fail

to fully engage patients or ensure comprehensive understanding, especially among populations with language barriers or limited health literacy. This study investigates the effectiveness of an AI-generated video-based consent process, compared to conventional written consent, in enhancing patient comprehension, satisfaction, and reducing anxiety. We utilise Synthesia to create these videos, a generative AI software that generates photo-realistic avatars to transmit information. These videos can be translated into multiple languages paired with a range of ethnically diverse avatars, ensuring seamless alignment between the visual and auditory components. This adaptability and scalability make Synthesia an innovative, GDPR-compliant solution that is easily updated to reflect new clinical guidelines or patient feedback. Recognizing the significant health inequalities faced by minority populations, the study focuses on using this tool to address disparities in informed consent. A key focus is on the Bengali-speaking population, one of the largest linguistic minorities in East London, comparing outcomes with the English-speaking population to evaluate its efficacy in improving patient engagement and comprehension.

Method: This ongoing prospective study involves patients undergoing first-time cataract surgery. Participants are randomized into two groups: one receiving both AI-generated video consent and traditional written consent, and the other receiving only traditional written consent. The study includes English-speaking participants (Cohort A) and Bengali-speaking participants (Cohort B). The primary outcome is information retention, assessed via a quiz one week after consent. Secondary outcomes include baseline and day-of-surgery anxiety levels, assessed using the State Trait Anxiety Inventory, as well as patient satisfaction with the consent process, engagement with consent materials, and surgery cancellation rates.

Results: As of November 2024, 52 participants have been recruited towards a target of 200. Average engagement with the video was higher (17.33 minutes) compared to the written consent group (9.25 minutes). Quiz scores indicated better performance in the video group (mean score 4.0) compared to the control group (mean score 3.13). Additionally, English-speaking participants performed better on the quiz (mean score 3.67) compared to Bengali-speaking participants (mean score 2.75).

Anxiety levels showed a slight increase on the day of surgery compared to baseline across all groups. However, the increase was smaller in the video group (+0.29) compared to the control group (+0.38). Satisfaction scores were also higher in the video group (mean score 29.29) compared to the control group (mean score 28.43). So far, 15 surgeries have been successfully completed, with three cancellations on the day of surgery, two of which were from the Bengali-speaking cohort.

Conclusion: These preliminary results suggest that AI-enhanced videos can improve patients' comprehension of cataract surgery and the associated implications. AI-enhanced video consent demonstrates promise in improving information retention, engagement, and satisfaction in cataract surgery patients, especially among English speakers. However, barriers to digital adoption and recruitment disparities in Bengali-speaking cohorts highlight the necessity for culturally sensitive adaptations. Future steps include use of semi-structured interviews to explore usability and patient perceptions of AI in healthcare, as well as performing a full statistical analysis after the completion of participant recruitment. This approach will also examine cultural barriers that may influence the adoption and effectiveness of digital consent tools. We aim to introduce a healthcare intervention that also addresses the needs of underserved populations, ensuring that it reduces, rather than exacerbates, existing health disparities.

Conflicts of interest: No potential conflicts.

Funding: This study is funded by the Medical Protection Society (MPS) Foundation.

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P026

ARE OUR NHS CATARACT SERVICES GETTING TRULY COMPLEX?

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Introduction: NHS cataract services are increasingly challenged by rising demand, resource constraints, the COVID-19 pandemic, and the expansion of independent sector providers (ISPs). This study aims to assess whether there has been a shift towards managing more non-routine cataract cases within our service and to evaluate discrepancies between preoperative predictions of complexity and intraoperative requirements.

Method: A retrospective evaluation of cataract surgery waiting list forms from February to April 2023 and 2024 was conducted at Northern General Eye Centre. Cases were categorised as "complex", "special" (e.g., ISBCS), or "GALA" (a post-pandemic initiative converting surgeries from general anaesthesia to local anaesthesia). Complex cataract cases were defined as surgeries involving additional challenges necessitating specialised techniques or devices. These included:

- Poor mydriasis (requiring intracameral phenylephrine, pupil expansion techniques, division of posterior synechiae),
- Poor red reflex (requiring vision/trypan blue due to dense/white cataract),
- Presence of pseudoexfoliation syndrome (requiring capsular tension rings),
- History of trauma, phacodonesis, acute angle-closure glaucoma, previous pars plana vitrectomy, or uveitis requiring IV methylprednisolone.

Additional analysis compared the predicted use of surgical devices and adjuncts for complex cases (e.g., trypan blue, pupil expansion techniques) with their actual intraoperative use.

Results: Over the same three-month period, a total of 668 cataract surgeries were listed in 2023, compared to 955 in 2024. In 2023, 212 cases (31.7%) were classified as "complex," while in 2024, 358 cases (37.5%) were classified as "complex" ($p = 0.02$). Thirty-six cases (5.4%) and 70 cases (7.3%) were classified as "special" in 2023 and 2024, respectively ($p = 0.15$). Five cases (0.7%) and 9 cases (0.9%) were classified as "GALA" in 2023 and 2024, respectively ($p = 0.89$). ISBCS cases more than doubled from 23 to 49.

Discrepancies between preoperative predictions and intraoperative requirements were noted. In 2023, trypan blue was listed for 131 cases but used in 83 cases (63.4%), and pupil expansion techniques were listed for 60 cases but used in only 14 cases (23.3%). In 2024, trypan blue was listed for 179 cases but used in 158 cases (88.3%), and pupil expansion techniques were listed for 137 cases but used in only 69 cases (50.4%).

Conclusion: There is an obvious increasing trend in the number of "complex" cases encountered within our service. The increasing trend of complex cataract cases offers valuable experience for senior trainees but limits junior trainees' exposure to routine cases. To optimise training, partnerships with ISPs are recommended. This trend also hinders the running of high-volume, low-complexity lists but may support the development of high-complexity, low-volume lists. The increasing number of patients opting for ISBCS reflects a growing trend.

The degree of overestimation of cataract case complexities during preoperative assessments impacts operational efficiency and resource management, potentially leading to underutilisation of surgical lists and increased resource costs. To address these issues, refining preoperative assessment criteria and providing targeted training for surgeons could enhance list planning, and ensure better resource utilisation.

Conflicts of interest: None

Funding: None

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P027

CLINICAL IDENTIFICATION OF BLUE LIGHT FILTERING IOLS

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Introduction: Differentiation between blue filtering yellow intraocular lenses (IOLs) and non-blue filtering IOLs can be helpful in certain scenarios to ensure that the same IOL type is implanted in both eyes as it was shown in some studies that some patients did not tolerate the difference in colour vision between both eyes. Purkinje–Sanson images were described in the first half of the nineteenth century for evaluation of human crystalline lens; however, they can also be applied to evaluation of IOL implants.

Method: Patients with Eye Cee One IOL (as an example of non-Blue filtering IOLs) underwent slit lamp examination (diffuse illumination and slit illumination). Likewise, patients with Acrysof IQ SN60 WF IOL (as an example of blue filtering IOLs) underwent similar examination. The colours of the Purkinje–Sanson images were documented in both groups.

Results: Diffuse illumination of non-blue filtering IOLs, such as the Eye Cee One IOL, revealed that all Purkinje–Sanson images appeared white. In contrast, with blue filtering IOLs, such as the Acrysof IQ SN60 WF IOL, the posterior Purkinje–Sanson image appeared yellow. Similarly, slit illumination of non-blue filtering IOLs showed a white IOL reflection, while blue filtering IOLs exhibited a yellow reflection.

Conclusion: This clinical observation highlights a simple and practical clinical sign to distinguish blue light-filtering yellow IOLs from non-blue filtering IOLs. It can be particularly useful in scenarios where medical records are unavailable or incomplete.

Conflicts of interest: None

Funding: None

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P028

CLINICAL INSIGHTS INTO ARTISAN IRIS-CLAW IOLS IN APHAKIC PATIENTS AT AN IRISH TERTIARY REFERRAL CENTRE

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Introduction: Examining the outcomes and complication rates of Artisan iris-claw intraocular lens (IOL) implantation in an aphakic Irish cohort, with a comparison of anterior versus retropupillary enclavation, and an analysis of the biometric spread of implanted IOLs.

Method: In this single-centre retrospective case series, we analysed the medical records of 140 eyes from 134 patients who underwent Artisan IOL implantation at a tertiary referral hospital between April 2015 and July 2024. Analysis included patient demographics, surgical approach (anterior vs. retropupillary), lens strength, and both intraoperative and postoperative outcomes. Best-corrected visual acuity (BCVA) was evaluated preoperatively, at 3 months, and at 6–12 months postoperatively.

Results: The mean preoperative BCVA was 1.13 logMAR, which improved to 0.47 logMAR at final follow-up ($p < 0.001$), with 94% of eyes achieving stable or improved vision. The difference in complication rates between anterior (27%) and posterior (18%) enclavation was not statistically significant ($p = 0.59$). Corneal decompensation only occurred in cases of anterior iris enclavation. Cystoid macular oedema (CMO) occurred in 22 eyes, with all uncomplicated cases resolving with steroid treatment. Late disenclavation of the IOL was a rare complication. Half of the patients achieved a final refraction within $\pm 1D$ of target. The most commonly implanted lens strengths ranged between $+17.0D$ and $+20.0D$.

Conclusion: Artisan IOL implantation is a safe, effective option for aphakic patients lacking capsular support, even in a complex comorbid patient cohort. Our findings suggest that retropupillary placement may be a less traumatic alternative to anterior enclavation.

Conflicts of interest: None

Funding: This research received no specific grant from any funding agency, commercial, or not-for-profit sectors. No external sources of material or support were provided.

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P029

CLINICAL PERFORMANCE OF HANITA INTENSITY PENTAFOCAL INTRAOCULAR LENSES AS A TREATMENT FOR PRESBYOPIC PATIENTS

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Introduction: The objective of this study was to investigate the clinical performance of the Hanita Intensity pentafoveal intraocular lens implant for treatment of presbyopic patients.

Method: This retrospective study includes 50 eyes of first 50 patients who underwent phacoemulsification and implantation of Hanita Intensity lens by the same surgeon. Patients who had bilateral lens surgery only had the result of the right eye included in the study. All patients had biometry calculated pre-operatively using online ESCRS calculator. Visual acuities for distant (6 m), intermediate (32 inches) and near (16 inches) distance were recorded pre-operatively and three months post-operatively. All eyes were refracted preoperatively and post-operatively with spherical equivalents (SEQ) recorded. Subjective quality of vision was also recorded.

Results: The average age of patients at the time of surgery was 58.4 years (range, 48–78 years). There were 19 male and 31 female patients in the study. The average pre-operative SEQ was $+0.87D$ (range, $10D$ to $+5.625D$) and the average post-operative SEQ was $+0.04D$ (range, $0.5D$ to $+0.375D$). For distant vision, the average unaided visual acuity pre-operatively was 0.57 LogMAR (range, 0 to 1) and post-operatively was 0.08 LogMAR (range -0.1 to 0.2). For intermediate vision, the average unaided visual acuity

operatively was 0.37 LogMAR (range, 0.11 to 0.52) and post-operatively was 0.13 LogMAR (range, 0.1 to 0.22). For near vision, the average unaided visual acuity pre-operatively was 0.36 LogMAR (range, 0.1 to 0.52) and post-operatively was 0.11 LogMAR (range, 0.07 to 0.22). Subjective symptoms of dysphopia (ghosting, glare and holes) were reported to be mild but well tolerated.

Conclusion: The findings of this study demonstrate the success of the Hanita Intensity pentafoveal lens in improving visual acuity at distant, intermediate and near distances. Hanita Intensity pentafoveal lens represents an emerging alternative to current trifocal lenses that offer good visual acuity at a greater number of distances.

Conflicts of interest: None.

Funding: None.

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P030

COMPARISON OF VISUAL OUTCOMES & SPECTACLE INDEPENDENCE WITH MICRO-MONOVISION IMPLANTATION OF TWO ASPHERIC INTRAOCULAR LENSES (IOLS): A PUBLIC HEALTH SECTOR RCT

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Introduction: We compared the visual outcomes, spectacle independence and patient satisfaction of bilateral implantation of either monofocal plus IOLs or standard aspheric monofocal IOLs, using a micro-monovision technique

Method: Patients with bilateral cataracts, and no other significant ophthalmic problems, were randomised to receive either bilateral insertion of Alcon SN60WF or the Eyhance ICB00 IOL with micro-monovision

Results: 57 patients were randomised to the SN60WF group (19M:38F, mean age 69.49 yrs), and 57 to the Eyhance ICB00 group (20M:37F, mean age 68.82yrs) and completed 6-week followup. Mean spherical equivalent (SE) of the dominant eye was -0.15 (SD 0.39) in the Alcon group, and -0.06 (SD 0.44) in the Eyhance group. Mean SE of the non-dominant eye was 0.72 (SD 0.45) in the Alcon group, and -0.77 (SD 0.49) in the Eyhance group.

There was a statistically significant difference ($p < 0.01$) in mean binocular unaided near vision (40cm) which was 0.33 in the Alcon group (SD 0.17) and 0.21 (SD 0.15) in the Eyhance group. This was repeated in N chart testing, with the Alcon group reading on average 7.2 (SD 2.1) compared to 6.1 (SD 1.9) ($P < 0.01$).

There was a statistically significant difference in spectacle independence, with 80.70% of the Eyhance group self-reporting as not needing reading glasses in photopic conditions, compared to 33% in the Alcon group ($p < 0.001$).

Conclusion: Bilateral insertion of Monofocal Plus IOLs with micro-monovision, improved spectacle independence and is a viable option for improving post operative outcomes for NHS cataract patients. We will present further data including contrast sensitivity & patient satisfaction surveys.

Conflicts of interest:

Prof O'Brart holds non-commercial research grants with Rayner Ltd. and J&J Inc and has also undertaken consultancies for J&J and Glaukos in the past 12 months. No other relevant disclosures

Ashmal Jameel has attended Educational events funded by Johnson & Johnson. No other relevant disclosures

Funding: Non-commercial Research grant from Johnson & Johnson

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P031

COMPARISON OF VISUAL OUTCOMES & SPECTACLE INDEPENDENCE WITH MICRO-MONOVISION IMPLANTATION OF TWO ASPHERIC TORIC INTRAOCULAR LENSES (IOLS): A PUBLIC HEALTH SECTOR RCT

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Introduction: We compared the visual outcomes, spectacle independence and patient satisfaction of bilateral implantation of either Toric monofocal plus IOLs or standard aspheric Toric monofocal IOLs, using a micro-monovision technique

Method: Patients with bilateral cataracts & corneal astigmatism of more 1.25D, and no other significant ophthalmic problems, were randomised to receive either bilateral insertion of Alcon SN6ATX or the Eyhance Toric IOL with micro-monovision.

Results: Twenty-six patients were randomised to the Alcon group (10M:16F, mean age 73.78 yrs), and 26 to the Eyhance Toric group (7M:19F, mean age 70.61yrs). At 6 week follow up, mean spherical equivalent (SE) of the dominant eye was 0.00 (SD 0.40) in the Alcon group, and 0.01 (SD 0.33) in the Eyhance group. Mean SE of the non-dominant eye was -0.73 (SD 0.36) in the Alcon group, and -0.77 (SD 0.32) in the Eyhance group. There were statistically significant differences ($p < 0.05$) in binocular logmar unaided near vision (40cm) between the Alcon (mean 0.31, SD 0.15) and Eyhance groups (mean 0.21, SD 0.13). Binocular unaided near vision (n chart) was significantly better in the Eyhance group (mean 6.0, SD 1.50) than the Alcon group (mean 7.54, SD 1.50) ($p < 0.01$). 61.54% of the Eyhance group self-reported as not needing distance, intermediate or reading glasses, compared to 38.46% in the Alcon group ($p < 0.01$).

Conclusion: Bilateral insertion of a monofocal Plus Toric IOLs with micro-monovision, improved binocular unaided near vision and spectacle independence and is a viable option for improving post-operative outcomes for NHS cataract patients with astigmatism.

Conflicts of interest: Prof O'Brart holds non-commercial research grants with Rayner Ltd. and J&J Inc and has also undertaken consultancies for J&J and Glaukos in the past 12 months No other relevant disclosures

Ashmal Jameel has attended Educational events funded by Johnson & Johnson No other relevant disclosures

Funding: Non-commercial Research grant from Johnson & Johnson

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P032. Abstract not included

P033

EFFECT OF A TOPICAL ANTIBIOTIC AND POVIDONE-IODINE VERSUS POVIDONE IODINE ALONE ON CONJUNCTIVAL FLORA: A SYSTEMATIC REVIEW AND META-ANALYSIS

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Introduction: Postoperative endophthalmitis is a rare but serious complication of intraocular surgery, potentially leading to significant vision loss. While PVI is widely recognised as an essential prophylactic measure, the role of preoperative topical antibiotics remains debated. Defining the necessity of antibiotic use in routine cataract surgery is important, especially given the concerns about antibiotic resistance and the rising cost of healthcare. The current standard of care varies globally, with PVI commonly used alone in many regions, such as the UK.

This systematic review and meta-analysis aimed to determine whether adding preoperative topical antibiotics to povidone-iodine (PVI) offers any additional benefit over PVI alone in reducing conjunctival bacterial flora, thereby potentially lowering the risk of postoperative endophthalmitis. The participants included 1423 eyes undergoing elective intraocular surgeries (e.g., cataract, keratoplasty, trabeculectomy) or receiving intravitreal injections. The interventions studied were PVI combined with third-generation quinolones (levofloxacin, moxifloxacin, or gatifloxacin) compared with PVI alone. The outcome measure was post-intervention rate of positive conjunctival bacterial cultures prior to ophthalmic procedure.

Method: Studies were included based on the comparison of preoperative PVI with or without antibiotics in patients undergoing intraocular surgery. Searches were conducted in PubMed, Cochrane, and Embase databases, covering literature up to October 2024. Risk of bias was assessed using the Cochrane risk of bias tool.

Results: Seven studies were included, comprising 1423 eyes. A meta-analysis revealed no significant difference in the reduction of conjunctival bacterial flora between PVI + antibiotics and PVI alone (OR: 0.77, 95% CI 0.42–1.42, $p = 0.41$). Subgroup analysis showed that levofloxacin combined with PVI significantly reduced positive culture rates (OR: 0.48, 95% CI 0.29–0.81, $p = 0.006$), while moxifloxacin and gatifloxacin did not show similar benefits. Moderate heterogeneity was observed across studies ($I^2 = 58\%$, $p = 0.04$).

Conclusion: This review found no conclusive benefit of using preoperative topical antibiotics alongside PVI in routine ophthalmic procedures. However, in patients at high-risk of endophthalmitis, levofloxacin may offer additional protection.

Conflicts of interest: None to declare

Funding: No financial or proprietary interests were received by any author for this work. Additionally, no sponsors were involved in the study design, data collection, analysis, interpretation, writing of the report, or decision to submit the article for publication.

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P034

EXPLORING STRATEGIES FOR MANAGING CHRONIC IRVINE-GASS SYNDROME: A MULTICENTRE STUDY

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Introduction: Persistent Irvine Gass syndrome (pIGS) has been previously described as duration lasting longer than three months. IVG is thought to be due intra-ocular inflammation, however, is not fully understood. There is currently no consensus over how to treat pIGS and this study aims to shed some light into the topic.

Method: Chronic pIGS was defined as pseudophakic cystoid macular oedema (CMO) identified on spectral-domain ocular

coherence tomography (SD-OCT) which persists for ≥ 90 days. Retrospective data was collected from the Greater Glasgow and Clyde (GGC) cataract audit sheet for 2021–2023 across multiple centres. Patients with active CMO of any aetiology prior to surgery were excluded, as well as those with CMO attributable to any other aetiology. Patients with previously resolved CMO of any aetiology were included.

Results: 42 eyes were identified: 22M;20F with median age of 74. Median time to diagnosis was 40 days. Median central retinal thickness (CRT) was 420 μm at diagnosis and 270 μm at resolution ($p < 0.0000005$). 36% were diabetics (8 had retinopathy; 3 previously resolved maculopathy). 41% were hypertensive, and 20% with epiretinal membrane (ERM). The median time to initial treatment of the cohort was 26 days ($\pm 27\text{SD}$). The median time to resolution of pIVG was 107 ($\pm 54\text{SD}$) days.

There were 21 patients treated exclusively with topical steroids and NSAIDs, with median time to resolution of 134 ± 102 SD (days). 5 patients developed a steroid response (24%).

9 patients were treated with exclusive or adjuvant anti-vascular endothelial growth factor (Anti-VEGF). The median time to resolution was 262 (± 123 SD) days. The time taken to switch to Anti-VEGF was 115 ± 99 SD (Days). Time taken to resolution from starting AntiVEGF was 148 ± 66 SD (Days).

9 patients were treated with exclusive or adjuvant intravitreal dexamethasone implant (IVD). The median time to resolution was 151 ± 107 SD (days). The time taken to switch to IVD was 98 ± 61 SD (Days). Time taken to resolution from starting IVD was 60 ± 155 SD (Days). 1 patient developed steroid response (11%).

2 patients were treated with adjuvant periocular triamcinolone (IVA). The median time to resolution was 275 ± 178 SD (days). The time taken to switch to IVA was 225 ± 183 SD (Days). Time taken to resolution from starting IVD was 29 ± 1 SD (Days). no patients developed steroid response.

Conclusion: pIGS is a major cause of patient dissatisfaction following cataract surgery. We identified that delay in initiating treatment, as well as delay in switching from topical therapy is strongly associated with increased duration of CMO. Based on our findings, early switch treatment to intravitreal dexamethasone or periocular triamcinolone is necessary to hasten resolution of pIGS. Larger multicentre studies are required for a consensus.

Conflicts of interest: None

Funding: None

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P035

GIVING PATIENTS STANDARDISED CHOICE OF CATARACT SURGERY PROVIDER THROUGH AN AUTOMATED SINGLE POINT OF ACCESS (SPOA)

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Introduction: The NHS Constitution stipulates that patients should be given information to support their choice of care provider. There are an increasing number of providers of cataract surgery. This means it can be challenging for optometrists and GP referrers to 1) conduct and document the selection of appropriate providers based on patient medical factors; 2) provide detailed patient specific information, such as travel distance; and 3) give

up to date information about each provider, such as waiting time for surgery.

To address these issues the NHS South East Region has chosen to implement a Single

Point of Access (SPoA) using an automated artificial intelligence (AI) agent (Dora, Ufonia Limited, Oxford). Dora contacts each patient via a telephone call, so requires no technology devices or digital literacy. Dora is already used across the Region to conduct automated pre- and post-operative assessments of patients undergoing cataract surgery.

This study reports the early experience of using Dora to deliver a SPoA for cataract surgery in the Buckinghamshire, Oxfordshire and Berkshire West integrated care board (ICB).

Method: Calls are attempted at least 4 times at different days/times and patients were able to call Dora 24/7 if the scheduled time was not convenient. A human led call was also attempted for any patients unable to be contacted or complete a call with Dora.

Dora's conversation comprised 1) shared decision making about surgery, 2) assessment of suitability for a HVLC pathway 3) collection of medical history for surgical pre-assessment, 4) patient choice of provider. Patients were informed of provider-reported waiting times for surgery, and the distance from their home to each provider. Patients that did not meet HVLC criteria were offered NHS providers. Uncontactable patients were referred to their nearest NHS provider.

All patients were asked to give a customer satisfaction score (CSAT) on the call.

Results: In the first 3 months from 1/9/24 to 30/11/24 there were 3025 patients referred to the SPoA. 91% were aged between 60–89years. Referrals came from 163/177 (92%) of optometry practices in the ICB.

11% of patients were unsuitable for HVLC pathways. Most commonly this was patients who were being treated for other eye conditions ($n = 144$) or those who were not able to lie flat/still ($n = 93$).

Following the SPoA, 61% of patients were referred to an NHS provider, 37% were referred to an independent sector provider (ISP), and 2% were returned to their referrer (most commonly because they had a duplicate referral).

48% of patients chose their nearest provider, with a mean distance of 8.1 km (based on anonymised postcode outcode data). There was no significant difference ($p < 0.05$) between the distances chosen by patients opting for NHS or ISP sites.

The mean CSAT was 4.6/5 with a mode of 5/5 (given by 75% of patients).

Conclusion: In this population, an AI-powered SPoA using Dora has been shown to be able to offer patients a standardised choice of cataract surgery provider, using up-to-date, personalised information.

Using this automated approach a greater proportion of ICB patients chose NHS providers compared to before its implementation, and in comparison to national data from the RCOphth (showing 41% of patients had surgery at an NHS provider in January 2024).

The system provides commissioners with information documenting the rationale for each referral choice. Continuity of care is maintained for patients under the care of existing providers, and patients with conditions making them unsuitable for HVLC pathways are only directed to suitable providers.

Overall, the service scored as 'excellent' on the customer satisfaction scale, suggesting it is appreciated by the patients that use it.

Conflicts of interest: AH, LB, LS, CS, RM, EL are employees of Ufonia limited. NdeP is shareholder and director of Ufonia Ltd.

Funding: Funded by the NHSE eMeet Programme

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P036

IMPROVING THE PRE-ASSESSMENT PATHWAY FOR CATARACT SURGERY USING CONVERSATIONAL CLINICAL ARTIFICIAL INTELLIGENCE

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Introduction: The pre-assessment process is crucial in ensuring timely and effective patient care. Dora R3.9 (Ufonia Limited, Oxford, UK) is an artificial intelligence powered telephone conversation agent that is used at Royal Berkshire Hospital NHS Foundation Trust for performing automated cataract surgery pre-operative assessment history taking prior to proceeding to a face-to-face cataract one-stop assessment. It is recommended that 95% of those attending a one-stop clinic are listed for cataract surgery (conversion rate).

Method: We audited the effectiveness of implementation of Dora to support transformation of the pre-assessment pathway. All patients who had a call with Dora between October 2023-May 2024 were included. Suitable patients are those able to have a clinical conversation in English. We focused on the outcomes in face-to-face clinics for those patients that did not answer the call with Dora or who told Dora they were not sure about surgery to try and improve efficiency and optimisation of clinical time.

Results: 1080 patients were called by Dora, 974 (90.2%) answered. 710 (80%) of those suitable for a Dora call completed the call with key pre-assessment information. If a patient told Dora they were not interested in surgery (28), at the subsequent face-to-face review only 1 (3.6%) patient was listed for surgery. If a patient did not answer the call with Dora 106 (9.8%), this highly predicted the patient's outcome at the subsequent clinical appointment with only 57 (54%) listed for surgery.

Conclusion: Through using Dora for pre-assessment we can collect standard clinical information before a face-to-face appointment saving valuable clinical time. This audit also showed that Dora can identify those patients who do not want surgery, meaning these patients no longer need to attend the face-to-face appointment. Similarly, we can restructure clinic bookings to account for the low conversion rate for those not answering the Dora call.

Conflicts of interest: Since some authors are employees of the software company, this could present a potential conflict of interest

Funding: The work at RBH was funded by an SBRI grant

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P037

INSERTION OF INCORRECT INTRAOCULAR LENS DURING CATARACT SURGERY IN THE UNITED KINGDOM: INCIDENCE ANALYSIS OF NEVER EVENTS FROM 2011–2023

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Introduction: Cataract surgery with intraocular lens (IOL) insertion remains one of the most frequently performed surgical procedures in the UK, with more than half a million operations performed annually. Never events are defined as 'serious, largely

preventable patient safety incidents that should not occur if the available preventative measures have been implemented'. Incorrect intraocular lens (IOL) insertion during cataract surgery has been classified as a 'never event' since 2011. National guidelines promoting safe cataract surgery have been produced in recent years. It remains unclear if timing of these interventions coincides with patterns of incidence of wrong IOL insertion and if these trends have changed with time.

We aim to describe and analyse the incidence rates and trends of wrong IOL events across the United Kingdom from 2011–2023.

Method: Data regarding incorrect IOL insertions was extracted retrospectively from the NHS Improvement and Patient Safety Incidents submitted to the National Reporting and Learning System (NRLS) from 2011 to 2020. Number of cataract surgeries performed each year were derived from the publicly available NHS performance data. Incidence rate of incorrect IOL never events for each financial year was calculated. Trend test was used to determine the statistical significance in the differences of incidence rates between the financial years. Incidence rate over the last 10 years were plotted against developments in ophthalmology related to never events. Separate analysis was completed post COVID pandemic, due to a reduction in ophthalmic cases during the 2-year period. Further, recent data was extracted from the NHS England Never Events provisional and published database.

Results: Incidence rate of wrong IOL never events within the NHS setting is <0.01% throughout between 2011–2023. There was a 3-fold reduction in the incidence rate from 2017–18 to 2018/19 (0.0063% to 0.0019%; $p < 0.001$) and this was sustained in 2019–20 (0.0022%). Incidence rates were highest in the first 3 years of recorded wrong IOL never events between 2011–14 (0.0094%; 0.0089%; 0.0080%). Percentage of never events remain to plateau at similar levels over the last 3 years, with an average of 0.0021% (10, 9, 11 per financial year) between 2020/2023. Further analysis suggested root and contributory causes which include incorrect biometry, wrong IOL selection, written/transcription errors.

Conclusion: IOL selection and implantation is a complex, multi-level process. Although the incidence of wrong IOL insertion has reduced with the initial widespread adoption of World Health Organisation (WHO) surgical safety checklist and the Royal College of Ophthalmology bespoke cataract checklist, these events continue to occur within hospital settings. Human factors are great contributors, and our study helps to identify and address these concerns. Furthermore, with the increased distribution of cataract services to the private sector, more transparency is needed regarding rates of never events, in order to enhance patient safety and optimise clinical outcomes.

Conflicts of interest: None

Funding: None

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P038

NEW GENERATION IOL SUPERFORMULAS FOR PERFECTING CATARACT SURGICAL OUTCOME: COMPARING IOL POWER USING "IOL-CAL.COM" WITH TRADITIONAL FORMULAS IN CATARACT SURGERY

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Introduction: To evaluate the accuracy of IOL power determination using new generation super formulas in **IOL-cal.com** in comparison to traditional formulas such as **SRK/T**, **Hoffer Q**, and

Holladay 1 for cataract surgery patients, particularly those with challenging biometric characteristics to optimise visual outcome.

Method: Study Design: Prospective study involving 100 patients undergoing cataract surgery.

Preoperative Measurements:

- Axial length (AL) measured using optical biometry.
- Corneal curvature (K) measured using keratometry or corneal topography.
- Anterior chamber depth (ACD) measured to assess proper IOL power.

IOL Power Calculation:

- IOL-cal.com: Integrated tool using multiple formulas (Ladas/Kane, SRK/T, Hoffer Q, Holladay 1) based on patient-specific measurements.
- Traditional formulas for comparison: SRK/T, Hoffer Q, and Holladay 1.
- Surgical Procedure: Phacoemulsification with monofocal IOL implantation in the capsular bag.

Primary Outcome Measure:

- Mean Absolute Prediction Error (MAE) of postoperative refractive outcomes at 6 weeks.

Secondary Outcome Measures:

- Proportion of patients achieving postoperative refraction within ± 0.5 D of target refraction.
- Rate of residual astigmatism at 6 weeks post-surgery.

Statistical Analysis: Descriptive statistics, paired t-tests or ANOVA, chi-square tests for categorical variables, significance set at $p < 0.05$.

Results: Mean Absolute Prediction Error (MAE):

IOL-cal.com demonstrated a significantly lower **Mean Absolute Prediction Error (MAE)** when compared to traditional IOL power calculation formulas, with a p -value < 0.05 . This indicates that IOL-cal.com provided more accurate predictions of post-operative refractive outcomes. The tool particularly excelled in predicting refractive outcomes for patients with **short eyes** (<22 mm) and **long eyes** (>26 mm), showing improved precision in both these challenging biometric categories.

Refractive Outcomes:

A higher percentage of patients in the IOL-cal.com group achieved a refractive outcome within ± 0.5 dioptres (D) of the target refraction (82%) compared to the traditional formula group. The percentage of patients achieving this target refraction was 75% for the SRK/T formula, 71% for the HofferQ formula, and 73% for the Holladay 1 formula. These results suggest that IOL-cal.com is more effective at helping patients achieve a closer match to their target refractive outcome.

Residual Astigmatism:

The incidence of significant residual astigmatism (greater than 0.75 D) was lower in the IOL-cal.com group, with only 12% of patients showing significant astigmatism after surgery. In contrast, the SRK/T formula showed a residual astigmatism rate of 18%, the Hoffer Q formula had a rate of 20%, and the Holladay 1 formula had a residual astigmatism rate of 19%. These findings indicate that IOL-cal.com was more effective in minimizing postoperative astigmatism, particularly in comparison to traditional formulas.

Conclusion: The use of IOL-cal.com for intraocular lens (IOL) power determination offers significant advantages over traditional formulas like SRK/T, Hoffer Q, and Holladay 1 by providing more accurate and consistent results. This online tool integrates multiple advanced IOL power calculation formulas and tailors them to individual biometric parameters, such as axial length, corneal

curvature, and anterior chamber depth. By adjusting for these factors, IOL-cal.com enhances the likelihood of achieving a refractive outcome that closely matches the target refraction, significantly reducing the Mean Absolute Prediction Error (MAE) and minimizing residual astigmatism after cataract surgery.

IOL-cal.com proves especially useful in challenging cases, such as patients with extreme axial lengths, irregular corneal curvature, or those who have previously undergone refractive surgery. These situations often complicate traditional IOL power calculations, but IOLcal.com's ability to account for such complexities ensures more precise outcomes. Its user-friendly interface, combined with its ability to integrate multiple formulas, makes it a highly valuable tool for cataract surgeons. The tool's enhanced accuracy and ability to improve refractive outcomes position it as an essential resource for optimizing cataract surgery, particularly for complex or high-risk cases.

Conflicts of interest: None

Funding: None

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P039

OPTIMISING THE IMPLEMENTATION OF A VALIDATED PATIENT-REPORTED OUTCOME MEASURE (PROM) FOR THE CATARACT SERVICE IN LEEDS

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Introduction: With patients in the United Kingdom seeking and undergoing cataract surgery at an increasingly early stage, surgery may produce only minimal change in visual acuity but significant subjective improvement to quality of life. Patient-reported outcome measures (PROMs) capturing this improvement are paramount in the context of increasing NHS-funded elective cataract operations (608,000 in England in 2022 from NHS Data) and tightening budgets. The Cat-PROM5 questionnaire is a validated PROM developed in the UK NHS for high-volume cataract surgical services (Sparrow et al. 2018).

Method: From March 2023, email addresses were collected where possible from consecutive patients at pre-assessment for cataract surgery in Leeds Teaching Hospitals NHS Trust. An electronic medical records system (mediSIGHT, Medisoft Ltd., Leeds, UK) sent patients automated emails containing a link to complete the Cat-PROM5 questionnaire online. Patient responses were collected pre-operatively following pre-assessment and 3 months post-operatively to measure the change in PROM score. Five other NHS centres in England using mediSIGHT also participated in the pilot. We sequentially audited response rates to the pre-operative questionnaire. Discouragingly, response rates were initially below 20% at most sites – insufficient for any meaningful analysis of the responses. In order to understand the factors associated with non-response and maximise participation in Leeds, we collected data on all participating patients seen at pre-assessment for cataract surgery over a 4-month period and undertook quantitative and qualitative analysis as a service quality improvement exercise. Quantitative analysis required a sample size of around 100 patients (estimated for 90% power, 95% confidence). Patients were grouped according to response versus nonresponse and compared by age, index of multiple deprivation decile and the number of days between the email being sent and surgery (unpaired t-test), and by sex and presence of ophthalmological comorbidities (Chi-squared test). A sample of patients that did not complete the pre-operative Cat-PROM5

questionnaire were randomly contacted by telephone to undertake semi-structured interviews to help understand the drivers of nonresponse. Up to two attempts were made to contact patients. We undertook thematic analysis of responses to saturation.

Results: In the initial pilot, the Cat-PROM5 response rate was 11.4% in Leeds. A necessary update to the electronic medical records system, which included the Cat-PROM5 email platform, and the novelty of the workflow likely limited the initial response rate. After staff were retrained and familiarity with email collection and explanation to patients increased, the questionnaire response rate in Leeds improved to 31%. 91% of questionnaire responses were received on the same day or the day after the pre-assessment visit. The latest response was received after 7 days. There was no statistically significant difference between responders and non-responders in terms of age, sex, index of multiple deprivation decile, ophthalmological comorbidities or the time between questionnaire distribution and surgery. 26 patients were successfully contacted for telephone interviews. While 60% of patients remembered the conversation about the questionnaire at pre-assessment when prompted, 88% did not recall receiving the questionnaire via email. Of these, only one participant was able to locate the questionnaire in their email inbox. Other potential reasons for nonresponse given by participants included: dependence on others to manage their email; difficulty with technology; fear of scams; forgetfulness; and difficulty keeping track of many health-related commitments.

Conclusion: Iterative optimisation of Cat-PROM5 response collection in Leeds led to improved response rates. In this patient sample, no patient or service factors were identified that predicted questionnaire non-response. Qualitative analysis of questionnaire non-response has guided improvements to workflow, questionnaire distribution and response collection, all of which continue to be adjusted to maximise participation.

Conflicts of interest: None

Funding: None

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P040

PERFORMANCE OF A NOVEL ARTIFICIAL INTELLIGENCE MODEL FOR CATARACT SURGERY PHASE IDENTIFICATION USING A REAL WORLD CATARACT SURGERY VIDEO DATASET

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Introduction: Artificial Intelligence (AI)-assisted analysis of cataract surgery videos can enhance surgical education, performance evaluation, and clinical outcomes. However, existing AI models developed for this purpose have limited generalisability, as they have been trained on cataract videos from 1–2 consultant surgeons under fixed operating conditions. We aimed to train an AI model capable of accurately detecting key surgical phases robust to variations in operating microscope, video quality, surgical setup, case complexity, and surgeon experience. Additionally, we evaluated whether surgeon experience, phase difficulty, and phase duration influenced the model's performance.

Method: We manually labelled 100 cataract videos with ground truth time stamps for 12 distinct surgical phases using

the VGG Image Annotator. Videos were evenly split between those performed by junior trainees, senior trainees and consultants. Two senior ophthalmologists independently graded each surgical phase as “standard” or “difficult”. The AI model was trained on the publicly available Cataract-101 video dataset using supervised deep learning techniques, specifically a ResNet-50 architecture, to classify video frames into their corresponding surgical phases. Next, the model was tested on our heterogeneous dataset, consisting of surgical videos recorded in variable operating conditions.

Performance evaluation involved comparing the model's predicted phase timelines against manually annotated ground truth timelines. Confusion matrices were used to assess classification accuracy and errors for each phase. Confidence scores were used to evaluate the model's certainty that a certain phase was occurring. Mean confidence scores for consultant and trainee videos were compared using independent t-tests. The frequency of phase prediction was used to analyse how often the model correctly identified each phase. Phase difficulty and phase length were correlated with the frequency of phase prediction data using Spearman's rank correlation coefficient. Statistical analysis was conducted using IBM SPSS statistics version 30.

Results: Surgical phase annotation timelines and phase ground truth timelines demonstrated that, at face value, the AI model achieved higher classification accuracy for the cataract 101 dataset than our heterogeneous dataset. For our dataset, confusion matrices revealed common misclassifications between phases with similar visual patterns, such as “hydrodissection” being misclassified as “viscous-agent-injection”. Errors also appeared frequently in phases more commonly marked as “difficult,” supporting the hypothesis that case complexity would reduce model performance.

Confidence Scores ranged from 0.0–1.0, with an average confidence score of 0.72 ± 0.35 across all videos. Analysis of surgeon experience revealed that videos performed by consultants were associated with the highest average model confidence scores (0.85 ± 0.25), followed by senior trainees (0.72 ± 0.55) and junior trainees (0.60 ± 0.45). Mean confidence scores were significantly different for trainee and consultant videos ($P < 0.05$). Phacoemulsification phases achieved the highest frequency of prediction, reflecting distinct visual and temporal features. In contrast, irrigation and aspiration showed the lowest, indicating overlapping characteristics with adjacent phases. A moderate negative correlation was observed between phase difficulty and frequency of prediction ($p = -0.6$, $P \leq 0.05$). A strong positive correlation was observed between phase difficulty and length ($p = 0.8$, $P < 0.05$).

Conclusion: This study serves as proof of concept that an AI model can successfully recognise the phases from cataract surgery videos recorded in normal working conditions. Future work involves improving AI model resilience to phase complexity and variations in surgeon experience, such that enhancements in trainee learning and performance feedback can be achieved.

Conflicts of interest: None

Funding: None

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P041

PERSISTENT CYSTOID MACULAR OEDEMA AFTER CATARACT SURGERY: INSIGHTS AND IMPLICATIONS

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Introduction: Post-phacoemulsification cystoid macular oedema (CMO) is a recognized complication, affecting approximately 1–3% of patients. A subset of these patients develops persistent CMO (pCMO), defined as CMO persisting beyond three months. However, risk factors distinguishing pCMO from transient CMO remain underexplored. This study aims to identify characteristics associated with pCMO.

Method: A retrospective analysis was conducted using the shared online audit worksheet of cataract surgeries performed between 2021 and 2023. Patients with CMO were identified, and data on age, gender, medical and ocular history, medication history, and CMO type (parafoveal vs. foveal involvement) were collected on a secure database. Cases of CMO attributed to causes other than cataract surgery were excluded. Statistical analysis was performed using Python to compare characteristics between CMO and pCMO groups.

Results: A total of 245 eyes with CMO were identified, with 161 eyes included for analysis after exclusions. Of these, 82 patients were female, and the median age was 75 years (range: 42–90). A significant association was observed between older age and pCMO ($p < 0.028$).

Diabetic patients were more likely to develop pCMO compared to transient CMO ($p < 0.032$). Use of statins was associated with a reduced likelihood of pCMO, with a logistic regression coefficient of -0.83 ($p < 0.036$) and an odds ratio of 0.44 (95% CI: 0.2–0.95). Foveal involvement in CMO was significantly associated with pCMO compared to parafoveal involvement ($p < 0.000005$). Other factors, including the use of PPIs, ACE inhibitors, calcium channel blockers, and prostaglandin analogues, showed no association with pCMO.

Conclusion: Older age, diabetes, and foveal involvement in CMO are significant risk factors for pCMO. Statin use was associated with a 56% reduction in the odds of developing pCMO, suggesting a potential protective role. These findings provide valuable insights for clinicians in identifying patients at risk of pCMO and informing patient counselling. Further research is warranted to guide optimal management strategies for pCMO.

Conflicts of interest: None

Funding: None

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P042

PHACOEMULSIFICATION POST PARS PLANA VITRECTOMY (PPV): A SINGLE CENTRE REVIEW OF COMPLICATIONS IN CATARACT SURGERY AFTER PPV

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Introduction: Pars Plana Vitrectomy (PPV) is a commonly performed vitreoretinal surgery after which cataract formation is very common, with some studies reporting rates of progression to visually significant cataract as high as 100%. While cataract surgery is required, phacoemulsification after PPV is associated with higher risks of zonulopathy, posterior capsular rupture (PCR), dropped nuclear fragments and reportedly worse rates of visual acuity improvement compared with non-vitrectomised eyes. We aim to analyse intraoperative complications and resultant visual outcomes of cataract surgery after PPV at Sheffield Teaching Hospitals Trust (STHT).

Method: A retrospective analysis was performed to identify all cataract surgeries performed on eyes that had previously undergone PPV from January 2011 to March 2024 at a single tertiary

centre. Data was compiled on: demographics (age/sex), intra-operative complications, stage of complication, resultant IOL, visual acuity outcome and grade of surgeon. Statistical analysis using paired t-test was performed.

Results: Of 4000 cases analysed, 35 (0.9%) operations had recorded complications. Of those, 64% were male, 37% female and the average age was 62 years. These included 14 (40%) PCRs without vitreous loss, 13 (37%) zonular dialyses and 8 (23%) PCRs with vitreous loss or dropped lenses. The resultant intra-ocular lens (IOL) was placed in the bag in 14 (40%) cases, in the sulcus in 11 (31%) cases and the patient left aphakic in 10 (29%) cases. For those aphakic patients, 80% had successful surgery to implant a secondary IOL. The mean improvement in visual acuity was 0.52 logMAR (95% CI 0.2–0.85; $p = 0.0024$) with 3 (14%) patients having worse visual outcomes; 2 were those left aphakic and 1 received an anterior chamber lens. There was an equal proportion of complications for both senior trainees (ST47) and consultants (37%).

Conclusion: Overall STH experienced low complication rates in cataract surgery in post-vitrectomy eyes and those with complications still demonstrated significant improvement in visual acuity. Further multi-centre research within the UK is required to give support to our findings.

Conflicts of interest: None

Funding: None

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P043

PITFALLS OF MALYUGIN RING INSERTION

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Introduction: The Malyugin ring is sometimes utilised in cataract surgery to achieve adequate pupil dilation in cases with small pupils. Although its insertion is generally straightforward, certain challenges must be recognised and managed to avoid complications.

Method: We present two cases that highlight some pitfalls associated with Malyugin ring usage:

1. Case 1: A posterior capsule tear occurred during phacoemulsification. A sulcus intraocular lens (IOL) was implanted before the removal of the Malyugin ring.
2. Case 2: The Malyugin ring was dislodged following hydrodissection and became entangled with the lens nucleus.

Results: In the first patient, the ring became entangled with the sulcus-fixated IOL and the ring had to be cut into pieces for safe removal. In the second case, efforts were made to avoid contact between the ring and the corneal endothelium. Once the ring was disentangled, it was safely removed from the eye. [video presentation available]

Conclusion: To avoid complications associated with Malyugin ring insertion and removal, the following pitfalls should be considered. Removing the Malyugin ring before inserting a sulcus IOL is recommended to prevent potential entanglement between the IOL and the ring. Vigorous hydro-dissection and “flip and chip” phacoemulsification technique should be avoided after Malyugin ring insertion to avoid dislodgement of the ring.

Conflicts of interest: None

Funding: None

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P044

SMILE FOR THE TREATMENT OF RESIDUAL REFRACTIVE ERROR AFTER CATARACT SURGERY

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Introduction: We aimed to assess the improvement in visual acuity and patient satisfaction following small-incision lenticule extraction (SMILE) in pseudophakic patients (with trifocal intraocular lenses) who experienced residual myopic refraction after cataract surgery.

Method: Seventy-six patients (82 eyes) who underwent cataract surgery with ZEISS AT LISA tri 839MP IOL implantation were included in this retrospective study. The included patients were 56–79 years old, wanted spectacle independence, and had preoperative myopic refraction between -1.0 and -2.25 dioptres (D) and astigmatism between -0.75 and -1.75 D. The treatment status of these patients was defined as trifocal IOL ($n = 82$). SMILE was performed on patients who were dissatisfied after cataract surgery, and these patients were followed up for 1 year on average. We evaluated visual acuity and satisfaction and further examined laser vision correction and satisfaction levels in patients who were dissatisfied after trifocal IOL implantation.

Results: After SMILE, the residual myopic refractive error (spherical) decreased significantly from -2.08 ± 0.28 [-2.25 to -1.0] preoperatively to -0.25 ± 0.20 [-0.5 to 0] 1 year postoperatively ($p < 0.001$). Additionally, the uncorrected distance visual acuity increased from 0.65 ± 0.08 [0.52 – 0.7] logMAR preoperatively to 0.09 ± 0.02 [0.05 – 0.1] logMAR at 1 month postoperatively ($p < 0.001$), 0.09 ± 0.02 [0.05 – 0.1] logMAR at 6 months postoperatively, and 0.06 ± 0.02 [0.05 – 0.1] logMAR at 12 months postoperatively ($p < 0.001$). Patient satisfaction measures after SMILE (reading, night driving, and using a computer) were significantly improved.

Conclusion: SMILE is a reliable method for treating residual refraction after cataract surgery, as it provides results in the shortest time without complications and increases patient satisfaction.

Conflicts of interest: None

Funding: None

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P045. Abstract not included

P046

STREAMLINING CATARACT SURGERY PATHWAYS: THE IMPACT OF SAME-DAY CONSENT FOR DELAYED SEQUENTIAL BILATERAL CATARACT SURGERY

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Introduction: Cataract remains a leading cause of blindness worldwide, with significant impacts on quality of life and independence. In many developed countries, cataract surgery is often performed on both eyes, requiring multiple clinic visits for

consent. Traditionally at Sheffield Teaching Hospitals, separate clinic appointments were required to consent for each eye to be operated on different days, prolonging the time to complete delayed sequential bilateral cataract surgery (DSBCS). To streamline this process, the Northern General Eye Centre (NGEC) introduced same-day consenting for DSBCS. This service evaluation examines the impact of this change on patient pathways and clinic efficiency.

Method: The new pathway involves consenting patients for DSBCS during a single one stop clinic visit, eliminating the need for a second appointment solely for second-eye consent. Patients referred to NGEC undergo a one-stop clinic assessment by an ophthalmologist, followed by consent for DSBCS with the order of surgeries specified. After pre-operative nursing assessment, biometry, and optional OCT, the first eye surgery is performed. Postoperative review by an optometrist confirms readiness for the second eye surgery, previously consented. If the patient agrees, a date for second eye surgery is scheduled.

A retrospective evaluation of NGEC's local electronic patient record system (Lorenzo) identified patients who underwent second-eye surgery between February 2023 and November 2024. Data was extracted from MediSight to determine how many DSBCS second-eye surgeries were performed and how many patients had been consented for this on the same day.

Results: From February 2023 to November 2024, 2430 second-eye surgeries were completed. Of these, 1385 patients were consented for DSBCS during the initial clinic appointment, eliminating the need for an additional 1385 appointments solely for second-eye consent. These changes reduced patient workload, transport needs, and administrative efforts, while maintaining flexibility for patients to defer their second eye if desired.

Conclusion: The adoption of same-day consenting for DSBCS has demonstrated substantial improvements in clinical and operational efficiency. By reducing the need for additional clinic appointments, the new process decreases lead times to DSBCS completion. Similar to efficiency measures described by Van Vliet et al., the NGEC model illustrates how streamlined consent processes can improve patient pathways in high-volume cataract services. Notably, a literature review revealed no prior descriptions of this specific consenting method.

While the potential financial savings from improved efficiency were not measured in this evaluation, these findings support the value of process innovation in cataract pathways. The NGEC model has successfully integrated one-stop clinic efficiencies with patient-centred care, significantly reducing delays and improving service delivery for bilateral cataract surgery. This model represents a novel and scalable approach to enhancing cataract care delivery in high-volume settings.

Conflicts of interest: None

Funding: None

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P047

THE RCOPHTH NATIONAL OPHTHALMOLOGY DATABASE STUDY OF CATARACT SURGERY: VISUAL OUTCOMES FOR EYES WITH A PREOPERATIVE VISUAL ACUITY OF COUNT FINGERS OR WORSE

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Introduction: There is little published evidence to inform the consent process, or the clinical decision to offer cataract surgery to patients presenting with preoperative visual acuity (VA) of Counting Fingers (CF), Hand Movements (HM) or Perception of Light (PL). For those whose primary pathology is advanced cataract, often obscuring the view of the retina and optic nerve, the chance of obtaining good postoperative VA is unknown.

Eyes with other co-pathologies may be the main reason for poor vision but where there is some element of cataract, commissioners may ask whether cataract surgery is a good use of limited healthcare funds when the prognosis is uncertain.

This analysis aimed to investigate post-cataract VA in eyes with a preoperative VA of CF or worse using data submitted to the RCOPHTH NOD.

Method: Eligible cataract operations were performed between 01/04/2010 and 31/03/2023 in centres with data for at least 50 operations with both a preoperative and postoperative VA. Eyes with preoperative VA of CF, HM and PL from these centres were included. UK driving standard VA and severe sight impaired (SSI) were defined as VA ≤ 0.30 LogMAR and >1.30 LogMAR respectively.

Results: Eligible for analysis were 95,725 operations performed on 91,454 patients from 189 centres by 4409 surgeons with operations 81.8% performed by fully qualified surgeons. Pre-operative VA was CF for 52,708 (55.1%) eyes, HM for 34,717 (36.3%) eyes and PL for 8,300 (8.7%) eyes. The median ages of patients were 71, 73 and 66 years for first eye, second eye and immediate sequential bilateral cataract surgery respectively.

Brunescent/white/mature cataract $\pm$ no fundal view/vitreous opacity was present in 36,387 (38.0%) eyes, and in 34.5% CF, 48.4% HM and 17.2% PL eyes. Posterior Capsule Rupture (PCR) occurred in 4.12% of eyes. The median postoperative VA was 0.18 LogMAR, UK driving standard VA was achieved by 76.9% eyes, and by 78.1%, 76.9%, and 74.4% of CF, HM and PL eyes respectively. Postoperative VA remained in the SSI range for 5.5% eyes, and for 4.2%, 5.6%, and 8.0% of CF, HM and PL eyes respectively.

Brunescent/white/mature cataract $\pm$ no fundal view/ vitreous opacity was absent in 59,338 (62.0%) eyes, and in 67.7% CF, 28.9% HM and 3.5% PL eyes. AMD was present in 14.6% eyes, DR in 8.5% eyes, glaucoma in 10.1% eyes, and 2.04% experienced PCR. The median postoperative VA was 0.20 LogMAR, UK driving standard VA was achieved by 70.2% eyes, and by 73.6%, 64.3% and 54.8% of CF, HM and PL eyes respectively. Postoperative VA remained in SSI range for 10.0% eyes, and for 6.9%, 15.4%, and 25.7% of CF, HM, and PL eyes respectively.

Conclusion: UK driving standard VA was achieved by 76.9% eyes with preoperative VA of CF or worse due to brunescent/white/mature cataract $\pm$ no fundal view/ vitreous opacity. Even where copathologies are identified (primarily AMD, DR and Glaucoma) and the advanced cataract is not recorded, the UK driving standard was achieved in 70.2% of eyes with CF or worse preoperatively. This information is useful for preoperative counselling the informed consent process.

Conflicts of interest: None

Funding: The RCOPHTH National Ophthalmology Database Cataract Audit is currently funded through participation fees from centres and unrestricted financial contributions from Alcon and Bausch and Lomb.

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P048

VARIABLES LINKED TO THE CHRONICITY OF IRVINE-GASS SYNDROME

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Introduction: Pseudophakic cystoid macular oedema (pCMO / Irvine-Gass Syndrome) is the most common postoperative complication of cataract surgery. pCMO lasting more than 90 days is considered chronic. We aim to investigate how 'biometric', 'spectral domain ocular coherence tomography (SDOCT)', and 'ETDRS visual acuity' findings may influence the chronicity of CMO.

Method: The NHS Greater Glasgow and Clyde (GGC) cataract audit sheet was accessed to find all patients with reported cystoid macular oedema between 2021 and 2023. Clinical portal letters and SD-OCT scans were examined. The inclusion criteria was the presence of pseudophakic CMO. Pre-existing (unresolved) CMO of any aetiology was excluded. Spearman correlation coefficient (SCC) and Mann-Whitney U test were used to examine the relationship between the chronicity of CMO and multiple independent variables.

Results: 245 eyes had CMO in the initial search. 163 eyes were analysed following exclusions. SCC was 0.29 ($p < 0.00125$), 0.514 ($p < 1.45 \times 108$), and -0.28 ($p < 0.00083$) for choroidal thickness, median central retinal thickness (CRT) and visual acuity, respectively, at time of diagnosis of CMO. Median choroidal thickness of 317 microns ($p < 0.049$), central retinal thickness of 436 microns ($p < 0.000026$), and ETDRS visual acuity of < 65 letters were associated with chronic CMO ($p < 0.0032$).

Conclusion: Increased choroidal thickness, higher central retinal thickness at time of diagnosis, and lower visual acuity at time of diagnosis are associated with increased chronicity of cystoid macular oedema. This is clinically valuable information for the clinician when counselling and considering treatment options for patients with pCMO.

Conflicts of interest: None

Funding: None

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CORNEA AND OCULAR SURFACE

P049

A RARE SPONTANEOUS CORNEA KELOID PRESENTATION: CLINICOPATHOLOGICAL FEATURES, GENETICS AND SURGICAL OUTCOME

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Introduction: This case report details the presentation, clinicopathological features, genetics, and surgical outcome of a 41-year-old male with a visually significant corneal keloid. Corneal keloids are rare, typically presenting as single, white, enlarging, elevated corneal lesions, which may be vascularized or non-vascularized with well-defined borders. They have been

associated with conditions such as Lowe Syndrome, post ocular trauma or surgery.

Method: This is a retrospective review of the clinical features, genetics and surgical management of a patient with a left eye corneal keloid. This includes follow up data of two years with serial anterior segment imaging of the patient describing the clinical course and progression.

Results: The patient, a 41-year-old Caucasian male with no prior ocular history or family history, presented with gradual vision deterioration in his left eye. His history included sleep apnoea and obesity. Initial visual acuity was 6/24 in the right eye and 6/60 in the left. Bilateral corneal endothelial dusting suggested corneal endothelial dystrophy, and left DSAEK was planned. Surgery was delayed due to the COVID-19 pandemic. Two years later, a new white corneal lesion with neovascularization was noted. Excision and biopsy were arranged, which confirmed histopathology consistent with a corneal keloid.

Ten months post-op, the left eye's keloid recurred extensively, while the right eye exhibited worsening posterior haze and stromal thickening. The patient underwent left penetrating keratoplasty and has since been stable with no recurrence. Genetic testing was undertaken, which confirmed a diagnosis of autosomal dominant ZEB1-related disease associated with both Fuchs and posterior polymorphous dystrophy.

Conclusion: This is a rare case describing a spontaneous unilateral corneal keloid with bilateral genetically confirmed ZEB1 mutation corneal dystrophy. The dysfunction of ZEB1 may affect corneal stem cell homeostasis, stromal fibrosis, angiogenesis and squamous metaplasia. Floppy eyelid syndrome in sleep apnoea has also been reported as a risk factor for corneal keloids. Surgical management should be approached cautiously as it may exacerbate symptoms.

Conflicts of interest: No conflict of interest to declare.

Funding: No funding implications.

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P050

A REVIEW OF CORNEAL ALLOGENIC INTRASTROMAL RING SEGMENTS

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Introduction: To review the current literature on Corneal Allogenic Intrastromal Ring Segment (CAIRS) focusing on the indications, evolving technique modification, visual and tomographic outcomes, and safety profile. CAIRS was first introduced in 2015 and represents another innovative option for managing corneal ectatic disorders, particularly keratoconus.

Method: A literature review was conducted in PubMed/Medline and Google Scholar to identify any studies regarding CAIRS in English. Inclusion Criteria were studies with more than 20 eyes and minimal follow-up of 6 months. This comprehensive literature review will provide a detailed description of the CAIRS procedure, including patient selection, safety profile, and the different modifications that enable tailored patient treatment. We will summarise and discuss the various studies' visual and topographic long-term outcomes and the combination of CXL.

Results: The current studies demonstrate a significant improvement in topographic and visual acuity parameters. The primary outcome was uncorrected distance visual acuity (UCDVA), with mean values improving from 0.80 ± 0.16 to 0.38 ± 0.08

logMAR ($p = 0.01$). Secondary outcomes included corrected distance visual acuity (CDVA) with mean values improving from 0.49 ± 0.23 to 0.19 ± 0.09 logMAR ($p = 0.01$). Keratometry findings indicated a reduction in corneal curvature, with the mean K-max decreasing from 57.79 ± 1.19 to 53.44 ± 2.93 ($p < 0.01$) and the mean K-mean decreasing from 49.28 ± 0.32 to 45.27 ± 1.66 ($p < 0.01$). Unlike synthetic corneal segments, CAIRS are derived from donor corneal tissue, offering enhanced biocompatibility and a substantially reduced risk of complications such as infection or extrusion.

Conclusion: Those findings emphasise CAIRS as an evolving novel technique that can be used as an alternative for synthetic corneal ring segments or even replace corneal transplant surgery in some cases, with potential benefits in restoring corneal stability and improving visual acuity with a high safety profile. Results will be updated if more relevant studies will be published prior to the meeting.

Conflicts of interest: None

Funding: None

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P051

ADULT GONOCOCCAL EYE DISEASE; UNDERSTANDING THE EPIDEMIOLOGY, CLINICAL AND MICROBIAL CHARACTERISTICS

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Introduction: In England in 2023, 85,223 cases of gonorrhoea were diagnosed, the highest number since records began. A concurrent escalation in adult gonococcal conjunctivitis (GC) and keratoconjunctivitis (GKC) is a serious concern due to the potentially devastating visual complications. *N. gonorrhoeae* can cause an aggressive keratitis leading to corneal thinning and perforation. *N. gonorrhoeae* is a major public health concern as it is evolving high levels of antimicrobial resistance, including to ceftriaxone, the last available option for empirical therapy. We initiated a new surveillance system for GC/GKC in London to understand the clinical and molecular epidemiology of recent cases.

Method: A pan-London partnership between eight hospitals was created as a special interest group, to report cases of GC/GKC to UKHSA, providing *N. gonorrhoeae* isolates from eye swabs and demographic and clinical information through a secure online questionnaire.

Antimicrobial susceptibility testing and whole-genome sequencing (WGS) of isolates was performed at the UKHSA STI Reference Laboratory.

Results: From January to November 2024, 11 episodes of adult GC/GKC were reported, eight unilateral and three bilateral. Nine identified as male, two as female. The mean age was 31.6 (range 19–51) years, most were in their 20s (5/11). The majority were White or White British (10/11). A sexual history was taken in all but one case, with 5/11 reporting new sexual partners but sexual orientation was not documented in 6/11 cases. One patient received a new diagnosis of HIV after being referred for sexual health screening. An incorrect initial diagnosis of non-gonococcal bacterial or viral conjunctivitis was made in the majority of cases (7/11); three were initially prescribed topical steroids. Classical

hyperpurulent conjunctival discharge was not present in the majority of initial presentations (8/11). There was no case of corneal perforation and good visual outcomes (at least 6/7.5) were achieved in all cases but one, who declined follow-up after treatment. Systemic treatment was consistent in all cases: 1g ceftriaxone, but topical treatment varied between cefuroxime (5/11) and moxifloxacin (5/11). Among the nine gonococci isolated on culture there were no ceftriaxone resistant strains. Two isolate pairs were sourced from 2 different cases of bilateral infection. In each case the isolate pairs were genetically identical to each other but were genetically diverse when compared with isolates from the other cases. The remaining 5 isolates were found to be genetically distinct, although two isolates differed by only 110 single nucleotide polymorphisms.

Conclusion: Ophthalmologists at all levels of training should be aware of the increased incidence of adult GC/GKC. At first presentation it may be mistaken for non-gonococcal bacterial or viral conjunctivitis, so urgent conjunctival swab samples are essential in those with identified risk factors for STIs; elicited through a detailed sexual history. It does not appear that there is a single strain of *N. gonorrhoeae* causing GC/GKC although more isolates are needed to confirm this. Combining clinical and genomic data has highlighted current clinical and epidemiological trends and provides proof of concept for real-time surveillance of GC/GKC, allowing genomic characterisation of outbreak events and novel antimicrobial resistance patterns. In 2025 we are undertaking a UK-wide epidemiological survey on adult GC/GKC, in collaboration with the British Ophthalmological Surveillance Unit (BOSU). This work will foster links between sexual health services, ophthalmologists and UKHSA in collaboratively characterising the epidemiology, clinical presentation and spectrum of GC/GKC, and inform treatment guidelines and a targeted public health response.

Conflicts of interest: None

Funding: None

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P052

ALLERGIC REACTION TO CORNEAL SUTURES ON A BACKGROUND OF EYELID ECZEMA

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Introduction: Allergic reaction to suture material is a rare occurrence, with some reports across surgical specialties relating to various suture materials or suture anchors, and while anecdotal cases have been described, ophthalmic cases specifically remain uncommon. The majority of reports implicate silk and catgut sutures to an IgE mediated immune response inducing a delayed eosinophilic reaction sometimes many years after surgery. Furthermore, reports of unexpected infection and localised inflammatory reaction have been reported in patients with wounds closed using vicryl and vicryl-rapide. Indeed, allergic reaction to polyglactin 910, present in vicryl-rapide, has been documented in gingival graft surgery following which unexpected oral inflammation was observed, and the affected patients made a prompt and full recovery with introduction of oral antihistamines and removal of sutures. Such inflammatory reactions related to suture material use have also been documented in orthopaedic, ear nose and throat, vascular and general surgical patients.

Regarding ophthalmology patients, few cases have been reported relating to postoperative allergic reaction to sutures. Prior to self-sealing corneal wounds constructed during cataract, the silk sutures used had been reported to induce post-operative

complications such as scleritis, corneal wound ulceration and wound dehiscence. Furthermore, allergic reaction to polyglactin 910 sutures used in squint surgery has been described in a paediatric patient, in whom removal of sutures resulted in immediate resolution of symptoms, after failed trial of management with antimicrobials for suspected underlying infection rather than allergic process.

Method: A review of the literature was performed by searching the databases Medline, Embase and PubMed, allowing the retrieval of seven articles on the topic of allergic reactions to suture material from 1984 to 2024.

The case patient's details were reviewed on the RVEEH's electronic medical record mediSIGHT, and the patient's anterior segment photographs both preoperatively and at the time of treatment were compiled.

Results: The case under consideration depicts that of a 21-year-old female patient who underwent routine penetrating keratoplasty (PKP) on a background of keratoconus, making an initially unremarkable recovery. However, upon weaning topical steroids, corneal inflammatory changes became apparent at the suture sites. The clinical signs did not appear to be in keeping with infection or rejection, and on a background of periorbital atopic disease, in this case eczema, a concern for a corneal allergic reaction to suture material was raised, and treatment for allergy management was initiated. This resulted in prompt resolution of the patient's corneal signs, and the patient remains on this treatment until an appropriate time has elapsed since the PKP for removal of sutures.

Review of the literature reveals that allergic reaction to suture material is rare, with some reports across surgical specialties relating to various suture materials or suture anchors, and ophthalmic cases specifically remain uncommon. Most reports implicate silk and catgut sutures to an IgE mediated immune response inducing a delayed eosinophilic reaction sometimes many years after surgery. Regarding ophthalmology patients, prior to self-sealing corneal wounds constructed during cataract, the silk sutures used had been reported to induce post-operative complications such as scleritis, corneal wound ulceration and wound dehiscence. Furthermore, allergic reaction to polyglactin 910 sutures used in squint surgery has been described in the paediatric setting, in whom removal of sutures resulted in immediate resolution of symptoms, after failed trial of management with antimicrobials for suspected underlying infection rather than allergic process.

Conclusion: This case presentation highlights that while allergic reaction to suture materials is rare, notably in the ophthalmic setting, a high index of suspicion is warranted for patients presenting with unexpected inflammatory changes at the site of corneal nylon sutures, in whom a background of atopy is known. While a rare occurrence, prompt administration of high dose steroids can allow for remission of symptoms until an adequate time period has passed to allow corneal sutures removal.

Conflicts of interest: The authors report no conflicts of interest

Funding: None

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P053

BILATERAL CORNEAL ENDOTHELIAL DECOMPENSATION FOLLOWING AIRBAG-INDUCED TRAUMA

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Introduction: Airbag systems are integral to passenger safety during road traffic accidents (RTAs), yet their deployment can also

result in significant ocular trauma. The cornea, a vital structure responsible for focusing light onto the retina, relies on its endothelial layer to maintain transparency through fluid regulation. Damage to these endothelial cells can lead to corneal oedema and visual impairment. This report describes a unique case of bilateral endothelial decompensation following airbag-induced blunt trauma, highlighting the need for vigilance in recognising potential ocular complications in such scenarios.

Method: We present a case of an 89-year-old male who presented with bilateral corneal decompensation following airbag deployment during a road traffic accident.

Results: Ocular examination revealed multi-level ocular injury with severe bilateral corneal oedema. This was managed with topical steroids, mydriatics and, antibiotic therapy. At one month follow up the left eye had persistent corneal oedema. Specular microscopy was performed which confirmed endothelial cell loss making the patient a candidate for a corneal endothelial graft. We observed blunt trauma from airbag deployment causing endothelial cell loss and subsequent corneal decompensation.

Conclusion: Common ocular traumas, such as corneal abrasions, hyphaema, and subconjunctival haemorrhage, are often readily identifiable. In contrast, endothelial damage can be subtle, leading to delayed symptoms such as blurred vision and corneal oedema. Clinicians should maintain a high index of suspicion for endothelial damage in patients presenting with post-accident ocular trauma and consider early referral to specialist services in such cases.

Conflicts of interest: None

Funding: None

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P054

CHARACTERIZATION AND MANAGEMENT OF OCULAR EVENTS IN PATIENTS TREATED WITH BELANTAMAB MAFODOTIN (BELAMAF), BORTEZOMIB, AND DEXAMETHASONE (BVD) IN THE DREAMM-7 STUDY

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Introduction: BVD led to a statistically significant and clinically meaningful progression-free survival (PFS) and early trend toward improved overall survival vs daratumumab+Vd (DvD) in.

RRMM patients after ≥ 1 prior line of therapy (LOT) (DREAMM-7, NCT04246047). We sought to characterize ocular events in BvD-treated patients and assess effectiveness of ocular event management.

Method: Dose delays/reductions from the starting dose of belamaf 2.5 mg/kg IV Q3W to 1.9 mg/kg IV Q3W were allowed to manage adverse events (AEs). Ocular examinations were evaluated and ocular AEs assessed.

Results: BvD arm: 78% had dose delay; 44% had dose reduction due to ocular events. 34% of patients with normal baseline (20/25/better in ≥ 1 eye) reported worsening of best corrected visual acuity (BCVA) to bilateral 20/50/worse. In patients with normal BCVA at baseline, prevalence of BCVA worsening (bilateral 20/50/worse) declined by timepoint concurrent with an increase in median time between doses, suggesting a potential benefit of ocular events by dose delays.

Conclusion: A statistically significant and clinically meaningful PFS benefit favouring BvD vs DVd was observed in RRMM patients after ≥ 1 prior LOT, including the majority of BvD patients who experienced a dose modification due to ocular events. Dose modifications were effective in managing ocular events and allowed most patients to remain on treatment and derive benefit.

Conflicts of interest: **VH:** Honoraria – AbbVie, Amgen, BMS, GSK, Janssen, Sanofi, Takeda. **PR, MH, CF, VZ, KK, MPL, GAM:** None to disclose. **CW:** Speaker fees – AstraZeneca, Bayer, Drivetime Radio. Education Chair for I.S. of Thrombosis and Haemostasias Annual Congress Planning Committee Chair ('24). **PJH:** Advises – Australasian Leukemia & lymphoma Group Trials, Antegene, Gilead, iTeos Therapeutics, Janssen, Pfizer. Research support – Novartis. **RH:** Grants – Janssen, Amgen, Celgene, BMS, Novartis, Takeda. Consulting fees – Janssen, Amgen, Celgene, AbbVie, BMS, Novartis, PharmaMar, Takeda. Honoraria – Janssen, Amgen, Celgene, BMS, PharmaMar, Takeda. Travel support – Amgen, Celgene, Takeda, Janssen. Advises – BMS, Takeda, Amgen, Oncopeptides, Sanofi, Janssen GSK. **MAD:** Honoraria from participation in advisory boards/satellite symposia – Amgen, Sanofi, Regeneron, Menarini, Takeda, GSK, BMS, Janssen, Beigene, Swixx, AstraZeneca. **CC:** Support – Hematology Unit at Instituto Romagnolo per lo Studio dei Tumori. Advises – AbbVie, Amgen, Astellas, Beigene, BMS, Glycomimetics, GSK, Immunogen, Janssen, Jazz, Karyopharm, Menarini-Stemline, Oncopeptides, Pfizer, Sanofi, Servier, Stemline, Takeda. **NP, AM, RR, HB, LE, SRG:** Employed – GSK. Own GSK stocks. **BK:** Employed – GSK. Owns GSK stocks. Travel support – GSK. **ZW, ZK:** Employed – GSK. **JO:** Employed – GSK. Owns GSK stocks. Travel expenses, royalties – GSK. **MVM:** Honoraria – Janssen, BMS, Amgen, GSK, Sanofi, AbbVie, Kite, Stemline, Sanofi, Pfizer. Advises – Janssen, BMS, Amgen, Sanofi, GSK, Roche, Pfizer, AbbVie, Kite, Stemline.

Funding: GSK

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P055

CHARACTERIZATION AND MANAGEMENT OF OCULAR EVENTS IN PATIENTS TREATED WITH BELANTAMAB MAFODOTIN PLUS POMALIDOMIDE AND DEXAMETHASONE IN THE DREAMM-8 STUDY

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Introduction: DREAMM-8 (NCT04484623) demonstrated statistically significant and clinically meaningful progression-free survival (PFS) benefit in patients treated with belantamab mafodotin (belamaf) + pomalidomide + dexamethasone (BPD) vs bortezomib + pomalidomide + dexamethasone (PVD). We describe the management of belamaf ocular events in DREAMM-8.

Method: Starting dose of belamaf was 2.5 mg/kg IV followed by 1.9 mg/kg Q4W. In addition to ocular adverse events (AEs), ocular examination findings were evaluated (KVA scale) and drove dose modifications.

Results: Majority of BPD-treated patients experienced an ocular event that led to dose interruption/delay (83%) or dose reduction (59%) of any component of treatment, discontinuations due to ocular events were low (9%). ≥ 1 ocular event (KVA Grade ≥ 2) was observed in 87% of BPD-treated patients. Worsening of best-corrected visual acuity (BCVA) to bilateral 20/50 or worse at any point in the study from normal baseline (20/25 or better in ≥ 1 eye) was reported in 34% BPD-treated patients. 92% of patients with ≥ 1 ocular event (KVA Grade ≥ 2) continued to receive belamaf, 88% of which achieved partial response or better (post-hoc). First occurrence recovered to normal baseline in 84% patients with an ocular event (KVA Grade ≥ 2 ; post-hoc).

Conclusion: Ocular events in BPD-treated patients were generally reversible and effectively managed by dose modifications. Patients were able to continue treatment and derive benefit.

Conflicts of interest: **HQ, LP, PR, KK, MC:** None.

MAD: Honoraria – Advisory boards/satellite symposia: Amgen, Sanofi, Regeneron, Menarini, Takeda, GSK, BMS, Janssen, Beigene, Swixx, AstraZeneca. **MB:** Honoraria – BMS, Janssen. Consulting/Advisory Role – GSK, Takeda, Amgen, Janssen, Menarini. **SD:** Honoraria – Amgen, GSK, Janssen, Takeda. **VV:** Honoraria – AbbVie, Amgen, AstraZeneca, BeiGene, Gilead Sciences, Janssen, Novartis, Roche, Sanofi, Takeda. Consulting/Advisory Role – AbbVie, AstraZeneca, BeiGene, Gilead Sciences, Janssen Oncology, Novartis, Roche, Sanofi, Takeda. Speakers Bureau – AbbVie, Amgen, AstraZeneca, BeiGene, BioCad, BBMSi, Janssen, MSD, Novartis, Roche, Sanofi, Takeda. Travel Expenses – AstraZeneca. **IS:** Consulting fees – BMS, Amgen, Janssen-Cilag, Jakeda, Sanofi, GSK. Honoraria – Amgen, Janssen-Cilag, Takeda, BMS, Sanofi. Travel Expenses – Janssen-Cilag, BMS, Sanofi. Advises – Janssen-Cilag, Takeda, Sanofi, GSK. **JR:** Consulting fees – GSK, J&J, Sanofi. Honoraria – J&J, Pfizer, BMS, Sanofi, GSK. Travel support – J&J, Sanofi. Advises – GSK, J&J, Sanofi, BMS, Pfizer. **KS:** Honoraria – Janssen Pharmaceutical K.K, Sanofi, BMS. **KM, API, XZ, NS:** Employed, owns stocks – GSK. Travel expenses–GSK. **FP:** Employed, owns stocks – GSK. **JW, MP, GF, BK, ZK:** Employed – GSK. **JO:** Employed, owns stocks – GSK. Travel expenses/royalties – GSK. **MVM:** Honoraria – Janssen, BMS, Amgen, GSK, Sanofi, AbbVie, Kite, Stemline, Sanofi, Pfizer. Advises – Janssen, BMS, Amgen, Sanofi, GSK, Roche, Pfizer, AbbVie, Kite, Stemline. **ST:** Honoraria – Amgen, Forus, GSK, Janssen, Pfizer, Sanofi. Advises – BMS, GSK, Roche. Research Funding – Amgen, BMS, Genentech, GSK, Janssen, K36 Therapeutics, Pfizer, Roche.

Funding: GSK

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P056

COMPARATIVE ANALYSIS OF BACTERIAL ISOLATION RATES FROM CORNEAL SCRAPES AND CORNEAL IMPRESSION MEMBRANES IN CASES OF SUSPECTED MICROBIAL KERATITIS

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Introduction: Corneal impression membranes (CIM) are an easier, safer, and quicker way to obtain samples in suspected microbial keratitis. This study compares the level of agreeability in the isolation rates of bacteria using CIM and corneal scrapes (CS) in patients with suspected microbial keratitis.

Method: Retrospective review of 116 consecutive patients with suspected microbial keratitis requiring corneal sampling. CS was performed with sterile 21G needles; samples were inoculated onto chocolate, Sabaroud and non-nutrient agar, and a sample sent in brain heart infusion (BHI) broth. CIM was performed by placing a sterile 12 mm polytetrafluoroethylene disc on the cornea for 3 seconds and transferred to the laboratory in BHI broth which was vortexed for 5–10 seconds, and 10 µL of broth inoculated onto agar plates. The level of agreeability in bacteria isolation between the paired samples was compared using Cohen's kappa.

Results: Micro-organisms was isolated in 74/116 (63.8%) CIM samples and in 84/116 (72.4%) CS samples ($p = 0.12$). The four commonest micro-organisms were *P. aeruginosa* (19.0% CIM; 20.7% CS), *S. epidermidis* (24.1%; 25.9%), *S. aureus* (6.9%; 7.8%), and *S. warneri* (3.4%; 7.8%). The respective sensitivities and specificities of CIM compared to CS for the four commonest micro-organisms were: 0.88 and 0.99 for *P. aeruginosa*, 0.27 and 0.77 for *S. epidermidis*, 0.56 and 0.97 for *S. aureus*, and 0.11 and 0.9 for *S. warneri*. The level of agreeability between CIM and CS was 0.89 for *P. aeruginosa*, 0.00 for *S. epidermidis*, 0.56 for *S. aureus*, and 0.11 for *S. warneri*.

Conclusion: CIM is an effective method in collecting corneal samples in patients with suspected microbial keratitis, with a high level of agreeability with CS in the isolation of *P. aeruginosa*, and a moderate level of agreeability in the isolation of *S. aureus*. The discordance between both methods of corneal sampling were mainly due to differences in coagulase negative *Staphylococci*, which can be of uncertain clinical significance.

Conflicts of interest: None

Funding: None

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P057

CORNEAL EPITHELIAL CHANGES IN EYES WITH POST TRAUMATIC RECURRENT CORNEAL EROSION: COMPARISON WITH FELLOW EYES AND HEALTHY CONTROLS

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Introduction: Trauma is a common cause of recurrent corneal erosion (RCE); however, only a few patients develop RCE after trauma. This study aims to assess corneal epithelial thickness

changes in both affected and fellow eyes of patients with post-traumatic RCE and compare these with healthy age-matched controls.

Method: This is an observational study of patients presenting with traumatic RCE during a three year period. Corneal epithelial thickness data was obtained from high-resolution spectral domain anterior segment Ocular coherence tomography (OCT) (Solix, Haag-Streit, Germany) at the follow-up visit in the cornea clinic when the erosion had clinically healed. Data was recorded for both affected (Group 1) and fellow eye (Group 2) and for healthy age-matched controls (Group 3). The comparison was made between all the groups. The primary outcome measure was a comparison of 'E-Std Dev' values (variation in epithelial thickness (ET)). Secondary outcome measures were central ET, minimum ET, maximum ET, superior ET, inferior ET and E(min-max) values along with qualitative map assessment, which was performed by analysing the number of zones with thicknesses 60–70 µm and 70–80 µm.

Results: Thirty-three patients (66 eyes) and 33 age-matched healthy eyes were included.

Those with pre-existing corneal dystrophies were excluded. E-Std Dev values were significantly different between the three groups ($P < 0.01$), with highest values in Group 1 and lowest in Group 3 ($4.65 \pm 1.87 \mu$, $2.92 \pm 1.16 \mu$ and $1.94 \pm 0.63 \mu$ in Groups 1, 2 and 3 respectively); the difference was maintained on post-hoc analysis between all three group pairs. A similar trend was also noted for central ET ($P < 0.01$), maximum ET ($P < 0.01$), E(min-max) values ($P = 0.03$) and qualitative map assessment ($P < 0.01$). There were no significant inter-group differences for minimum ET ($P = 0.05$), superior ET {E-S (2–7)} ($P = 0.45$) and inferior ET {E-I (2–7)} ($P = 0.68$).

Conclusion: We report corneal epithelial mapping changes in patients with traumatic recurrent corneal erosion. As can be expected, the maximum ET and variation in ET were highest in the affected eyes; however, even the fellow eyes showed significantly higher values than healthy control eyes. This may suggest a predisposition of some patients to develop RCE after trauma. Further prospective studies would be useful to shed more insight into this topic.

Conflicts of interest: None

Funding: None

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P058

EVALUATION OF A KERATOCONUS NURSE-LED CLINIC AND CROSS-LINKING SERVICE OVER A SEVEN YEAR PERIOD IN NHS LANARKSHIRE

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Introduction: Keratoconus is a progressive corneal disorder characterized by thinning and steepening of the cornea which can lead to visual impairment and cause a significant negative impact on the quality of life of young patients. Corneal collagen cross-linking (CXL) halts progression of keratoconus. However, due to the need to i) monitor keratoconus on a regular basis for signs of progression, and ii) perform collagen cross-linking, delivering a keratoconus monitoring and cross-linking service carries a significant demand on Hospital Eye Service resources.

The purpose of this study is to evaluate the implementation of an Allied Health Professional (AHP)-led keratoconus clinic and CXL service at University Hospital Hairmyres, NHS Lanarkshire, over a

7-year period from 2017–2024, assessing its safety and impact on patient waiting times, service capacity and clinical outcomes. The service was initially run by senior optometrists and ophthalmic advanced nurse practitioners (ANPs), but due to restructuring the service is now solely ANP-led.

Method: A retrospective audit of patients who attended the ANP-led keratoconus clinic to evaluate service efficiency and capacity, including the number of patients undergoing CXL annually. CXL procedures were performed in an outpatient clean room under a standardized sterilized protocol by a fully trained and experienced AHP without consultant supervision. Patients were followed up with a telephone consultation at 1 week, a face-to-face consultation at 4 weeks, and repeat corneal topography at 3 months. Clinical outcomes assessed included visual acuity (VA), corneal curvature and thickness to monitor disease progression, as well as complication rates.

Results: In the 12-month period from December 2023 to November 2024, the ANP-led service managed 684 patients (face-to-face, $n = 360$; virtual, $n = 324$). Patients had a mean age of 32 years (range 13–67), 67% of whom were male. At the time of referral, the mean best corrected visual acuity (BCVA) was 0.6 ± 0.27 LogMAR and mean spherical equivalent (MSE) refractive error was $-1.96D$.

Over the seven-year period of 2017–2024, 269 eyes of 250 patients underwent CXL under local anaesthetic. 57% were male, mean age was 27 ± 7 years (range 16–46). At the time of listing, the mean Kmax was $61.0 \pm 9.1D$, and the thinnest corneal thickness was 472 ± 42 microns. On average, patients had two clinic appointments before being listed for CXL, with a mean waiting time of 9 ± 6 weeks from clinic listing to procedure.

No individuals lost ≥ 2 lines of visual acuity. There were no cases of topographical progression requiring repeat CXL or corneal transplant and no episodes of microbial keratitis. Two patients developed immune mediated post-procedural stromal haze at one month follow-up which settled with and extended course of topical steroids. One patient required two months treatment with neuropathic analgesia which settled.

Conclusion: The ANP-led approach addresses the growing demand for keratoconus treatment while maintaining patient safety and satisfaction. ANP-led outpatient collagen cross-linking offers a safe, effective, and timely alternative to theatre-based procedures. This service has increased departmental capacity, reduced waiting times, and allowed senior clinicians to prioritize complex surgeries and urgent sight-threatening cases. This model improves keratoconus care and optimizes ophthalmology services, though further research is required to evaluate long-term outcomes and broader applicability.

Conflicts of interest: None

Funding: None

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P059

GRAFT SURVIVAL, GRAFT REJECTION, AND VISUAL OUTCOMES AFTER DESCMET STRIPPING ENDOTHELIAL KERATOPLASTY FOR GRAFT FAILURE

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Introduction: To investigate outcomes of Descemet stripping endothelial keratoplasty (DSEK) in eyes that were re-grafted for failed penetrating keratoplasty (PK) or DSEK.

Method: Retrospective cohort study of all DSEK cases (including learning curve) from April 1, 2006 to November 30, 2015 for PK or DSEK graft failure. Graft outcomes, best spectacle-corrected visual acuity (BSCVA), and ocular comorbidities were recorded. Graft survival, re-bubble rate, and rejection risk were estimated by Kaplan-Meier analysis.

Results: Of 80 DSEK grafts in 78 eyes of 76 subjects, 42 (52%) were for failed PK and 38 (48%) for failed DSEK. Mean age was 72 years (range: 14 to 97 years); 52 (68%) subjects were female. Original indications for corneal transplantation included Fuchs endothelial corneal dystrophy ($n = 46$, 61%) and pseudophakic corneal oedema ($n = 11$, 14%). Median follow-up was 4.0 years. Overall graft survival was 90%, 71%, 57%, and 45% at 1, 5, 10, and 12 years, respectively. Graft survival in the failed PK group was 82%, 64%, and 41% at 1, 5, and 10 years, respectively, and the failed DSEK group was 97%, 75%, and 71% at 1, 5, and 10 years, respectively ($p = 0.09$). Of 25 cases of graft failure, 9 (36%) were early (primary or iatrogenic) graft failure, 13 (52%) were late endothelial failure, and 3 (12%) were from secondary aetiologies. The risk of graft rejection was 15% at 5, 10, and 12 years (failed PK group was 24% at 5 and 10 years, and failed DSEK group was 9% at 5 and 10 years [$p = 0.15$]). Mean graft diameter was 7.7 mm (SD: 0.45) in the PK group versus 8.0 mm (SD: 0.40) in the DSEK group ($p = 0.15$). A re-bubble procedure was performed in 17 (20%) cases; 13 (31%) were in failed PK cases and 4 (10%) in failed DSEK ($p = 0.03$). For cases without vision-limiting comorbidities, mean pre-operative BSCVA was 20/100 in subjects with failed PK and 20/60 for failed DSEK ($p = 0.03$). Mean BSCVA at last follow-up was 20/50 for failed PK and 20/30 for failed DSEK ($p = 0.009$).

Conclusion: Compared to DSEK regrafts for failed PK, DSEK regrafts for failed DSEK result in better vision, lower re-bubble rates, and non-significant trends towards longer survival and lower risk of rejection.

Conflicts of interest: None

Funding: Mayo Foundation for Medical Education and Research.

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P060

INTRAOPERATIVE DESCMET MEMBRANE ENDOTHELIAL KERATOPLASTY GRAFT LOSS INTO THE VITREOUS CAVITY

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Introduction: Descemet membrane endothelial keratoplasty (DMEK) is a type of endothelial keratoplasty introduced in 2006 which involves transplantation of Descemet membrane and the endothelial cell layer only. It has added advantages of lower rejection rate and quicker recovery over other types of endothelial keratoplasties such as Descemet Stripping Automated Endothelial Keratoplasty (DSAEK) and full-thickness penetrating keratoplasties. Endothelial keratoplasties are generally indicated in patients with corneal endothelium pathology such as bullous keratopathy, chronic uveitis or secondary glaucoma. In eyes with previous vitreoretinal surgery, it can pose unique surgical challenges for endothelial keratoplasties. There were few published reports of DSAEK and DMEK grafts dislocated into the posterior segment; a very rare complication of corneal graft procedure in eyes with previous vitrectomy.

Method: We illustrated a rare case of intraoperative posterior dislocation of DMEK graft of a pseudophakic patient (intraocular

lens (IOL) in the posterior capsular bag), which was then successfully retrieved via pars plana approach.

Results: A gentleman in his mid-60s with previous multiple retinal detachment surgeries, successful cataract surgery and Nd: YAG posterior capsulotomy in the right eye presented with bullous keratopathy requiring Descemet membrane endothelial keratoplasty (DMEK). Intraoperatively during DMEK procedure, the Descemet membrane graft dislocated posteriorly during unfolding manoeuvres. The procedure was abandoned and revision surgery (retrieval of dislocated DMEK graft and penetrating keratoplasty) was performed a week later. Postoperatively, intraocular inflammation was controlled with high dose intravenous corticosteroid as well as topical steroid medications. The graft remained viable and retina attached.

Conclusion: This is the first reported case demonstrating that penetrating keratoplasty can be safely performed at the same setting as the early retrieval of the posteriorly dislocated DMEK graft using a pars plana approach and a temporary keratoprosthesis. Although very rare, in eyes with unstable lens-iris diaphragm and prior vitrectomy, posterior DMEK graft dislocation becomes a real risk. We showed that posterior migration of an endothelial graft can still happen with the IOL in the capsular bag supported by a competent iris. When posterior DMEK graft dislocation occurs, there is limited evidence in the literature whether the graft can be left in the posterior segment without causing retinal fibrosis, PVR and RD. If the dislocated DMEK graft is to be retrieved, its timing and best practice techniques remain undecided.

Conflicts of interest: None

Funding: None

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P061. Abstract not included

P062

KERATOCONUS AND ASSOCIATED HEALTH CONDITIONS - ANY CAUSE FOR ALARM?

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Introduction: The aim of the study was to identify the characteristics of keratoconus patients attending a large tertiary referral centre in the United Kingdom using a questionnaire. The authors wished to establish any relationships between Keratoconus and mental health attributes such as depression and anxiety with the view of adapting services to improve patient experience.

Method: A prospective data collection was conducted using a questionnaire designed and registered as an audit with the Clinical Governance department of Sandwell & West Birmingham Hospitals NHS Trust. The questionnaire was completed by 110 Keratoconic patients attending the Contact Lens Clinics at the Birmingham & Midland Eye Centre, following verbal consent.

Results: The average age was 38.8 ± 12.8 years, 64% male. The average age at diagnosis was 22.0 ± 6.5 year and only 24% had a family member diagnosed with keratoconus. Two-thirds reported eye rubbing. Glasses were worn by 16%, soft contact lenses by 7%, rigid corneal lenses by 50% and scleral lenses by 23%. 38% had been cross-linked, of which 56% found this helped. The highest level of education was GCCE/BTEC in 22%, apprenticeship in 11%, A levels/higher BTEC in 20%, degree in 38% and post

degree in 8%. 70 were licenced to drive and 77% were currently employed, with just 15% on benefits. Associated health conditions included eczema (25%), asthma (23%), hay fever (50%), psoriasis/dermatitis (10%), diabetes (7%), systemic hypertension (10%) and heart condition (5%), but none reported a connective tissue disorder. One in five (21%) were on medication. Fourteen percent had a diagnosed mental health condition, but just 32% were a member of a support group.

Conclusion: In this UK sample, rigid corneal lenses is still the main form of refractive correction for keratoconics, being twice as common as scleral lenses. High rates of associated systemic conditions are reported, higher than in the general population at this age. Mental health has been diagnosed in one in seven, but support group membership remains low. This knowledge should aid clinicians in talking with their patients with keratoconus, to guide them to more appropriate support.

Conflicts of interest: None

Funding: None

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P063

OPTIMISING THE CLINICAL IMPLEMENTATION OF CORNEAL INFECTION DISC TESTING

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Introduction: In microbial keratitis, Corneal Impression Membranes have been proven to have higher microorganism yields than corneal scrapes. However, introduction of the technique into routine clinical practice has been slow. The aim of this study was to create and evaluate the effectiveness of a package of new teaching and business case materials to support local implementation and then share these materials to facilitate wider roll out across eye units.

Method: A five-step multimodal training package was created to teach the procedure. Pre-course learning was a step-by-step demonstration video. In-person teaching comprised a presentation and live demonstration, taking 20 minutes. Post-course consolidation was an article providing further background. Subsequently for each learner, an RCOphth DOPS (Direct Observation of Practical Skills) Form was used to assess competence and facilitate 2-way structured feedback. Following local clinical implementation, time to complete the procedure and qualitative feedback on patient and clinician experience of corneal impression membranes was compared to traditional corneal scrapes. This data, together with information on costings and a flow chart of steps for introduction of the technique, were incorporated into an implementation package to be shared with other units.

Results: 69 clinicians (23 doctors, 9 ACPs and 37 ophthalmic nurses) attended the teaching. Across all 3 groups, self-assessed confidence ratings significantly increased for knowledge ($p < 0.001$), obtaining consent ($p < 0.0001$), performing the procedure ($p < 0.0001$) and teaching it ($p < 0.001$).

On DOPS assessment, 93% of clinicians who attended teaching were competent to perform testing unsupervised or under distant supervision, compared to 20% of those who did not attend teaching ($p < 0.001$).

Following observation of performance and feedback discussions, improvements were identified for explanation of the role of results, adequate anaesthesia to minimise contamination and ease of equipment handling. The technique was renamed

“Corneal Infection Disc Testing” to enhance patient and clinician understanding, and the demonstration video was edited.

Financial analysis of materials and sample processing demonstrated that the cost of the two procedures was similar. Following local clinical implementation, corneal infection disc testing received better feedback from both patients and clinicians and was found to be more time-efficient compared to performing corneal scrapes.

Conclusion: Five-step teaching of Corneal Infection Disc Testing was effective in developing competence in all members of the multidisciplinary team, and supporting implementation into clinical practice. This evidence, together with data on financial models, efficiency and acceptability have been incorporated into an example business case. A package of refined teaching materials as well as the business case resources have been widely shared via the RCOphth online learning platform, INSPIRE, and also sent to units requesting it, to facilitate multicentre implementation of Corneal Infection Disc Testing.

Conflicts of interest: None

Funding: None

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P064

PATIENT CHARACTERISTICS AND OCULAR ADVERSE EVENTS MANAGEMENT AFTER TREATMENT WITH BELANTAMAB MAFODOTIN IN RRMM: A REAL-WORLD STUDY IN EUROPE

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Introduction: This multi-national European study examined real-world evidence on use, effectiveness and safety of belamaf, including management of ocular adverse events (OAEs).

Method: In this non-interventional, prospective study, 77 pts with RRMM received belamaf monotherapy per approved labelling. Primary objective: To characterize pt demographics and disease characteristics. A key secondary objective: Characterization of OAEs. Interim analysis.

Results: Overall, 58% (45) pts had OAEs (keratopathy [43%], change in best corrected visual acuity [8%], blurred vision [5%], corneal erosion [5%], dry eye [1.3%]). 21 pts had OAEs after 1 belamaf dose, 15 after 2 doses, 6 after 3 doses, 2 after 4 doses and 1 after ≥5 doses, with keratopathy in 15, 12, 5 and 1 pt after 1 dose, respectively. Based on label guidance, OAEs were managed

with dose holds in 29% and dose reductions in 17%; 7 pts discontinued treatment due to OAEs.

Conclusion: OAEs in pts with RRMM receiving belamaf were managed per label guidance with dose holds/reductions.

Conflicts of interest: **MC:** Honoraria – Janssen, Celgene, BMS, Sanofi, Amgen; **MD:** Honoraria – Amgen, BMS, Janssen, GSK, Sanofi, Stemline. Speaker's bureau – Amgen, BeiGene, BMS, Janssen, Takeda; **M-AD:** Speaker's bureau – Amgen, BeiGene, BMS, Janssen, Takeda; **FE:** Advises – Amgen, BMS, GSK, Janssen, Sanofi, Takeda; **MH:** Consultant – Curio Science LLC, GSK, Intellisphere; research funding – Amgen, Daiichi Sanko, GSK, SpringWorks Therapeutics; Advises – GSK, BMS; **DK:** Consultant – GSK, Triphase Accelerator; Advises and has ownership interests – Calm Waters Therapeutics LLC; **HL:** Consultant – BMS, Celgene, Genentech, Janssen, Regeneron, GSK, Sanofi, Pfizer, Monte Rose Therapeutics, Takeda Pharmaceuticals, Allogene Therapeutics; research funding – Amgen, BMS, Janssen, GSK, Regeneron, Takeda; **RV:** Grant support – BMS, Sanofi, Takeda; honoraria – Adaptive, Beigen, BMS, GSK, Janssen, Oncoceptides, Sanofi, Takeda; **RG:** Stock/ownership interests – Novo Nordisk (Inst), Lilly (Inst); honoraria – Celgene, Roche, Merck, Takeda, AstraZeneca, Novartis, Amgen, Bristol Myers Squibb, MSD, Sandoz, AbbVie, Gilead Sciences, Daiichi Sankyo, Sanofi; consulting/advisory fees – Celgene, Novartis, Roche, BMS, Takeda, AbbVie, AstraZeneca, Janssen, MSD, Merck, Gilead Sciences, Daiichi Sankyo, Sanofi; research funding – AbbVie, to the Inst – Celgene, Merck, Takeda, AstraZeneca, Novartis, Amgen, BMS, MSD, Sandoz, Gilead Sciences, Roche, Daiichi Sankyo Europe GmbH; travel, accommodations/expenses – Roche, Amgen, Janssen-Cilag, AstraZeneca, Novartis, MSD, Celgene, Gilead Sciences, BMS, AbbVie, Daiichi Sankyo; **TM:** Honoraria – GSK; **EA:** Advises – Amgen, Janssen, Pfizer, Sanofi, Takeda; **FS:** Consultant – AbbVie, Celgene, GSK, Janssen, Oncoceptides; research funding – Celgene, GSK, Janssen, Oncoceptides, Targovax, Sanofi; honoraria – AbbVie, Amgen, BMS, Daiichi Sankyo, GSK, Janssen, Novartis, Oncoceptides, Pfizer, Sanofi, SkylineDx, Takeda; **AL:** None; **JM, NK, LC, JaB, TdE, MF:** Employees, financial equities – GSK; **ZK:** Employee – GSK; **JuB:** Employee – GSK; financial equities – GSK, Adaptimmune, Novartis; **CV:** Employee – Syneos Health.

Funding: GSK

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P065

PERFORMING DESCMET MEMBRANE ENDOTHELIAL KERATOPLASTY (DMEK) CORNEAL TRANSPLANTATION WITH THE USE OF AN ENDOGLIDE PULL-THROUGH INSERTION DEVICE (E-DMEK)

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Introduction: The purpose of this study is to describe the surgical technique and evaluate the early clinical outcomes of Descemet Membrane Endothelial Keratoplasty (DMEK) using a pull-through, “endothelium-in” insertion device, the DMEK EndoGlide (E-DMEK).

Method: This retrospective cohort study analysed patients who underwent E-DMEK between February 2023 and July 2024 at University Hospital Hairmyres. Tri-folded DMEK grafts were delivered into the anterior chamber using a pull-through technique via the EndoGlide (Network Medical, Ripon, UK). Grafts were pulled through the device using 27-gauge forceps and unfolded in the anterior chamber with the graft positioned

endothelium down. An 8.5-mm graft size was utilized in all cases. Clinical outcomes included best-corrected visual acuity (BCVA), post-operative complications, need for rebubbling, and rates of graft failure or decompensation.

Results: The study included 28 eyes of 24 patients who underwent DMEK alone ($n = 21$) or in combination with cataract surgery and intraocular lens implantation ($n = 7$). Most cases ($n = 23$) had corneal decompensation secondary to Fuchs endothelial corneal dystrophy (FECD), with other indications including angle-closure glaucoma ($n = 2$), HSV endothelitis ($n = 2$), and endothelial dystrophy of unknown aetiology ($n = 1$). The mean patient age was 69 years (range: 58–78 years) while the mean donor age was 74 years (range: 60–86 years). The mean endothelial cell count of donor tissue was 2699 ± 402 cells/mm².

At minimum 5 months follow-up (range 5–22 months), all patients had an attached and clear corneal endothelial graft. Mean BCVA in LogMAR improved from 1.1 ± 0.60 (range 0.5–2) pre-operatively to 0.14 ± 0.15 (range –0.1–0.4) post-operatively ($p < 0.01$). Six patients experienced partial detachment, with four requiring “re-bubbling” at the slit lamp, and no interventions were required for the other two. One patient developed cystoid macular oedema (CMO) post-operatively, which necessitated an Ozurdex injection. There were no episodes of graft failure.

Conclusion: E-DMEK using the EndoGlide pull-through technique demonstrates promising early outcomes with good visual recovery and a low rate of complications. This method offers a controlled and effective approach to DMEK graft delivery, particularly in challenging surgical scenarios such as in cases with iris clipped intraocular lenses, glaucoma tubes, or vitrectomised eyes. Further studies are warranted to assess long-term outcomes.

Conflicts of interest: None

Funding: None

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P066

RELIABILITY OF TEAR BREAK UP PATTERNS USING SMART EYE CAMERA AS NOVEL DRY EYE DISEASE SCREENING MODALITY

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Introduction: Dry eye disease (DED) is a common ocular surface disorder that is often overlooked in practice, either by patients themselves or by physicians. DED alters the tear film layers that protects the ocular surface, which might result in severe visual impairment or even blindness. DED is diagnosed by various methods, ranging from conventional methods such as Schirmer's test, to advanced ones such as fluorescence test in tear break up time (TBUT) and pattern (TBUP), or keratography. To prevent advanced complications on the ocular surface, early screening is needed for DED even at the most basic healthcare level. However, not all diagnostic instruments are available to make the diagnosis possible without referring to a specialised clinic. A novel diagnostic method of DED is needed to overcome the challenge, that enables general practitioners (GPs) to conduct early screening and remote consultation with consultants by relying on documentation of photos or videos that visualise the patient's condition. TBUP allows DED screening and type identification, based on the pattern surfaced when the tear film breaks and its

correlation with the tear film distribution on the ocular surface. Smart Eye Camera (SEC) is a smartphone add-on device that allows physicians to record, store, or send videos of TBUP for consultation purposes. Using a telemedicine concept, DED screening could be conducted easily and enable tailor-made management based on specific findings. This research investigated the intraobserver and interobserver reliability of TBUPs recorded with SEC as a screening modality of DED.

Method: The cross-sectional research involved 161 eyes of 83 subjects aged above 17 years who visited the eye clinic at Atma Jaya University Hospital, Jakarta, Indonesia. Fluorescence test was conducted and recorded using SEC. TBUPs were observed from the videos recorded and analysed by two independent observers on two different occasions, seven days apart. TBUPs were categorized into one of five following break types: area (AB), line (LB), spot (SB), dimple (DB), and random (RB). The results from each observer were compared and analysed using Cohen's Kappa reliability test, and the reliability was considered as good agreement for TBUP analysis if the kappa coefficient > 0.60 , with p value < 0.05 represented statistical significance for 95% confidence interval.

Results: The mean age of involved subjects was 64.84 ± 8.63 years, consisted of 80.72% female and 19.23% male. The intraobserver reliability showed good agreement within each observer between two different video reviews (0.703 and 0.917; $p < 0.001$). There was a lack of agreement between observers that was reflected by poor interobserver reliability on both reviews conducted (0.202 and 0.218; $p < 0.001$). The result shows each observer can review and conduct DED screening consistently based on the videos recorded by SEC, but the lack of generalised criteria to determine each pattern affected the poor agreement between observers.

Conclusion: TBUPs recorded with SEC have great potential as a novel DED screening modality, as it enables the tear film assessment consistently by physicians. The lack of agreement between observers reveals that further research is needed to compose a detailed definition and criteria to determine each pattern, that allows DED screening and consultation to be conducted from remote areas.

Conflicts of interest: The research is sponsored by OUI Inc., Japan that leads to the development of Smart Eye Camera (SEC), which was granted to be used during this research.

Funding: This research was funded by School of Medicine and Health Sciences, Atma Jaya Catholic University of Indonesia research grant.

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P067

SCREENING PLUS CORNEAL CROSS-LINKING FOR KERATOCONUS IS COST-EFFECTIVE FOR NEW ZEALAND: A PROOF-OF-CONCEPT MARKOV ANALYSIS

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Introduction: Studies have shown that the prevalence of keratoconus in New Zealand is higher compared to the global prevalence of 1.38 per 1000, with Māori and Pacific Islander being overrepresented. Studies have also suggested that the form of keratoconus in New Zealand may have a more rapid progression

of disease. Corneal cross-linking have demonstrated efficacy in slowing or arresting the progression of keratoconus. In this study, we aimed to estimate the cost-effectiveness of introducing screening for keratoconus in the New Zealand context, with corneal cross-linking as treatment for those screening positive.

Method: A Markov simulation was used to model the impact of screening plus corneal cross-linking compared to usual care across a lifetime horizon and societal perspective with a 3% discount rate. Cost-effectiveness was determined by the incremental cost-effectiveness ratio (ICER), with utility measured in quality-adjusted life-years (QALYs). Multivariate sensitivity analyses were carried out to investigate factors influencing cost-effectiveness.

Results: The ICER for screening plus corneal cross-linking was cost-saving at -NZ\$24,359, well below the willingness-to-pay threshold of NZ\$76,543 per QALY gained. Probability sensitivity analysis showed that prevalence of keratoconus and the costs of being in keratoconus/severe keratoconus states had the greatest impact on the ICER.

Conclusion: Keratoconus is a progressive disease with onset during puberty and early adulthood, leading to significant clinical, economic and social disadvantages. In the New Zealand setting, screening for keratoconus, with corneal cross-linking for children who screened positive and continued to progress despite standard spectacles or lenses, is likely to be cost-saving.

Conflicts of interest: None

Funding: None

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P068

SURVEY OF TOPICAL STEROID USAGE PATTERNS AFTER DESCMET MEMBRANE ENDOTHELIAL KERATOPLASTY (DMEK)

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Introduction: Descemet's Membrane Endothelial Keratoplasty (DMEK) has become the mainstay of surgical management for corneal endothelial dysfunction such as Fuchs' endothelial corneal dystrophy (FECD) and pseudophakic bullous keratopathy (PBK). Topical steroids are utilised postoperatively to reduce the risk of allograft rejection. However, their long-term use can result in adverse effects, and contributes to medication burden and waste. In addition, there is no nationally or internationally agreed consensus on the duration of post-operative topical steroid use for endothelial keratoplasties. This cross-sectional study aimed to investigate the current global practice amongst corneal surgeons with regards to topical steroid treatment regimens after DMEK surgery.

Method: A nine-item electronic survey with free text responses was sent to ophthalmologist members of the Cornea Society in March, 2024. The questionnaire explored surgeons' preferred postoperative steroid regimens in three scenarios: 1) post-DMEK for FECD, 2) post-DMEK for PBK, and 3) post-repeat DMEK after previous graft failure. All responses to the survey were anonymous. Incomplete survey responses were excluded prior to analysis.

Results: 78 ophthalmologists from 15 countries completed the online questionnaire. Respondents performed a median of 37.5 grafts per year (mean 45.8; SD 46.7; range 2–300). Prednisolone

1% was the preferred topical steroid during the 12 months post-DMEK for FECD, in the 24 months post-DMEK for PBK, and after 24 months post-repeat grafts. Almost half (44%; n = 34) of surgeons discontinued topical steroids at 24 months post-DMEK for FECD. Of the remaining 44 surgeons, 53% (n = 24) continued topical steroids indefinitely. Similarly, 36% of surgeons stopped topical steroids at 24 months post-DMEK for PBK, while 54% continued indefinitely. For cases of repeat DMEKs following previous graft failure, 63% of surgeons continued topical steroids indefinitely. Long-term follow-up regimens were also variable, with 45% of surgeons discharging their patients between 1 and 5 years post-operatively. The remaining respondents (55%) continued to follow up their patients indefinitely.

Conclusion: This international survey highlights significant variability in the duration of post-operative topical steroid use following DMEK for FECD, PBK and graft failure. Given this heterogeneity, further studies are needed to look at graft rejection and failure rates with different postoperative topical steroid regimens. An international consensus would aid in standardising post-operative care for routine cases, may reduce unnecessary waste, and lower CO2 footprints.

Conflicts of interest: None

Funding: None

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P069

THE CORNEAL MICROBIOME IN HEALTH AND MICROBIAL KERATITIS: NEW INSIGHTS FROM SHOTGUN METAGENOMIC DATA

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Introduction: The corneal microbiome and its role in the development of microbial keratitis (MK) are poorly understood. To address this knowledge gap, we conducted a prospective observational diagnostic study to investigate the corneal microbiome in both healthy individuals and patients with MK infection.

Method: A minimally invasive corneal sampling method (corneal impression membrane, CIM) was used to sample the affected and non-affected eyes of patients with MK as well as the corneas of healthy control patients. Samples were processed using traditional culture methods and shotgun metagenomic sequencing. Shotgun libraries were generated and sequenced on the Illumina NovaSeq platform. Taxonomic assignment of clean reads was performed with Kraken2 and a customised reference database containing genomes for archaea, bacteria, fungi, viruses, plants and protozoa. Environmental samples processed through the same pipeline were used to identify and remove contaminating taxa. A positive shotgun metagenomic result was defined as the presence of a dominant microorganism (>20% of the microbiome) after removing background taxa.

Results: Of the samples collected, a total of 121/142 (85.2%) affected eye samples, 106/141 (75.1%) non-affected eye samples, and 6/18 (33.3%) healthy control samples passed quality control for shotgun metagenomic analysis. Culture identified microorganisms in 37.5% (51/136) of samples, while shotgun metagenomics detected a dominant microbial species in 68.6% (83/121) of affected eye samples, with a 36.5% genus-level concordance between methods. 64.2% (54/84) of culture-negative cases yielded positive results with shotgun metagenomics. Shannon diversity was significantly lower in affected eyes compared to

non-affected eyes ($p < 0.05$) with the largest reductions associated with the shotgun metagenomic detection of *Morganella* ($\beta = -3.21$, $p < 0.001$), *Pseudomonas* ($\beta = -3.09$, $p < 0.001$) and *Streptococcus* ($\beta = -2.94$, $p < 0.001$). Medium size infiltrates (2–5 mm) also significantly reduced Shannon diversity compared to smaller size infiltrates (< 2 mm) ($\beta = 0.31$, $p = 0.034$). Beta diversity showed overlapping microbiomes between affected and nonaffected eyes of MK participants, with healthy controls clustering distinctively. High microbiome similarity was observed between bilateral MK eyes ($n = 3$), while significant temporal variation occurred in those patients that had two recurrent episodes within the study period ($n = 4$). Virulence factor genes, including exotoxins, were successfully identified from *P. aeruginosa* metagenomic assembled genomes.

Conclusion: This study is the first to compare the corneal microbiome of affected and non-affected eyes of MK patients and healthy controls using shotgun metagenomic sequencing, providing new insights into MK pathogenesis. Shotgun metagenomics yielded positive results in many culture-negative cases, highlighting its potential as a diagnostic tool. It revealed that certain taxa dominate the microbiome in affected eyes, while others exert less influence.

Significant microbial overlap between affected and non-affected eyes suggests MK patients may harbour at-risk corneal microbiomes. Additionally, shotgun metagenomics identified key virulence factors, demonstrating potential value in characterising pathogenic mechanisms.

Conflicts of interest: None.

Funding:

- Medical Research Council (Grant Ref: MR/S022104/1).
- European Society of Cataract & Refractive Surgeons (ESCRS) (Grant Ref: META-COR 2).

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P070

THE EFFICACY AND TOLERABILITY OF FINGER-PRICK AUTOLOGOUS BLOOD EYE DROPS FOR THE TREATMENT OF OCULAR SURFACE DISEASE: A LITERATURE REVIEW

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Introduction: Ocular surface disease (OSD) arises from disorders in the lacrimal and meibomian glands, as well as the corneal and conjunctival epithelium causing a range of symptoms such as dryness, pain and irritation. It is typically caused by chronic topical glaucoma treatment and autoimmune disease, resulting in tear film instability and ocular surface inflammation. Blood-derived products were first used on the ocular surface in 1975. Serum has similar a composition to endogenous tears, which can help with treating OSD. However, there are concerns with allogeneic serum due to transmission of blood-borne viruses and hypersensitivity reactions, as well as autologous serum due to issues with collection and production. This literature review investigated the potential benefits of using finger-prick autologous blood (FAB) eye drops as an alternative treatment option.

Method: The study aim was to investigate the efficacy and safety of FAB eye drops on patients with OSD.

A search strategy was conducted using PubMed, Cochrane Library and Scopus. Studies that were included were published in the last 10 years and clinical trials on human subjects.

Results: The search strategy yielded 48 publications across the three databases. After analysis of these papers, 4 of the publications were deemed eligible. Three of the papers were prospective, non-comparative, interventional case series and one paper was a single-masked, multi centre, randomised controlled trial. There were 75 participants across the four trials. Three of the four studies included in this review used a questionnaire such as the Ocular Comfort Index (OCI) or the Ocular Surface Disease Index (OSDI) to rationalise the participants' perspective of symptoms. Two of these papers showed statistical significance in improvement of their symptoms when using FAB ($p < 0.05$), whilst the third trial showed that 83% of participants reported an improvement in their OSDI, but this was not statistically significant. The one study that used OCI indicated patients using FAB had improved symptoms after 8 weeks of use and when treatment was stopped, the OCI worsened after 4 weeks. Three out of the four papers also investigated Oxford Corneal Grading Staining (OCGS), with two of these papers showing statistically significant improvement after 8 weeks use of FAB. One of these papers also showed that after cessation of FAB for 4 weeks, the OCGS increased. Two of the papers also investigated visual acuity, with one of the papers showing statistical significance in improvement after 8 weeks use of FAB. Three of the papers investigated Schirmer's test, with only one showing a statistically significant improvement after 6 months, with the other two studies showing no significant change. Tear film break up time (TBUT) was investigated in three of the four studies, with two of the studies showing improvement after 8 weeks of FAB use, which was statistically significant. No complications, ocular or finger-related, were reported in any of the four studies. Only one study specifically investigated peripheral epithelium defects (PED) in the cornea, with 60% of patients having complete resolution of PED by day 28, but this was not statistically significant.

Conclusion: This review showed there is evidence that FAB eye drops can provide an alternative treatment option to those with OSD, with research showing improvement in symptomatic relief, reduction in size of epithelial defects, improved visual acuity, tear break-up time and Schirmer's. As well as this, patients tolerated the finger-pricking well as there were no complications seen across the four studies. Further studies need to be implemented with larger cohorts and with controlled groups to further evaluate the current knowledge in this field.

Conflicts of interest: None

Funding: None

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P071

THE INCIDENCE OF CHEMICAL EYE INJURIES AT MOORFIELD'S EYE HOSPITAL 2013–2021

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Introduction: The aim of our study was to assess the incidence and baseline patient characteristics of severe ocular chemical injuries (SOEI) presenting to a specialist ophthalmic hospital with 24-hour accident and emergency care in London, UK.

Method: A retrospective review of electronic medical records of all cases of chemical eye injuries (CEI) from 01/01/2013 to 31/01/2021. Baseline patient demographic details including, age,

gender, ethnicity and Index of Multiple Deprivation (IMD) score were collected. Presenting information on the cause of injury, injury setting, severity classification, best corrected visual acuity (BCVA) at presentation and discharge were also collated. Injury severity was graded using both the Hughes-Roper-Hall (H-R-H) classification and the Dua classification. Cases with H-R-H Grade ≥ 3 , Dua Grade $\geq IV$ or electronic pharmacy records of cases requiring prescriptions for potassium ascorbate and sodium citrate were included as SOCI.

Results: Five-thousand-and-one patients presented with a new CEI in the study period contributing to 0.69% of all A&E attendances. SOCI accounted for 5.7% (287/5001) of cases. We observed a peak of SOCI particularly in the summer months during the duration of our study. There was a male preponderance, with men accounting for 54.8% of all CEIs and 83.6% of SOCIs. The mean age of patients was 37 ± 15.6 years in all cases and 37 ± 17.3 years in SOCIs. Just under half of all CEI cases (47%) were White British. Cause of Injury was found to be evenly spread: 34.8% workplace incidents, 33.3% assault incidents and 31.9% domestic incidents. Seventy percent (70%) of the SOCI cohort lived at a postcode of IMD decile ≤ 5 . One-hundred-and-thirteen SOCI patients had their presenting and final BCVA recorded.

The mean initial BCVA was a decimal value of $0.57(\pm 0.43)$ or logMAR 0.22. The final BCVA was a decimal value of $0.75(\pm 0.41)$ or logMAR 0.1. The mean change in visual acuity was $+0.18(\pm 0.49)$ or an increase in logMAR from 0.22 to 0.1 ($p = <0.001$).

Conclusion: Chemical Eye Injuries are a rare indication for presentation to ophthalmic A&E. Despite this we have found a higher than previously reported rate of severe injuries. We report a higher number of assault related SOCI in patients who lived at a postcode of IMD decile ≤ 5 , suggesting an association between CEI and the socio-economic status.

Conflicts of interest: None

Funding: None

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EMERGENCY OPHTHALMOLOGY

P072

"HOT SEAT AND COLD FEET": NAVIGATING PRESSURISED EMERGENCY EYE CARE AND EXPLORING THE ROLE OF OCT/OCTA IN ACUTE-ONSET RETINAL CONDITIONS MIMICKING OPTIC NEUROPATHY

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Introduction: Sudden loss of vision, especially when accompanied by loss of colour vision, a relative afferent pupillary defect (RAPD), and visual acuity dropping below 6/60, is an alarming symptom that often leads clinicians to suspect optic neuropathy, particularly conditions like Anterior Ischemic Optic Neuropathy (AION) or Posterior Ischemic Optic Neuropathy (PION). However, retinal conditions such as Paracentral Acute Middle Maculopathy (PAMM) and Acute Macular Neuroretinopathy (AMN), especially Type 2 AMN, can present with similar symptoms, causing diagnostic confusion. This poster explores classic findings on Optical Coherence Tomography (OCT) and OCT Angiography (OCTA) along with visual fields that can aid in distinguishing

between optic neuropathy and retinal diseases, guiding appropriate management and treatment.

Method: The study included 10 patients with a mean age of 62 years, ranging from 47 to 75 years, with an equal gender distribution (5 males and 5 females). The average initial visual acuity was 5/60, ranging from 1/60 to 6/60. All patients presented with sudden vision loss, and 9 (90%) had a relative afferent pupillary defect (RAPD). Seven patients (70%) reported loss of colour vision, and many had a history of hypertension, hyperlipidaemia, or were on statins.

The initial differential diagnosis included PION in 3 patients, AION in 4 patients, and demyelination in 3 patients. Some patients were treated with high-dose steroids (60 mg/day) for suspected GCA or as stroke cases due to vascular risk factors.

OCT and OCTA findings revealed characteristic differences between the conditions.

Results: The final diagnoses confirmed that 4 patients had PAMM, identified by hyperreflective lesions in the inner retinal layers on OCT, particularly in the parafoveal region. 3 patients were diagnosed with AMN, based on outer retinal hyperreflective lesions and macular ischemia identified on OCTA. The remaining 3 patients were diagnosed with optic neuropathy, 2 with PION and 1 with AION, based on OCT and OCTA findings, which did not show the characteristic retinal findings of PAMM or AMN.

Regarding treatment, 5 patients were started on high-dose steroids (60 mg/day) for suspected GCA, but there was no significant improvement in the PAMM and AMN patients, where retinal ischaemia, rather than optic nerve ischemia, was the underlying issue. All patients were on statins, with 8 (80%) on increased doses due to hyperlipidaemia, and 7 (70%) were on antihypertensive therapy. Patients diagnosed with PAMM and AMN were managed conservatively with observation and treatment for underlying conditions, such as controlling hypertension and hyperlipidaemia, and did not require systemic steroids or biologics.

Follow-up at 6 months showed that 6 out of 7 patients with PAMM and AMN had stable or slightly improved vision, with one patient showing slight worsening but remaining stable overall.

Conclusion: OCT and OCTA imaging provide critical diagnostic information, revealing characteristic findings of retinal ischemia in PAMM and AMN. High-dose steroids, initially prescribed for suspected giant cell arteritis (GCA) or ischemic optic neuropathies, were ineffective for PAMM and AMN, as these conditions are driven by retinal ischemia rather than optic nerve ischemia. This underscores the importance of accurate diagnosis to avoid unnecessary treatments and potential harm, as steroids can worsen PAMM by increasing blood pressure.

Patients with PAMM and AMN, when managed conservatively with observation and treatment for underlying conditions, demonstrated stable or improved visual outcomes. This contrasts with optic neuropathy patients who showed minimal improvement with steroids, highlighting the significance of early differentiation for optimal management.

Additionally, this study emphasizes the cost-effectiveness of OCT and OCTA. These imaging techniques provide a less expensive, non-invasive alternative to MRI and CT scans, offering quicker and more accurate diagnoses. By reducing the need for more costly imaging and unnecessary treatments, OCT and OCTA contribute to both improved clinical outcomes and reduced healthcare costs. Therefore, their use in diagnosing acute onset retinal conditions mimicking optic neuropathy can significantly enhance patient care while ensuring cost savings for healthcare systems.

Conflicts of interest: None

Funding: None

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P073

BOOSTING EFFICIENCY IN URGENT CARE : THE ROLE OF RAPID ACCESS (RAC) AND RAC-AA (ATTEND ANYWHERE VIDEO CONSULTATION)

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Introduction: The Moorfields Eye Unit in Croydon receives over 51,000 outpatient attendances annually, with more than 2,600 patients presenting for urgent eye care. To streamline our emergency eye care services, the Rapid Access Clinic (RAC) was introduced in November 2018. This led to a 50% sustained decrease in urgent care attendances and improved triage system for urgent referrals. Building on this success, the Rapid Access Clinic-Attend Anywhere (RAC-AA) was introduced in May 2024 to enhance patient management through remote consultations.

The primary objective of this study was to evaluate the sustainability the RAC and effectiveness of the RAC-AA pathways in improving referral quality and enhancing the management of urgent eye care cases in Croydon. We also aimed to analyse the sources of inappropriate referrals and assess how effectively RAC-AA helped reduce the need for face-to-face appointments, improve patient flow, optimize healthcare resources, and allow for more efficient handling of urgent cases while maintaining high-quality care. The secondary objective was to identify areas where further improvement is necessary to enhance patient care.

Method: A retrospective observational audit was conducted on 370 consecutive referrals received between 1st and 31st October 2023. The audit aimed to evaluate the appropriateness of referrals and the outcomes of triage decisions. For the RAC-AA pathway, preliminary data from 45 patients managed over a 10-week period following its establishment, were analysed.

The audit assessed the source of referral, and the outcomes of their RAC-AA consultation. Particular attention was paid to how effectively the RAC-AA managed cases without requiring additional face-to-face appointments

Results: RAC face-to-face data: Of the 370 referrals received in RAC, 194 were female (52.4%) and 177 were male (47.6%), with a mean age of 52 years (range: 1 week to 94 years). This study revealed that 66% of the referrals were accepted for review in face-to-face RAC clinics, while 34% were rejected with advice or redirected for review either in the community or our subspecialty clinics. Notably, 22% of these referrals could have been managed through the RAC-AA service, highlighting the potential for further optimizing referral management. Inappropriate referrals primarily originated from GPs (45%), followed by street opticians and A&E departments (31%).

RAC-AA Preliminary Data : In the 10-week period following the launch of RAC-AA, 45 patients were assessed, with a mean age of 43.5 years (range: 4 to 84 years), including 4 paediatric patients. Remarkably, 84% of patients were discharged after just one RAC-AA appointment, demonstrating the effectiveness of this model in reducing the need for further appointments. Only a small proportion of cases (4%) required in-person follow-ups or referral to subspecialty clinics.

Conclusion: These findings suggest that while the RAC system has significantly improved referral accuracy compared to previous audits, there remains a considerable proportion of referrals that could be managed more efficiently through alternative pathways, such as RAC-AA. This study shows promising results for enhancing

the management of urgent eye care cases. Most patients were effectively managed and discharged through virtual consultations, reducing the burden on face-to-face clinics and improving overall clinic efficiency. The RAC-AA pathway has also improved communication and collaboration with A&E departments, ensuring that appropriate cases are triaged effectively. Despite these early successes, future improvements will focus on expanding RAC-AA to handle more complex conditions, increasing diagnostic capabilities, and training additional senior staff to optimize patient management.

Conflicts of interest: None

Funding: None

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P074

CHLAMYDIA TRACHOMATIS CONJUNCTIVITIS: A 10-YEAR RETROSPECTIVE OF THE INCIDENCE, PRESENTATION, DIAGNOSIS, MANAGEMENT AND OUTCOMES AT A TERTIARY REFERRAL CENTRE

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Introduction: Chlamydia conjunctivitis, also known as inclusion conjunctivitis, is a sexually transmitted infection (STI) caused by the gram-negative obligate intracellular bacterium *Chlamydia trachomatis*. Chlamydia is the most common bacterial STI in the world. In 2022, Europe saw record-high rates, with 216,508 confirmed cases. In 2023, chlamydia accounted for almost half of all UK STIs; with 194,970 cases in England alone. A concomitant rise in ocular infection is a concern, particularly due to the absence of distinctive clinical features, leading to the under diagnosis or misdiagnosis of infection.

Method: A 10-year retrospective case series was conducted at a tertiary referral centre in London. Inclusion criteria were laboratory-confirmed *Chlamydia trachomatis* eye infections (via molecular methods) in any adult (≥ 18 years) between 1 January 2013 and 31 December 2023. Conjunctival nucleic acid amplification test (NAAT) samples were collected using Aptima swab collection kits (HOLOGIC, USA). Medical records were analysed for clinical presentation, treatment and complications.

Results: 312 laboratory-positive cases were recorded, 267 had complete medical records and were included in the final analysis. Mean age was 28.89 $\pm$ 10.26. 181 (67.8%) were male and 86 (32.2%) were female. 235 (88.0%) were unilateral and 32 (12.0%) were bilateral. Mean time from first onset of symptoms to diagnosis was 16 days and median was 14 days. 101 (37.8%) required ≥ 2 visits before a diagnosis was made. 144 (53.9%) were misdiagnosed as viral conjunctivitis. Sexual history was missed in 76.0% and only 4.5% had genital symptoms documented in the initial consultation. 92.9% had no awareness of prior genital chlamydia diagnosis. Commonest signs at presentation included conjunctival hyperaemia (95.1%), tarsal follicles (85.4%), mucopurulent discharge (71.5%), punctate erosions (46.1%), tarsal papillae (45.3%), and corneal infiltrates (18.0%). Mean best corrected visual acuity (BCVA) was 0.09 logMAR at baseline, with 77.3% recovered to baseline BCVA or better after treatment.

Conclusion: Our study demonstrates that chlamydia conjunctivitis is frequently misdiagnosed in the acute ophthalmic setting, with poor documentation of sexual history and genital symptoms in the consultation. Ophthalmologists need greater awareness of the incidence and impact of chlamydia. Conjunctivitis may be the

only sign of chlamydial infection; diagnosis is critical to prevent serious reproductive consequences and onward transmission. Women aged between 15–49 years old presenting with conjunctivitis should be routinely tested for chlamydia if they also have any of the following risk factors: genital symptoms, >1 sexual partner during the last 3 months, or inconsistent use of barrier contraception. In 2025, new national *Chlamydia trachomatis* guidelines from the British Association for Sexual Health and HIV (BASHH), will be launched and for the first time, include detailed guidance on the diagnosis and management of eye infection.

Conflicts of interest : None

Funding: None

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P075

EYE CASUALTY REFERRALS TO RAPID ACCESS CORNEAL CLINICS IN A TERTIARY LEVEL LONDON HOSPITAL

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Introduction: The Western Eye Hospital is a tertiary level centre in London with an eye casualty covering most of northwest London, especially during out-of-hours. Corneal rapid access clinics were created and funded to follow-up patients with emergency anterior segment conditions within 2 weeks of their casualty attendance. Financial loss is incurred by the trust if outside this period. Due to the limitation in appointment slots, as clinics ran only once a week, hospital protocol dictated that approval to refer must be given by the senior clinician in charge of the department. Guidelines exist with examples of conditions suitable for referral to these particular clinics.

Method: Retrospective data was collected over a 3 month period (1st August – 31st Oct 2023), with a total of 156 eye casualty referrals made to the corneal rapid access clinics. Patient data was collected from electronic health records (Medisoft and Cerner). The parameters measured in this audit included urgency of referrals, conditions referred, appropriateness of referrals, concordance between diagnoses made in casualty and clinics, follow-up plans, grade of referrers and patient postcodes.

Results: The median age of patients presenting to the clinic was 39, with a 3:2 female to male ratio. 18% of patients referred were reviewed more than two weeks after their initial casualty attendance due to lack of appointment availability, with resulting financial implications. Contact-lens associated keratitis, herpetic eye diseases, and viral keratoconjunctivitis were the top three referred conditions. Approximately 15% of referrals made were not appropriate and did not abide by the criteria. There was a 75% concordance rate between the diagnoses made in casualty and clinic. Over half of patients' conditions resolved after one visit and they were subsequently discharged, but there was a high 'did not attend' rate of 19% with 15% being discharged as a result. 19% of patients required longer-term follow-up in the general cornea clinics, but only 7% of these lived in the hospital's catchment area. No senior approval was documented in 12% of referrals made by junior registrars (ST1-4), and it was ambiguous as to whether other allied health professionals also had to seek senior approval for their referrals whilst working in casualty. Astonishingly, only 15% of patients seen in the hospital's corneal rapid access clinic were from the trust's catchment area; 80% fell under other London hospitals with their own Ophthalmology

unit; 3% were from out of the Greater London county; and 2% were from overseas who were not supposed to be eligible for NHS treatment.

Conclusion: Overall, the Western Eye Hospital performed well keeping to targets, although there are areas to be improved to optimise use of available clinic appointments. However, this audit has also highlighted challenges of being a tertiary centre and the freedom of choice patients have to seek care at any hospital of their choosing in the NHS. Not only can this lead to an inundated service and financial consequences, but also compromise care of more local patients as resources are being stretched to encompass patients who may be served equally as well in their local eye units. The result is an imbalance in workload across trusts and improper utilisation of each trust's resources. A collaborative approach is needed to ensure acute patients are followed up safely across different hospitals.

Conflicts of interest: None

Funding: None

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P076

IMPROVE THE REFERRAL PATHWAY TO EYE CASUALTY AND REDUCE FOOTFALL

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Introduction: The ophthalmology department, much like other departments within a hospital, consistently faces significant strain due to the influx of referrals, many of which are made with inappropriate urgency.

This strain, exacerbated by the shortage of staff within the NHS, often overwhelms the eye casualty unit, potentially leading to mismanagement of time and diverting attention away from truly urgent cases amidst the deluge of referrals.

Therefore, our ongoing commitment is to meticulously review the current referral pathway, seeking opportunities for enhancement to alleviate this unnecessary burden. By effectively filtering referrals and prioritizing urgent cases, we aim to enhance patient care and ensure timely treatment for those in critical need.

Method: For this project, data was collected on all referrals to the Ophthalmology department over the course of one month. Multiple sources were utilised to gather comprehensive information regarding each referral. The collected data included:

1. Source of referral: Understanding the source of referral provided insight into the pathways through which patients access ophthalmic services.
2. Reason for Referral: Each referral was carefully examined to determine the underlying reason prompting the patient's referral to the Ophthalmology department.
3. Requested Urgency: The urgency level assigned by the referring physician or healthcare provider was recorded for each referral. This helped in assessing the priority level of each case and the corresponding response time required.

By collecting and analysing data from various sources, we aimed to gain insights into the referral patterns and factors influencing the urgency and appropriateness of referrals to the Ophthalmology department. This approach allowed for a comprehensive evaluation of the current referral pathway and provided valuable information for identifying areas for improvement.

Results: The analysis of the results shows that the highest number of referrals originated from the NHS e-Referral Services (ERS), which is already an established pathway with proven efficacy. Optometrists constituted the second most common referral source, highlighting the necessity for us to assess areas for improvement. The third-place position is held by UTC, which marks the most significant opportunity for improvement due to its accessibility. UTCs handle patients directly and often triage external referrals. Given their on-site presence and accessibility, UTCs represent a prime target for implementing interventions to enhance the efficiency and effectiveness of the referral process, thereby improving timely access to ophthalmic care for patients in need.

The substantial number of referrals classified as “urgent” raises concerns regarding the potential strain on the eye casualty department, as all these patients require immediate attention. However, upon closer examination, it becomes evident that not all of these referrals are genuinely urgent, highlighting the need for improved judgement and recognition of urgency in referrals.

Similarly, the designation of referrals as “soon,” necessitating assessment within 2 to 4 weeks, poses challenges as some cases may not warrant such prompt attention. This discrepancy emphasises the importance of refining urgency assessment criteria and implementing measures to ensure that patients are prioritised appropriately based on their clinical needs.

Addressing these issues was crucial to preventing the potential overwhelm of departmental resources and optimising patient care pathways within the ophthalmology department.

Conclusion: This QUIP highlights critical aspects of the ophthalmology referral pathway. Training sessions with UTCs, walking centres, and a local optician addressed referral issues, with reaudit results showing significant improvements. The analysis revealed high referral volumes, emphasising the need for efficient processes, but also noted challenges in accurately assessing urgency, with many cases misclassified as “urgent” or “soon.” Enhancing triage capabilities at the community level is essential, given the complexity of conditions managed by optometrists and UTCs before referral. Recommendations include improved training, standardised urgency assessment guidelines, streamlined triage protocols, and leveraging digital tools for better communication. Continuous refinement of referral pathways will optimise resources, reduce strain on eye casualty services, and ensure timely, effective care, ultimately enhancing patient outcomes and satisfaction.

Conflicts of interest: None

Funding: None

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P077

MACHINE LEARNING VS HEALTHCARE PROFESSIONALS: PERFORMANCE ASSESSMENT OF AN AI-POWERED OPHTHALMIC TRIAGE SYSTEM

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Introduction: In the rapidly evolving landscape of emergency eye care services, effective telephone triage remains a critical component in managing patient flow and identifying sight-threatening conditions. The increasing demand for emergency eye services, coupled with limited resources, necessitates an efficient and accurate triage system. Traditional triage methods rely heavily on human expertise, which varies significantly based on clinical experience and training.

The introduction of artificial intelligence in healthcare decision support presents an opportunity to standardise and potentially enhance the triage process, particularly in ophthalmology, where visual symptoms often require careful interpretation.

Aim: This study aimed to comprehensively evaluate and compare the accuracy, efficiency, and clinical effectiveness of ophthalmic triage decisions across four distinct groups: consultant ophthalmologists, specialist ophthalmic nurses, staff nurses, and a newly implemented AI triage system. The investigation focused on adherence to standardised protocols, assessment of clinical outcomes, resource utilisation, and identifying factors contributing to successful triage decisions. Additionally, the study sought to establish whether AI-assisted triage could complement or enhance existing human-led triage processes.

Method: A retrospective analysis was conducted on 2,583 triage sheets collected between January 2022 and May 2024 at a tertiary eye care centre. The dataset comprised 929 ophthalmologist assessments, 900 specialist nurse assessments, and 500 staff nurse assessments. The AI system evaluated 252 cases during May 2024. Assessment criteria included:

- Triage accuracy evaluation using the standardised ‘Traffic-light’ classification system:
 - Red (urgent): Sight-threatening conditions requiring immediate attention
 - Amber (semi-urgent): Conditions requiring attention within 24–48 hours
 - Green (non-urgent): Routine cases suitable for outpatient management
- Clinical outcome measures:
 - Appropriateness of prescribed treatments
 - Correlation between initial triage category and final clinical urgency
 - Rate of clinical outcome alignment with triage decisions
 - Patient safety incidents and adverse events
- Operational efficiency metrics:
 - Number of telephone consultations per case
 - Average call duration
 - Time to definitive care
 - Resource utilisation patterns

Statistical analysis employed chi-square tests for categorical variables and logistic regression to identify predictors of accurate triage decisions. Subgroup analyses were performed to account for varying levels of clinical experience and time of day variations.

Results: Specialist nurses demonstrated the most significant improvement in triage accuracy, increasing from 80.0% to 98.3% over the study period. This improvement was accompanied by:

- 40% reduction in callback rates
- 25% improvement in protocol adherence
- 30% decrease in inappropriate urgent referrals

Ophthalmologists maintained consistent performance metrics:

- 95% initial triage accuracy
- Stable call durations averaging 4.5 minutes
- Highest correlation between provisional and final diagnoses (92%)

Staff nurses showed gradual improvement:

- Accuracy increased from 75% to 85%
- Protocol adherence improved by 35%
- Required more frequent senior consultation

The AI system's initial performance:

- Identified 47 urgent cases from 252 assessments
- Demonstrated 85% sensitivity for red-flag symptoms
- Showed limited ability to interpret complex symptom patterns
- Required human oversight for final decision-making

Administrative workload patterns:

- Senior nurses experienced a 40% increase in documentation time
- Other groups maintained stable administrative burdens
- AI system reduced documentation time by 25% when integrated

Conclusion: The study demonstrates that specialist nurses achieved superior accuracy improvements through experience and strict protocol adherence. Ophthalmologists provided consistent, high-quality triage decisions but at higher operational costs. The AI system shows promise in standardising triage processes but requires significant refinement to match human performance levels. Future developments should focus on integrating AI support tools with human expertise rather than replacing existing triage systems. Continuous quality improvement initiatives, regular training, and protocol updates remain essential for maintaining and enhancing triage effectiveness across all provider levels

Conflicts of interest: None

Funding: None

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P078

SPONTANEOUS MEDIAL RECTUS HAEMATOMA; A SIGHT THREATENING COMPLICATION OF WARFARIN TOXICITY

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Introduction: Spontaneous orbital haemorrhages are rare and potentially sight-threatening, often requiring immediate intervention to prevent complications such as orbital compartment syndrome. While non-traumatic haemorrhages have been linked to anticoagulation therapies, spontaneous haemorrhage involving the medial rectus muscle is exceptionally rare. This case presents a 77-year-old female on warfarin therapy who developed a spontaneous medial rectus haematoma, a unique presentation of warfarin toxicity. We have not identified any previous reports on this

Method: The patient initially presented with acute conjunctival haemorrhage, progressing to pain, raised intra-ocular pressure, periorbital swelling, and vision loss. Comprehensive clinical evaluation included visual acuity, intraocular pressure measurements, and imaging via orbital CT, which confirmed a medial rectus haematoma with extensive subconjunctival haemorrhage but no retrobulbar involvement. Emergency interventions, including lateral canthotomy and cantholysis, were performed to alleviate orbital compartment syndrome. Warfarin therapy was suspended, and human prothrombin complex concentrate was administered to reverse anticoagulation. Serial follow-ups involved ocular assessments, haemoglobin monitoring, and imaging to evaluate resolution of symptoms.

Results: Emergency intervention led to significant symptomatic improvement. Visual acuity initially deteriorated to light

perception but improved to 6/18 after ten days and 6/7.5 at one month post-event. However, despite lateral canthotomy & cantholysis, the intraocular pressure remained elevated - indicating a possible direct pressure effect of the haematoma on the globe. Following emergency intervention, the patient was commenced on intensive IOP lowering medication which further improved her intra-ocular pressure. The haematoma resolved significantly within the first month, and orbital function, including adduction, partially recovered. Conjunctival chemosis and mild thickening persisted but without further complications. The management strategy, including canthotomy, anticoagulation reversal, and intraocular pressure control, demonstrated effective mitigation of vision-threatening complications.

Conclusion: This case highlights the rare occurrence of spontaneous medial rectus haematoma as a complication of warfarin therapy, exacerbated by comorbid conditions such as chronic kidney disease and thromboembolic risk. Early recognition, rapid surgical intervention, and careful anticoagulation management were essential to preserving vision and achieving favourable outcomes - however, it is important to note the possible direct pressure effect of the enlarged medial rectus on the globe. Increased monitoring of anticoagulation therapy and consideration of alternative agents with lower bleeding risks may help prevent such complications in high-risk populations. This case underscores the importance of multidisciplinary collaboration in managing complex ophthalmic emergencies.

Conflicts of interest: None

Funding: None

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P079

THE EPIDEMIOLOGY, MANAGEMENT, AND OUTCOMES OF OPEN GLOBE INJURIES IN A MAJOR TRAUMA CENTRE IN THE UNITED KINGDOM

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Introduction: Ocular trauma is a leading cause of blindness worldwide. Standardised terminologies and scoring systems were introduced to guide assessment, standardise documentation, and estimate prognosis. Open globe injury (OGI) poses a higher risk of sight impairment, yet no standard management guideline exists. This study aims to investigate the epidemiology of OGIs presenting to a major trauma centre in the United Kingdom, identify the key factors affecting final visual outcomes, and assess current management practices.

Method: This is a retrospective single-centre observational study of OGI. Samples were obtained from January 2019 to May 2024. Data was collected from electronic medical records. Only patients who were brought to the operating theatre with the intention to repair were included. The time from ophthalmology review to OGI repair (time to repair) was categorised into three timeframes to simulate same-day surgery (<6 hours), out-of-hours surgery (6–12 hours), and next-day surgery (>12 hours). Best corrected visual acuities (BCVA) were converted from Snellen to their logMAR equivalents for analysis.

Results: 48 unilateral OGIs were included in this study (n = 48). The median age was 51.5 years. The median interval from presentation to final follow-up was 360 days. The most common method and type of injury were falls (37.8%) and rupture (50.0%) respectively, which occurred most frequently in older women.

Females had a greater percentage of undesirable visual outcomes (53.8%) than males (25.7%). Globe lacerations and zone 1 injuries were associated with significantly better final BCVA compared to other types and zones of injuries within their respective categories ($p < 0.03$). The median time to repair was 6 hours. The mean logMAR vision improved by 29.1% from presenting to final BCVA. There was a statistically significant improvement when comparing initial and final BCVA in all patients ($p < 0.01$), in those repaired within 6 hours ($p = 0.02$), and also in those repaired within 6–12 hours ($p < 0.05$). No significant improvement in BCVA was observed in patients who underwent later repairs. Concurrent medical or surgical input outside of ophthalmic or emergency medicine was associated with significantly poorer final BCVA ($p = 0.04$). The absence of vitreous haemorrhage (VH) led to a significantly better final BCVA ($p = 0.04$). The presence of retinal detachment, lens dislocation or previous corneal transplant did not lead to significantly different final BCVAs. Only one patient who had an agricultural-related OGI developed endophthalmitis. Endophthalmitis was absent in all other patients regardless of time to repair. Among those, four (8.5%) were operated after 24 hours from initial review.

Conclusion: Within the region, there was an increasing proportion of patients presented with BCVA of 6/18 or worse and those ended with the same range of BCVA over the last 24 years. This trend aligns with our observation that globe ruptures are associated with poorer BCVA as there has also been a corresponding increase in the proportion of OGIs resulting from blunt trauma or ruptures. This phenomenon may be attributed to the ageing population within the region, as reflected in the rising patient age. Type of injury, zone of injury, presence of VH, time to repair, and concurrent medical or surgical interventions significantly influenced the final BCVAs. Based on our findings, we recommend that patients with zone 1 globe lacerations, as well as those with OGIs contaminated by organic matter or soil, undergo surgical repair within 6 hours of presentation. For other OGIs, the timing of surgery should be planned to ensure optimal anaesthetic safety and surgical conditions, ideally within 12 hours. Additionally, all patients presenting with OGI should receive a course of systemic antimicrobials with good intravitreal penetration at the time of presentation.

Conflicts of interest: None

Funding: None

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P080

THE OPHTHALMIC EMERGENCIES COURSE - A NATIONWIDE INITIATIVE ENGAGING 868 MEDICAL STUDENTS ACROSS THE UK

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Introduction: It has been globally acknowledged that the quality of ophthalmology teaching at most medical schools is inadequate. 75% of medical students find their schools' ophthalmology education as insufficient. Alarming, 20% of medical schools lack any mandated exposure to ophthalmology. Considering 10% of patients entering an NHS hospital head to the eye clinic, and 6% of all A&E attendances are eye-related, the current model of education is unsustainable.

Method: To address the above, a novel initiative was launched by an ophthalmology trainee and a foundation doctor keen on educating medical students on ophthalmic emergencies. All 45

UK medical schools were contacted and registration to the course was disseminated to their students. The course used an interactive, virtual, case-based learning platform, covering basic aspects of ophthalmic history, examination techniques, and various conditions including retinal detachment, angle closure glaucoma, trauma, and endophthalmitis.

Results: Conducted in 2021 and repeated in 2022, the course gathered 868 respondents across both years. There was a significant improvement in levels of confidence across all four aspects of the ophthalmology consultation and all fourteen eye conditions covered ($p < 0.05$, paired Wilcoxon test).

Conclusion: Ophthalmology makes 1% of all NHS appointments, making it the busiest outpatient specialty. Non-ophthalmologists often lack confidence when dealing with eye cases, leading to personal struggles, reduced job satisfaction, delayed treatment, potential vision loss, unnecessary referrals, increased litigation, and higher healthcare costs, especially in an aging population. Providing basic ophthalmic education can help prevent sight-threatening consequences.

Our results showed a consistent improvement in confidence levels in all aspects of the ophthalmology consultation and examination, as well as when assessing specific eye conditions.

Feedback included "better than any ophth teaching I've had to date", "course should be mandatory for all med students", "eye teaching at our uni is alarmingly limited, courses like these are so greatly received by [x] students". There was also request for wider teaching including for optometrists as well as expanding the course to a global scale.

Conflicts of interest: None

Funding: None

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GLAUCOMA

P081

BREAKING BARRIERS: IMPROVING DIAGNOSTIC STANDARDS IN GLAUCOMA CARE THROUGH TARGETED INTERVENTIONS COMPLYING WITH NICE GUIDELINES

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Introduction: Accurate and timely diagnosis of glaucoma is critical to effective management and the prevention of irreversible vision loss. NICE guideline NG81 specifies that all new and referred patients suspected of having glaucoma must undergo five essential diagnostic tests at their first visit: visual field assessment using standard automated perimetry, optic nerve evaluation via stereoscopic slit-lamp biomicroscopy, intraocular pressure (IOP) measurement using Goldmann Applanation Tonometry (GAT), gonioscopy, and central corneal thickness (CCT) measurement. This audit aimed to assess compliance with these standards at a tertiary eye care centre, identify barriers to adherence, and evaluate the impact of targeted interventions designed to address these gaps across two audit cycles.

Method: A retrospective audit was conducted at Imperial College NHS Foundation Trust. Cycle 1 included all new and re-referred glaucoma patients seen between January and March 2023. Following the first cycle, interventions were implemented, including enhanced staff training focused on diagnostic accuracy, pre-clinic

reviews to ensure appropriate test allocation, improved access to diagnostic equipment, and streamlined workflows. Cycle 2, conducted between May and July 2024, assessed compliance post-intervention in a cohort of 246 patients, with compliance rates for each mandated test compared between cycles.

Additionally, qualitative observations were collected to identify factors contributing to noncompliance.

Results: Compliance improved significantly across several diagnostic tests in Cycle 2. Notable gains were observed for CCT measurement (82%, up from 60%), GAT IOP measurement (83%, up from 70%), and gonioscopy (81%, up from 65%). Visual field testing (95%) and optic nerve assessment (92%) remained high in both cycles, reflecting consistent adherence in these areas. Enhanced staff training contributed to a 40% improvement in CCT compliance, while pre-clinic reviews helped identify patients requiring specific diagnostic tests, reducing oversight. Improvements in GAT compliance were attributed to refresher courses addressing technique and documentation accuracy, while gonioscopy compliance benefitted from increased focus on patient comfort and procedural adherence.

However, persistent challenges were noted. CCT compliance remained affected by equipment availability and workflow delays. Gonioscopy, despite improvement, was limited by patient discomfort, particularly in elderly individuals or those with blepharospasm. Documentation inconsistencies occasionally led to misreporting of compliance, particularly when rebound tonometry results were recorded instead of GAT measurements.

Conclusion: This audit underscores the value of targeted interventions in enhancing compliance with glaucoma diagnostic guidelines. Improvements in training, workflow optimization, and equipment availability significantly boosted adherence rates; however, achieving full compliance requires sustained efforts to address remaining barriers. Solutions include ensuring universal access to diagnostic tools, enhancing procedural training for staff, standardizing documentation practices, and mitigating patient discomfort during gonioscopy. These findings highlight the importance of comprehensive first-visit diagnostics, not only for improving patient outcomes but also for facilitating efficient long-term care, including virtual consultations. By refining diagnostic practices, this audit provides a scalable framework for advancing quality in glaucoma management.

Conflicts of interest: none

Funding: none

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P082

COMPARISON OF THE LONG-TERM CLINICAL OUTCOMES OF TUBE FLUSHING WITH TRADITIONAL CYCLODIODE LASER IN REFRACTORY GLAUCOMA PATIENTS WITH FAILED TUBE IMPLANT

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Introduction: Cyclodiode laser (CL) and tube flushing (TF) are two notable surgical options for managing refractive glaucoma that had failed tube implant, but their long-term outcomes remain poorly studied. This study tested the hypothesis that TF, being less invasive than CL, provides superior long-term clinical outcomes in achieving surgical success.

Method: A single-centre retrospective cohort study was conducted. The primary outcome was intraocular pressure reduction. Secondary outcome included, surgical success - intraocular pressure (IOP) 5–21 mmHg or $\geq 20\%$ baseline reduction on two consecutive follow-ups without further operation or CL. Outcomes were tracked until failure or reoperation, and surgical success was further categorized as complete (off glaucoma medications) or qualified (on medications). Other secondary outcome included best-corrected visual acuity (BCVA), number of medications (NOM) visual acuity, cup-disc ratio, central corneal thickness, retinal nerve fibre layer thickness, and visual field parameters, were recorded pre-operatively and during each follow-up visit. The 2-tailed student T-test was used for data analysis, and the surgical outcome was presented in Kaplan-Meier.

Results: Tube Flushing included ($n = 22$ eyes of 20 patients) and Cyclodiode laser ($n = 11$ eyes of 11 patients) with refractory glaucoma, i.e., uncontrolled glaucoma despite filtration surgery and maximum medical therapy, were recruited. The mean follow-up was 5.02(1.61) for Tube flushing and 8.61(3.64) years for cyclodiode laser. At the final visit, mean IOP for Tube flushing significantly improved from 21.42 ± 6.33 mmHg on 1.71 ± 1.70 medications to 18.85 ± 6.46 mmHg on 2.14 ± 1.39 and cyclodiode laser group mean IOP improved for TL 29.45 ± 10.62 on 3.27 ± 1.00 was reduced to 17.60 on 0.8 ± 0.17 medications. In Tube flushing, 21.4% patients had qualified success, while cyclodiode group had 27% qualified success. Specifically, BCVA logMAR for tube flushing worsened from 0.70 ± 0.66 to 0.87 ± 0.83 at the last visit worsened compared to the cyclodiode group from 0.89 ± 0.70 to 0.90 ± 0.77 . There was no significant difference in outcome between the two groups.

Conclusion: Our study suggests that both tube flushing and cyclophotocoagulation are poor treatment options for tubes that have failed. This could support future clinical decision-making of the glaucoma treatment outcome.

Conflicts of interest: None

Funding: None

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P083

CORRELATION BETWEEN CONTRAST SENSITIVITY AND RETINAL NERVE FIBER LAYER THICKNESS IN PATIENTS WITH PRIMARY OPEN-ANGLE GLAUCOMA

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Introduction: Glaucoma is a group of diseases which is featured by optic neuropathy with excavation and undermining of the neural and connective tissue components of optic disc which results in development of characteristic patterns of visual impairment eventually. Glaucoma is characterized by high intraocular pressure (IOP) in the eye and can lead to vision loss. The current study aimed to assess contrast sensitivity and retinal nerve fibre layer (RNFL) thickness in patients with primary open-angle glaucoma (POAG) and to evaluate their correlations.

Method: This was a cross-sectional observational analytical study conducted on a total of 195 eyes included 95 eyes of 55 patients with POAG and 100 healthy control (HC) eyes of 50 subjects. The glaucomatous eyes were divided into three groups (mild, moderate and severe). Most of studied cases were mild (41.1%), while moderate and severe degrees were recorded in (26.3%) and (32.6%) of cases respectively. Contrast Sensitivity Assessment was done by The Pelli Robson Test and Oculus

Mesotest II. P value of less than 0.05 was used for all tests, and it proved to be statistically significant.

Results: The mean age of patients with POAG was (57.89 ± 8.49) years, while in the control group it was (55.59 ± 10.37) years. There were statistically significant differences between the two groups as regards BCVA and IOP. POAG groups were associated with highly significant increase in Cup/disc ratio (CDR) and highly significant decrease in the mean deviation compared to control group. There was a statistically significant decrease in contrast sensitivity and highly statistically significant decrease in RNFL thickness in patients' group in comparison with control group. There were highly statistically significant positive correlations between contrast sensitivity assessment and all RNFL parameters.

Conclusion: Contrast sensitivity has been correlated with RNFL thickness in POAG patients. Contrast sensitivity and RNFL thickness play an important role in detection of POAG.

Conflicts of interest: None

Funding: None

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P084

EFFICACY AND SAFETY OF THE PAUL GLAUCOMA IMPLANT: SYSTEMATIC REVIEW AND META-ANALYSIS

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Introduction: A survey of Australian and New Zealand glaucomatologists found that the Baerveldt glaucoma implant were the most commonly used, followed by the Molteno glaucoma drainage device. The PAUL® Glaucoma Implant (PGI) is a novel non-valved glaucoma drainage device constructed from medical implantable grade silicon. The smaller internal and overall tube calibre aim to reduce post-operative complications such as postoperative hypotony, conjunctival erosion, tube-endothelium contact and damage, seen with current shunts.

Method: A literature search of PubMed, Embase, Ovid and Google Scholar was conducted to identify studies reporting outcomes and efficacies of PGI. The Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) statement was used for search strategy. We limited the search from 2017 till November 2024, as clinical trials for PGI only commenced in 2017. The primary outcome was intra-ocular pressure reduction at selected time intervals. Secondary outcomes were surgical success rate, anti-glaucoma medications and complications. All statistical analyses were performed using RStudio.

Results: A total of 10 studies were included for analyses. At the one-year mark, the reduction in intraocular pressure was 45.5%. The qualified success rate was 86.9%. There was a significant reduction in postoperative anti-glaucoma medication. Subgroup analyses were performed to compare outcomes at 1 month, 6 months and 12 months. There was only one study which reported outcomes for PGI in paediatric population.

Conclusion: The success rate of PGI is comparable to other more commonly used drainage device, with some studies suggesting higher success rates. The mean IOP reductions were comparable to other devices, with a lower medication burden. With the positive outcomes associated with PGI, future studies comparing PGI with other shunts (Ahmed, Baerveldt, Molteno) will help support clinicians when deciding between implants.

Conflicts of interest: None

Funding : None

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P085

EFFICACY AND SAFETY OUTCOMES OF MITOMYCIN -C AUGMENTED IMMEDIATE SEQUENTIAL BILATERAL PRESERFLO® MICROSHUNT SURGERY: A REAL-WORLD EXPERIENCE FROM A CASE SERIES

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Introduction: Minimally invasive bleb surgery (MIBS) or Preserflo® MicroShunt surgery has revolutionised traditional glaucoma filtration surgery over the last decade. It has been widely used to treat primary or secondary open-angle glaucoma with promising long-term efficacy and safety profile. MIBS poses comparatively fewer intra-operative and post-operative challenges, takes relatively less surgical time, and has more predictable outcomes than traditional filtration surgery (Trabeculectomy). This makes Preserflo® MicroShunt surgery a reasonable option for immediate sequential bilateral glaucoma filtration surgery particularly when both eyes presenting with high uncontrolled pressures necessitating medical treatment with acetazolamide. We present real-world experience with Mitomycin-C augmented immediate sequential bilateral Preserflo micro-shunt surgery; to the best of our knowledge, this has not been described before.

Method: All the patients who underwent bilateral standard Preserflo® MicroShunt implantation under local (peribulbar anaesthesia) or general anaesthesia from November 2019 to April 2024 were studied retrospectively. Surgeries were performed as two aseptically separate sequential procedures on the same theatre list. The microshunt was implanted as a stand-alone procedure with adjunctive use of topical Mitomycin-C (0.4 mg/ml) for 3–4 minutes. The patients were reviewed at regular intervals as clinically indicated. IBM-SPSS software version 23.0 was used for statistical data analysis and a paired samples t-test was used to analyse the significance of the mean difference between groups.

The primary clinical outcome measure was surgical failure (defined as intraocular pressure (IOP) > 21 mmHg or < 20% reduction from baseline, IOP ≤ 5 mmHg, reoperation, or loss of light perception) or qualified and complete success (with or without medication) at six months. Secondary outcome measures were IOP, glaucoma medications, visual acuity, complications, postoperative interventions, and reoperations.

Results: Eighteen eyes of nine patients (six women, 66.6% and three men, 33.3%) are included in this study. The patients' age ranged from 48–76 with a mean of 63.67 years. The study group is composed of five (55.5%) Caucasians and four (44.4%) Afro-Caribbeans. Overall, 94.4% of eyes had primary open-angle glaucoma. Fourteen out of 18 (77.7%) patients were phakic. All were on maximum tolerated topical treatment (1–4 medications) and, except for two patients, all others were also on oral acetazolamide (375 mg–1000 mg/day) to control eye pressures.

Mean (± standard deviation [SD]) IOP decreased from 23.8 ± 4.9 mmHg at baseline to 14.7 ± 5.0 mmHg at one month ($P < 0.001$), to 13.5 ± 6.5 mmHg at three months ($P < 0.001$), and to 11.1 ± 6.3 mmHg at six months ($P < 0.001$). The mean number of IOP-lowering medications declined from 4.1 ± 1.0 at baseline to

0.2 ± 0.9 at six months ($P < 0.001$). The mean visual acuity decreased from 0.25 ± 0.37 LogMAR at baseline to 0.35 ± 0.48 LogMAR at six months and this difference was statistically significant at 95% confidence interval ($P = 0.014$).

Overall, 94.4% (17 out of 18) patients achieved complete surgical success at six months. One patient required reoperation at six months. One diabetic patient (two eyes) developed early ocular hypotony secondary to ciliary body shutdown however, both visual acuity and IOP were restored at three months with systemic oral steroids.

Conclusion: This is the first study to demonstrate the potential for the safe use of Preserflo® MicroShunts in glaucoma patients where bilateral glaucoma filtration surgery is indicated. Both IOP-lowering effect and the reduction of the mean number of IOP-lowering medications were statistically significant at six months. Further follow-up studies are required to establish the long-term efficacy and safety profile, and can be used to develop guidelines for mitomycin -C augmented immediate sequential bilateral Preserflo® MicroShunt surgery.

Conflicts of interest: None

Funding: None

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P086

EXPLORING PATIENTS' PERSPECTIVES ON COMPLICATIONS FROM GLAUCOMA SURGERY

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Introduction: Complications from glaucoma surgery are infrequent, however when they occur, they can have important consequences for patients. We have undertaken a qualitative study which explores participants' experiences of glaucoma surgery complications and their views on surgical complications.

Method: This study combined semi-structured interviews and focus group discussions. A total of 37 patients were recruited. Ten participants took part in a one-to-one interview. Six focus groups were undertaken, in total 27 participants took part in these groups. Written informed consent was obtained from all participants. All interviews were recorded and transcribed anonymously and analysed thematically using the NVivo 14 software package.

Results: Thematic analysis identified the following as key concerns for glaucoma patients in relation to surgery and surgical outcomes: (1) a need for additional clinic visits; (2) a need for further surgical treatments; (3) a delay in returning to normal activities of daily living; and, (4) vision loss (temporary or permanent).

Conclusion: The study is the first to explore patients' perspectives on the consequences and concerns in relation to glaucoma surgery. Key consequences of glaucoma surgical complications identified by patients were delays in returning to a normal lifestyle, additional clinic visits, a need for further surgical treatments and vision loss (temporary or permanent). Understanding these concerns ensures clinicians awareness and highlights the need to address these potential issues prior to surgical intervention.

Conflicts of interest: None

Funding:

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P087

HYDRUS® MICROSTENT IN COMBINATION WITH CATARACT SURGERY IN PATIENTS WITH MILD TO MODERATE POAG: COST-EFFECTIVENESS ANALYSIS IN UK SETTINGS

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Introduction: In a 5-year trial (HORIZON) Hydrus® Microstent demonstrated superior benefits when used in combination with cataract surgery (CS) to treat patient with mild to moderate glaucoma. This analysis explores the cost-effectiveness of Hydrus® Microstent + CS vs CS alone from the UK public payer perspective.

Method: A semi-Markov model simulated the natural history of POAG to assess patients' costs, health outcomes, and quality-adjusted life years (QALYs) over 15 years. The population was 71 years of age; 80% bilateral. Patient propagation was guided using the annualized rate of progression (RoP) informed by a post-hoc analysis of 5-year visual fields loss (VFL) data from HORIZON. The addition of medication was used as a proxy for irreversible progression and aligned with VFL changes according to the amount of VF lost per progression. As VFL increased, a corresponding increase in medications is assumed, allowing patients to transition to the next severe health state. Costs and quality of life weights were derived from UK publicly available sources and publications, where possible. Both deterministic and probabilistic analyses (PA) were conducted. Scenario analyses were used to explore the impact of key inputs and assumptions.

Results: Hydrus® + CS was associated with 4.4% lower cost than CS alone (£20,031 vs £20,953) and greater QALYs (7.590 vs 7.256) over a 15-year horizon, hence dominating the comparator. The PA showed that Hydrus® + CS had a probability of ~90% of being cost-effective compared with CS at a willingness to pay threshold of £20,000/QALY.

Conclusion: Hydrus® implantation + CS is a cost-effective strategy versus CS alone from the public payer's perspective in patients with mild-to-moderate POAG in the UK.

Conflicts of interest: Gus Gazzard, Anthony King, and Douglas Rhee received consulting fees relating to this research. Sergey Muratov is an employee of IQVIA, the company contracted by Alcon to conduct the research. DO'B is an employee of Alcon, the research sponsor.

Funding: The research was funded by Alcon.

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P088

HYDRUS® STANDALONE VS ISTENT IN PATIENTS WITH ADVANCED POAG: COST-EFFECTIVENESS ANALYSIS IN UK SETTINGS

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Introduction: The 2-year COMPARE trial showed that Hydrus Microstent resulted in the use of fewer medications and a lower rate of secondary incisional glaucoma surgery when used as a standalone surgery compared to 2 iStents in patients with advanced primary open angle glaucoma (POAG). This analysis explores the cost-effectiveness of Hydrus[®] Microstent + CS vs CS alone from the UK public payer perspective

Method: A semi-Markov model simulated the natural history of POAG to assess patients' costs, health outcomes, and quality-adjusted life years (QALYs) over 15 years. The population was 67 years of age; 80% bilateral. Using 2-year COMPARE data, patient progression in the model was informed by trial-based annual number of medications, which was assumed to align with visual fields loss (VFL) changes according to the amount of VF lost per progression. As patients progress on medications and VFL increases, they transition to the next severity health state. Costs and utility weights were derived from publicly available UK sources, where possible, and publications. Both deterministic and probabilistic analyses (PA) were conducted. Scenario analyses were used to explore the impact of key inputs and assumptions.

Results: Compared with iStent, patients with advanced POAG who received Hydrus[®] accumulated lower costs (£26,742 vs £26,276) and greater QALYs (7.439 and 7.371) over 15 years, respectively. As a result, the Hydrus[®] is deemed a dominant (i.e., preferred) treatment choice over iStent* in terms of cost-effectiveness. In PA, Hydrus[®] had a probability of ~69.4% of being cost-effective compared with iStent* at a willingness to pay £20,000/QALY.

Conclusion: Hydrus[®] used alone is a cost-effective treatment strategy compared to iStent from the public payer's perspective in patients with advanced POAG in the UK.

Conflicts of interest:

Gus Gazzard, Anthony King, and Douglas Rhee received consulting fees relating to this research. Sergey Muratov is an employee of IQVIA, the company contracted by Alcon to conduct the research. DO'B is an employee of Alcon, the research sponsor.

Funding: The study was funded by Alcon

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P089

IDENTIFYING BLIND GLAUCOMA PATIENTS WITH A SEMI-AUTOMATED ALGORITHM: A CROSS-SECTIONAL SURVEY REVEALING MISSED CERTIFICATION OF VISUAL IMPAIRMENT

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Introduction: In the UK, severely sight impaired (blind) patients are offered a certificate of visual impairment (CVI), which entitles them to social benefits and support in the community. Registration requires referral from a consultant ophthalmologist, and eligibility guidance emphasises that decisions are clinical despite providing suggested visual acuity and perimetry cut-offs determining sight impairment. Many eligible patients do not receive a CVI; some due

to personal choice by many due to lack of consideration or communication by their doctor. A significant source of error lies in the subjectivity of decision-making based on perimetry plots, and while standardised criteria exist to improve assessment, these are generally too labour-intensive for routine use.

Method: Here, we undertook a retrospective cross-sectional survey of glaucoma patients attending a tertiary referral clinic over a period of 1 year. We designed a semi-automated screening protocol based on raw visual acuity (expressed in logMAR format) and Hodapp-Parrish-Anderson criteria implemented for patients' perimetry plots by an online computer vision application, Glaucoma Field Defect Classifier (GFDC). Severely sight impaired patients were analysed further to explore whether there were recurrent reasons for missed registration. To validate GFDC, 168 consecutive glaucoma patients' perimetry plots were inputted to the web application, and expert human researchers were tasked with classifying the same plots to calculate an accuracy score.

Results: 892 sight impaired and 64 severely sight impaired (blind) patients were identified from a total cohort of 5,620. Of the 64 blind patients, 7 (11%) were misclassified by the semi-automated algorithm, and 36 (56%) were already registered for a CVI. 21 of 57 eligible patients (37%) were not CVI registered. Reasons for missed registration varied, but five common themes emerged on deeper analysis: administrative failure, lack of consent, reversible visual impairment, frailty and co-morbidity, and mental health diagnoses. All misclassification cases were due to omission of data during implementation—the system did not make a single false positive classification. In the validation experiment trialling GFDC's ability to classify visual field defects based on Hodapp-Parrish-Anderson criteria, the web-application reached the same conclusion as expert human researchers in 168 of 168 consecutive glaucomatous perimetry plots: an accuracy of 100%.

Conclusion: Objective visual acuity and perimetry criteria can successfully identify blind patients eligible for social support, and implementation of a web application facilitates efficient use of these criteria to facilitate semiautomated screening of large cohorts. In this study, some 37% of blind glaucoma patients were not CVI-registered, sometimes due to concurrent illnesses conferring eligibility for alternative forms of support, but likely also due to subjectivity inherent in certification procedures. By synthesising national guidelines into an objective and semi-automated algorithm, we provide a semi-automated screening schemata which could be used to prompt ophthalmologists to consider registration or used as an alternative referral mechanism. This screening may reduce the number of vulnerable patients missing out on crucial support in the community, and improve the equity and fairness of care provision.

Conflicts of interest:

RB has received consultancy fees/honoraria from Thea, Santen, and AbbVie; and is eligible for payment in receipt for provision of CVIs as a consultant ophthalmologist. AJT and RS have received funding from HealthSense for machine learning research.

Funding: None

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P090

MINIMALLY INVASIVE TUBE SURGERY (MITS) - A NOVEL METHOD IN GLAUCOMA DRAINAGE DEVICE IMPLANTATION

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Introduction: We present a novel technique to inserting a glaucoma drainage device eyePlate300. The plasticity of the eyePlate300 is such that it can be folded and inserted without creating a large conjunctival peritomy. The senior surgeon (Faisal Ahmed) created a novel technique of minimally invasive tube surgery (MITS) during the COVID-19 lockdown.

The conjunctival peritomy is fashioned with a radial incision in the superotemporal quadrant, blunt dissection is performed posteriorly & circumferentially to allow space for the folded eyePlate300 to unfold onto bare sclera 10 mm from the limbus. The superior and lateral recti are identified, the eyePlate300 is sutured to the sclera. The tube is sized, stented and tied-off with a 7/0 vicryl before being inserted into the anterior chamber. The peritomy is sutured closed.

Method: We performed a retrospective analysis to assess the efficacy of the MITS technique using the eyePlate300 in a tertiary teaching eye hospital in London with 12 months follow up.

Thirteen patients were included who underwent the MITS technique. Primary outcome was complete success: defined as 'IOP > 5 mmHg and < 21 mmHg and off-drops. Secondary outcomes were best corrected visual acuity (BCVA) at 12 months (LogMAR), complications and number of drops at 12 months.

Results: Twelve patients (92%) had significantly lower IOP and one patient (8%) achieved the same IOP after 12 months with reduced number of medications. Mean pre-operative IOP was 35.69 mmHg (20–53 mmHg). Mean 12 months post-operative IOP was 11.08 mmHg (5–20 mmHg). Mean reduction in IOP was 24.62 mmHg $\pm$ 2.863 ($p < 0.001$).

Mean pre-operative drops was 3.462 (3–4) significantly reducing to 0.85 (0–3) at 12 months ($p < 0.001$). And at 1 year nine patients were on 1 drop or lower.

Mean BCVA (LogMAR) pre-operatively was 1.354 (0.0–2.4). Mean BCVA at 12 months was 0.90 (0.0–2.4). No intraoperative complications were recorded and no further IOP lowering surgeries were required.

Conclusion: The MITS technique facilitates the implantation of a glaucoma drainage device without the need for a large a limbal peritomy, employing a smaller incision in the conjunctiva to minimise trauma. Preliminary results from the surgery suggest that it may effectively and safely reduce IOP.

Conflicts of interest: No conflicts of interest

Funding: No funding

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P091

MORE THAN MEETS THE EYE: CATARACT SURGERY'S EVOLVING ROLE IN GLAUCOMA TREATMENT

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Introduction: Cataract surgery is a cornerstone of ophthalmic care, offering not only vision restoration but also intraocular pressure (IOP) reduction in glaucoma patients. For primary open-angle glaucoma (POAG), phacoemulsification alone has shown efficacy in reducing IOP, particularly in mild cases. However, adjunctive procedures such as minimally invasive glaucoma surgeries (MIGS), including iStent, Hydrus Microstent, and GATT, may provide enhanced outcomes in patients with moderate to advanced disease. This review explores the role of cataract surgery as a platform for managing coexisting glaucoma and the comparative benefits of combining MIGS with phacoemulsification.

Method: A systematic literature review was performed using Ovid Medline and Embase to identify studies evaluating the impact of cataract surgery, with and without MIGS, on IOP reduction, medication use, and safety in patients with POAG. Data were analysed to assess the relative advantages of standalone phacoemulsification versus combined approaches in achieving better clinical outcomes.

Results: Phacoemulsification alone effectively reduces IOP in patients with mild glaucoma.

- In the **cataract surgery-only (CS)** group, IOP decreased from 20.93 ± 1.28 mmHg to 17.79 ± 2.50 mmHg at 24 months, accompanied by a modest reduction in medication use from 1.03 ± 0.19 to 0.76 ± 0.69 .
- For patients undergoing **Hydrus Microstent combined with cataract surgery (HMS+CS)**, IOP reduction was more pronounced, from 25.5 ± 3.0 mmHg to 17.4 ± 3.7 mmHg, with a significant decline in medication use from 1.7 ± 0.9 to 0.3 ± 0.8 .
- Similarly, **Phaco-GATT** demonstrated superior outcomes, with IOP decreasing from 29 ± 2.88 mmHg to 12 ± 2.77 mmHg, and complete cessation of medications in many cases.

The addition of MIGS to cataract surgery provided enhanced IOP control and a greater reduction in the reliance on medications, particularly in advanced glaucoma cases.

Conclusion: Cataract surgery not only improves visual function but also offers a valuable opportunity to address elevated IOP in patients with coexisting glaucoma. While standalone phacoemulsification is sufficient for mild POAG, adjunctive MIGS procedures significantly enhance outcomes in patients with moderate to advanced disease. These findings support a tailored approach to cataract surgery, integrating glaucoma management strategies to optimize patient outcomes. Further research should explore the long-term benefits and safety of combined procedures, ensuring comprehensive care for this patient population.

Conflicts of interest: None

Funding: None

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P092

MORTALITY FOLLOWING GLAUCOMA SURGERY IN OCTOGENARIANS WITH PRIMARY OPEN ANGLE GLAUCOMA

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Introduction: This study aims to quantify the mortality rate following glaucoma surgery in octogenarian patients with primary open-angle glaucoma (POAG) and identify risk factors associated with mortality in this age group.

Method: A retrospective case series was conducted involving 115 patients aged 80 and over who underwent glaucoma surgery at the Birmingham Midland Eye Centre between January 2015 and December 2019. Demographic data, systemic comorbidities, and ophthalmic parameters, including intraocular pressure, visual acuity, and frailty scores, were collected and analysed.

Results: The all-cause mortality incidence for patients over 80 undergoing glaucoma surgery is 66.86 deaths per 1000 person-years. On univariable analysis, pre-operative risk factors[hazard ratio [HR], 95% Confidence interval (CI)] for increased mortality include age,[1.099(1.013,1.192)], clinical frailty score [1.250(1.036,1.507)], COPD

[2.187(1.252,3.818)], dementia [2.397(1.075, 5.343)], myocardial Infarction [2.392 (1.208, 4.737)], respiratory system disease[1.974, (1.179, 3.305)], and solid tumours [7.344 (2.170, 24.849)]. Visual acuity was the only ophthalmic factor significantly associated [1.7453 (1.046, 2.902)] with mortality, while other glaucoma-related parameters, such as intraocular pressure and retinal nerve fibre layer thickness, were non-significant. On multivariable analysis, Afro-Caribbean became significant [0.0469(0.224, 0.979)], and all univariable risk factors except for clinical frailty score and dementia. In terms of glaucoma surgery, cyclodiode laser is the only operation that had a significant hazard ratio [2.271 (1.083,4.762], likely due to selection bias of severely ill patients.

Conclusion: The overall five-year survival rate post-surgery was 72.01%, with a nine-year survival rate of 43.30. Clinical frailty scale should be included and assessed for all elderly patients considered for glaucoma surgery. Assessing systemic health in elderly glaucoma patients is crucial for optimizing surgical outcomes and potentially improving survival in this population.

Conflicts of interest: None

Funding: None

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P093

OUTCOMES OF GLAUCOMA REFERRAL FILTERING SERVICES BASED IN PRIMARY CARE OPTOMETRY IN ENGLAND FROM 2021 TO 2024

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Introduction: Glaucoma referral filtering services are long established and advocated by NICE (National Institute for Health and Care Excellence) and GIRFT (Getting It Right First Time) to reduce the burden of false positive referrals on Hospital Eye Services (HES). However, these services are not universally commissioned across England.

While there have been previous evaluations of such services, these have focussed on single localities and larger scale multi-locality outcome data is required. We review the outcomes of two levels of filtering pathways delivered through Primary Eyecare Services: The Glaucoma Repeat Readings Service (GRRS), commissioned within 26 Integrated Care Boards (ICBs), and the Glaucoma Enhanced Referral Service (GERS), commissioned within 11 ICBs.

Method: Glaucoma filtering services allow additional measurements and/or investigation to be undertaken by primary care optometrists to determine whether a glaucoma referral to HES is required.

GRRS is the first level of referral filtering following a routine eye examination. It allows repeat intraocular pressure (IOP) measurement by applanation tonometry (AT) and repeat visual field (VF) on some pathways, when IOP by non-applanation tonometry is ≥ 24 mmHg and optic disc appearance is healthy.

GERS allows a higher level of case finding for patients with non-urgent signs suggestive of glaucoma or ocular hypertension (OHT), typically identified at a sight test. Delivered by primary care optometrists with additional glaucoma accreditation, the assessment includes identification of risk factors, AT, assessment of the anterior chamber, optic nerve head, and VF.

Primary outcomes for both services are discharge to routine sight testing or referral to HES with appropriate urgency and clinical report.

Outcomes for all GERS and GRRS episodes delivered by Primary Eyecare Services via primary care optometry practices from April 2021 to March 2024 were analysed using the clinical data and outcomes recorded through the IT platform.

Results: 20,112 patients entered GRRS from 1st April 2021 to 31st March 2024 (mean age 64.8 ± 13.4 yrs, 53% female, 47% male). 13,783 patients (68.5%) were assessed within the service and discharged, 11,803 (58.7%) following first assessment and a further 1980 (9.8%) following second assessment. The remaining 6329 (31.5%) patients were referred to secondary care, 11.3% following first assessment and 20.2% following second assessment. 14,376 patients entered GERS from 1st April 2021 to 31st March 2024 (mean age 65.2 ± 13.6 yrs, 53% female, 47% male). 49.2% (7080/14,376) of patients were discharged to routine follow up, avoiding a referral to secondary care. The remaining 50.8% (7296/14,376) were referred to secondary care, 45.3% routinely to glaucoma service, 4.1% urgently to glaucoma service and 1.4% referred to a speciality other than glaucoma.

Conclusion: Confirmation of initial measurements using GRRS standardised techniques and protocols results in deflection of single spurious measures of raised IOP or VF that would otherwise be referred into HES. The GERS pathway takes the logical next step in case-finding through structured investigation when there is evidence of optic nerve head change at the sight test.

Due to the additional clinical work, commissioning costs are higher in GERS than GRRS. When initial sight test findings do not include potentially glaucomatous optic nerve head changes the GRR pathway is a cost-effective filter, deflecting a high percentage of potential referrals. However, where optic neuropathy is suspected, analysis of a large number of GERS episodes shows that this more sophisticated case-finding approach is also an effective filter. GERS reduces demand for initial assessment in specialist ophthalmology services by around a half.

Both GRRs and GERS provide an effective reduction of false positive referrals to the HES, with a $\approx 50\%$ deflection rate for GERS, and $\approx 70\%$ for GRRS. There is substantial potential to improve health system efficiency through consistent commissioning of GRRS and GERS across England.

Conflicts of interest: None

Funding: None

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P094

PACS PLUS CRITERIA: A RETROSPECTIVE AUDIT REVIEW OF PATIENTS TREATED WITH YAG PERIPHERAL IRIDOTOMY PRE AND POST RCOPHTH GUIDELINE IMPLEMENTATION

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Introduction: Publication of the Royal College Guideline of Management of Angle Closure Glaucoma in June 2022, gave clear guidance on the management of patients at low risk of sight loss from glaucoma or pathology related to acute angle-closure in people with occluded drainage angles. Those with presumed occluded angles should only be referred on the basis of elevated IOP, glaucoma, or with risk factors constituting "PACS PLUS".

Management of narrow angles in primary angle closure suspects (PACS) with laser peripheral iridotomy (LPI) is advised in patients with PACS plus additional risk factor(s) such as an "only eye", a family history of significant angle closure disease, high hypermetropia, diabetes or other conditions necessitating regular pupil dilation, use of antidepressants anticholinergic medications, and those living or working in remote locations. In patients without additional risk factors LPI is not recommended.

Our retrospective cohort review of patients undergoing LPI in a tertiary university hospital eye service, aimed to identify if

patients undergoing elective LPI met the criteria for PACS plus requiring laser treatment pre and post guideline implementation.

Method: A retrospective audit review of all patients who received a LPI in 2021 and in 2023 in the Eye Department at Queen Elizabeth University Hospital Birmingham, UK. All laser procedures performed were documented in a paper laser logbook. The hospital laser logbook was reviewed to identify all patients that underwent LPIs between 01/01/2021–31/12/2021, and post RCOphth guideline introduction between 01/01/2023–31/12/2023. Electronic patient records for corresponding patients were identified and reviewed to determine the diagnosis PACS and presence of “Plus” criteria.

Data collection included age, sex, ethnicity, only eye status, high hypermetropia, use of antidepressants or anticholinergic medication, family history of angle closure disease, need for repeated dilation, vulnerable adult status and limited access to healthcare.

For the purpose of this audit, diagnosed PACS patients treated with LPI were included in data analysis. Other diagnoses of Primary Angle Closure (PAC), Primary Angle Closure Glaucoma (PACG) and Acute Angle Closure (AAC) in patients receiving LPI were excluded from data analysis.

Results: Forty eight patients with gonioscopy-confirmed diagnosis of PACS were treated with YAG PI in 2023, compared with seventeen patients treated in 2021. The mean age of patients presenting with angle closure disease was 65.3 years in 2021 and 59.3 years in 2023. Gender preponderance in both years gave similar results with female majority (61%) compared with male (39%) patient cohort.

Prior to PACS “Plus” guideline implementation in 2021, 8 (47.1%) patients had no “Plus” features compared with 11 (22.9%) in 2023 post guideline practice. Thirty seven (77.1%) patients met PACS Plus criteria in 2023 with twenty six (70.3%) patients having 1 “Plus” feature, 10 (27%) had 2 “Plus” features and 1 (2.7%) patient had 3 “Plus” features. Consistently, the “Plus” criteria was met in 2023 cohort due to the use of antidepressant or anticholinergic medication in twenty four patients (64.9%), diagnosis of diabetes mellitus or requiring frequent dilation in eleven patients (29.7%) and/or having an “only” eye (16.2%).

Conclusion: A demonstrable improvement in practice of identifying patients at risk of acute angle-closure was demonstrated in review of practice with LPIs in suitable PACS Plus patients pre and post RCOphth guideline publication in 2022.

Implementation of this guideline has shown improved PACS patient selection for LPI, avoiding overtreatment and optimising service resources.

Notably, the majority of confirmed PACS Plus patients met this criteria due to use of antidepressant or anticholinergic medication, increasing the risk of angle closure disease. It is therefore imperative that a careful and detailed medication history is taken to enable safe and accurate patient diagnosis, and guide appropriate management.

Conflicts of interest: None

Funding: None

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P095

REAL WORLD LONG-TERM OUTCOMES OF LASER PERIPHERAL IRIDOTOMY FOR ACUTE PRIMARY ANGLE CLOSURE: THE LEICESTER EXPERIENCE

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Introduction: Acute primary angle closure (APAC) is an ophthalmological emergency. There is limited published data on the long term visual and intraocular pressure (IOP) outcomes for APAC treated by laser peripheral iridotomy (PI). We conducted a real-world study to explore these outcomes.

Method: We conducted a retrospective cohort study at Leicester Royal Infirmary. Patients diagnosed with APAC with a minimum of 12 months follow-up were included. The main outcome measure was IOP, at presentation and final visit, statistically analysed using the paired t-test. Secondary outcome measures included final visual acuity (VA), final visual field index (VFI), cup-disc-ratio (CDR) at presentation, proportion developing primary angle closure glaucoma (PACG), and subsequent cataract surgery or clear lens extraction.

Results: 51 eyes of 47 patients were included, as four patients had bilateral APAC. 29 patients were female, 18 were male. Mean age at presentation was 66 years (SD: 12 years; range: 34–90). Mean follow-up was 2.3 years (SD 1.0; range: 1.0–4.8). Mean IOP was 52 mmHg (SD: 15) at presentation, and 16 mmHg (SD: 8) at final visit, representing a statistically significant mean reduction of 36 mmHg (95% confidence interval: 31–41; $P < 0.0001$). 15 patients (32%) were diagnosed with primary angle closure (PAC), and 32 (68%) with PACG. 26 patients went on to have lens extraction during their follow up with an average of time 247 days (range 8882) after presentation. 5 patients had cyclodiode treatment with an average time of 234 days (range 35–508) with one patient requiring it repeated 392 days after the first session. 9 patients had further glaucoma surgery: Ab-interno Canaloplasty (2), Trabeculectomy (2) Miniject (1), selective laser trabeculoplasty (1), Aqueous tube shunt (1). Based on visual acuity alone in the affected eye, 9 patients met the criteria for severe sight impairment at the final visit. Mean final VA was 0.64 (SD: 0.84; $N = 49$). Mean final VFI was 73% (SD: 28%; $N = 36$). 18 patients (60%) had CDR 0.6 or greater at presentation. CDR at presentation was inversely correlated with long-term VFI ($R^2 = 0.52$; $F = 19$; $P = 0.0003$). 28 eyes (55%) underwent subsequent cataract surgery or clear lens extraction during the follow-up period.

Conclusion: Our study demonstrated that laser peripheral iridotomy was associated with significant long-term reduction of IOP by 36 mmHg on average. CDR at presentation was inversely correlated with long-term VFI

Conflicts of interest: None

Funding: Sohaib Rufai is funded by an NIHR Academic Clinical Lectureship (CL-2022-11-001).

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P096. Abstract not included

P097

SIMULATION-BASED TRAINING ENHANCES CONFIDENCE IN GLAUCOMA LASER PROCEDURES (PI AND SLT)

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Introduction: Peripheral iridotomy (PI) and selective laser trabeculoplasty (SLT) are essential procedures in the management of glaucoma, requiring both technical precision and confidence. Simulation-based training has proven to be a valuable tool for enhancing skill acquisition in ophthalmology. This study assessed the efficacy of simulation training on the confidence of London-based ophthalmic specialty trainees in years 1 and 2 in performing PI and SLT.

Method: Fifteen trainees completed a blended learning program that included e-learning modules and videos demonstrating the PI and SLT procedures. Trainees practiced both procedures on SimuEYE Ophthalmic Surgical Training Models during an in-person session under consultant supervision. Confidence levels were recorded on a 1–5 Likert scale before and after training. Descriptive statistics were used to summarize the data.

Results: For PI, pre-course confidence averaged 1.20 ± 0.14 , rising to 4.13 ± 0.13 post-course ($p < 0.0001$, paired students t-test). For SLT, pre-course confidence increased from 1.20 ± 0.14 to 3.47 ± 0.24 post-course ($p < 0.0001$, paired students t-test). Notably, 12 out of 15 trainees had no prior laser simulation training experience. All trainees expressed a desire for additional simulation-based laser training opportunities.

Conclusion: Simulation-based training, including e-learning, instructional videos, and hands-on practice, significantly improved confidence in performing PI and SLT among early-stage ophthalmology trainees. These results underscore the importance of simulation in building clinical competency, promoting patient safety, and ensuring better outcomes in glaucoma management.

Conflicts of interest: None

Funding: None

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P098

THE FUTURE OF GLAUCOMA: ESSENTIAL THEMES FROM THE PATIENT PERSPECTIVE

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Introduction: Glaucoma management has evolved significantly with greater utilisation of virtual and technician-led clinics, allied health professionals and technological innovations. It is essential that changes in how care is delivered are patient-informed. We sought to understand the patient perspective on the future of glaucoma patient care.

Method: Focus groups and one-to-one interviews were conducted with patients, recruited through a range of channels, including from databases of patients who have expressed an interest in participating in healthcare research, via social media channels and healthcare forums to ensure a spectrum of demographics, disease aetiology, severity and patient expertise (specifically to ensure the majority were non-expert patients). Interviews were qualitative in nature using an Interpretative Phenomenological Analysis approach to identify common themes and differences across age groups and ethnicities.

Results: Twenty-six patients were included in the study: 18 participating across three focus groups, and eight via one-to-one interviews. Seven of the participants were from higher-risk ethnic backgrounds for glaucoma.

Six key themes emerged as priorities to ensure positive care experiences:

- Accurate, targeted testing: Greater emphasis on the need for regular testing, with targeted messaging amongst those at greatest risk of sight loss.

- Confidence in testing process: trust in the accuracy of a diagnosis, with more specific advice aligned to individual circumstances and time to ask questions.
- Knowing the available options: empowering joint decision making and alleviating anxiety throughout their disease management (not only at the point of making decisions), with a clear explanation and rationale for the options available to them now and potentially available to them in the future (including surgical and nonsurgical options).
- Joined-up care: Clarity on decision pathways and who is responsible for what, so that factors important to patients are communicated without repetition and key stakeholders have the right information to recommend the best treatments.
- Ownership: Clarity on which clinician has clinical ownership – this could be the community optometrist, Consultant or another member of the multi-disciplinary team. In addition, patients want to know what **they** can be doing outside of the clinical touchpoints, with more information and tools to help them self-advocate with the knowledge that they are in direct contact with the expert leading their care.
- Patient-friendly communication: Clear explanations on how glaucoma and treatments like eye drops can impact their daily life to empower patients to access relevant material and support, including non-written formats.

Patients with a diagnosis of glaucoma who were from high-risk groups (due to ethnicity or family history) and lower socioeconomic groups highlighted the need for greater focus on targeting high-risk groups with specific and relevant information to improve engagement and self-advocacy, as well as improved awareness amongst front-line healthcare professionals such as opticians to ensure targeted screening is performed.

Conclusion: This study highlights patients' expectations of their glaucoma care and tangible ways in which their experience could be improved. The data gathered, although qualitative in nature, is structured and can be deployed to build patients' confidence in the system and support engagement. Patients value a stronger sense of 'agency' around the management of their long-term condition. Higher-risk groups need to be engaged at every stage in order to reduce risk of sight loss from glaucoma.

Conflicts of interest:

Kay Fisher was a paid Consultant to Santen who funded the study.

Simon Harris is an employee of Santen UK & Ireland who funded the study.

Funding: The research was funded by Santen UK & Ireland.

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P099

THE REAL WORLD MANAGEMENT AND TWO YEAR OUTCOMES OF PATIENTS UNDERGOING CYCLODIODE LASER

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Introduction: Cyclodiode laser is traditionally reserved for patients with poor visual potential and painful ocular symptoms. Increasingly it is seen as a primary treatment option in eyes with good visual potential in patients who are not candidates for surgery or who struggle with the burden of topical treatment. We wanted to examine the use of cyclodiode laser in patients in a single centre in the UK, the indication for its use and whether any major complications occurred as a result.

Method: A retrospective computer search was carried out over 2 years from December 2022 to December 2024 and the notes of all adult patients who had undergone cyclodiode laser were reviewed. Four main outcomes were reviewed:

1. Patients were undergoing cyclodiode for correct indications: no perception of light (NPL) and reporting pain; NPL and reporting no pain but needing intraocular pressure (IOP) control; poor visual potential (6/60 – PL) and reporting pain; poor visual potential and reporting no pain but needing IOP control; good visual potential (>6/60) requiring IOP control without pain.
2. For patients with good visual potential - whether low power settings were being employed for their treatment. As per the UK National Cyclodiode Laser Survey (2011) we consider this to be 1500 mv and 2000 ms
3. Rate and type of complications (phthisis, inflammation or decreased vision/wipe out)
4. Mean IOP reduction in patients for each indication

Results: 125 adult patients underwent cyclodiode laser between December 2022 and December 2024. 99% of those patients had a correct indication for treatment. Those undergoing cyclodiode laser with good visual potential had low power settings used in 93.5% of cases. 96% of cases had no major complications following treatment. 5 patients with poor visual potential progressed to NPL following treatment and one patient with good visual potential progressed to hand movements. The main subtype of glaucoma requiring cyclodiode laser was neovascular glaucoma. The largest cohort requiring cyclodiode laser was in eyes with poor visual potential (6/60-PL) who reported no pain, but needed IOP control in 29.6%. This was followed by 25.6% of cases who had poor visual potential and pain and 25% of cases who had NPL vision and pain. The largest mean IOP reduction was 27.5 mmHg and this was in patients with poor visual potential, reporting pain. Painful NPL eyes had a mean reduction of 18.7 mmHg. Those with good visual potential needing IOP control had a mean reduction of 8 mmHg.

Conclusion: The largest reductions in IOP were seen in patients with poor visual potential which could be attributed to the more aggressive settings utilised. No patients who had good visual potential suffered from wipeout of vision, although one patient did progress from 6/48 to Hand Movements vision. 5 patients progressed to no perception of light who already had poor visual potential. We conclude that cyclodiode laser need not be reserved for solely painful blind eyes and can be safely utilised in patients with poor and even good visual potential to control both pain and intraocular pressure. For patients with good visual potential we recommend the use of more gentle settings, this is with the understanding that this comes with more modest reductions in IOP.

Conflicts of interest: None

Funding: None

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P100

THE USE OF PRESERFLO® MICROSHUNT IN APHAKIC GLAUCOMA

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Introduction: The Preserflo® MicroShunt is a novel approach to glaucoma filtration surgery by combining a minimally invasive device with significant intraocular pressure (IOP) lowering benefits. The procedure is associated with fewer post-surgical complications than trabeculectomy as a scleral flap, sclerostomy and peripheral iridectomy are not required. Results have shown the IOP lowering benefit in Preserflo® MicroShunt to be comparable to that of trabeculectomy; however, lower rates of post-operative complications, specifically hypotony have been demonstrated.

Filtration surgery for aphakic glaucoma is associated with higher complication and failure rates, with vitreous incarceration in glaucoma drainage devices an important cause of failure. Studies of Preserflo® MicroShunt have included patients predominantly with primary open angle glaucoma as well as other secondary causes of glaucoma, but to the best of our knowledge, there are no specific case reports or studies describing the use of the Preserflo® MicroShunt in aphakic glaucoma as a unique subgroup.

Method: Four eyes underwent implantation of Preserflo® MicroShunt for the management of aphakic glaucoma under local (peribulbar anaesthetic) or general anaesthetic between March 2020 and December 2023. A standard Preserflo® MicroShunt implantation technique was performed with topical Mitomycin-C 0.4 mg/ml for four minutes, with a 25-gauge bimanual anterior vitrectomy at the start of all cases. Follow up was as clinically indicated, with data collected at these various time points. All glaucoma medications were stopped immediately after surgery and Dexamethasone 0.1% preservative free (pf) every two hours and Chloramphenicol 0.5% pf four times daily commenced to the operated. Dexamethasone 0.1% pf was gradually tapered to stop over a period of three months and Chloramphenicol 0.5% pf was stopped after two weeks.

Results: Mean pre-operative IOP was 24.8 mmHg (17 mmHg – 35 mmHg) with all patients on four topical IOP lowering medications. Average length of follow up was 2.6 years (1 year – 4 years). At one year the mean IOP was 11.8 mmHg (10 mmHg – 14 mmHg). Intraocular pressures of 14 mmHg or less were achieved in 3/4 eyes without IOP lowering medication at all time points. None required IOP lowering medication at one year and to date. Visual acuity was maintained at baseline in all cases. There was no evidence of visual field or optic disc progression post-operatively and no serious intra-operative or post-operative complications were recorded.

Conclusion: Traditional filtration surgery in aphakic glaucoma poses many challenges with a high failure and complication rates. The Preserflo® MicroShunt has shown promising results in primary open glaucoma but to date has not been reported on for the surgical management of aphakic glaucoma. We describe the use of Preserflo® MicroShunt in the surgical management of aphakic glaucoma with promising results. Anterior vitrectomy as a surgical adjunct may be important in reducing failure rates due to vitreous incarceration in the setting of aphakia. Further studies of Preserflo® MicroShunt are warranted to evaluate its long-term safety and efficacy in the surgical management of aphakic glaucoma.

Conflicts of interest: None

Funding: None

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P101

USE OF MACHINE LEARNING IN THE DETECTION AND PREDICTION OF GLAUCOMA PROGRESSION: A SYSTEMATIC REVIEW

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Introduction: Glaucoma is the leading cause of irreversible blindness worldwide. Effective assessment of glaucoma progression could facilitate timely interventions and improve prognosis. This systematic review aims to evaluate the use of machine learning (ML) in the detection and prediction of longitudinal glaucoma progression.

Method: As outlined in the protocol (PROSPERO CRD42021272728), an online search of MEDLINE, EMBASE, Cochrane Controlled Register of Trials, and ClinicalTrials.gov was conducted on 31.12.2022. Papers addressing the review's primary outcome were included: ML algorithm detection or prediction of Glaucomatous Optic Neuropathy progression. Outcome data collected included ML algorithm type, maximum sensitivity, specificity, and area under the receiver operating characteristic curve (AUROC), with the QUADAS-2 tool used to assess risk of bias.

Results: Of the 577 records initially identified, 40 studies with over 156,000 eyes were included. Overall, 30 studies included traditional ML methods, and 13 studies included representational methods. 28 studies used supervised ML, whilst 12 studies used unsupervised ML. 24 studies performed classification analysis, 13 studies included regression analysis and 3 studies included both classification and regression analysis.

11 studies evaluated structural glaucoma data, including retinal nerve fibre layer thickness. Six studies detected glaucoma progression, with mean pooled maximum sensitivity, specificity, and AUROC of 68.9% (range 36.9–86.0%), 80.5% (49.094.0%) and 0.80 (0.60–0.91), respectively. Five studies predicted glaucoma progression, with mean pooled maximum sensitivity, specificity, and AUROC of 84.9% (range 81.0–93.0%), 73.5% (59.0–91.7%) and 0.82 (0.71–0.90), respectively. 23 studies evaluated functional visual field data. 10 studies detected glaucoma progression, with mean pooled maximum sensitivity, specificity, and AUROC of 74.8% (range 3.8–95.0%), 94.3% (90.0–98.1%) and 0.89 (0.82–0.94), respectively. Four studies predicted glaucoma progression, with mean pooled maximum sensitivity, specificity, and AUROC of 52.2% (range 20.0–72.0%), 72.3% (49.095.0%) and 0.73 (0.44–0.86), respectively.

Eight studies combined structural and functional data. Two studies detected glaucoma progression and six studies predicted glaucoma progression, with mean pooled maximum sensitivity, specificity, and AUROC of 66.6% (range 44.2–83.7%), 90.0% (85.0–100%) and 0.79 (0.61–0.88), respectively.

Conclusion: This systematic review found that ML algorithm detection and prediction of longitudinal glaucoma progression is highly accurate. However, there is significant heterogeneity in ML methodologies. Standardization is needed for robust evidence of the clinical utility of ML.

Conflicts of interest: None

Funding: None

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MEDICAL RETINA

P102

"IS YAG BETTER THAN STAB?": COMPARING OBJECTIVE, SUBJECTIVE VISION AND OCT ON SUB-THRESHOLD MACULAR GRID LASER (USING PASCAL) VERSUS ANTIVEGF FOR TREATMENT CRVO

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Introduction: Central Retinal Vein Occlusion (CRVO) can lead to significant vision impairment due to macular oedema and retinal ischemia. The traditional approach to managing macular oedema in CRVO has included anti-VEGF therapy, but recent studies have highlighted the potential role of sub-threshold macular grid laser (using Pascal). This audit aims to compare the efficacy of sub-threshold macular grid laser with anti-VEGF injections in CRVO treatment, focusing on both objective and subjective vision outcomes, OCT findings, and comparing the use of YAG laser versus stab laser for treatment.

Method: A retrospective audit was performed involving 30 patients with CRVO. The patients were divided into two treatment groups:

1. **Sub-threshold Pascal macular grid laser** (15 patients)
2. **Anti-VEGF therapy** (15 patients)

Inclusion criteria included patients with persistent macular oedema after CRVO. Exclusion criteria included other retinal conditions or severe diabetic retinopathy.

Follow-up was performed for 6 months. Visual acuity was measured using Snellen charts, and subjective improvements in vision were recorded. OCT was used to assess macular oedema, retinal thickness, and other retinal changes.

Results:

1. Visual Acuity:

- **Anti-VEGF group:** The mean visual acuity improved from 20/200 at baseline to 20/80 at 6 months. However, only 60% of patients in this group reported subjective improvement in vision.
- **Sub-threshold Pascal macular grid laser group:** The mean visual acuity improved from 20/200 to 20/100 at 6 months, with 80% of patients showing subjective improvement in vision. Notably, the patients who received the subthreshold laser reported higher levels of satisfaction with their vision in daily tasks, particularly in reading and near-vision activities.

2. OCT Findings:

- **Anti-VEGF group:** OCT showed a significant reduction in macular oedema in 80% of the patients, with 70% showing near-normal macular thickness by the 6-month follow-up.
- **Sub-threshold Pascal macular grid laser group:** OCT showed a more consistent reduction in macular oedema, with 90% of patients showing complete or partial resolution of oedema. The retinal thickness improved more rapidly in the laser group, and the improvement in retinal structure was sustained at follow-up.

3. Subjective Outcomes:

- **Anti-VEGF group:** Visual function improvement was reported by 60% of patients, with some patients experiencing difficulty with near-vision tasks despite the improvement in macular oedema.
- **Sub-threshold Pascal macular grid laser group:** 80% of patients in the laser group reported significant subjective improvements, particularly in visual clarity and comfort. These patients also experienced less visual distortion and better daily functionality, highlighting the benefit of laser treatment in improving overall visual quality.

4. Use of YAG Laser vs. IVI:

- **Laser:** 10 patients in the sub-threshold grid laser group were treated with YAG laser. These patients reported a higher level of comfort and fewer adverse effects, including a lower incidence of retinal haemorrhages. Visual outcomes were comparable to those of patients treated with anti-VEGF therapy, with a faster resolution of macular oedema

Complications:

- **Anti-VEGF group:** Two patients developed transient intraocular pressure increases, which were managed with topical medications. No other significant complications were reported.
- **Sub-threshold Pascal macular grid laser group:** The YAG laser group had no significant complications, while mild retinal haemorrhages were observed in two patients who received the anti-VEGF. These resolved without further issues, and no cases of retinal damage or macular atrophy occurred.

Conclusion: Anti-VEGF therapy showed better outcomes in terms of reducing macular oedema and improving visual acuity, but the sub-threshold Pascal macular grid laser treatment group demonstrated superior patient satisfaction and long-term structural improvements. Patients in the laser group had better subjective visual function, including improved near vision and reduced visual distortions, which is a critical aspect of daily living. The laser treatment, particularly with PASCAL yellow/ YAG laser, was associated with fewer complications and better overall visual quality compared to the stab laser.

The laser demonstrated a clear advantage in patient comfort and fewer adverse effects compared to the stab laser, making it the preferred choice for sub-threshold macular grid laser in CRVO.

Conflicts of interest: None

Funding: None

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P103

THE GROWING THREAT OF LASER-INDUCED RETINAL INJURIES: A CASE SERIES IN ADOLESCENTS

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Introduction: Laser light retinopathy is becoming more common, especially with the widespread availability of laser devices sold online. Many of these devices, often marketed as toys, are mislabelled and accessible to children, despite the potential for

causing retinal damage and permanent vision loss. Laser pointer retinopathy occurs when handheld lasers are directed at the eyes, causing retinal injury. Symptoms may include blurry vision or scotomas. Laser pointer induced retinal injuries can vary in their morphological presentation. Common findings include subretinal or intraretinal haemorrhages, retinal oedema, RPE scarring, and foveal granularity. In some cases, patients may develop vitreous or chorioretinal haemorrhages, perifoveal drusenoid-like deposits or pigment clumps, and ring-shaped hypopigmented lesions in the fovea. More severe injuries can result in macular holes or, rarely, choroidal neovascularization. In many cases, patients may present with pigment changes and chorioretinal scars, particularly in the foveal region, which are indicative of laser-induced maculopathy.

A 2016 UK survey found that 80% of those affected were under 20, with many cases going unreported, particularly if the injuries occur in unsupervised children or with improperly acquired lasers. The risk of eye damage from laser pointers increases based on several factors, including the wavelength of the laser, its power output, the duration of exposure, and the size of the laser spot.

Method:

Case series

Case 1

A 17-year-old male presented to the casualty clinic with a history of reduced visual acuity in his left eye. The unaided visual acuity was 0.50 in the left eye. The patient reported that a laser gun had been aimed at his left eye by a friend for approximately 2 minutes one year ago. The patient had no other medical conditions or previous ocular issues.

On clinical examination, fundoscopy revealed a normal optic disc in both eyes, with a healthy macula in the right eye. However, RPE changes were noted in the left eye. Optical coherence tomography (OCT) revealed photoreceptor and RPE damage in the left eye, which was consistent with laser-induced maculopathy. The patient was subsequently referred to the medical retina clinic for further management.

Case 2

A 16-year-old male presented to the retina clinic after an incident in which a laser pointer was aimed directly at his right eye. On clinical examination of the posterior segment, fundoscopy revealed a submacular haemorrhage and scarring in the right eye. OCT findings showed a pigment epithelial detachment (PED), subretinal fluid (SRF), and breaks in the Bruch's membrane. A fundus photo was also taken, which confirmed these findings. These features were consistent with photic retinopathy and secondary CNV. The patient was scheduled for fluorescein angiography (FFA) for further evaluation and management.

Discussion: Laser-induced maculopathy can result in permanent retinal damage and visual impairment, often without immediate symptoms. The varying manifestations of laser-related retinal damage, such as subretinal haemorrhages, retinal oedema, and RPE scarring, illustrate the complexity of these injuries. OCT imaging plays a crucial role in diagnosing and assessing the extent of damage. In both cases, OCT was instrumental in confirming the retinal changes, guiding management decisions, and providing evidence of laser-induced injury. Although treatment options are limited and self-resolution is often observed, continuous monitoring and patient education are key components in the management of these injuries. In both cases, the affected individuals were young, and the injuries occurred due to the misuse of laser devices, emphasizing the need for increased public awareness and safety measures.

Conclusion: Laser-related eye injuries, including maculopathy and photic retinopathy, are preventable causes of vision loss. Timely intervention and regular follow-up are essential for managing these injuries, particularly when retinal damage is evident. These cases underscore the importance of early diagnosis, patient education, and the need for greater regulation surrounding the sale and use of laser devices.

Conflicts of interest: None.

Funding: None

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P104

"WHY AM EYE NOT A CRITERIA": THIS PROPOSAL SEEKS TO ADDRESS A CRITICAL GAP IN THE CURRENT NHS GUIDELINES FOR CGM/GLUCOSE MONITOR PRESCRIPTION IN DIABETIC RETINOPATHY

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Introduction:

- Meta-Analysis to address and bridge the gap in current NHS Guidelines on the Use of Continuous Glucose Monitors (CGM) in Type 1 and Type 2 Diabetes with Significant Retinopathy
- The current NHS England guidelines allow the use of CGM for patients with Type 1 diabetes who meet certain criteria such as:
 - Frequent episodes of severe hypoglycaemia or hypoglycaemia unawareness.
 - Frequent hospital admissions for hypoglycaemia.
 - Pregnant women with Type 1 diabetes.
 - Children and young people with Type 1 diabetes who may benefit from continuous monitoring.

Type 2 diabetes patients may also be eligible for CGM, particularly those who have difficulty managing their blood glucose or are on insulin therapy, though this may require a referral to a specialist.

Method: This meta-analysis includes randomised controlled trials (RCTs), cohort studies, and observational studies from 2015 to 2023 involving T1DM and T2DM patients with significant retinopathy. A total of 12 studies were selected, with sample of 1200 participants (600 T1DM and 600 T2DM). Data focused on three key outcome measures:

1. Visual outcomes: Retinopathy progression and visual acuity changes.
2. Glycaemic control: HbA1c levels, time in range, glucose variability.
3. Lifestyle changes: Dietary adherence, physical activity, and quality of life

Inclusion Criteria

- Adults with T1DM or T2DM and significant retinopathy as determined by ophthalmological examination.
- CGM use as part of diabetes management.

Results:

- The studies included a mix of RCTs (5 studies) and cohort studies (7 studies), totalling 1200 participants, with an average age of 58 years and diabetes duration of 15 years. The mean baseline HbA1c across studies was 8.3%.
- **Retinopathy Progression:** CGM was associated with a 30% reduction in the risk of retinopathy progression compared to conventional blood glucose monitoring ($p = 0.02$). This reduction was particularly significant in patients with

moderate to severe retinopathy, supporting previous findings from the Diabetes Control and Complications Trial (DCCT) and the UK Prospective Diabetes Study (UKPDS), which demonstrated that tight glycaemic control slows retinopathy progression in T1DM and T2DM patients (DCCT Research Group, 1993; UKPDS Group, 1998).

- **Visual Acuity:** Those with HbA1c levels below 7% showed a trend toward stabilization or slight improvement in visual acuity ($p = 0.08$), suggesting that effective glycaemic management through CGM may help preserve vision in individuals with DR (Wong et al., 2016).
- **Retinal Imaging:** Retinal imaging revealed improvements in retinal microvascular health in 18% of CGM users, compared to only 8% in the control group ($p = 0.05$), supporting the hypothesis that tighter glucose control helps prevent further retinal damage (Simo et al., 2014).

Glycaemic Control:

- **HbA1c:** The use of CGM resulted in a significant reduction in HbA1c (mean reduction of 0.9%), improving glycaemic control (Heinemann et al., 2015).
- **Time in Range:** CGM users spent 71% of the day within the target glucose range (70–180 mg/dL) compared to 55% in the control group ($p < 0.001$), aligning with findings from other studies indicating that CGM increases time in target range and reduces glucose variability (Bergenstal et al., 2018).
- **Glucose Variability:** CGM users showed a 14% reduction in glucose variability (measured by standard deviation), highlighting the role of CGM in minimizing glycaemic fluctuations ($p = 0.03$) (Rodríguez et al., 2017).

Lifestyle Changes:

- **Dietary Habits:** Patients using CGM reported better dietary adherence, with 65% following recommended dietary guidelines, compared to 52% in the control group ($p = 0.01$). This is consistent with evidence that CGM promotes better diabetes self-management by providing real-time feedback on the effects of food on blood glucose levels (Vigersky et al., 2012).
- **Physical Activity:** CGM users increased their weekly physical activity by 22%, compared to 12% in those using SMBG ($p = 0.04$).
- **Quality of Life:** Patients using CGM reported improved quality of life, including reduced anxiety about hypoglycaemia and better diabetes management ($p = 0.03$), supporting findings that CGM can positively impact mental well-being (Bergenstal et al., 2018).

Conclusion: The meta-analysis suggests extended CGM use, ophthalmologist collaboration and patient education.

Conflicts of interest: None

Funding: None

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P105

1-YEAR OUTCOMES OF TREATMENT-RESISTANT NEOVASCULAR AGE-RELATED MACULAR DEGENERATION SWITCHED FROM AFLIBERCEPT 2 MG TO FARICIMAB

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Introduction: While previous studies have demonstrated the efficacy of Faricimab in treatment-naïve neovascular age-related macular degeneration (nAMD) patients, its outcomes in patients switched from Aflibercept are less understood. This study aims to assess 1 year clinical, anatomical and functional outcomes of switching to Faricimab in patients undergoing Aflibercept intravitreal injections for nAMD with suboptimal response.

Method: Retrospective case series review of patients switched from Aflibercept 2 mg to Faricimab due to persistent suboptimal response to Aflibercept (persistent presence of intraretinal (IRF) or subretinal fluid (SRF) on OCT despite receiving six consecutive 4 or 6-weekly Aflibercept injections). Patients were loaded with Faricimab then extended on a treat and extend protocol.

Results: 81 eyes of 68 patients were included. There was no change in mean visual acuity pre switch (65 letters) compared to 1 year post switch (65 letters), $p = 0.63$. All patients pre switch had IRF and/or SRF on OCT. 1 year post switch 37/78 (47%) of eyes were dry. There was a significant increase in the mean injection interval between injections 1 year post switch (6.8 weeks) compared to pre switch (4.4 weeks), $p = 0.0001$.

Conclusion: Among patients with treatment-resistant nAMD, switching from Aflibercept to Faricimab may serve as a safe and effective option. Switching to Faricimab showed significant anatomical improvements on OCT and eyes were able to be extended to less frequent injections at 1 year.

Conflicts of interest: None

Funding: None

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P106

A RETROSPECTIVE AUDIT TO SHOW THE IMPACT OF INTRODUCING A LONG ACTING STEROID INTO THE DMO TREATMENT PATHWAY IN THE EYE DEPARTMENT OF A DISTRICT GENERAL INFIRMARY

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Introduction: A retrospective audit to show the impact of introducing a long acting steroid, Fluocinolone acetonide intravitreal implant 0.19 mg (FAC) into the Diabetic Macular Oedema (DMO) treatment pathway in the Eye Department of a District General Infirmary - with its effect on patient outcomes and the service, focusing on clinical benefits and cost/resource impact.

In this retrospective audit we have analysed 21 eyes from 17 patients treated with FAC for diabetic macular oedema at a District General Infirmary between 2022 and 2024, to demonstrate how it's introduction further improved the efficacy and safety of the treatment of patients with DMO. The audit analysed budgetary and capacity outcomes to quantify the outcomes on the NHS.

Method: Data from Electronic Medical Records were identified for all patients receiving FAC for chronic DMO between 2022 and 2024 at the Ophthalmology Department. Best corrected visual acuity (BCVA) with ETDRS chart and Central Macula Thickness (CMT) were recorded at each visit. The incidence of adverse events including patients requiring treatment for elevated intraocular pressure were recorded as well.

The beneficial impact of introducing this long-acting steroid implant on the whole service was also analysed.

Results: A total of 21 eyes of 17 patients were evaluated. (Ratio of 8:9 M:F, 76% Type II diabetes). Mean age at baseline was 66.0y. 23.5% of patients were treated bilaterally. Patients had previously received an average of 9.0 anti-VegF injections and 3 Dexamethasone intravitreal implant injections prior to their FAC

Mean BCVA improvement was 5 letters (ETDRS) from 59 letters (ETDRS) to 64 letters (ETDRS) after first FAC implant with 17.6% of patients gaining 22–23 letters (ETDRS) of vision. 14.2% of patients were reporting loss of vision, but none of them lost more than 3 letters (ETDRS). Mean reduction of CMT following the FAC implant - 549 μ m to 394 μ m, a 28% reduction.

19% of patients had an increase in IOP over 10 mmHg from baseline. These were successfully managed with IOP lowering drops. 14.2% of patients in the study required additional IOP lowering drops (9.5% on prior to FAC implant). No additional surgical or laser procedures were necessary.

In addition, patients with earlier switch to FAC implant showed more significant improvement of BCVA, average of 10 letters, while the whole audit group showed average of 5 letters improvement. Early switchers showed also more significant reduction in CMT, 41% (568 to 333 μ m) compared to average 28% reduction in CMT for the whole audit group.

Conclusion: This audit demonstrates improvement and maintenance of vision along with significant reductions in CMT in patients that have been previously treated and found to be insufficiently responsive to available therapies. The safety outcomes demonstrate that FAC implant can be a well-tolerated long-term method to manage disease activity in these patients.

Furthermore, we have been able to demonstrate additional gains in BCVA in those patients who were switched earlier to FAC implant, i.e. after receiving an average of 3 anti-VegF injections and a minimum of 1 Dexamethasone intravitreal implant. There were also fewer reported cases with elevated IOP in those early treated patients.

Earlier introduction of FAC in the DMO treatment pathway for suitable patients demonstrates a clear benefit to the trust. It enabled not only clinical outcome gains, but also budgetary and capacity gains to the NHS. Patients required fewer intravitreal injections, fewer monitoring and management visits within our macular service.

Conflicts of interest:

Support to attend Conference

Sponsorship to attend RCO Congress 2025 from Alimera Sciences Ltd.

Funding: None

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P107

A REVIEW OF ORAL NUTRITIONAL SUPPLEMENTS COMMERCIALY AVAILABLE IN THE UK FOR AGE-RELATED MACULAR DEGENERATION

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Introduction: Dry age-related macular degeneration (AMD) is a sight-threatening condition, accounting for 8.7% of global blindness and constituting the leading cause of irreversible blindness in developed countries. There are currently limited options in the UK to prevent progression of dry AMD, or its conversion to wet AMD. Patients with intermediate or advanced

dry AMD and/or wet AMD should be advised to consider taking AREDS2 formulation oral nutritional supplements. The AREDS2 formulation contains: 80 mg of zinc, 2 mg of copper, 500 mg of vitamin C, 400 IU of vitamin E, 10 mg of lutein and 2 mg zeaxanthin. The latter two ingredients are xanthophyll carotenoids with antioxidant activity, which are selectively uptaken into the retina. AREDS2 supplementation has been shown to reduce the risk of progression from intermediate to advanced AMD by 25%. There is currently no published research reviewing the various commercially available oral nutritional supplements for AMD in the UK.

Method: A review of all commercially available oral nutritional supplement products in the UK for AMD was undertaken. Products were identified by searching in-store and online retail websites within the UK.

Data for 24 products were extracted, thus enabling the identification and comparison of: ingredients, AREDS2 formulation compliance, number of tablets required per day, cost per day and vegetarian status.

Results: Whilst 83% ($n = 20$) of products contained AREDS2 formulation ingredients, the majority of products failed to provide sufficient dosages of each individual ingredient. Only two products contained the correct dosages to be considered 'AREDS2 formulation-compliant': Bausch & Lomb PreserVision by Bausch & Lomb, and MacuSan A2 by MacuSan. The most common reason for failing to meet the AREDS2 formulation threshold was containing insufficient zinc, with many products containing only 10 mg zinc, compared to the 80 mg dosage recommended in the AREDS2 formulation. 60% ($n = 12$) of the 20 AREDS2-containing products were vegetarian. The mean cost per day for these 20 products was £0.64. Interestingly, 4 of the 20 products contained another xanthophyll carotenoid called meso-zeaxanthin.

Conclusion: Patients with intermediate or advanced dry AMD and/or wet AMD should be advised to consider taking AREDS2 formulation oral nutritional supplements, however, it is important to be aware that not all oral nutritional supplements in the UK are AREDS2 formulation-compliant.

There is also wide variety between products regarding cost, daily tablet requirement and vegetarian status, which may influence product suitability.

Further investigation is required into meso-zeaxanthin to evaluate its efficacy in the prevention of AMD progression; as the most prevalent carotenoid at the fovea, and with more significant antioxidant activity and chemical stability than lutein, meso-zeaxanthin may prove to be a powerful tool in the armamentarium against AMD in the future.

Conflicts of interest: None

Funding: None

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P108

A TWIN-CENTRE STUDY EVALUATING THE EFFICACY OF GRID/ FOCAL PASCAL SUBTHRESHOLD LASER WITH VISION, CRT, RECURRENCE RATE AND PROMS IN CENTRAL SEROUS RETINOPATHY (CSR)

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Introduction: Central Serous Retinopathy (CSR) is a common retinal condition that primarily affects young individuals,

characterised by fluid accumulation beneath the retina, leading to blurred vision and potential long-term retinal damage. Traditionally, CSR has been managed through observation, with treatments like photodynamic therapy (PDT) and laser therapy carrying the risk of central vision damage, particularly with repeated use.

This study investigates the efficacy of sub-threshold PASCAL laser therapy in CSR, comparing outcomes across grid, focal, and combined laser treatments. Additionally, the study explores whether this treatment provides a solution to the visual disturbances and anxiety related to potential vision loss in young patients, addressing both clinical outcomes and patient quality of life.

The study will also suggest an appropriate technique of using grid format of subthreshold laser with Landmark settings on the PASCAL laser.

Method: Inclusion: 50 patients from each centre with confirmed diagnosis of CSR on FFA/ICG

Exclusion: AMD/ other macular, retinal and Optic nerve disease were excluded to reduce bias in outcome

Methodology: All patients with confirmed diagnosis had Vision, OCT done at every visit- Baseline, 1, 3 and 6–12 months. It also included a short PROMS questionnaire at the end of treatment. Patients were treated with focal, grid and combined approach on PASCAL laser and results were subgrouped. Focal treatment was applied to point of leak on dye test and grid was applied to the pachychoroid seen on OCT and funduscopy imaging

PROMs explored subjective vision, visual symptoms post laser and satisfaction level on a scale of 1–10. Hypertension, use of steroids and type of personality were explored in the study.

Technique of laser: Grid approach combining the OCT for pachychoroid spectrum and Colour fundus to visualise dilated choroid. Will be described with fundus photos in future if approved. The use of landmark laser in sub threshold will be elaborated in detail.

Results: Our findings suggest that subthreshold PASCAL laser therapy is a safe and effective alternative, offering significant benefits for patients with chronic or recurrent CSR by minimizing the risk to central vision.

The results show that sub-threshold laser treatment led to significant reduction in subretinal fluid, as confirmed by OCT imaging. At 6 months, 85% of patients demonstrated complete or near-complete resolution of subretinal fluid. Visual acuity improved in 78% of patients, with a mean improvement of 2 lines on the Early Treatment Diabetic Retinopathy Study (ETDRS) chart. Recurrence rates were lower with only 12% of patients experiencing a reoccurrence of CSR within the follow-up period. No significant retinal damage or vision-threatening complications were observed in any patient.

PROMS reinforced the above outcome.

The results also explore and make suggestion on the correct use of grid and focal laser in this pachychoroid spectrum disease. It emphasises on the correct technique of use of sub threshold grid laser using colour fundus and OCT scans.

Conclusion: PASCAL sub threshold laser using the modality of focal and grid laser can be used to address CSR effectively.

Sub-threshold laser therapy is a promising and safe treatment option for patients with chronic or recurrent CSR. It provides a non-invasive approach that offers significant improvement in retinal fluid resolution and visual acuity, with a low risk of complications. Further long-term studies are needed to confirm the durability of these results and to refine treatment protocols for optimal patient outcomes.

Conflicts of interest: None

Funding: None

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P109

A YEAR OF REAL-WORLD EXPERIENCE WITH WIDEFIELD OPTICAL COHERENCE TOMOGRAPHY AS AN ALTERNATIVE TO FLUORESCIN ANGIOGRAPHY FOR DIABETIC RETINOPATHY ASSESSMENT

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Introduction: Widefield Optical Coherence Tomography Angiography (WF-OCTA) is an innovative imaging modality that provides high-resolution, detailed images of the retina and retinal vasculature. Unlike traditional Fluorescein Angiography (FA), WF-OCTA is non-invasive and does not require intravascular dye injection.

This study aims to evaluate the effectiveness and reliability of WF-OCTA as an alternative to FA for detecting and monitoring the progression of Diabetic Retinopathy (DR) over a one-year period.

Method: This retrospective, cross-sectional, descriptive study analysed data from 144 patients with diabetic retinopathy who underwent WF-OCTA imaging during DR screening clinics in a large NHS trust between January 2024 and December 2024. The specific indications for WF-OCTA, findings, and the necessity for further Fluorescein Angiography were documented.

Results: 218 eyes of 144 patients were imaged using WF-OCTA. Of these, 75.4% did not need further FA and WF-OCTA was sufficient to guide following management. This was particularly beneficial for 10.4% patients for whom FA was contraindicated or was unsuccessful (due to pregnancy, dye allergy, leaking cannula, etc.).

In 18.8% of all eyes WF-OCTA confirmed suspected neovascularisation, which allowed for prompt referral for an urgent pan retinal coagulation (PRP). WF-OCTA supported lack of active inflammation or NVE in 48.6% of eyes, meaning they continued to be monitored in clinic, without the need for invasive FA.

24.6% of patients required additional FA: 9.2% due to poor quality of WF-OCTA image, and 15.7% to further visualise the retinal vessels.

Conclusion: WF-OCTA proved to be an effective and reliable tool for monitoring of Diabetic Retinopathy (DR) in this real-world setting. While fluorescein angiography remains the gold standard for assessing PDR, WF-OCTA can significantly reduce the reliance on that invasive procedure in the context of the DR screening clinics.

Conflicts of interest: Tariq Aslam: Travel grants, lecture fees, research grants, advisory boards for Novartis, Bayer, Topcon, Canon, Heidelberg, Roche.

Funding: None

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P110

ADDRESSING RISK FACTORS FOR AGE-RELATED MACULAR DEGENERATION (AMD) IN CURRENT PATIENTS AND FUTURE GENERATIONS THROUGH SPECIALISED WET AMD CLINICS

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Introduction: Rapid access AMD clinics and frequent patient follow-ups fostered strong rapport with both patients and their

first-degree relatives. This environment not only promotes ongoing care for individuals with wet AMD but also serves as an essential platform for educating family members about preventive measures. By sharing knowledge on risk factors, early symptoms, and the significance of regular eye examinations, families can be empowered to take proactive steps in safeguarding their ocular health and potentially mitigating the effects of age-related macular degeneration in future generations

Method: A prospective study was conducted over one year at a district general hospital's one-stop AMD clinic. Patients with wet AMD and their first-degree relatives were educated about the disease process, risk factors, and preventive measures to reduce deterioration. During follow-ups, data were gathered on the preventive measures adopted by relatives. A total of 140 patients (mean age 82.4) requiring treatment were included, with an average of 4.6 visits. First-degree relatives included siblings, with 20% accompanying patients (mean age 78) and 18% not accompanying them (mean age 74.7). Children accompanied 70% of patients (mean age 61), while 26% did not (mean age 58).

Results:

- 70% of patients were ex-smokers, with a cessation period of 10–20 years.
- 6% of patients, with a mean age of 82, continued to smoke.
- 24% of patients either quit smoking, gradually reduced their intake, or participated in counselling to quit.
- 90% maintained a regular lutein-rich diet and used supplements.

First-degree relatives:

- 36% quit smoking.
- 7% regularly monitored their central vision using an Amsler grid.
- 76% adopted a healthy diet.
- 60% received cardiovascular assessments from their GPs and had eye examinations by optometrists.
- 72% sought additional information about lutein and zeaxanthin supplements through the internet or health shops.

Conclusion: AMD clinics can serve as platforms to reduce risk factors among patients and their family members, potentially preventing or delaying the onset of AMD in future generations.

Conflicts of interest: None

Funding: None

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P111

ADULT VITELLIFORM MACULAR DEGENERATION: A NEW UNITED KINGDOM PERSPECTIVE

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Introduction: With emergence of new imaging technology adult vitelliform macular degeneration (AVMD) is being diagnosed with increasing frequency over the last two decades. The condition is often mistaken for wet age-related macular degeneration (ARMD) when the vitelliform material starts scrambling showing pseudo fluid in optical coherence tomography (OCT) scans. In addition, in some AVMD patients the vitelliform lesions in fact do develop choroidal neovascularisation (CNV) requiring urgent intravitreal

injections. We aim to highlight the prevalence, natural progression and treatment options for patients presenting with AVMD.

Method: We have retrospectively analysed 123 eyes of 82 patients of AVMD presented to this department over the last 4 years.

Results: We identified 44 females and 38 males, 41 patients with bilateral involvement at presentation.

The clinical findings varied from a classical sub foveal yellow lesion to partial empty spaces within the lesion, to complete atrophic scar following resorption of all the colloid material. The presenting Visual acuity in the worse eye also varied from 6/9 to HM according to the stage of degeneration they presented with. Thirty-nine patients were asymptomatic and AVMD was an incidental finding in one of their eyes. We have obtained serial optical coherence tomography (OCT) scan to monitor the progress whenever possible.

All patients had OCT, optical coherence tomographic angiography (OCTA), and fundus autofluorescence (FAF). Some patients in addition underwent fundus fluorescein angiography (FFA) and indocyanine green angiography (ICG) when a choroidal neovascular membrane (CNVM) was suspected.

Sixty nine of 82 patients did not need any active interventions. Fourteen patients were already taking antioxidant supplementations.

Thirteen patients have developed significant subretinal fluid (SRF) and were treated with intravitreal anti-vascular endothelial growth factor (VEGF) injections. On further reviewing of the OCTA scans and FFA only 10 of 13 showed active CNVM. The other 3 patients only had passive subretinal fluid with no evidence of CNVM. Therefore only 12 % of AVMD patients seem to have developed wet ARMD in our series.

Response to the anti-VEGF injections was suboptimal in AVMD patients with CNV compared to a case of conventional wet ARMD. This unfavourable response was most likely due to pre-existing long standing atrophic and degenerative changes. Contrary to earlier publications, based on family history alone we did not find any strong evidence of autosomal dominance inheritance therefore we prefer to call it a degenerative rather than dystrophic condition.

Conclusion: With increasing use of OCT imaging by optometrists and primary care physicians more patients are being diagnosed with macular pathology, AVMD is one such condition. Most of these patients do not need any active interventions or long term follow ups in hospital clinics. Our study shows only 12% of AVMD patients go on to develop CNV requiring intravitreal anti-VEGF injections. As our cohort shows it may not be unusual to mistake passive fluid lakes within vitelliform lesions for active wet ARMD.

It is important for clinicians to be aware of the possibility of AVMD with passive subretinal fluid and not to resort to unnecessary long term intravitreal anti-VEGF injections. As we have shown even in cases of AVMD with secondary CNV the prognosis is poor for long term retention of central vision.

Conflicts of interest: None

Funding: None

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P112

ADULTS SICKLE CELL RETINOPATHY: PREVALENCE, PRESENTATIONS AND TREATMENT OUTCOMES

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Introduction: Sickle cell disease affects predominantly African/Afro-Caribbean descendants, less in Arabic/Eastern ethnicities. In

2021 population censor, 5.5% of England ethnicity is African/Afro-Caribbean with 0.4% haemoglobinopathies registration (from 2018 UK Public health record). To date, there is very limited published literature on sickle retinopathy especially in the guidance of disease management.

With the recent years of the availability of advanced diagnostic tools (ultra-widfield photography, OCT) and treatments options of laser photocoagulation, anti-VEGF, we are interested to investigate sickle retinopathy management outcomes from our local sickle retinopathy cohort

Purpose: To report adults' sickle retinopathy prevalence, referral pathway, disease severities and treatment outcomes in a single-centre eye clinic.

Method: All sickle patients attended our eye clinic had tests included vision test, IOP check, OCT scan, ultra-widfield photography and dilated fundoscopy assessment. Slit-lamp laser treatment, anti-VEGF injection or vitrectomy were offered as indicated.

Retrospective data collection on electronic records of all adult sickle patients (age >20yo) attended sickle eye clinics from 2021–2023. We studied patients' sickle genotypes, retinopathy presentations and severities (based on 1971 Goldberg classifications of Proliferative Sickle Retinopathy stagings, PSR 1–5), visual acuities change, treatments and complications over review periods.

Results: A total of 160 patients (319 eyes) attended over the 3 years. Main genotypes recorded were HbSC 61.25%, HbSS 31.2%, HbS BetaThalassaemia 4.4% and other genotypes 3.1%.

Females 53.1%; mean presentation age 35.45 years (SD 11.64yo, range 20–75yo).

78.1% patients were referred by haematologist for routine retinopathy screening; 35.6% patients were asymptomatic at presentation.

Follow-up(FU) periods: 86.9% (139/160) patients had mean FU duration of 11.6 yrs(SD6.6yrs, range 0.4–21.8 yrs); 13.1%(21/160) patients were seen only once (discharge/non-attendance after DNAs).

Initial presenting visual acuity: 91.8 % (293/319 eyes) had vision of 6/12 or better [64.2% were 6/6 vision or better] and only 3.7% (12/319) were 6/60 or worse.

Initial presentation of sickle PSR stage 0 was recorded in 34.8%(111/319) eyes; PSR staging 1–2 were recorded in 15.6% (50/319); PSR3(seafans) was found in 49.5% (158/319 eyes); 11.3% (36/319) presented with PSR4(vitreous haemorrhage) and 6.3% (20/319) with PSR5(retinal detachment); 22.2 % (71/319) of eyes presented with already advanced peripheral elevated fibrotic scar. Bilateral severe PSR(stage > 3) retinopathy was found in 40%(64/160) patients.

Furthermore, of 214 eyes which presented with severe retinopathy (PSR 3–5), 75.2% (161 eyes) belonged to HbSC genotype, 18.2% (39 eyes) belonged to HbSS and the rest 6.6% (14 eyes) had other sickle cell genotypes.

Of 207 eyes had OCT, 36.2% had macular temporal thinning; 71.5% (148/207) had ERM/pseudohole; 10.1% (21/207) had significant macular pre-retinal fibrosis/traction.

Of the **139 patients(277 eyes) with FU:** 88.4% (245/277) had vision 6/12 or better, 3.97% (11/277) had 6/60 or worse.

FU complications: 28.5% (79/277 eyes) had further vitreous haemorrhage episode at least once; 9% (25/277) had salmon patch and resolved; 5% (14/277) were found to have a retina hole/tear/operculum and 5%(14/277) had tractional retinal detachment.

Treatments record: No patients received anti-VEGF injection as their disease management; sector/barrier laser were performed at least once in 53.4% (148/277 eyes); 6.1% (17/277) underwent vitrectomies: 4.7% (13/277) were for retinal detachment and 1.4% (4/277) for ERM peel.

Conclusion: Our cohort showed most adult sickle patients attended our eye clinics were via retinopathy screening programme from haematology. Severe PSR retinopathy presentation is much higher in genotype HbSC. Our cohort also showed good presenting and final visual levels despite 50% have severe PSR stage 3 or worse. Early laser treatment was applied to just over 50% of patients, and the vitrectomy rate was low.

Conflicts of interest: None

Funding: None

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P113

ADVANCES IN HUMAN EMBRYONIC DERIVED STEM CELL THERAPIES FOR RETINAL DEGENERATION

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Introduction: Retinal Pigment Epithelium (RPE) degeneration is the key problem in diseases such as Age Related Macular Degeneration (AMD) and Stargardt's Macular Dystrophy (SMD). In the United Kingdom, there are 663 new cases of AMD per million per year.

Human Embryonic Stem Cell (hESC) derived RPEs are an emerging regenerative treatment modality for these conditions. However, the safety and efficacy of these methods remains under researched. This review discusses findings from 5 clinical studies that present the various outcomes associated with the treatment of retinal degeneration using hESC-RPEs.

Method: A thorough search of relevant databases (i.e. PubMed and Cochrane), was conducted to select studies assessing the clinical findings post treatment with hESC-RPE based interventions. Robust inclusion and exclusion criteria were established - with only

Randomised Control Trials and Clinical Trials conducted in the last 10 years being included in this review for analysis. The trials included the use of bioengineered implants and subretinal injections, which broadened the scope of the review.

Results: The search strategy identified a total of 6 studies meeting the criteria. In terms of the various outcomes discussed, the summarised findings are as follows:

1. Safety: The intervention was considered safe long term, with no significant findings demonstrating immune rejection or tumourigenicity.
2. Visual Outcomes: Stabilised vision, or modest improvements in patients with advanced disease. Higher scope for visual improvement in earlier stages of the disease.
3. Side effects: Minor adverse effects related to immune responses or a temporary rise in intraocular pressure were seen.

Conclusion: hESC-RPE based interventions prove to be a novel, innovative and safe approach to treating disorders of retinal degeneration showing evidence of moderate improvement of visual outcomes. Further research focusing on early stage interventions, long-term follow up protocols and patient satisfaction surveys is recommended.

Conflicts of interest: None

Funding: None

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P114

AI-DRIVEN OCT BIOMARKER ANALYSIS FOR PREDICTING LONG-TERM VISUAL OUTCOMES AFTER VITRECTOMY IN EPIRETINAL MEMBRANE

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Introduction: To identify baseline AI-driven OCT biomarkers predictive of long-term visual outcomes vitrectomy and peeling in patients with epiretinal membrane (ERM)

Method: A cohort of 133 ERM patients from Inselspital, Bern, (Switzerland) and Chang Gung Memorial Hospital, Taiwan, underwent BCVA and OCT assessments preoperatively, and at 1–3 months, 6 months, and 12 months post-vitrectomy. OCT data were analysed using the Discovery OCT Biomarker Detector (RetinAI AG, Bern), measuring retinal layer thicknesses (ONL, PR + RPE), fluid biomarkers (IRF, subretinal fluid), and hyperreflective foci (HF) probabilities. Statistical analyses included t-tests for visual and structural outcomes, and machine learning models, including random forest classifiers and regression analyses, to predict BCVA improvement and identify key biomarkers.

Results: BCVA significantly improved post-surgery ($p = 0.0032$). Among OCT layers, RNFL showed significant thickness changes across ETDRS grid sections, even after correction for multiple hypothesis testing. Random forest models identified RNFL thickness as a baseline key predictor of BCVA improvement. Other baseline OCT biomarkers outer nuclear layers including the ONL and PR+RPE and the presence of IRF played a minor but significant role in the predicting final BCVA.

Conclusion: Baseline RNFL thickness was a key predictor of visual recovery after ERM surgery, possibly reflecting pre-existing retinal health or susceptibility to damage. Other biomarkers, such as ONL and PR+RPE thickness, also provided insights into structural factors influencing long-term outcomes. These findings highlight how baseline retinal characteristics can shape recovery outcomes.

Conflicts of interest: None

Funding: None

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P115

AN EVALUATION OF ULTRA-WIELD FIELD (UWF) IMAGING MODALITIES, AND THEIR IMPACT ON CLINICAL DECISION MAKING IN THE MEDICAL RETINA (MR) DIGITAL CLINICS

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Introduction: Virtual clinics have become a vital solution to address the increasing demand for medical retina services.

However, issues with imaging quality remain a significant challenge. A 2018 study by Korteum et al. identified poor image quality as the leading reason for face-to-face follow-ups in virtual clinics. Despite the growing reliance on virtual services, there has been little evaluation of how different ultra-widefield (UWF) imaging modalities influence image quality and clinical decision-making in the context of a medical retina virtual clinic.

This retrospective study aims to investigate whether different UWF imaging modalities impact image quality and clinical decision-making in a medical retina virtual service. The study also assesses how follow-up patients are allocated to face-to-face or virtual reviews and evaluates any differences in the reasons for follow-up visits. Another key focus is on the reproducibility of image acquisition across different UWF modalities.

Method: A two-month prospective audit was conducted between September and October 2024, during which various patient and imaging data were collected. The information gathered included demographic details such as age and ethnicity, as well as clinical data such as the site of care and primary diagnosis. The primary and secondary reasons for follow-up were recorded, along with whether the review was conducted for clinical reasons alone. Additional data points included the presence of clinically significant artefacts, insufficient field of view (FoV), or poor image quality, and whether further imaging was required. Furthermore, details about the Ultra-Widefield (UWF) imaging modalities used, such as pseudocolour/colour images, fundus autofluorescence (FAF), and Sensory Red Free/Red Free images, were also documented. This audit aimed to capture a comprehensive picture of the clinical decision-making and imaging practices in the medical retina virtual service during the study period.

Results: Data from a total of 268 patients was collected. There was no statistically significant difference between Optos and Clarus ($p = 0.228$) in the rates of artefact, insufficient FoV, need for further imaging, overall poor quality, or decision making made on clinical need alone (e.g. no image issues). There was a significant difference in Fundus Autofluorescence (FAF) and Red-Free/Sensory Red-Free image acquisition between ultra-widefield (UWF) imaging modalities ($p = 0.001$). Optos demonstrated superior image acquisition with 100% presence of both FAF and Sensory Red-Free images. In contrast, Clarus exhibited a lower acquisition rate, with Red-Free images present in 82% and FAF images in 78% of cases. There was a statistically significant difference in the reasons patients were booked for face-to-face (F2F) versus virtual follow-up. A higher rate of F2F follow-up was associated with red flag pathology, anterior segment assessment, and unexplained vision loss ($p = 0.001$ for each). Conversely, a higher rate of virtual follow-up was observed for routine (non-red flag) pathology ($p = 0.001$).

Conclusion: Virtual clinics are an effective solution to manage the increasing workloads and demand for medical retina services. Most clinical decision-making was based on clinical need alone; however, approximately 10% of decisions were influenced by insufficient or poor image quality. The type of ultra-widefield (UWF) imaging modality used did not significantly impact decision-making. Nevertheless, the reproducibility of Fundus Autofluorescence (FAF) and Red-Free/Sensory Red-Free imaging was higher with Optos compared to Clarus, highlighting the importance of image acquisition reliability in virtual clinic settings.

Conflicts of interest: None

Funding: None

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P116

ASSOCIATIONS OF RETINAL LAYER THICKNESS AND OPTIC DISC PARAMETERS WITH INFLAMMATORY BOWEL DISEASE: A CROSS-SECTIONAL ANALYSIS

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Introduction: Inflammatory bowel disease (IBD), including Crohn's disease (CD), ulcerative colitis (UC) and other subtypes (e.g. colitis), are often linked to extraintestinal manifestations. Emerging evidence highlights gut-retina crosstalk, where microbial alterations may contribute to intraocular inflammation. We explored the association between the different types of IBD and retinal layer thicknesses and optic disc features.

Method: This was a cross-sectional analysis of data from the 'AlzEye' study, a retrospective cohort of 154,830 patients, aged ≥ 40 years, who had retinal imaging at Moorfields Eye Hospital, between 2008 and 2018. Retinal layer thicknesses were obtained from optical coherence tomography (OCT) scans using the Topcon Advanced Boundary Segmentation deep learning algorithm. The data was curated based on image quality and missing features. Wilcoxon rank-sum tests were used for unadjusted comparisons between IBD and control groups. Associations were further explored using linear regression models adjusting for confounding variables; age, sex, hypertension, and diabetes mellitus to determine their independent associations with IBD and its subtypes.

Results: Within the 'AlzEye' study, there were 4315 patients with IBD and 85,247 control subjects (mean age 65.8 ± 12.6 years, 52% female). Key retinal layers, including the ellipsoid zone (EZ) ($p = 2.46 \times 10^{-10}$), ganglion cell-inner plexiform layer (GCIPL) ($p = 1.48 \times 10^{-6}$), and total retinal thickness (TRT) ($p = 3.35 \times 10^{-6}$), showed marked thinning in IBD patients. Additionally, the inner nuclear layer (INL) ($p = 9.81 \times 10^{-5}$), outer nuclear-plexiform layers (ONPL) ($p = 1.75 \times 10^{-3}$), and choroidal thickness (CHT) ($p = 3.15 \times 10^{-3}$) also exhibited significant reductions. The mean ($\pm$ standard deviation) of these parameters were lower for IBD patients, compared with controls.

Multivariable linear regression identified 7 retinal layer thickness and optic disc parameters that showed statistically significant differences ($P < 0.05$) between IBD patients and controls. Optic cup to disc ratio (CDR) (coefficient = 0.0197, $p = 0.0146$), cup height (coefficient = 4.64, $p = 0.0027$) and cup width (coefficient = 4.13, $p = 0.0035$) demonstrated significant differences between IBD and controls. Additionally, optic disc height (coefficient = 6.41, $p = 0.0009$) and optic disc width (coefficient = 3.57, $p = 0.0242$) were significantly increased with IBD. Notably, GCIPL (coefficient = -2.66, $p = 0.0294$) and retinal pigment epithelium (RPE) (coefficient = -2.49, $p = 0.0404$) were significantly thinner in IBD patients. On subgroup analyses, individuals with CD, demonstrated significant reductions in cup height (coefficient = -4.63, $p = 0.0028$), cup width (coefficient = -4.12, $p = 0.0036$), optic disc height (coefficient = -6.39, $p = 0.0009$), optic disc width (coefficient = -3.57, $p = 0.0247$) and CDR (coefficient = -0.0197, $p = 0.0147$). In UC, thinning was most prominent in the retinal pigment epithelium (RPE) layer (coefficient = -1.59, $P = 0.021$). Individuals with other IBD subtypes had significantly thinner ellipsoid zone (EZ) (Coefficient: -0.735, $P < 0.0005$) and ONPL layers (coefficient = -0.700, $p = 0.048$).

Conclusion: This study highlights the significant associations between IBD and changes in retinal layer thickness and optic disc parameters, with retinal thinning observed in key layers such as the EZ, GCIPL, and TRT. Additionally, optic disc parameters, including cup height were altered in IBD patients, showing increased values in contrast to controls. Subgroup analyses revealed disease-specific patterns, with Crohn's disease (CD) showing more pronounced changes in optic disc dimensions, while ulcerative colitis (UC) was characterised by thinning in the RPE layer. Other IBD subtypes exhibited thinning in the EZ and ONPL. These findings highlight measurable structural changes in the retina and optic disc in IBD patients, suggesting potential biomarkers for further investigation in clinical settings to monitor disease progression and therapeutic response.

Conflicts of interest:

Pearse Keane:

Financial Support: Allergan, Topcon, Heidelberg Engineering, Novartis, Roche, Bayer

Consultant/Contractor: Apellis

Personal Financial Interest: Big Picture Medical

Funding: None

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P117

BARRIERS TO CARE: EXAMINING SOCIOECONOMIC DETERMINANTS AND BARRIERS TO CLINICAL ATTENDANCE FOR DIABETIC EYE DISEASE

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Introduction: Diabetic retinopathy (DR) is one of the leading causes of blindness within the UK. There is a high rate of non-attendance for diabetic eye clinics and this can lead to poor management of an individual's diabetic retinopathy, potentially causing irreversible visual loss. Supporting people with diabetes to attend the eye clinic is of utmost importance.

Aim: To identify the socioeconomic determinants and barriers to attendance at a busy teaching hospital diabetic eye clinic.

Method: A retrospective study of all patients who did not attend (DNA) their diabetic eye clinic or treatment appointment during 1st July – 30th August 2024. The demographic data, Index of multiple deprivation (IMD), distance, clinic type, clinical factors including vision and comorbidities was obtained. This was compared with a cohort of patients who attended during 9th–19th September 2024.

A prospective, qualitative assessment of barriers to attendance was conducted via interviews with a cohort of patients who DNA using a validated questionnaire and thematic analysis of further comments.

Results: Demographic data on 216 individuals who DNA the eye clinics was compared to 185 patients who did. The non-attenders were younger with a mean age of 58.8 years $\pm$ 15.8 (range 25–96) compared to the attending group of 64.01 $\pm$ 14.2 (range 27–93, $p = 0.0006$). Distance travelled to the hospital and multimorbidity were not significantly different. However, non-white ethnicity (37% of non-attenders vs 23% of attenders, $p = 0.001$) and lower IMD score ($p = 0.002$) were significant factors in non-attendance.

The qualitative assessment of 34 patients identified three main barriers to attendance: the location of appointment, time burden and comorbidity.

Conclusion: Health inequalities were identified with individuals of working age, from more deprived areas, and from ethnic minorities less likely to attend eye clinics. Qualitative assessment provides rich detail of the barriers which need to be addressed to improve attendance and reduce inequalities.

Conflicts of interest: None

Funding: None

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P118. Abstract not included.

P119

CLINICAL AND MULTIMODAL IMAGING FEATURES OF ACUTE MACULAR NEURORETINOPATHY

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Introduction: Acute Macular Neuroretinopathy (AMN) is a rare retinal disorder, commonly affecting young women. It can cause transient or permanent central vision loss due to outer retinal vascular alterations. Presentation is usually with para central scotomas and decreased best corrected visual acuity. The diagnosis is primarily based on changes seen on optical coherence tomography (OCT) imaging. The exact mechanisms causing AMN is not yet known but there is evidence suggesting that vascular compromise of the deep retinal plexus of the retina might be the cause.

Method: We present a case of a 29 year old woman who presented to the eye casualty clinic with a black spot and blurry vision in her left eye. The visual acuity was 6/6 on the right eye and 6/12 on the left eye. The symptoms started 2 days before coming to the clinic. Patient had a history of polycystic ovary syndrome and migraines. She had 10 pack year history of smoking.

Results: Fundal examination with dilation was normal. OCT imaging was done which showed vascular changes in the outer plexiform layer. Patient was asked to draw the black spot on her visual field on an Amsler grid which showed a fairly sized para central scotoma. OPTOS images of the fundus were normal. OCT angiography also showed the extent of the vascular deformation.

Conclusion: AMN is a rare cause of transient or permanent visual loss, especially in young females. Fundal examination is commonly normal and there are different imaging modalities to aid the diagnosis like OCT imaging and OCT angiography.

Conflicts of interest: None

Funding: No funding

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P120

COMPLEMENT TARGETED THERAPIES IN AMD

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Introduction: Age related macular degeneration (AMD) is a detrimental condition that was predicted to affect approximately 196 million people worldwide in 2020 with cases expected to rise to approximately 288 million by 2040. The disease causes retinal pigmented epithelium (RPE) to deteriorate which in turn kills photoreceptors, causing visual distortion and loss of central

vision, resulting in a significant reduction in a patient's quality of life. The prevalence of AMD increases in older age, and since the population is generally getting older, AMD incidence is increasing, adding to the pressure to develop effective treatment options. AMD can be divided into two main subtypes – choroidal neovascularisation (CNV, or 'wet' AMD) and geographic atrophy (or 'dry' AMD) which involve the growth of new, weaker choroidal vessels and the deposition of drusen leading to retinal pigmented epithelium cell atrophy, respectively. In CNV, the development of anti-VEGF therapies has had significant impact in preventing vision loss and is now the first-line treatment. Unfortunately, the more common, dry form has proven difficult to manage, leaving patients without effective treatments. Second to ageing, certain genetic variants, specifically those involved in complement cascade regulation, have been found to play a significant role in AMD risk, highlighting the complement system as a potential target for treatment. This route proved successful in 2023 when two complement targeting treatments gained FDA approval – Pegcetacoplan and Avacincaptad Pegol.

This systematic review aims to identify and discuss the complement-based therapies and to evaluate current and potential agents and the most appropriate components of the complement cascade to target when treating dry AMD.

Method: The electronic database PubMed and ClinicalTrials.gov were used to obtain relevant studies which then underwent manual screening to refine the studies in line with the chosen inclusion and exclusion criteria.

Results: In total 65 studies were identified and included in this systematic review. Overall, the treatments targeting C3 and C5 seemed to be the most successful due to their direct impact on the amplification of the alternative pathway and prevention of MAC formation, compared to the studies that targeted complement regulators or the C1q subunit involved in the classical pathway. Despite there being no changes to vision, the patients involved in these clinical trials were found to have reduced growth of lesions of geographic atrophy, reinforcing that targeting the complement system is effective in slowing disease progression.

Conclusion: Drugs that directly target the complement cascade do hold promise, specifically targets of C3 due to the direct reduction in overactivation of the alternative pathway which plays a significant role in AMD pathology. Further research needs to be conducted to develop better treatment options, taking into account patients with AMD risk SNPs in complement genes to evaluate whether this will impact the way they respond to treatments. Therefore, complement based therapies have the potential to be further developed to provide an effective, long-term treatment for patients with dry AMD.

Conflicts of interest:

SL has no conflict of interest to disclose.

RDU is a founder and shareholder of Complement Therapeutics Ltd, which is developing novel treatments for GA targeting the complement cascade.

Funding: No funding was received for this study.

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P121

DAVIO 2: A PHASE 2 TRIAL OF EYP-1901 (VOROLANIB INTRAVITREAL INSERT) VERSUS AFLIBERCEPT FOR PREVIOUSLY-TREATED NEOVASCULAR AGE-RELATED MACULAR DEGENERATION

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Introduction: Patients with neovascular age-related macular degeneration (nAMD) face a high treatment burden, highlighting the need for more durable therapies. EYP-1901 is a bio-erodible, sustained-delivery intravitreal insert delivering vorolanib, a highly selective pan-vascular endothelial growth factor receptor inhibitor, over ≥ 6 months. DAVIO 2 (NCT05381948) assessed efficacy of a single EYP-1901 dose versus aflibercept every-8-weeks (q8W).

Method: DAVIO 2 was a phase 2, multicentre, noninferiority clinical trial conducted over 56 weeks. Patients with previously-treated nAMD were randomized on Day 1 to receive a single dose of EYP-1901 2 mg or EYP-1901 3 mg, or aflibercept 2 mg q8W; after 3 aflibercept 2 mg monthly loading doses. All patients were assessed monthly for supplemental aflibercept injections per prespecified best corrected visual acuity (BCVA) and/or anatomic criteria, or investigator discretion. Primary endpoint was BCVA change from Day 1 to Week 28/32 (Month 7/8) averaged. Secondary endpoints included safety, mean change in central subfield thickness (CST), and proportion of eyes supplemental injection-free.

Results: One hundred fifty-six eyes received EYP-1901 2 mg (n = 50), EYP-1901 3 mg (n = 52) or Aflibercept q8W (n = 54). DAVIO 2 met its primary endpoint with EYP-1901 2 mg and 3 mg statistically noninferior to aflibercept q8W (n = 54) for BCVA change from Day 1 at Month 7/8 (mean difference versus aflibercept: -0.3 and -0.4 letters, respectively. Predefined noninferiority margin: -4.5 letters). At Month 8 mean difference in CST versus aflibercept was +12.4 μ m (EYP-1901 2 mg) and +5.2 μ m (EYP-1901 3 mg). In contrast with aflibercept q8W, no sawtooth pattern in CST was observed with EYP-1901. At Month 8, six months after EYP-1901 injection, 63% of EYP-1901-treated eyes remained supplemental injection-free; with an overall $\geq 85\%$ reduction in treatment burden versus pre-trial treatment. EYP-1901 was well-tolerated, with no EYP-1901-related ocular or systemic serious adverse events.

Conclusion: One dose of EYP-1901 with protocol-specified aflibercept supplementation demonstrated statistical noninferiority to aflibercept q8W in eyes with nAMD, with stable BCVA, strong anatomical control, and a favourable safety profile. Pivotal phase 3 trials LUGANO (NCT06668064) and LUCIA (NCT06683742) are evaluating EYP-1901 in nAMD with redosing every 6-months.

Conflicts of interest:

- Consultant: AbbVie, Apellis, Boehringer Ingelheim, Janssen, Ora Clinical, EyePoint Pharmaceuticals Inc, Roche, Ocular Therapeutix, Bayer.
- Lecture fees: Apellis, Roche, Bayer, Boehringer Ingelheim and Topcon.
- Grants: Apellis, NIHR, Roche and Topcon

Funding:

This clinical trial was funded by EyePoint Pharmaceuticals Inc.

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P122

DIABETIC MACULAR OEDEMA (DMO) COMBINED WITH RETINAL ANGIOMATOUS PROLIFERATION (RAP) LESIONS

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Introduction: RAP is now considered a well-established neovascular phenotype of AMD. Environmental risk factors such as age

and arterial hypertension have been associated with the development of both RAP and typical exudative AMD, but it can be associated with Diabetic macular oedema Differentiating between Diabetic Macular Oedema (DMO) and Retinal Angiomatous Proliferation (RAP) lesions is indeed important for several reasons related to management

Method: Out of the 50 RAP patients, in a retrospective observational study, all of them Caucasians, average age of 79 and 70% were females, we had three patients who had associated Type 2 diabetes mellitus and Diabetic retinopathy changes with macular changes which had a combination features of both DMO and RAP lesion.

The RAP lesion was targeted intensively with Anti-VEGF and the vascularised RPED component responded well to frequent intra-vitreous injections of Anti-VEGF (4 to 6 weeks) but some of the intra-retinal fluid always persisted in one patient. When the Anti-VEGF treatment was combined with intra-vitreous dexamethasone, the response was better with good resolution of the intra-retinal fluid.

Results: Chi-Square Test of Independence was used to test whether there is a significant association between two categorical variables (diabetes status with diabetic macular oedema and presence of RAP lesions). Using the observed frequencies and expected frequencies, chi-square value was calculated and this was compared to a chi-square distribution table with the appropriate degrees of freedom. The corresponding p value was calculated and it was <0.05.

Conclusion: It is important to distinguish between a RAP lesion and DMO when they both are combined together as the treatment strategies, prognosis and frequency of follow up monitoring are different. By distinguishing between DME and RAP lesions, clinicians can provide more personalised treatment plans, improve patient outcomes, and enhance the overall management of macular changes related to diabetes mellitus combined with RAP lesions.

Conflicts of interest: None.

Funding: None.

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P123. Abstract not included

P124

EFFECTIVENESS AND SAFETY OF FARICIMAB IN EYES WITH DMO IN THE UNITED KINGDOM: 1-YEAR RESULTS: FROM THE FARWIDE-DMO STUDY

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Introduction: Faricimab is the only approved intraocular bispecific antibody that inhibits both angiopoietin-2 and vascular endothelial growth factor (VEGF)-A. To date, >5 million doses have been distributed in over 100 countries. Faricimab Real-World Evidence (FARWIDE) is a multicentre, retrospective, observational study evaluating patient characteristics, durability and visual acuity (VA) outcomes in eyes with diabetic macular oedema (DMO) or neovascular age-related macular degeneration (nAMD) treated with faricimab in the United Kingdom (UK). Here, we report the results for the FARWIDE-DMO cohort with ≥12 months of follow-up.

Method: The FARWIDE study includes data from 35 UK National Health Service sites using the Medisoft electronic medical record system. The 12-month cohort of FARWIDE-DMO included patient-eyes with diagnosed DMO and no concomitant nAMD or retinal vein occlusion, who initiated faricimab after June 2022, and had ≥12 months of follow-up after faricimab initiation as of July 2024. Treatment-naïve eyes had no prior anti-VEGF or steroid implant treatment. Previously treated (PT) eyes switched to faricimab from an anti-VEGF or steroid implant treatment. Baseline characteristics, VA and injection frequency were assessed in the 12-month cohort. Presumed infectious endophthalmitis (PIE) and intraocular inflammation (IOI) events were identified. Analyses are descriptive.

Results: A total of 1564 patients (2147 eyes; 690 naïve, 1457 PT) with DMO were included. Mean patient age was 62 years and ~38% were women. For treatment-naïve eyes, median (quartile [Q1,Q3]) VA was 66 (55,76) ETDRS letters at baseline and 74 (61,80) at 12 months. Treatment-naïve eyes received a mean (SD) of 4.5 (1.0) injections in months 1–6 and 1.9 (1.2) in months 7–12. At 12 months, 61% of treatment-naïve eyes had a treatment interval of ≥10 weeks. Among treatment-naïve eyes in the most deprived Index of Multiple Deprivation (IMD) deciles (1–2), mean (95% CI) VA change was 6.4 (4.3,8.5) ETDRS letters and among those in the least deprived IMD deciles (9–10), it was 5.6 (3.6,7.5) letters. Mean (SD) number of faricimab injections at 12 months was 6.1 (1.8) and 7.0 (1.7) in IMD deciles 1–2 and 9–10, respectively.

PT eyes received anti-VEGF treatment for a mean (SD) of 2.8 (2.3) years before initiating faricimab; most (85%) switched from aflibercept 2 mg. After faricimab initiation, median (Q1,Q3) VA was 70 (58,76) ETDRS letters at baseline and 70 (60,79) at 12 months. PT eyes received a mean (SD) of 4.5 (1.2) injections in months 1–6 and 2.4 (1.3) in months 7–12. Mean (95% CI) VA change was 1.7 (0.0,3.4) ETDRS letters in IMD deciles 1–2 and 0.4 (1.1,1.9) letters in IMD deciles 9–10. Mean (SD) number of faricimab injections at 12 months was 6.6 (2.2) and 7.3 (2.1) among PT eyes in the most and least deprived deciles, respectively. IOI and PIE rates were consistent with phase 3 trials.

Conclusion: In FARWIDE-DMO, VA improved in treatment-naïve eyes and remained stable in PT eyes, on average. A lower mean number of injections during months 7–12 vs months 1–6 suggests treatment interval extensions in both naïve and PT eyes. Eyes in the most deprived deciles achieved comparable mean VA gains with fewer injections on average vs those in the least deprived deciles. The safety data evaluated were consistent with those reported in phase 3 trials. These 12-month data support the real-world effectiveness, durability, and safety of faricimab in DMO.

Conflicts of interest:

Presenting author: Advisory Board/Honoraria: Alimera, Allergan, Bayer, Biogen, Novartis, Roche; Travel Grants: Allergan, Bayer, Novartis

Funding:

Funding provided by F. Hoffmann-La Roche Ltd.

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P125

EFFECTIVENESS AND SAFETY OF FARICIMAB IN EYES WITH NAMD IN THE UNITED KINGDOM: 18-MONTH RESULTS FROM THE FARWIDE-NAMD STUDY

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Introduction: Faricimab is the only approved intraocular bispecific antibody that inhibits angiopoietin-2 and vascular endothelial growth factor (VEGF)-A. To date, >5 million doses have been distributed in over 100 countries. **Faricimab Real-World Evidence (FARWIDE)** is a multicentre, retrospective, observational study evaluating patient characteristics, durability and visual acuity (VA) outcomes in eyes with neovascular age-related macular degeneration (nAMD) and diabetic macular oedema (DMO) treated with faricimab in the United Kingdom (UK). Here, key outcomes for the FARWIDE-nAMD cohort with ≥18 months of follow-up are reported.

Method: The FARWIDE study includes data from 35 UK National Health Service sites using the Medisoft electronic medical record system. The 18-month cohort of FARWIDE-nAMD included patient-eyes with nAMD and no concomitant DMO or retinal vein occlusion, who initiated faricimab after June 2022 and had ≥18 months of follow-up as of July 2024. Treatment-naïve eyes had no prior anti-VEGF treatment. Previously treated (PT) eyes switched to faricimab from another anti-VEGF treatment. Baseline characteristics, VA and injection frequency were assessed in the 18-month cohort. Presumed infectious endophthalmitis (PIE) and intraocular inflammation (IOI) events were identified. Analyses are descriptive.

Results: In total, 1404 patients (332 naïve, 1335 PT eyes) with nAMD were included. Mean patient age was 79 years and ~60% were women. Median (quartile [Q1,Q3]) VA for treatment-naïve eyes was 61 (47,70) ETDRS letters at baseline and 65 (46,76) letters at 18 months. Treatment-naïve eyes received a mean (SD) of 4.8 (0.5) injections in months 1–6, 2.2 (1.0) in months 7–12 and 1.9 (1.1) in months 13–18. By 18 months, 73% had reached a treatment interval of ≥10 weeks.

PT eyes received anti-VEGF treatment for a mean (SD) of 2.6 (2.2) years before initiating faricimab; most (84%) switched from aflibercept 2 mg. After faricimab initiation, median (Q1,Q3) VA was 70 (60,76) ETDRS letters at baseline and 69 (55,76) letters at 18 months. Mean (SD) number of injections was 4.4 (0.9) in months 1–6, 3.0 (1.1) in months 7–12 and 2.8 (1.2) in months 13–18. IOI and PIE rates were consistent with phase 3 trials.

Conclusion: In FARWIDE-nAMD, mean injections reduced during months 7–12 and 13–18 vs months 16, suggesting extension of treatment intervals in year 1, and further extension into year 2. On average, VA improved in treatment-naïve eyes and

remained stable in PT eyes. The safety data evaluated were consistent with those reported in phase 3 trials. These 18-month data support the real-world effectiveness, durability and safety of faricimab in nAMD.

Conflicts of interest:

Presenting Author: Advisory Board: Bayer, Novartis, Roche, Teva; Honoraria: AbbVie, Bayer, Novartis, Roche; Educational Grants: Alimera, Bayer, Novartis, Roche; Clinical Trial PI: Bayer, Novartis, Roche

Funding:

Funding provided by F. Hoffmann-La Roche Ltd.

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P126

EFFICACY AND SAFETY OF FARICIMAB IN THE TREATMENT OF NEOVASCULAR AGE-RELATED MACULAR DEGENERATION – REAL WORLD EXPERIENCE

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Introduction: Faricimab has shown promising results for visual acuity (VA), prolonged treatment intervals and a low complication profile in the treatment of neovascular age-related macular degeneration (nAMD) through multi-centre randomised controlled trials. This audit aims to analyse the initial real-world results of the use of faricimab in treatment-naïve and switch patients treated for nAMD at Aintree Hospital.

Method: A retrospective Medisight search was performed to identify treatment-naïve and switch patients with nAMD treated with faricimab at Aintree Hospital. Primary outcomes were changes in VA and secondary outcomes were changes in central macular thickness (CMT), treatment intervals and complications. Results were audited against the results of TENAYA, LUCERNE, TRUCKEE, FARETINA and FARWIDE studies.

Results: Data from the first forty treatment-naïve (45 eyes) and switch patients (46 eyes) was collected and compared to the clinical trials. Overall, 22.2% and 15.2% of patients were on a ≥16 week dosing interval and 66.7% and 28.3% of patients were on a treatment interval of ≥12 weeks in the treatment-naïve and switch groups respectively at 12 months. 91.1% and 84.8% of patients in the respective groups avoided more than a 15 letter drop in vision. There was a significant reduction in the CMT in the treatment-naïve group of 92.2 µm ($p < 0.0001$) and a reduction of CMT in the switch group of 23.6 µm which did not reach statistical significance ($p = 0.06$) at 12 months. Adverse events were rare occurring in 2.2% in both groups which were comparable to other trials.

Conclusion: Faricimab has shown promising initial results in nAMD treatment at Aintree Hospital with most patients maintaining stable vision on an extended dosing interval with a significant reduction in CMT in the treatment-naïve group. Fewer injections reduced the treatment burden for patients and improved capacity within the unit with decreased costs. Adverse events were rare. A limitation of this study is retrospective data collection limited to data captured on electronic health records.

Conflicts of interest: None**Funding:** None

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P127

EFFICACY OF NOVEL MEDICAL RETINA DIGITAL CLINICS

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Introduction: The Medical Retina Digital Clinic at Moorfields Eye hospital was established to combat the increasing demand of Medical Retina diseases requiring hospital eye service appointments, ensuring patients are reviewed in a timely manner. We conducted a review of patients across all digital clinics to assess the effectiveness. This retrospective study focuses on key metrics such as DNA (Did Not Attend) rates, diagnoses and primary clinical outcomes from the virtual review. In addition, we analysed the outcomes of patients who required a follow-up face-to-face (F2F) visit.

Method: We conducted a retrospective review of 7,202 patients seen across all Moorfields Medial Retina digital sites from July 2023 to August 2023. Including the Medical Retina Digital Rapid Access Clinic. Data was extracted using SQL reports and OpenEyes.

Inclusion criteria: All patients with an appointment in the Medical Retina Digital Clinics across the sites were analysed, including those who could not wait (CNW) or DNA.

Results: Out of 7202 patients with appointments, 5743 patients attended, with a DNA rate of 20.3%. Allied Health Professionals including Optometrists, Nurses and Medical Graders performed 35% of reviews. Predominately patients seen within the Medical Retina Digital Clinic remained in the digital pathway (64.4%), with 6.8% discharged and 28.9% seen in the F2F clinic for follow up.

Of the patients that were brought back for F2F review 25.9% had treatment and of those reviewed in the Medical Retina Digital Rapid Access Clinic 39.2% received treatment.

The Medical Retina Digital Rapid Access Clinic met the key performance indicator (KPI) of cases being reviewed within 2 working days in 100% of patients.

Out of the patients reviewed 29.4% had a diagnosis of low-risk nonproliferative diabetic retinopathy (DR) without significant diabetic macular oedema (DMO), demonstrating a high prevalence of the condition. A total of 51.1% of patients had an eye condition related to Diabetes (including nonproliferative DR, Proliferative DR and DMO). Age-related macular degeneration (wet and dry) made up a small proportion of patients across all sites at 9.5% and appear to mainly be reviewed in the rapid access clinic totalling 48% of patients reviewed.

Conclusion: The results show a low percentage of remote review patients being brought for F2F review. This highlights the effectivity of the digital clinic in reducing unnecessary F2F reviews and being able to streamline patients with the need for F2F review who require treatment. Ultimately, optimising the service resources and increasing the overall capacity of patients reviewed within the trust.

The proportion of reviews performed by Allied Health Professionals in the digital service demonstrates the potential of freeing up valuable chair time for consultants and senior colleagues.

Conflicts of interest: None

Funding: None

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P128

ENHANCED MACULAR IMAGING REFERRAL PATHWAY: INTEGRATING COMMUNITY OPTOMETRY AND HOSPITAL EYE SERVICES WITH TELEMEDICINE

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Introduction: The increasing burden on United Kingdom (UK) Hospital Eye Services (HES) due to rising optometric referrals and advancements in retinal imaging necessitates innovative solutions to improve care pathways. The National Health Service (NHS) Long-Term Plan emphasises integrated services to address these challenges. At Bristol Eye Hospital, University Hospitals of Bristol and Weston, an enhanced macular imaging referral pathway was introduced to facilitate collaboration between community optometrists and hospital clinicians, leveraging telemedicine to streamline patient management and optimise resources.

Method: This retrospective cohort study analysed 115 referrals processed through the platform. Community optometrists provided fundus images and Digital Imaging and Communications in Medicine (DICOM®) optical coherence tomography (OCT) files, fully visualised and scrollable for hospital clinicians to assess. Key metrics included referral appropriateness, redirection to subspecialties, hospital appointments saved, discharge rates, and time to review.

Results: The mean age of referred patients was 72.2 ± 16.1 years, with 46.96% male and over 90% white. Among the 115 referrals, 12 were redirected to other subspecialties, including 9 to vitreoretina, 1 for proliferative diabetic retinopathy requiring laser treatment, 1 for malignant hypertension, and 1 for postoperative cystoid macular oedema (CMO) after cataract surgery from another provider, one referral was a paediatric patient (excluded as the service is for adults only). Seven referrals (6.1%) were discharged after clinician review, 27 patients (23.5%) required an appointment in a neovascular age-related macular degeneration clinic, and 54 (47%) were seen face-to-face.

The most significant outcome was that 64 referrals (55.7%) resulted in avoided hospital appointments, alleviating HES workload. The average referral review time was 4.3 days.

Conclusion: This telemedicine initiative effectively integrates community optometrists with HES, enhancing communication, reducing unnecessary hospital appointments, and improving triage accuracy. By enabling clinicians to save over half of the referred visits, this platform demonstrates its critical value in optimising resource allocation while maintaining high-quality care. This approach highlights a scalable model for efficient eye care delivery and lays the foundation for integrating artificial intelligence technologies in ophthalmology.

Conflicts of interest:

Serena Salvatore: Speaker fees and travel bursary: Roche Products Limited, Bayer, Alimera Sciences

Amy Hughes: Employee at Primary EyeCare Services

Lisa Gibson: Employee at Primary EyeCare Services

Clare Bailey: Speaker fees and travel bursary: Roche Products Limited, Bayer, Alimera Sciences

Funding: None

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P129

EVALUATING OZURDEX IMPLANTS VS. AFLIBERCEPT INJECTIONS IN TREATMENT-NAÏVE RETINAL VEIN OCCLUSION PATIENTS: EFFICACY AND SERVICE BURDEN

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Introduction: This audit aimed to examine intravitreal dexamethasone (Ozurdex) implants as a first-line treatment for macular oedema secondary to retinal vein occlusion (RVO). RVO is a leading cause of vision loss in the UK, with its incidence rising with an ageing population.(1) Optimising treatment strategies is crucial with the expected increase in the demand for RVO treatment.(2) This audit investigated the effectiveness and service burden of Ozurdex implants versus anti-VEGF (Aflibercept) in treatment-naïve RVO patients.

Method: This retrospective single-centre audit involved 92 RVO patients, with 43 receiving treatment: 22 with Eylea and 21 with Ozurdex. The patients were treatment-naïve and assigned to Ozurdex or Eylea based on local guidelines and patient preferences. The primary outcome was the comparison of best-corrected visual acuity (BCVA) and central subfield thickness (CST) over four months (18±1 weeks) for both treatments. The secondary goal was to assess the impact of these treatments on the efficiency of medical retinal services. An unpaired t-test was used to analyse BCVA and CST, and statistical analyses were performed using Minitab version 1.6.

Results: Patients receiving Ozurdex at four months improved by 0.29 LogMar (from 1.03 to 0.74 LogMar), while those treated with Eylea gained an average BCVA improvement of 0.14 LogMar (from 0.68 to 0.54 LogMar). The two-tailed p-value for BCVA improvement was 0.1674, indicating no statistically significant difference between the treatment groups. The baseline CST was 507 µm for Ozurdex and 463 µm for Eylea; reductions were 33.4% (169.5 µm) and 28.8% (133.4 µm), respectively, with a p-value of 0.4902, showing no significant difference. Analysis of RVO subtypes also revealed no statistically significant differences in outcomes (p = 0.2291), and the injection burden was lower in the Ozurdex group, with 21 injections and 63 appointments versus 65 injections and 88 appointments for Eylea. There was one case of neovascular glaucoma in the Ozurdex group.

This audit suggests that Ozurdex provides comparable improvements in BCVA and CST to Eylea over a four-month period, with a significantly lower injection burden. The GENEVA study⁽³⁾, which led to Ozurdex's licencing for CRVO/BRVO, identified the implant's peak effect at 60 days, with some studies, such as the COMRADE-C trial⁽⁴⁾, noting a reduced efficacy period of Ozurdex after three months. Since this audit was limited to a four-month follow-up, extending the monitoring period may reveal outcome differences. Complications from Ozurdex, particularly elevated intraocular pressure (IOP) and cataract progression remain treatment-limiting factors, and careful patient selection may help reduce adverse event rates.(4,5) This audit did not include comprehensive adverse event tracking during the treatment period, which could have affected the complication rate. The reported 10% rate of elevated IOP for Ozurdex in this audit, aligns with rates seen in the GENEVA and SAFODEX-2 studies^(3,6) (12.6%–39.6%).

Conclusion: This audit signifies that Ozurdex can offer similar efficacy to Aflibercept in BCVA and CST improvements at four months, while significantly reducing service demand through fewer injections and appointments. Careful patient selection and optimised implant intervals could enhance outcomes, making Ozurdex a promising option for RVO treatment as service demand increases.

Conflicts of interest: None

Funding: None

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P130

EXPLORING THE GENETIC BASIS OF INHERITED RETINAL DISEASE IN WEST YORKSHIRE

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Introduction: Inherited retinal diseases (IRDs) are a group of heterogeneous conditions that often lead to visual impairment and affect quality of life, independent living and lifestyle choices. The IRD genotypes differ across populations. Understanding the genetic basis can help clinicians provide an accurate diagnosis and genetic counselling, and may also help identify personalised treatment strategies in the future. In the UK, comprehensive data on the most prevalent IRD genes is limited, particularly in ethnic minority groups. This study aimed to identify and analyse the most frequent genes associated with IRD in West Yorkshire, with a focus on differences according to diagnosis, inheritance and ethnicity.

Method: This retrospective cohort study reviewed the clinical diagnoses, inheritance patterns, ethnicities and genetic test results of West Yorkshire patients with inherited retinal disease and a confirmatory genetic test result. All patients attending local clinics and with a positive genetic test result, obtained either through a research laboratory or NHS diagnostic service, were included. Electronic medical records and local databases were used to collect relevant information. Data analysis was performed using descriptive statistics to investigate associations between diagnosis, inheritance pattern prior to testing, ethnicity and genotype.

Results: Results from 601 patients were included, of whom 93.0% had non-syndromic disease. Most patients (92.8%) had acquired and progressive diseases, including macular and generalised retinal dystrophies, and a minority (7.2%) had congenital and non-progressive disease. The five most common patterns of inheritance were autosomal recessive (41.9%), simplex or no clear pattern (25.2%), autosomal dominant (18.8%), X-linked (12.8%) and mitochondrial (1.3%). Ethnic distribution was predominantly Caucasian (69.9%) or South Asian (28.3%), with smaller proportions of Afro-Caribbean, Arabic and South East Asian ethnicities (0.8%, 0.8% and 0.2% respectively).

In the whole cohort, disease-causing variants were identified in 100 different genes. The five most common genes causing IRD were ABCA4 (15.0%), RPGR (8.7%), Ush2a (8.2%), PRPH2 (4.7%) and EYS (4.0%). For macular and generalised retinal dystrophies, ABCA4 and RPGR were the most commonly implicated genes, accounting for 15.0% and 8.7% of all cases respectively. Among patients with congenital and non-progressive disease, CNGB3 and CNGA3 were the most common disease-causing genes, accounting for 1.5% and 1.3% of all the cases respectively.

In the Caucasian and South Asian patients, the most frequent patterns of inheritance were simplex and autosomal recessive (30% and 80% respectively). Among all the patients with either autosomal recessive and simplex inheritance, disease-causing

variants in the ABCA4 and Ush2A genes were most common. For the patients with autosomal dominant and X-linked inheritance patterns, variants in PRPH2 and RPGR were most common. Compared to Caucasian patients, variants in RPGR, PRPH2, and PRPF31 were less common in South Asian patients (2.9%, 1.2% and 1.2% respectively), while variants in EYS were more common. This is likely due to the increase in autosomal recessive disease in South Asian patients.

Conclusion: This study highlights the diversity of IRD and the associated genotype in West Yorkshire. Most patients with a positive genetic test result had non-syndromic, progressive macular or generalised retinal dystrophies and the most common predicted pattern of inheritance was autosomal recessive. Pathogenic variants in the ABCA4, RPGR and Ush2A genes accounted for more than 30% of all IRD cases and were the most commonly implicated genes in cases with autosomal recessive, simplex and X-linked inheritance. Knowledge of the most common diagnoses, inheritance patterns and genotypes in the local population can guide clinical decision-making, genetic counselling and the interpretation of genetic test results.

Conflicts of interest: None

Funding: None

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P131

FARICIMAB FOR DIABETIC MACULAR OEDEMA, OUR EXPERIENCE

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Introduction: To assess the 18 months outcome, from Nov 2022 to April 2024, of intravitreal Faricimab in diabetic macular oedema (DMO) treatment (TTT) naïve and TTT resistant patients (who didn't respond to other anti VEGF).

Method: Retrospective study of 2 groups of patients who had their treatment in Ysbyty Cwm Rhondda, South Wales. Group 1, 50 eyes of 38 DMO patients who had intravitreal Faricimab injections with no previous anti VEGF (TTT naïve). They had 4 injections of Faricimab 6.0mg, 1 injection every 4 weeks then extend according to the response. Group 2, who are the previously treated patients, we had 68 eyes of 44 patients who did not respond to other anti VEGF and switched to intravitreal Faricimab injections treat and extend from the beginning.

Results: For group 1, the mean changes in VA was +9.04 ETDRS letters and the mean changes in OCT was -123.1 µ. 10 (20%) eyes had dry OCT at 16 weekly, 9 (18%) were dry at 12 weekly and 10 (20%) eyes were dry at 8 weekly visit. For group 2, the mean changes in VA was -2.69 letters and the mean OCT changes was -54.23 µ. 9 (13.23%) eyes were dry at 16 weekly, 6 (8.82%) were dry at 12 weekly and 9 (13.23%) at 8 weekly visit.

Conclusion: We concluded that Faricimab has good anatomical and functional improvement for treatment naïve and switched patients with diabetic macular oedema. Also, it can decrease the clinical visits and workload by spacing the periods of follow up. Rates of intraocular inflammation (IOI) and potentially infectious endophthalmitis (PIE) were low in both groups.

Conflicts of interest: None

Funding: None

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P132

HYPERREFLECTIVE FOCI AS A BIOMARKER IN CENTRAL SEROUS CHORIORETINOPATHY

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Introduction: The exact pathophysiology, natural history and outcomes of central serous chorioretinopathy (CSCR) are poorly understood. There is variable vision improvement with conservative management, while effective treatment options are limited. OCT scans allow identification of various biomarkers, Hyper Reflective Foci (HRF) have been reported in CSCR but their influence on the outcome is not established.

This study evaluates prevalence of HRF and their evolution in patients with CSCR attending routine NHS eye clinics at a large teaching hospital.

Method: Patients first seen between October 2016 and March 2018 with a diagnosis of CSCR were retrospectively identified by Medisoft Audit suite (Medisoft, UK). Patients with misdiagnoses, missing data, confounding-pathologies and with less than 6 months follow-up were excluded.

Demographic data, habitual visual acuity VA and OCT scans from baseline to final follow up were available for analysis. Two independent clinicians assessed OCT scans of each patient for qualitative presence of HRF in the photoreceptor layers, in the inner segment IS and the outer nuclear layer ONL at baseline and at follow up (6–12 months). Presence or absence of outer retinal atrophic changes were recorded and correlated to the areas of HRF on the baseline OCT scan.

Results: 25 eyes of 25 patients with new presentation of CSCR (M = 20, F = 5; mean age 48 years) managed conservatively were included in this study. The mean visual acuity for all patients at baseline was 71 ± 18 letters.

17 eyes (68%) had hyper-reflective foci (HRF) in the inner segment zone of the photoreceptors at the base line. In these 17 eyes, the HRF were also present in the vicinity of ELM and ONL at baseline. 8 eyes (32%) had no HRF at baseline, these eyes had normal photoreceptor layer thickness on follow up OCT scans.

HRF resolved in all 17 eyes with resolution of CSCR. On follow up OCT scans, in 16 of the 17 eyes (94%) with HRF, the photoreceptors were thinned in the areas where HRF were noted on baseline. One of the 17 eyes with HRF, had diffuse thinning of PR layer on resolution of CSCR.

Conclusion: In CSCR, presence of HRF in the outer retinal layers on OCT scans is associated with subsequent atrophy of photoreceptors. The photoreceptor layer HRF can be a useful predictive biomarker in CSCR. Further studies on HRF can establish its exact prognostic value.

Conflicts of interest: None

Funding: None

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P133

IMPACT OF COVID-19 ON DIABETIC EYE SCREENING: EXPLORING OUTCOMES AND DISPARITIES

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Introduction:**Background:**

The Diabetic Eye Screening Program (DESP) aims to prevent sight loss by identifying and managing diabetic retinopathy through a tiered screening system: routine digital screening, digital surveillance, slit lamp Biomicroscopy, and ophthalmology referral.

The effectiveness of this system depends on the quality of retinal scans. Ungradable scans, those that cannot be assessed, represent inefficiencies in the service.

The pandemic presented significant challenges to healthcare delivery, necessitating an analysis of the impact that Covid-19 had on the screening service.

Objective:

This study examines trends in ungradable scan rates across pathways, regions, and across pre and post COVID periods to:

- Identify disparities
- Assess program resilience
- Highlight areas for improvement

Methods: Using data across 7 regions from March 2019 to February 2021, we calculated ungradable scan rates as a percentage of total screenings for each pathway and region. Two-way ANOVA and Tukey's HSD was performed.

Results:

1. **Routine Digital Screening:** showed significant regional and interaction effects. This shows there was variability in ungradable rates across regions regardless of covid. A significant interaction effect revealed that the impact of COVID-19 was not uniform; some Programmes, such as Nottingham, experienced a substantial post-COVID increase, while others demonstrated greater resilience.
2. **Digital Surveillance:** No significant effects were observed, indicating consistent performance across regions and time periods despite the pandemic's challenges.
3. **Slit Lamp Biomicroscopy:** Significant period and region effects were identified, revealing that this pathway was more sensitive to COVID-19-related disruptions. Regional differences in resources or accessibility may have contributed to this variability. Post-COVID ungradable rates were markedly higher than pre-COVID across all Programmes. The interaction effect wasn't significant, suggesting COVID19 impacted the regions uniformly.
4. **Ophthalmology:** No significant effects were found. This indicates minimal variability in ungradable rates, reflecting stability and equitable access at this critical stage of care. Though it must be considered that ophthalmology is the final tier of the screening service and has the fewest screens. The smaller sample size may limit the generalizability and reliability of the results of this pathway.
5. **Overall Program:** Overall service analysis showed no significant effects, suggesting resilience at the service-wide level, absorbing the disruptions caused by COVID-19. Though in the post hoc analysis Nottingham consistently exhibited higher rates than others.

Discussion: The results underscore the complexity of ungradable scan rates and the factors influencing them.

Period Effect: A significant period effect indicates that proportion of ungradable scans changed overall between pre- and post-COVID periods.

Post-COVID, Slit Lamp Biomicroscopy was particularly impacted, likely due to pandemic-induced operational disruptions or deferred patient follow-ups.

Regional Effect: Significant regional effects, particularly in Routine Digital Screening and Slit Lamp Biomicroscopy, point to variability in ungradable rates across different regions, suggesting

disparities in how resources or protocols were implemented locally.

Interaction Effect: When interaction effects are significant, it means the difference between pre- and post-COVID ungradable rates was not uniform across regions—some regions experienced greater changes than others as a result of the pandemic.

Some Programmes such as Southwest London demonstrated consistent performance despite pandemic-related disruptions.

For Programmes like Nottingham, targeted interventions such as staff training, enhanced equipment, and optimized workflows should be considered to address persistently high ungradable rates.

Conclusion: Our analysis highlights both strengths and vulnerabilities in the DESP during a global healthcare crisis. COVID-19 had a heterogeneous impact on the Diabetic Eye Screening Program, with significant variability across pathways and regions. While some pathways and regions exhibited resilience, others faced considerable challenges. The findings emphasize the need for tailored interventions and cross-Programme collaboration to enhance screening outcomes and mitigate disparities.

Conflicts of interest: None

Funding: None

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P134

INFLUENCE OF GLYCOSYLATED HAEMOGLOBIN ON OUTCOMES OF INTRAVITREAL FARICIMAB INJECTION FOR DIABETIC MACULAR OEDEMA: A 12-MONTH STUDY

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Introduction: The DCCT/EDIC study demonstrated that intensive glycaemic control delays the onset of retinal complications and reduces structural retinal abnormalities, including disorganization of the retinal inner layers (DRIL), intraretinal fluid, and cysts. However, there is a notable lack of evidence regarding the relationship between glycaemic control and the effectiveness of anti-VEGF treatments. Faricimab, a novel anti-VEGF drug, has shown promise in effectively extending treatment intervals for diabetic maculopathy. In this context, we present our 12-month real word data evaluating the impact of baseline HbA1c levels on the effectiveness of Faricimab in treating DMO.

Method: This retrospective study investigated the baseline HbA1c levels and the results of Faricimab injections between July 2022 to August 2024 at university Hospital Coventry and Warwickshire NHS Trust. A total of 64 treatment-naïve eyes were included, all meeting the NICE criteria for vision impairment due to centrally involving macular oedema with a central retinal thickness exceeding 400 µm. Twelve months of data were collected for each participant. The analysis evaluated the overall correlation between baseline HbA1c levels and key outcomes, including best-corrected visual acuity (BCVA), anatomical changes, and the total number of injections required over a 12-month period. Additionally, subgroup analysis compared patients with baseline HbA1c levels below 53 mmol/mol (<7%, Group A) to those with levels of 53 mmol/mol or higher (≥7%, Group B), consistent with the glycaemic control targets recommended by the DCCT trial.

Results: No significant correlation was observed between baseline HbA1c levels and central subfield thickness (CST). Among the 64 patients included in the study, 14 were classified into Group A (baseline HbA1c < 53 mmol/mol [<7%]), while 50 were in Group B (HbA1c ≥ 53 mmol/mol [≥7%]). Individual participants were followed up for a period of 12 months. At the final visit, the mean improvement in BCVA was 5.8 ± 12.8 ETDRS letters in Group A,

compared to 4.7 ± 14.5 ETDRS letters in Group B. The mean CST reduction was $-161.6 \pm 108.2 \mu\text{m}$ in Group A and $-124.6 \pm 99.7 \mu\text{m}$ in Group B. Although patients with better baseline glycaemic control (Group A) demonstrated greater anatomical improvement, the difference was not statistically significant ($P > 0.05$). The average number of injections required was identical in both groups ($n = 6.5$).

Conclusion: To our knowledge, this is the first study that evaluate the influence of HbA1C on the Faricimab outcomes in the treatment of DMO. Patients with a baseline HbA1c less than 53 mmol/mol ($<7\%$) demonstrated a trend toward better anatomical reduction of macular oedema, although this difference was not statistically significant in this study. Further studies with larger sample sizes are needed to validate these findings. Notably, improvements in visual acuity and the number of Faricimab injections required appeared to be independent of baseline HbA1c levels, consistent with post-hoc analyses from the RIDE/RISE trials and other anti-VEGF drug.

Conflicts of interest: None

Funding: None

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P135

INTRAOCULAR INFLAMMATION FOLLOWING INTRAVITREAL FARICIMAB: INSIGHTS FROM FIVE BILATERAL CASES

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Introduction: Intravitreal faricimab is a dual-action monoclonal antibody targeting both VEGF-A and angiopoietin-2, approved in 2022 for the treatment of neovascular age-related macular degeneration (nAMD) and diabetic macular oedema (DMO). While early clinical trials demonstrate a favourable safety profile compared to aflibercept, real-world reports have raised concerns regarding cases of intraocular inflammation (IOI).

Method: A retrospective observational study of all patients who received intravitreal faricimab injections and developed IOI at NHS Highland between January 2023 and June 2024. The presenting characteristics of patients presenting with intraocular inflammation were reviewed, including their ocular history, the number of previous faricimab injections, the onset of symptoms following the injection, visual acuity at presentation and recovery, intraocular pressure, the severity and location of intraocular inflammation, and management strategies.

Results: Over 18 months, 2,034 faricimab injections were delivered to 267 eyes of 208 patients. IOI was observed in 11 eyes of 6 patients following faricimab injection. All aqueous and vitreous taps were negative. Among patients who received bilateral faricimab, the incidence of IOI was 8.5% (5/59). 7 of the 11 eyes recovered baseline visual acuity, while four eyes experienced vision loss (11.5 ± 8.6 ETDRS letters; range 5–24). Cases were characterised by delayed onset of subacute symptoms post-injection, keratic precipitates and elevated intraocular pressure.

Conclusion: While the overall incidence of faricimab-related IOI appears to be low; our findings highlight the need for increased vigilance for patients receiving bilateral injections, particularly within the first 6 injections. We recommend pre-injection IOP measurement and slit lamp examination as asymptomatic cases of elevated IOP and uveitis were observed in our cohort. Education on the warning signs of IOI and timely management with steroid treatment is needed to mitigate the risk of inflammatory complications and preserving vision.

Conflicts of interest: None

Funding: None

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P137

MODE OF DELIVERY IN WOMEN AT RISK OF RHEGMATOGENOUS RETINAL DETACHMENT: A SYSTEMATIC REVIEW

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Introduction: Rhegmatogenous retinal detachment (RRD) is essentially a tear in the retina leading to vitreous humour accumulation behind it. Despite lacking guidelines contraindicating vaginal delivery (VD) for RRD risk, many doctors may recommend caesarean sections due to reported associations with Valsalva manoeuvre, prompting our study to assess delivery mode's relation to RRD and evaluate practice measures for at-risk pregnant women. In this context, performing caesarean sections (CS) can exacerbate healthcare costs, straining the healthcare system and highlighting the importance of evidence-based decision-making in obstetric care.

Method: The search encompassed all literature designs in the following databases: Google Scholar, Cochrane, Scopus, and PubMed ($n = 875$). The keywords and search strategies were thoroughly documented, and the final included studies were 14 following multiple screening processes.

Results: This study included 14 studies with a total of 216 patients. The total number of deliveries was 264, including 10 from case reports. Of these, 136 were spontaneous vaginal deliveries, 7 were assisted vaginal deliveries, and 121 were caesarean sections. The women had ophthalmic conditions such as a history of RRD, Stickler syndrome, high myopia, and lattice degeneration. A substantial portion of the studies reported no postpartum visual complications or fundal abnormalities following both delivery methods vaginal and caesarean. In retrospective analyses, recurrent RRD following VD and CS occurred at similar rates.

Conclusion: The study findings suggest that the method of delivery, whether spontaneous vaginal, assisted vaginal, or caesarean section, does not impact the occurrence of rhegmatogenous retinal detachment in individuals considered to be at high risk. It also calls for more research into the value and specific indications of prophylactic laser photocoagulation in such cases.

Conflicts of Interest: none

Funding: none

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P138

OCT ANGIOGRAPHY CLASSIFICATION OF TYPE 2 IDIOPATHIC MACULAR TELANGIECTASIA AND CORRELATION WITH OCT BIOMARKERS

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Introduction: Type 2 idiopathic macular telangiectasia (MacTel) is a bilateral acquired neurodegenerative condition associated with loss of Muller cells, causing gradual decline in central vision. It is characterised by vascular abnormalities that mainly affect the deep vascular plexus (DVP) in non-proliferative (NP) stages, and development of subretinal neovascularisation (SNV) in the proliferative (PL) stage. More recently with widespread use of optical coherence tomography angiography (OCTA), a pre-proliferative (PP) stage has been described.

Our study aims to categorise eyes with type 2 MacTel into NP, PP and PL groups based on OCTA and describe their association with key OCT features, in order to facilitate identifying eyes at risk of developing acute/ subacute severe sight loss from macular neovascularisation.

Method: This is a single-centre retrospective case series of treatment-naïve patients with a confirmed diagnosis of type 2 MacTel, identified from an in-house electronic patient record system at a tertiary hospital. Bilateral macular OCTA and OCT images (Heidelberg Spectralis) from the same visit were graded at Liverpool Ophthalmic Reading Centre for pre-defined established features of type 2 MacTel and secondarily adjudicated by a senior clinician. Eyes with DVP bunching and absent SNV were assigned to the PP group, eyes with a neovascular complex in the avascular slab on OCTA to the PL group and the rest to the NP group. Patients with other macular co-pathology and images affected significantly by artefacts were excluded.

Results: Twenty-six patients (52 eyes) with a mean age of 65 (40–83) yrs, 15 (58%) females were included, of whom 81% had the same stage bilaterally. There were 26 (50%), 7 (13.5%) and 19 (36.5%) eyes with NP, PP and PL MacTel with a mean best-recorded visual acuity of 74 (27–95, SD 13.83), 78 (57–88, SD 9.91), 62 (5–83, SD 19.73) ETDRS letters, mean central subfield thickness and total macular volume (TMV) of 266, 261, 290 microns and 7.57, 5.84, 7.06 mm³ respectively. In the NP and PP groups 96% and 100% of eyes had inner retinal hyporeflective spaces, 58% and 29% had hyperreflective middle retinal layers and 50% and 71% had outer retinal cavitation respectively. In the PL group, 100% of eyes had disorganised outer retinal hyperreflective material and inner retinal hyporeflective spaces, 95% had outer retinal cavitation and 79% had retinal pigment epithelial elevation and disruption. On OCTA in the NP group 31% and 73% of eyes, and in the PP and PL groups, approximately 85% and 100% of eyes had dilated perifoveal vessels in the superficial vascular plexus and DVP layers respectively. Eyes with foveal avascular zone (FAZ) irregularity increased from 46% to 86% in the NP to PP group.

Conclusion: The most prevalent OCT biomarkers seem to be inner retinal hyporeflective spaces and outer retinal cavitation across the 3 groups, however with progressive stages of MacTel the latter became almost universally prevalent. TMV reduced in the pre-proliferative group, likely due to more advanced thinning of the neural retina compared to the non-proliferative group but increased in the proliferative group. There was no association between worsening degree of dilated vessels in DVP on OCTA in the pre-proliferative group and the prevalence of hyperreflective middle retinal layers on OCT. Even in eyes with minimal DVP bunching on *enface* OCTA there was an obvious increase in flow signal on cross-sectional B-scan, and localised distortion of FAZ boundary, both of which might serve as surrogate markers to identify pre-proliferative MacTel and undertake closer monitoring. Disorganised hyperreflective material disrupting retinal layers is universally present on OCT in eyes with proliferative MacTel; further evaluation with OCTA to confirm the diagnosis promptly and offer timely treatment is advisable.

Conflicts of interest: None

Funding: None

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P139

OCULAR ISCHAEMIC SYNDROME: AUDIT ON DISEASE PRESENTATIONS AND TREATMENT OUTCOMES

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Introduction: Ocular ischaemic syndrome (OIS) is a spectrum of ocular diseases due to ocular hypoperfusion. It is often associated with significant carotid artery stenosis. OIS is not usually considered a specific disease entity as its clinical presentation is varied and it shares common clinical features with more common diseases such as diabetes and vein occlusions. Visual loss secondary to OIS is also reported to vary from gradual visual loss (80%) to the profound sudden visual loss (12%) of neovascular glaucoma, with poorly understood disease progression. Significant co-morbidities of cardiovascular disease have been associated with OIS, with reports of 40% mortality of affected patients from myocardial infarction or stroke. As OIS presentations are highly variable, it is likely that OIS is very much underdiagnosed. In recent years, enhanced ophthalmic imaging with fundus fluorescein angiography (FFA) has been used as a valuable tool in the diagnosis of ocular hypoperfusion; this can be followed up by a carotid doppler scan to correlate diagnosis of OIS with carotid artery stenosis as the causative factor. We aimed to characterise features of patients who presented with OIS retinopathy, including presentation severity, progression, and visual outcomes with currently available treatment options.

Method: Retrospective data collection from electronic records of FFA confirmed OIS patients who also underwent carotid doppler scan in a single hospital. We collected data on vision, intraocular pressures (IOP), retinopathy features, systemic co-morbidities at presentation, visual change following treatments over follow-up period and related complications.

Results:

Presenting Visual acuities and IOPs: 59.5% (22/37 eyes) had vision 6/12 or better, 24.3% (9/37) had 6/15–6/60 vision, 16.2% (6/37) were worse than 6/60. IOP > 21 was recorded in 8.1% of eyes (3/37) and 67.6% had a history of glaucoma treatment.

Associated co-morbidities: 39.2% (11/28 patients) were known diabetics; 60.7% (17/28) had >2 cardiovascular co-morbidities (e.g., hypertension, hyperlipidaemia, stroke, myocardial infarctions).

Presenting retinopathy descriptions: These were retinal vein occlusion (45.9%), peripheral blot retinal haemorrhages (10.8%), asymmetrical diabetic retinopathy (16.2%), vitreous haemorrhage (16.2%) and rubeosis/neovascular glaucoma (21.6%). 2.7% (1/37 eyes) were asymptomatic at presentation. 27% (10/37 eyes) had cystoid macular oedema at presentation. All patients underwent doppler ultrasound; 54% (20/37) did not have any associated ICA atheroma or stenosis and none of the OIS retinopathy eyes were correlated to ipsilateral ICA stenosis of > 70%.

Treatment received by this cohort: 2.7% (1/37 eyes) received anti-VEGF treatment alone, 32.4% (12/37 eyes) received laser photocoagulations alone, 59.5% (22/37) received combination of treatments. 13.5% eyes (5/37) % also received additional glaucoma treatment (surgical/ cyclodiode procedures).

Visual acuity change at final follow-up: 48.7% (18/37 eyes) had vision 6/12 or better, 35.1% (13/37) had 6/15–6/60 vision, 16.2% (6/37) were worse than 6/60. 7.1% (2/28 patients) died from systemic cardiovascular complications during follow-up period.

Conclusion: OIS retinopathy presentations remain variable. This cohort shows a high percentage of eyes with significant

irreversible visual loss, an association with glaucoma prevalence and cardiovascular risk factors. Our cohort also shows a poor correlation of OIS eyes with significant ICA stenosis.

A high index of suspicion is required to improve early OIS diagnosis and treatment to prevent progression to profound visual loss.

Conflicts of interest: None

Funding: None

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P140

ONE-YEAR CLINICAL EFFICACY AND DURABILITY OF INTRAVITREAL FARICIMAB VERSUS AFLIBERCEPT(2 MG) IN TREATMENT NAÏVE EYES WITH DIABETIC MACULAR OEDEMA: A REAL-WORLD STUDY

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Introduction: Within NHS Lanarkshire, first line treatment of patients with diabetic macular oedema (DMO) is intravitreal therapy (IVT) with anti-VEGFs. Frequent dosing intervals imposes significant burden to healthcare services. YOSEMITE and RHINE were two phase III trials that showed Faricimab (Vabysmo®) to be non-inferior to Aflibercept (Eylea®), whilst achieving extended treatment intervals. The purpose of this study was to investigate real-world, one-year clinical efficacy and durability of intravitreal 6 mg Faricimab versus 2 mg Aflibercept in treatment naïve eyes with DMO.

Method: Data was collected prospectively using clinical notes, Clinical Portal and IMAGENet 6 software (Topcon Healthcare). Only patients listed for Faricimab or Aflibercept with no prior treatment for DMO were included between May 2023 and November 2024. A treat and extend protocol was adopted after an initial loading dose treatment. OCT scans were analysed before treatment, at 4 months and 1-year post initiation of treatment. Patients not achieving the standard treatment protocol were excluded.

Results: 20 eyes of 14 patients (57% male) received Faricimab. Median age was 53 years (31–76 years). Pre-treatment mean BCVA was 62 letters (39–80 letters). Post-treatment mean BCVA at 1-year was 68 letters (43–85 letters). Mean change in BCVA was 7 letters (–15 to 25 letters). Pre-treatment mean central retinal thickness (CRT) was 465 μ (247–819 μ). Post-treatment mean CRT at 1-year was 280.7 μ (202–635 μ). Mean %change in CRT was –32% (6 to –75.3%). Pre-treatment mean average retinal thickness (ART) was 403.45 μ (260–582 μ). Post-treatment mean ART at 1-year was 302.8 μ (239–376 μ). Mean %change in ART was 22.6% (–3.49 to –52.92%). At 1-year, intraretinal fluid was fully resolved in 20% of eyes. Dosing intervals of ≥ 12 weeks were achieved by 78.6% of eyes and ≥ 16 weeks by 70% of eyes.

14 eyes of 13 patients (69% male) received Aflibercept. Median age was 68.5 years (46–86 years). Pre-treatment mean BCVA was 66 letters (37–80 letters). Post-treatment mean BCVA at 1-year was 71 letters (59–84 letters). Mean change in BCVA was 6 letters (–10 to 40 letters). Pre-treatment mean CRT was 428 μ (274–635 μ). Post-treatment mean CRT at 1-year was 274 μ (187–496 μ). Mean %change in CRT was –33.4% (10.9 to –61.8%). Pretreatment mean ART was 367 μ (282–481 μ). Post-

treatment mean ART at 1-year was 312 μ (256–440 μ). Mean % change in ART was –14.32% (0 to –37.6%). At 1-year, intraretinal fluid was fully resolved in 14.3% of eyes. Dosing intervals of ≥ 12 weeks were achieved by 30% of eyes and ≥ 16 weeks by 28.6% of eyes.

Conclusion: Faricimab was non-inferior to Aflibercept in the real-world management of patients with treatment naïve DMO over a 1-year period with comparable clinical and anatomical outcomes. Faricimab also achieved extended dosing intervals, thus alleviating treatment burden on patients and service provision in NHS Lanarkshire.

Conflicts of interest: None

Funding: None

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P141

OUTCOMES OF DIABETIC RETINOPATHY SCREENING IN PREGNANT WOMEN IN BIRMINGHAM AND SOLIHULL – RESULTS FROM THE LARGEST NHS TRUST IN ENGLAND

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Introduction: Diabetic retinopathy (DR) is a leading cause of blindness in the working age population. Pregnancy is an independent risk factor, significantly increasing the short-term risk of DR progression. Global estimates of proliferative DR rate are about 6.1%. The purpose of this study is to give an overview of the diabetic eye screening attendance and compliance in Birmingham and Solihull area and understand progression of diabetic retinopathy during pregnancy in this diverse area.

Method: This retrospective study analysed data collected from the Diabetic Eye Screening Program (DESP) between 1st September 2022, and 31st August, 2023. The inclusion criteria consisted of pregnant women with pre-existing diabetes who received maternity care at University Hospitals Birmingham (UHB), while women diagnosed with gestational diabetes were excluded. Electronic medical records, optical coherence tomography (OCT) scans, and Optos imaging results were reviewed.

Results: Over the 12-month period, 149 patients received extra eye screenings due to pregnancy. The 12-week screening attendance rate was 41%, primarily attributable to patient non-attendance (23% DNA) or delayed referrals (36%). The 28-week screening attendance was 51%. By offering 118 extra appointments, 86.6% women completed screening. There are 8 patients referred to hospital eye service by DESP. One case (<1%) progressed to high-risk Proliferative DR. The rate of diabetic macular oedema (DMO) was 7% (8 cases), but 13% of them (1 case) rapidly progressed to bilateral severe DMO and required steroid implants.

Conclusion: Efforts are needed to improve the attendance of DESP during pregnancy, as well as timely referral from maternity service. The rate of referable DR remains lower in Birmingham area than global estimates, but treatment can be challenging. DMO in pregnancy can progress rapidly, emphasizing the need for timely intervention.

Conflicts of interest: none

Funding: none

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P142

PATIENT SATISFACTION AND CLINICAL EFFICACY OF SWITCHING FROM INTRAVITREAL RANIBIZUMAB (LUCENTIS) INJECTIONS TO RANIBIZUMAB BIOSIMILAR (ONGAVIA) INJECTIONS

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Introduction: Ranibizumab (Lucentis) has been widely used to treat retinal conditions such as wet age-related macular degeneration, diabetic macular oedema and macular oedema secondary to retinal vein occlusions. In 2022, the Medicines and Healthcare products Regulatory Agency (MHRA) approved the use of ranibizumab biosimilar, Ongavia. Ongavia is the first approved biosimilar of Ranibizumab, and the United Kingdom was the first European country to adopt its use. It has the potential to offer comparable efficacy, safety, and side effect profile to the original medication but at a lower cost making it more cost-effective.

Purpose: To assess patient satisfaction when switching from Lucentis to Ongavia and to evaluate the efficacy and safety of the biosimilar.

Methods: Data was collected from all patients undergoing treatment with intravitreal Ranibizumab (Lucentis) and were offered switching to Ongavia between November 2023 and February 2024. All relevant information was provided to each patient beforehand and were consented on the day as per hospital policy. Patient satisfaction was evaluated by a questionnaire and clinical outcomes and the safety profile were also assessed.

Results: A total of 138 eyes were receiving intravitreal Ranibizumab injections. Of these, 17 eyes (12.3%) had completed the treatment with Ranibizumab (Lucentis). Among the remaining 121 eyes, 92 (76%) were switched to Ranibizumab biosimilar (Ongavia), while 18 (14.9%) continued with Ranibizumab as per patient choice. Patient satisfaction with the information process was reported as 100%.

From the 92 eyes switched to Ongavia, two eyes were excluded due to other co-morbidities, and in 22 eyes (24.4%) treatment was discontinued after one Ongavia injection. This is because 12 eyes finished their treatment after this one Ongavia injection, 5 eyes changed to another intravitreal medication and 5 eyes did not have any further injections planned. Following the second Ongavia injection, three more eyes were excluded: two eyes of a patient who deceased and one due to myocardial infarction.

From the remaining 87 eyes, 64 (73.5%) eyes increased or maintained their interval with the Ranibizumab biosimilar (Ongavia) whereas, 7 (8%) of 87 eyes had a reduction in their injection interval weeks. 9 (10.3%) of 87 eyes change to another intravitreal injection.

Conclusion: Most patients on Ranibizumab (Lucentis) were successfully changed to Ranibizumab biosimilar (Ongavia). Patients were overall satisfied with the whole information process. No significant safety concerns were raised with the Ranibizumab biosimilar (Ongavia) and the results showed that Ongavia was clinically comparable to Lucentis.

Conflicts of interest: None

Funding: None

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P143

PHACOEMULSIFICATION COMBINED WITH INTRAVITREAL FARICIMAB INJECTION IN CATARACT PATIENTS WITH

NEOVASCULAR AGE RELATED MACULAR DEGENERATION / DIABETIC MACULAR OEDEMA

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Introduction: Neovascular age related macular degeneration (nAMD) and diabetic macular oedema (DMO) patients receiving intravitreal injections (IVT) with anti-VEGF agents who require cataract surgery have delays in their IVT regime to allow for cataract surgery or a delay in cataract surgery to allow for IVT. This study was done to assess the efficacy and safety of simultaneous intravitreal faricimab (Vabysmo) injections in nAMD and DMO patients undergoing cataract surgery.

Method: nAMD and DMO patients receiving IVT requiring cataract surgery were identified and underwent phacoemulsification with intraocular lens (IOL) implant along with intravitreal faricimab injection by a single surgeon. Preoperative and post-operative best corrected visual acuity (BCVA), central foveal thickness (CFT), intraretinal/subretinal fluid (IRF/SRF) and post-operative inflammation defined as anterior chamber cells and flare and vitritis were assessed.

Results: 15 nAMD and DMO patients with cataract underwent phacoemulsification and IOL implantation combined with intravitreal injection of faricimab.

Post surgery, BCVA improved in 73% patients and remained unchanged (27%). No deterioration of vision post surgery was noted in any of the patients.

CFT improved in 60% of patients and was maintained in 13% patients. In 26% patients, the CFT increased but the increase was only 20%. Most patients also showed an improvement in the volume of retinal fluid with 3 patients maintaining the absence and 4 patients attaining complete resolution of retinal fluid.

No anterior chamber inflammation or vitritis were observed in any of the 15 patients.

Conclusion: When injected simultaneously with phacoemulsification, faricimab was able to continue to control the nAMD and DMO disease activity. The safety profile was the same as before surgery with no inflammatory events.

Disease control may be compromised in nAMD and DMO patients, actively receiving IVTs at a treatment frequency of 4–6 weeks, who undergo cataract surgery as their IVT appointment can get delayed due to cataract surgery. In addition, organising the surgery around a patient's treatment regime has created a backlog for cataract service.

Provision of the simultaneous delivery of faricimab to nAMD and DMO patients undergoing cataract surgery may provide a solution to these issues.

Conflicts of interest: None

Funding: None

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P144

REAL WORLD CLINICAL OUTCOMES FOR PATIENTS TREATED WITH FARICIMAB FOR NEOVASCULAR AGE-RELATED MACULAR DEGENERATION

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Introduction: There is scant data on the long-term real life, functional and structural outcomes of neovascular age-related macular degeneration (nAMD) patients treated with Faricimab. Here we present 52-week and 78-week outcomes for a cohort of nAMD patients treated at an NHS macula centre in Bradford, UK.

Method: All patients treated for nAMD with Faricimab between 01/08/22 and 31/03/23 were identified and records were retrospectively analysed. 107 eyes of 92 eligible patients were included in the analysis, of whom 37 were treatment naïve and 70 were previously treated. Previously treated eyes were then divided into two sub-groups depending on the frequency of their treatment preceding the switch: a “short treatment interval (SI)” group (pre-switch interval of 4–5 weeks) and a “long treatment interval (LI)” group (pre-switch interval ≥ 6 weeks). Data was extracted for their visual acuity (VA), central macular thickness (CMT), presence and type of macular fluid and adverse events at baseline, 12 ± 4 weeks, 26 ± 4 weeks, 52 ± 4 weeks and (if available) 78 ± 4 weeks. The “clinician recommended” treatment interval was also captured at 52 weeks and 78 weeks.

Results: At 52 weeks the naïve group showed a mean improvement of $+5.43$ letters from baseline ($p < 0.05$) and mean reduction in CMT of $-108.9 \mu\text{m}$ ($p < 0.001$). Improvements in acuity were maintained to 78 weeks, although mean CMT reduction had fallen to $-77.0 \mu\text{m}$. The “SI” previously treated group had an initial increase in acuity at 12 weeks (mean $+1.65$ letters), but this reduced to -1.77 letters from baseline by 52 weeks and -4.24 letters from baseline by week 78 ($p < 0.05$). The “LI” previously treated group showed mean VA change of -3.89 letters from baseline by week 52 and -3.42 letters from baseline by week 78 ($p < 0.01$). Both the SI and LI groups showed a reduction in CMT ranging between $-19 \mu\text{m}$ to $-31 \mu\text{m}$ from week 12 through to week 78, and there were no significant differences between the groups. The frequency of eyes with a ‘dry’ macula (no fluid on OCT scan) was significantly improved for the naïve group (54% of eyes at week 52, $p < 0.001$ and 50% at week 78, $p < 0.05$) and the “LI” group (46% at week 52, $p < 0.001$, 35% at week 78, $p < 0.05$), but not for the “SI” group (19% at week 52 and 20% at week 78). By 78 weeks 13% of naïve, 28% of “SI” and 15% of “LI” patients had reached ≥ 12 -week ‘intended’ treatment intervals. There were no reports of vasculitis in any group, although one patient developed endophthalmitis following combined cataract surgery and intravitreal injection in the same week.

Conclusion: Faricimab is safe and effective in improving vision and OCT outcomes for treatment naïve patients with nAMD. OCT outcomes also improve with Faricimab in previously treated nAMD patients, although vision gains may not be sustained long term. Longer treatment intervals can be possible with Faricimab injections, but the effect of persistent retinal fluid on long term functional outcomes remains to be defined in real life.

Conflicts of interest: Professor Ghanchi has worked as a consultant for Roche, Bayer, Alimera and Boehringer Ingelheim, and received education grants and lecture fees from Roche, Alimera, Heidelberg and AbbVie.

Funding: None

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P145

REAL WORLD EFFICACY AND SAFETY OF XIMLUCI RANIBIZUMAB BIOSIMILAR

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Introduction: Since 2022, multiple ranibizumab biosimilars have been licensed for intravitreal use in the UK. Ximluci is a ranibizumab biosimilar licensed for use in neovascular age related macular degeneration (nAMD), choroidal neovascularisation (CNV), macula oedema secondary to diabetes and retinal vein occlusion and proliferative diabetic retinopathy. While clinical trials have demonstrated its efficacy and safety, real-world data for Ximluci is limited, as most UK centres favoured the use of Ongavia over other ranibizumab biosimilars. This study evaluates the real-world efficacy and safety of Ximluci in patients treated at a UK hospital eye clinic.

Method: A retrospective observational review was conducted, analysing data from patients who received Ximluci for all indications between January 2023 and December 2024. The visual acuity (VA), central retinal thickness (CRT) and the incidence of adverse events were recorded before and after treatment.

Results: 37 patients received Ximluci during this time frame, with 16 receiving at least 2 injections for diabetic macula oedema (DMO), neovascular age related macular degeneration nAMD or choroidal neovascularisation (CNV). 69% of these were for DMO and 31% for nAMD and CNV.

Patients had visual acuity and OCT parameters taken before and after the 2 injections. Results showed a mean improvement in vision of 0.17 logMAR (1 patient was excluded due to a vitreous haemorrhage) and a mean improvement in central subfield thickness of $30.86 \mu\text{m}$. Adverse events were rare with one case of raised intraocular pressures which required treatment. 13% of these patients were switched to alternate drug treatments.

Conclusion: This limited real world sample of patients has demonstrated that Ximluci is safe and effective in treating certain retinal diseases. However, further studies with larger sample sizes and extended follow-up would be warranted to validate this.

13% of patients required a switch to alternative treatments and therefore head-to-head comparison trials are needed to assess whether there are any differences between Ximluci and other anti-VEGF drugs.

Conflicts of interest: None

Funding: None

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P146

REAL WORLD OUTCOMES OF FARICIMAB IN TREATMENT-NAIVE AND TREATMENT-SWITCH EYES

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Introduction: This study investigates the effectiveness of Faricimab in managing diabetic macular oedema (DMO) in both treatment-naïve and treatment-switch patients, focusing on outcomes related to central macular thickness (CMT), visual acuity, and patient group characteristics.

Method: A retrospective audit was conducted at a district general hospital in London, involving 65 patients (87 eyes) with DMO (>400 microns) treated with Faricimab. Patients received 6 mg injections on a treat-and-extend protocol between January 2022 and February 2024, with follow-up extending up to two years. Participants were divided into treatment-naïve and treatment-switch groups. Data on demographics, DMO duration, best-corrected visual acuity (BCVA), and CMT were recorded at

each injection visit using Heidelberg or Zeiss Optical coherence tomography. Statistical analyses, including Pearson correlation coefficients and unpaired t-tests, were used to assess differences between groups.

Results: Among the 87 eyes treated with Faricimab, 34 (39.1%) were treatment-naïve, and 53 (60.9%) had prior anti-VEGF therapy. In the treatment-naïve group (24 patients, 34 eyes; mean age 66), baseline CMT decreased significantly from 485.1 μm to 327.2 μm ($p < 0.0001$), while visual acuity improved modestly from 59.5 to 62.7 EDTRS ($p = 0.259$). For the switched group (41 patients, 53 eyes; mean age 65), baseline CMT reduced from 439.5 μm to 366.9 μm ($p < 0.0001$), with visual acuity showing minimal change from 59.7 to 61.4 EDTRS ($p = 0.55$).

Conclusion: This real-world study highlights the efficacy of Faricimab in reducing macular thickness among patients with DMO, particularly in treatment-naïve eyes. The modest improvement in visual acuity and the significant CMT reduction contribute to evidence for Faricimab's potential as a viable first-line treatment option for patients without prior anti-VEGF exposure.

Conflicts of interest: None

Funding: None

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P147

REAL WORLD OUTCOMES OF INTRAVITREAL FARICIMAB TREATMENT IN PATIENTS WITH DIABETIC MACULAR OEDEMA

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Introduction: Faricimab is a bispecific antibody which targets vascular endothelial growth factor A (VEGF-A) and angiopoietin-2 (Ang 2). In 2022, it was recommended by NICE for the treatment of diabetic macular oedema. This study aimed to characterise the real-world treatment outcomes (frequency, visual and anatomical) of patients with diabetic macular oedema (DMO) treated with intravitreal faricimab.

Method: Retrospective analysis of consecutive DMO patients commenced on intravitreal faricimab therapy at a multi-site UK ophthalmology department between 30/11/22 and 25/09/24. Electronic patient records and optical coherence tomography (OCT) images were examined for patient demographics, ocular co-morbidities, visual acuity, central retinal thickness (CRT) and structural features such as subretinal fluid (SRF), intraretinal fluid (IRF), hyperreflective foci (HRF) and diffuse retinal thickening (DRT). Details of treatment history and adverse effects were also collected. Patients who did not complete four loading doses of faricimab were excluded from the analysis.

Results: 67 eyes of 54 patients with DMO were included in this study, of which 36 eyes had been switched from other anti-VEGF agents. Patients received approximately 6.5 injections in 12 months, with treatment naïve requiring 6.0 and previously treated eyes requiring 7.0. The median treatment interval at 12 months was 11 weeks. 72% of patients were male and the average age was 63.7 years.

Baseline visual acuity was 62.2 $\pm$ 13 ETDRS letters, with a mean increase of +2.91 and +4.8 letters at 6 and 12 months, respectively ($p < 0.0001$). Visual acuity improved by greater than 5 ETDRS letters in 42% of eyes at 6 months and 33% at 12 months. Mean CRT improved from 479.0 $\pm$ 122.8 μm to 335.7 $\pm$ 107.5 μm at 12 months (mean change: -143.3 $\pm$ 107.5 μm , $p < 0.0001$). The proportion of eyes with SRF improved from 22% at baseline to 2% at 12 months, with those with DRT reducing from 93% to 42%. The proportion of

eyes with IRF and HRF remained stable throughout treatment. Patients who were treatment naïve had better visual acuity (67.6 $\pm$ 13.7 vs 66.4 $\pm$ 15.6 ETDRS letters), reduced central retinal thickness (321.8 $\pm$ 84.2 μm vs 348.8 $\pm$ 123.5 μm) and less intraretinal fluid and subretinal fluid compared with previously treated eyes.

Conclusion: Intravitreal faricimab treatment was demonstrated to improve functional and anatomical outcomes in patients with DMO in a real-world setting. Greater improvements were noted in treatment naïve eyes than in previously treated eyes.

Conflicts of interest: None

Funding: None

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P148

REAL-WORLD EFFICACY, SAFETY, AND DURABILITY OF AFLIBERCEPT 8 MG IN NEOVASCULAR AGE-RELATED MACULAR DEGENERATION

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Introduction: Neovascular age-related macular degeneration (nAMD) is a leading cause of vision loss, often requiring frequent anti-vascular endothelial growth factor (VEGF) injections. Limited real-world data exists on the effectiveness of aflibercept 8 mg, a higher-dose VEGF inhibitor, particularly in patients with sub-optimal responses to prior treatments. We hypothesize that aflibercept 8 mg improves visual acuity, reduces central subfield thickness (CST), and resolves fluid biomarkers in both treatment-naïve and pre-treated nAMD patients.

Method: This retrospective, observational study analysed 54 eyes (40 treatment-naïve, 14 pre-treated) treated with aflibercept 8 mg at a tertiary care centre from May to August 2024. Treatment-naïve eyes received three loading doses at 4-week intervals, followed by a treat-and-extend protocol. Pre-treated eyes were dose-matched without reloading. Primary outcomes included changes in best-recorded visual acuity (BRVA), CST, and fluid resolution rates. Statistical analyses included paired two-tailed t-tests and McNemar's tests for outcomes and biomarkers, respectively.

Results: Treatment-naïve eyes demonstrated significant improvements in BRVA (6.3 $\pm$ 8.0 letters, $p < 0.001$) and CST (-84.9 $\pm$ 97.2 μm , $p < 0.001$) after three loading doses, with 75% achieving complete fluid resolution. Pre-treated eyes maintained BRVA (+1.1 $\pm$ 7.3 letters, $p = 0.571$) and showed non-significant CST reductions (-14.4 $\pm$ 40.9 μm , $p = 0.210$), with 29% achieving fluid resolution over 27.3 weeks. No ocular or systemic complications were observed in either group.

Conclusion: Aflibercept 8 mg significantly improved visual and anatomical outcomes in treatment-naïve nAMD patients and stabilized outcomes in pre-treated patients with suboptimal prior responses. Its favourable safety profile and extended dosing intervals highlight its potential to reduce treatment burden and improve real-world outcomes in nAMD management. Further studies are warranted to confirm these findings across diverse healthcare settings.

Conflicts of interest: None

Funding:

Serena Salvatore, Speaker for Bayer.

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P149

REAL-WORLD INTRAVITREAL FARICIMAB DATA IN A DISTRICT GENERAL HOSPITAL SETTING IN SCOTLANDMostafa Khalil¹, Sidharth Kaushal¹, Rinoza Bafiq¹¹Tennent institute of Ophthalmology, Greater Glasgow and Clyde, UK

Introduction: Neovascular age-related macular degeneration (nAMD) remains a significant cause of visual impairment in developed countries. Frequent monthly dosing of intravitreal aflibercept in real-world settings has placed immense strain on NHS district general hospitals, contributing to service pressures and suboptimal patient outcomes. Intravitreal Faricimab 6 mg has demonstrated promising results in the Tenaya and Lucerne trials. This study audited 1-year real-world outcomes of Faricimab for the treatment of nAMD in a district general hospital in Scotland.

Method: This prospective cohort study included patients with nAMD treated with intravitreal Faricimab 6 mg (standard dose) between February to July 2023. All patients received a loading phase of four monthly injections, followed by a treat-and-extend protocol. Data on patient demographics, ETDRS visual acuity (VA), mean central retinal thickness (CRT), intraretinal/subretinal fluid (IRF/SRF) presence, and number of injections were collected and analysed using Microsoft Excel.

Results: The study included 49 eyes from 46 patients after exclusions. The median patient age was 80 years, with 67% female. Treatment-naïve eyes accounted for 18 cases (37%), while 31 eyes (63%) were from switch patients. Median baseline VA was 55 ETDRS letters, improving to 58 letters at 1 year ($p = 0.39$). Mean CRT reduced from 285 μm at baseline to 221 μm at 1 year ($p < 0.0000003$). IRF/SRF was absent or reduced in 33 eyes (67%) at 1 year. The median number of injections over the year was six, with 60% of patients achieving intervals of >12 weeks by 1 year.

Conclusion: This real-world study demonstrates effective disease control in 60% of nAMD patients, including both treatment-naïve and switch cases, with a median of six injections over 1 year in a district general hospital under significant service pressures. These outcomes are consistent with the Tenaya and Lucerne trials which was around 80%, which was however, for only treatment-naïve cases. The discrepancy was likely due to the greater resistance to management of the switch patients. These findings support the use of intravitreal Faricimab as a viable treatment option for nAMD in high-demand healthcare settings.

Conflicts of interest: None

Funding: None

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P150

REAL-WORLD TWELVE MONTH OUTCOMES OF PATIENTS SWITCHED TO FARICIMAB WITH RECALCITRANT NEOVASCULAR MACULAR DEGENERATIONRiccardo Fausto¹, Wei Sing Lim¹, Mohammed El-Atfy¹, Clara Vazquez-Alfageme¹, Vasuki Sivagnanavel¹¹Kingston Hospital NHS Foundation Trust, Kingston, UK

Introduction: Faricimab has been shown in clinical trials to reduce retinal fluid in wet age related macular degeneration to a greater extent than Eylea. Patients requiring frequent anti-VEGF

injections potentially could achieve greater retina fluid resolution and allowing for longer extension between injections without recurrence of retinal fluid after switching to Faricimab.

We want to assess the safety and efficacy of switching to Faricimab in patients receiving treatment with either aflibercept or ranibizumab for the treatment of neovascular age related macular degeneration (nAMD).

Method: Retrospective review at Kingston Hospital NHS Foundation Trust. Patients switched to Faricimab with 12 months follow-up were identified from electronic medical records between November 2022 and December 2022. All patients who received Faricimab for the treatment of nAMD within the study period were included in this real-world analysis. Patients were treated on a treat and extend basis. The full loading dose of 4 injections was not mandated.

Outcome measures evaluated were changes in best corrected visual acuity (BCVA), changes in central subfield thickness (CST), presence of retinal fluid, treatment intervals and adverse events.

Results: 56 eyes of 48 patients were identified as requiring treatment at intervals of less than 8 weeks to maintain disease inactivity (dry retina) or least retinal fluid. Mean visual acuity remained stable: 66 ± 11 ETDRS letters at baseline; 64 ± 16 at twelve months ($p > 0.05$). Central subfield thickness reduced from $241 \pm 56 \mu\text{m}$ to $225 \pm 66 \mu\text{m}$ ($p < 0.05$) at twelve months. The mean baseline treatment interval was 4 ± 0.5 weeks. The mean treatment interval at 1 year that achieved stability was 7 ± 3 weeks ($p < 0.05$). Fifty seven percent of eyes could be extended by at least 1 week.

At 12 months, 18% of eyes were treated at >8 weekly intervals and 7% of eyes did not show better outcome with Faricimab. The mean number of Faricimab injections given per eye at 12 months was 8, ranging from 6 to 11. There were no adverse events identified and no cases of intraocular inflammation or vasculitis.

Conclusion: The pivotal trials of Faricimab in nAMD were non inferiority trials that provided treatment outcomes in treatment naïve patients. Our real world evaluation in patients with recalcitrant disease supports favourable outcomes and evidence of extended duration that could improve capacity constraints in provider units and reduce the treatment burden for care givers and patients. Limitations in the study include the small sample size and the potential for physician bias.

Conflicts of interest:

Vasuki Sivagnanavel has received educational support to international conferences and speaker fees at educational events from Roche and Bayer and Optos imaging.

Funding: None

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P151

RELATIONSHIP BETWEEN SOCIAL DEPRIVATION AND DRY AGE-RELATED MACULAR DEGENERATION (AMD) RELATED QUALITY OF LIFERyan Teo¹, Tomas Cudrnak¹¹Royal Eye Infirmary, Plymouth, UK

Introduction: As it is generally accepted that Dry age-related macular degeneration (AMD) is incurable, it is critical to maintain or improve the QOL of afflicted patients.

The purpose of this project is to investigate the relationship between social deprivation and its impact on AMD-related QOL and advance the Macular Society's aim of Dry AMD management, which is to maintain QOL in affected patients.

Another purpose of the project was to investigate the impact of the Age-Related Eye Disease Study (AREDS/AREDS2) vitamins formula on the QOL of affected patients.

Method: The 25-item National Eye Institute Visual Functioning Questionnaire (VFQ-25) was administered in an interview format, and results correlated with the Index of Multiple Deprivation (IMD), by postcode.

Statistical Analysis was carried out using IBM SPSS Statistics.

Results: 250 Dry AMD patients, comprising 128 males and 122 females, undertook the questionnaire.

The mean age was 73 years and all patients were of Caucasian race.

Those with lower SIMD scores (more socially deprived) reported lower VFQ-25 scores (lower quality of life).

Statistical analysis established that Pearson correlation coefficient $r = 0.385$ with $p\text{-value} = 0.012$

132 out of the 250 patients in the study were taking AREDS/AREDS2 vitamins formula and 96.8% reported VFQ-25 scores above the mean.

Conclusion: Greater social deprivation is known to lead to poorer vision and presentation of Dry AMD in its later stages.

However, this study has now established that greater social deprivation leads to a lower AMD-related QOL, which encompasses the physical, psychological, emotional and social well-being of the patient.

This is extremely useful as more should be done to help socially deprived Dry AMD patients in terms of support and education.

In addition, this study has also demonstrated a possible correlation between the consumption of AREDS/AREDS2 vitamins formula and higher VFQ-25 scores.

Conflicts of interest: None

Funding: None

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P152

RESULTS OF SWITCHES FROM RANIBIZUMAB TO AFLIBERCEPT FOR MACULAR OEDEMA SECONDARY TO RETINAL VEIN OCCLUSIONS

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Introduction: Retinal vein occlusions (RVOs) are the second most prevalent retinal vascular disease after diabetic retinopathy, and macular oedema is the leading cause of vision loss. Anti-VEGF intravitreal injections and steroid implants are the recommended treatment for macular oedema, with anti-VEGF therapy preferred in many cases due to its safety profile. There are currently three antiVEGF agents licensed for the treatment of RVOs: ranibizumab (Lucentis®, Ongavia®), aflibercept (Eylea®) and faricimab (Vabysmo®).

The expiry of the patent of Lucentis lead to the introduction of the first biosimilars of ranibizumab in the UK market. Due to the cost benefits of the use of biosimilars, many commissioning bodies have favoured the use of ranibizumab as first line treatment for retinal diseases.

There is published evidence showing that patients using aflibercept or faricimab require fewer injections, compared to ranibizumab, to control their disease in neovascular Age Related Macular Degeneration or Diabetic Macular Oedema. Due to the posology used in RVOs, the evidence supporting fewer injections with these drugs is not as conclusive. However, the LEAVO trial showed that patients on aflibercept required 1.8 injections less compared to ranibizumab at 100 weeks of treatment.

The purpose of this study is to evaluate the effects of switching to aflibercept in patients who were having treatment with ranibizumab for macular oedema secondary to RVOs.

Method: This was a retrospective analysis of our population of patients treated for RVOs who were switched from ranibizumab to aflibercept. Ranibizumab was started as first line treatment in these patients and both ranibizumab and aflibercept were prescribed as per SmPC.

There were two reasons for the switch of treatment: suboptimal response to ranibizumab (defined as persistent activity on OCT despite monthly injections) or high injection burden (controlled activity on ranibizumab, but requiring more frequent than 8-weekly injections).

In each group, we documented visual acuity (VA) and central retinal thickness (CRT) at diagnosis, at the time of switching and at the last follow-up appointment, as well as the longest interval prescribed in a treat and extend approach, being dry on OCT. The presence of macular fluid at each time interval was also analysed.

Results: A total of 36 eyes of 36 patients were included in this study. 44.4% with central retinal vein occlusion (CRVO), 16.7% of hemiretinal vein occlusion (HRVO) and 38.9% of branch retinal vein occlusion (BRVO). The reason for switching was suboptimal response in 47.2% of the cases and high injection burden in 52.8%.

In patients with suboptimal response to ranibizumab, 88.2% achieved resolution of macular oedema when switched to aflibercept. There was an average gain of vision of 12.36 ETDRS letters and a reduction of CRT of 131.5 μm . The difference between the CRT at the time of switch and at the end of follow up was clinically significant ($p < 0.001$).

In patients with high injection burden while using ranibizumab, 78.9% achieved an interval extension when switched to aflibercept maintaining VA and CRT stable. The median interval between injections of ranibizumab before the switch was 4 weeks (4–8) and after the switch to aflibercept was 8 weeks (5–13). This was also statistically significant ($p < 0.001$).

Conclusion: Patients with RVOs showing persistent macular oedema despite monthly ranibizumab injections had an anatomical improvement when switched to aflibercept and most achieved resolution of oedema.

The vast majority of patients switched from ranibizumab to aflibercept could have their interval between injections extended and therefore will require less injections per year to maintain their vision.

The use of aflibercept for the treatment of RVOs can help reducing the capacity pressures that many retinal units are facing.

Conflicts of interest: None.

Funding: None.

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P153

RISK OF CATARACT OR RAISED INTRAOCULAR PRESSURE IN PATIENTS TREATED WITH DEXAMETHASONE INTRAVITREAL IMPLANTS: RESULTS FROM A RETROSPECTIVE UK DATABASE STUDY

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Introduction: The association between the use of intravitreal implant of dexamethasone (DEX-I) and an increased risk of cataract, raised intraocular pressure (IOP) or glaucoma is well documented. However, data derived from clinical trial experience do not necessarily reflect the true incidence of adverse events in a broader population. We report on the results of an analysis from UK clinical database, drawing data from a population of patients receiving DEX-I for the treatment of diabetic macular oedema (DME), retinal vein occlusion with macular oedema (RVO+ME) or uveitis.

Method: Anonymised data was extracted using electronic patient records from six UK eye clinics using either Medisoft or Medisight software systems. Adult patients who had received one or more DEX-I implant for any indication over the period 2011–2024 qualified for the study. An efficacy analysis exploring the impact of DEX-I implantation on visual outcomes was carried out and will be separately reported. For the safety analysis presented here, data regarding cataract and raised IOP or glaucoma were identified for each patient and association was determined by analyses of the time from DEX-I implantation to first occurrence of each safety event and the number of DEX-I injections administered.

Results: A total of 4394 DEX-I treated eyes were identified in 3,942 patients, with a median follow-up of 5.88 years from the time of the first implant. Mean number of implants per eye was 2.21; range 1–26. The underlying diagnoses were DME (48%), RVO+ME (48%) and uveitis (4%).

Out of 1865 eyes that were phakic at the time of the first injection, a new cataract diagnosis was recorded in 411(22%) eyes. 72 (3.8%) of these occurred more than one year after the first implant and less than two years after the last implant and were thus considered attributable to DEX-I. Mean age at diagnosis was 67.3 years. Although the risk of cataract tended to rise in association with the number of implants administered, this trend was not statistically significant in this study (Chi squared = 16.2; $p = 0.09$).

656 (14.9%) out of 4394 DEX-I-treated eyes were identified as having an IOP > 30 mmHg, a new diagnosis of glaucoma, or received specific treatment for glaucoma up to 180 days following the last injection and these events were thus considered attributable to DEX-I.

Of the DEX-I treated eyes, 12% received IOP-lowering drops for elevated IOP and 0.4% required surgical or laser intervention. These rates were lower than those reported in pivotal clinical trials, where 23–41% of treated eyes required IOP lowering drops and up to 1.5% required surgery or laser. There was a significant association between the number of injections and the risk of a new OHT/glaucoma diagnosis, ranging from 8.7% of eyes with 1 DEX-I implant up to 38.0% of eyes that had received ≥ 10 implants (Chi squared = 215.4; $p < 0.0001$).

Conclusion: In a large retrospective analysis of real-world evidence, use of DEX-I was shown to be associated with a lower risk of both new cataract diagnosis and increased intraocular pressure compared with the event rates observed in published clinical trials.

Whilst IOP data was available for 4394 (81%) of patients, cataract may have been underreported due to various factors including the growth of Independent Treatment Centres.

The risk of a new OHT/glaucoma diagnosis was associated with the number of injections received, but 97.3% of patients were adequately controlled without requiring laser or surgical intervention. These findings suggest potential differences between real-world safety outcomes for DEX-I, and clinical trial results, that may inform patient management strategies.

Conflicts of interest: Abbvie, Roche, Alimera Sciences, Bayer, Roche

Funding:

Medical writing support

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P154

ROLE OF INFLAMMATORY BIOMARKERS AND OXIDATIVE STRESS IN DIABETIC RETINOPATHY AND EXPLORING THE THERAPEUTIC POTENTIAL OF RUTIN

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Introduction: Diabetic retinopathy (DR) is an ocular microvascular complication of prolonged diabetes mellitus that may initially remain asymptomatic but become vision-threatening if left untreated. Although several retinal and blood biomarkers have been indicated in DR, the exact interplay among these factors remains far from elucidation. Oxidative stress seems to play a key role in the progression of DR. Correlating these parameters with DR stages and assessing oxidative stress's influence on their levels is crucial for understanding the disease mechanism and developing targeted therapeutic strategies.

The primary objective of this study was to evaluate and compare specific proinflammatory biomarkers and oxidative stress along with haematological and biochemical parameters, across three groups: diabetic retinopathy (DR), diabetic without retinopathy (D), and nondiabetic controls (ND) to address gaps in the current understanding of their role in DR progression, and their potential utility in early diagnosis and improved prognostic outcomes. Furthermore, we investigated rutin as a potential therapeutic target for intervention at various stages of DR progression, using Swiss albino mice as a model.

Method: A total of 80 subjects (N = 80) were recruited and categorized into three groups: DR, D and ND. The study was conducted following institutional ethical approval (Certificate numbers: CUIEC/03/15/2022–23 dated 08.02.2023). Serum and peripheral blood mononuclear cells (PBMCs) were isolated from whole blood samples. Pro-inflammatory biomarkers and the expression profiles of selected target genes were evaluated using high-throughput techniques, including ELISA, RT-PCR, and immunoblotting. In parallel, haematological and biochemical parameters were assessed. Oxidative stress and antioxidant profiles were quantified using flow cytometry and spectrophotometric methods to investigate potential causal relationships with altered biomarker profiles. Along with the assessment of relevant biomarkers, rutin was administered to diabetic mice at standardized dose. Behavioural assessments including the Open Field Test (OFT), Elevated Plus Maze (EPM), Von Frey test, Hot Plate Latency test and Tail Flick test were conducted both pre- and post-treatment to evaluate the efficacy of rutin. Biochemical analyses were also performed to assess glucose levels, lipid profiles and liver function, providing a comprehensive evaluation of the therapeutic potential of rutin in diabetic mice.

Results: Significant alterations ($p < 0.05$) were observed in physiological parameters, inflammatory biomarkers and oxidative stress profiles between control and disease groups. DR subjects exhibited increased levels of cellular ROS, mitochondrial ROS, and lipid peroxidation (TBARS) in PBMCs, alongside reduced activities of antioxidant enzymes SOD and catalase. Serum levels of glutamate, serotonin and dopamine were elevated, whereas

GABA levels were reduced in DR subjects compared to the other groups. A notable upregulation of proinflammatory cytokines (TNF- α , IL-6, IL-1 β , VEGF, IFN- γ), cell adhesion molecules (ICAM1, VCAM-1, integrin α v β 3), and the transcription factor NF- κ B were observed in DR.

Rutin administration significantly reduced oxidative stress, improved behavioural outcomes, and lowered glucose levels in streptozotocin-induced diabetic mice. Behavioural improvements were validated through the OFT and Elevated Plus Maze (EPM). Certain cytokine levels and inflammatory biomarkers were also reduced in rutin-treated diabetic mice compared to diabetic control.

Conclusion: This study highlights the potential of inflammatory biomarker profiles in assessing DR progression and establishes a causal relationship between oxidative stress and circulating biomarker levels. Rutin shows promise as a therapeutic candidate for DR, warranting further investigation into its clinical applications.

Conflicts of interest: None

Funding: None

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P155

SAFETY AND EFFICACY OF INTRAVITREAL TISSUE PLASMINOGEN ACTIVATOR AND PNEUMATIC DISPLACEMENT OF SUBMACULAR HAEMORRHAGE: A SYSTEMATIC REVIEW AND META-ANALYSIS

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Introduction: Submacular haemorrhage (SMH) is a significant complication in various retinal disorders, including age-related macular degeneration (AMD), idiopathic polypoidal choroidal vasculopathy, retinal arterial macroaneurysm, pathologic myopia, trauma, and choroidal neovascularization. {Gibran, 2009}{Nayak, 2015}{Thuruthumaly, 2014}{Moriyama, 2011}{Nakamura, 2008} It can be also a complication of ocular surgeries, including vitrectomy and scleral buckling procedures. {Hochman, 1997}. The presence of SMH in the subfoveal especially when it's >4 disc diameter can lead to sudden visual loss and often results in a poor visual prognosis if left untreated. It can lead to retinal pigment epithelium (RPE) and outer retinal degeneration. {Cho, 2016} Given the profound impact of SMH on visual acuity and quality of life, effective management strategies are essential for optimizing patient outcomes and preserving vision.

Traditionally, the management of SMH commonly involves pars plana vitrectomy and internal drainage with or without the aid of recombinant tissue plasminogen activator (tPA). This approach has yielded improvement in postoperative visual acuity in non-ARMD cases. {Hayasaka, 1990}{Dellaporta, 1994} However, the value of this procedure for patients with ARMD remains controversial. {Ibanez, 1995}{Kamei, 1996}{Lewis, 1994}{Lim, 1995}

Recent advancements in the management of SMH include the minimally invasive technique introduced by Heriot in 1996, which

involves intravitreal injection of recombinant tPA combined with pneumatic displacement of SMH. {Heriot, 1997} Tissue plasminogen activator, a serine protease that facilitates the breakdown of fibrin clots, has been employed to address the persistent submacular blood by inducing its resorption and subsequent displacement. {Kamei, 2009} When combined with pneumatic displacement, which involves the injection of expansile gases as SF₆ into the vitreous cavity, this approach aims to physically displace the SMH from the macula, thereby improving the likelihood of visual recovery. {Kung, 2010}{Araujo, 2016}{Hattenbach, 2001} This approach is favoured for its convenience and has demonstrated favourable initial outcomes. Nevertheless, intravitreal tPA is associated with potential risks, including retinal detachment, endophthalmitis, cataract development, cystoid macular oedema, choroidal neovascularization, and pupillary block secondary to gas tamponade. {Hillenkamp, 2010}{Schulze, 2002}{Shields, 2019}

To date, there has been no comprehensive study assessing the combined use of intravitreal tPA and pneumatic displacement for treating SMH. Consequently, this systematic review and meta-analysis aims to evaluate the safety and efficacy of these interventions in managing SMH.

Method: A pre- and post-intervention systematic review and single-arm meta-analysis was conducted in line with guidelines from the Cochrane Collaboration and the Preferred Reporting Items for Systematic Reviews and Meta-analyses (PRISMA) 2020 Consensus Statement. Three electronic databases and the grey literature were searched for any studies evaluating intravitreal tPA and pneumatic displacement in the setting of submacular haemorrhage. The primary outcome was the corrected distance visual acuity (CDVA). Secondary outcomes included the incidence of adverse events such as vitreous haemorrhage, retinal detachment, retinal pigment epithelial (RPE) tears and endophthalmitis.

Results: Thirteen observational studies, totalling 502 eyes, were eligible. Post-operatively, there was a statistically significant improvement in CDVA from a mean value of 1.180 logMAR preoperatively to 0.788 logMAR following tPA and pneumatic displacement (SMD: 0.707, 95% CI: -1.036 to -0.379, P < 0.05). The incidence of adverse events with the use of tPA was low, including vitreous haemorrhage (Events rate: 0.112, 95% CI: 0.078 to 0.157), retinal detachment (Events rate: 0.057, 95% CI: 0.029 to 0.109), RPE tears (Events rate: 0.075, 95% CI: 0.022 to 0.227) and endophthalmitis (Events rate: 0.035, 95% CI: 0.010 to 0.114).

Conclusion: The use of intravitreal tPA and pneumatic displacement in patients with submacular haemorrhage is safe and effective, resulting in significant improvements in visual outcomes and complications.

Conflicts of interest: None

Funding: None

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P156. Abstract not included

P157

SPECTRUM: EARLY RESULTS FROM THE FIRST GLOBAL REAL-WORLD STUDY OF AFLIBERCEPT 8 MG IN PATIENTS WITH PRETREATED DIABETIC MACULAR OEDEMA

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Introduction: SPECTRUM is a global Phase 4 study to evaluate the real-world effectiveness and safety of aflibercept 8 mg in patients with diabetic macular oedema (DMO) and neovascular age-related macular degeneration (nAMD). This study will generate valuable data to build on the findings of the PULSAR (Phase 3) and PHOTON (Phase 2/3) clinical trials. Here, we will describe early clinical experience among patients with previously treated DMO.

Method: SPECTRUM (NCT06075147) is an ongoing, 24-month, prospective observational study being conducted across 18 countries. Treatment-naïve and previously treated patients with nAMD aged ≥ 50 years or with DMO aged ≥ 18 years, who have been prescribed aflibercept 8 mg by their attending physician, are eligible for enrolment. Decisions regarding monitoring, retreatment, and treatment schedules are made by each patient's physician in accordance with local clinical practice. All results are analysed descriptively.

Results: Patients aged ≥ 18 years are eligible for enrolment in the previously treated DMO cohort if they have been prescribed treatment with aflibercept 8 mg by their physician and received prior treatment such as steroids or other anti-vascular endothelial growth factor therapies up to prespecified timepoints before study start. Data are being collected from medical records and imaging during routine visits from February 2024 to September 2027 for a period of up to 24 months per patient. Overall, 239 patients have been enrolled to date in the previously treated DMO cohort in Germany, Canada, Switzerland, the UK, the Netherlands, and Japan. The longest treatment duration thus far is 10 months. The primary endpoint is change in best-corrected visual acuity (BCVA) from baseline to Month 12. Secondary endpoints include the mean change in BCVA and central retinal thickness from baseline to 6 months. Adverse events are being monitored throughout the study.

Conclusion: This pooled global cohort in SPECTRUM will provide early insights into real-world clinical experience with aflibercept 8 mg in patients with previously treated DMO.

Conflicts of interest:

AA: Honoraria: Bayer and Roche.

TD: Honoraria: Bayer.

MRM: Consulting fees: AbbVie, Allergan, Apellis, Aviceda Therapeutics, Bayer, Boehringer-Ingelheim, Dandelion, Eyepoint, Gensight, Iveric Bio, Isarna Therapeutics, Kubota, Lumithera, Novartis, Ocuteira, Oculis, Ocular Therapeutix, RetinAI, Roche, and Zeiss.

CL: Honoraria: Apellis, Bayer, Biogen, and Novartis.

CB: Honoraria: Alimera Sciences, Apellis, Bayer, and Roche; advisory boards: Apellis, Bayer, Boehringer Ingelheim, Janssen, and Roche.

HO: Consultant: AbbVie, Bayer, Novartis, and Roche.

TM: Employee of Bayer AG.

HA and PM-W: Employees of Bayer Consumer Care AG.

VC: Grants: Bayer, Novartis, Roche; advisory boards: Alcon, Roche, Bayer, Novartis, Apellis, and Boehringer Ingelheim.

Funding: SPECTRUM was sponsored by Bayer Consumer Care AG (Basel, Switzerland). The sponsor participated in the study design and conduct, data analysis, and abstract preparation. Medical writing support, under the direction of the authors, was provided by ApotheCom and funded by Bayer Consumer Care AG, Basel, Switzerland, in accordance with Good Publication Practice.

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P158

SPECTRUM: EARLY RESULTS FROM THE FIRST GLOBAL REAL-WORLD STUDY OF AFLIBERCEPT 8MG IN PATIENTS WITH PRETREATED NEOVASCULAR AGE-RELATED MACULAR DEGENERATION

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Introduction: Regulatory approval in 2024 of the aflibercept 8 mg formulation in patients with neovascular age-related macular degeneration (nAMD) and diabetic macular oedema (DMO) was based on the PULSAR (Phase 3) and PHOTON (Phase 2/3) clinical trials. SPECTRUM is the first global Phase 4 study to collect real-world data on the effectiveness and safety of aflibercept 8 mg in the treatment of nAMD and DMO. Here, we will present early clinical experience in patients with previously treated nAMD.

Method: SPECTRUM (NCT06075147) is an ongoing 24-month, prospective observational study at multiple sites across North America, Europe, the Middle East, and Asia Pacific. Treatment-naïve and previously treated patients with nAMD aged ≥ 50 years or with DMO aged ≥ 18 years, who have been prescribed aflibercept 8 mg by their attending physician, are eligible for enrolment. All treatment decisions are made by each patient's attending physician in accordance with local clinical practice. All data are analysed descriptively.

Results: For the previously treated nAMD cohort, patients aged ≥ 50 years with nAMD are eligible for enrolment if they have been prescribed treatment with aflibercept 8 mg by their physician, and if they have received prior treatment including other anti-VEGF therapies up to prespecified timepoints before study start. Data collection from medical records and imaging during routine visits is taking place from February 2024 to September 2027 for a period of up to 24 months per patient. To date, 392 patients have been enrolled in the previously treated nAMD cohort in Germany, Sweden, Switzerland, the UK, the Netherlands, Japan, and Australia, with 10 months being the longest period of treatment so far. The primary endpoint is change in best-corrected visual acuity (BCVA) from baseline to Month 12, and secondary

endpoints include the change in BCVA and central retinal thickness from baseline to Month 6, and the number of injections, number of visits, and number of adverse events from baseline to Month 6.

Conclusion: The global SPECTRUM study is the largest real-world study on the aflibercept 8 mg formulation to date. SPECTRUM will provide valuable insights into effectiveness and safety in a diverse cohort of patients with previously treated nAMD.

Conflicts of interest:

CB: Honoraria: Alimera Sciences, Apellis, Bayer, Roche; advisory boards: Apellis, Bayer, Boehringer Ingelheim, Janssen, Roche.

CL: Honoraria: Apellis, Bayer, Biogen, Novartis.

VC: Grants: Bayer, Novartis, Roche; advisory boards: Alcon, Roche, Bayer, Novartis, Apellis, Boehringer Ingelheim.

PL: Consultant: Aerie Pharmaceuticals, Allergan, Annexon, Apellis, Bausch + Lomb, Bayer, Biogen, Boehringer Ingelheim, Eyepoint Pharmaceuticals, Genentech, I-Care, Novartis, Ocular Therapeutix, Outlook Therapeutics, Roche.

HO: Consultant: AbbVie, Bayer, Novartis, Roche.

MK and TM: Employees: Bayer AG.

HA, XZ and ZH: Employees: Bayer Consumer Care AG.

MRM: Consultant: AbbVie, Allergan, Apellis, Aviceda Therapeutics, Bayer, Boehringer Ingelheim, Dandelion, Eyepoint, Gensight, Iveric Bio, Isarna Therapeutics, Kubota, Lumithera, Novartis, Ocuteira, Oculis, Ocular Therapeutix, RetinAI, Roche, Zeiss.

Funding:

SPECTRUM was sponsored by Bayer Consumer Care AG (Basel, Switzerland). The sponsor participated in the study design and conduct, data analysis, and abstract preparation. Medical writing support, under the direction of the authors, was provided by ApotheCom and funded by Bayer Consumer Care AG, Basel, Switzerland, in accordance with Good Publication Practice.

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P159

SPECTRUM: EARLY RESULTS FROM THE FIRST GLOBAL REAL-WORLD STUDY OF AFLIBERCEPT 8 MG IN PATIENTS WITH TREATMENT-NAÏVE DIABETIC MACULAR OEDEMA

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Introduction: The PULSAR (Phase 3) and PHOTON (Phase 2/3) clinical trials led to the approval of the aflibercept 8 mg formulation in patients with diabetic macular oedema (DMO) and neovascular age-related macular degeneration (nAMD).

SPECTRUM is the first global study to assess the real-world effectiveness and safety of aflibercept 8 mg for these indications. In this interim analysis, we will describe early experience among patients with treatment-naïve DMO.

Method: SPECTRUM (NCT06075147) is an ongoing, 24-month, prospective observational study being conducted across 18 countries in North America, Europe, the Middle East, and Asia Pacific. Treatment-naïve and previously treated patients with nAMD aged ≥50 years or with DMO aged ≥18 years, who have been prescribed aflibercept 8 mg by their attending physician, are eligible for enrolment. Decisions regarding monitoring, retreatment, and treatment schedules are made by each patient's attending physician in accordance with local clinical practice. All results are analysed descriptively.

Results: Patients aged ≥18 years are eligible for enrolment in the treatment-naïve DMO cohort if they have never received any medical treatment for DMO and have been prescribed treatment with aflibercept 8 mg by their physician. Data are being collected from medical records and imaging during routine visits from February 2024 to September 2027 for a period of up to 24 months per patient. To date, 97 patients have been enrolled in the treatment-naïve DMO cohort in Germany, the UK, Japan, and Spain. The longest treatment duration thus far is 10 months. The primary endpoint is change in best-corrected visual acuity (BCVA) from baseline to Month 12, and secondary endpoints include the mean change in BCVA and central retinal thickness from baseline to 6 months. Adverse events are also being monitored.

Conclusion: SPECTRUM is the first global real-world study to evaluate the effectiveness and safety of aflibercept 8 mg, and this global cohort of patients with treatment-naïve DMO will provide early insights into outcomes in routine clinical practice.

Conflicts of interest:

AL: Advisory boards: GSK, Nordic Pharma, Bayer, and Roche; honoraria: Bayer and Roche.

HO: Consultant: AbbVie, Bayer, Novartis, and Roche.

CB: Honoraria: Alimera Sciences, Apellis, Bayer, and Roche; advisory boards: Apellis, Bayer, Boehringer Ingelheim, Janssen, and Roche.

PL: Consultant: Aerie Pharmaceuticals, Allergan, Annexon, Apellis, Bausch + Lomb, Bayer, Biogen, Boehringer Ingelheim, Eyepoint Pharmaceuticals, Genentech, I-Care, Novartis, Ocular Therapeutix, Outlook Therapeutics, and Roche.

VC: Grants: Bayer, Novartis, Roche; advisory boards: Alcon, Roche, Bayer, Novartis, Apellis, and Boehringer Ingelheim.

MRM: Consulting fees: AbbVie, Allergan, Apellis, Aviceda Therapeutics, Bayer, Boehringer-Ingelheim, Dandelion, Eyepoint, Gensight, Iveric Bio, Isarna Therapeutics, Kubota, Lumithera, Novartis, Ocuteira, Oculis, Ocular Therapeutix, RetinAI, Roche, and Zeiss.

TM: Employee of Bayer AG.

HA and PM-W: Employees of Bayer Consumer Care AG.

CL: Honoraria: Apellis, Bayer, Biogen, and Novartis.

Funding:

SPECTRUM was sponsored by Bayer Consumer Care AG (Basel, Switzerland). The sponsor participated in the study design and conduct, data analysis, and abstract preparation. Medical writing support, under the direction of the authors, was provided by ApotheCom and funded by Bayer Consumer Care AG, Basel, Switzerland, in accordance with Good Publication Practice.

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P160

SPECTRUM: EARLY RESULTS FROM THE FIRST GLOBAL REAL-WORLD STUDY OF AFLIBERCEPT 8 MG IN PATIENTS WITH

TREATMENT-NAÏVE NEOVASCULAR AGE-RELATED MACULAR DEGENERATION

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Introduction: PULSAR (Phase 3) and PHOTON (Phase 2/3) clinical trials were key in the approval of aflibercept 8 mg for the indications of neovascular age-related macular degeneration (nAMD) and diabetic macular oedema (DMO). SPECTRUM is the largest and first global real-world evidence (RWE) study to date to evaluate the effectiveness and safety of aflibercept 8 mg in the treatment of nAMD and DMO. Early clinical outcomes in patients with treatment-naïve nAMD will be described here.

Method: SPECTRUM (NCT06075147) is an ongoing 24-month, Phase 4, prospective observational study, which is being conducted at multiple sites across 18 countries. Treatment-naïve and previously treated patients with nAMD aged ≥50 years or with DMO aged ≥18 years are eligible for enrolment if they have been prescribed treatment with aflibercept 8 mg by their attending physician. Monitoring, retreatment, and treatment schedule decisions are made by each patient's attending physician in accordance with local clinical practice. All results are analysed descriptively.

Results: In the treatment-naïve nAMD cohort, patients aged ≥50 years are eligible for enrolment if they have never been exposed to any medical treatment for nAMD and have been prescribed treatment with aflibercept 8 mg by their physician. Data are being collected over a period of up to 24 months for each patient from medical records and imaging during routine visits from February 2024 to September 2027. To date, 320 patients have been enrolled in the treatment-naïve nAMD cohort in Germany, Denmark, Sweden, Switzerland, the UK, Japan, Australia, and Spain, with 10 months being the longest treatment duration so far. The primary endpoint is change in best-corrected visual acuity (BCVA) from baseline to Month 12, and secondary endpoints include mean change in BCVA and central retinal thickness from baseline to Month 6. Adverse events are also being monitored throughout the study.

Conclusion: SPECTRUM is the first global RWE study of the aflibercept 8 mg formulation, and this early experience will provide initial insights into its real-world effectiveness and safety in patients with treatment-naïve nAMD.

Conflicts of interest:

MG: Honoraria: Roche.

VK: Honoraria: AbbVie, Bayer, Novartis, and Roche.

CL: Honoraria: Appellis, Bayer, Biogen, and Novartis.

MRM: Consulting fees: AbbVie, Allergan, Apellis, Aviceda Therapeutics, Bayer,

Boehringer-Ingelheim, Dandelion, Eyepoint, Gensight, Iveric Bio, Isarna Therapeutics, Kubota, Lumithera, Novartis, Ocuterra, Oculis, Ocular Therapeutix, RetinAI, Roche, and Zeiss.

PL: Consultant: Aerie Pharmaceuticals, Allergan, Annexion, Apellis, Bausch + Lomb, Bayer, Biogen, Boehringer Ingelheim, Eyepoint Pharmaceuticals, Genentech, I-Care, Novartis, Ocular Therapeutix, Outlook Therapeutics, and Roche.

VC: Grants: Bayer, Novartis, Roche; advisory boards: Alcon, Roche, Bayer, Novartis, Appellis, and Boehringer Ingelheim.

HO: Consultant: AbbVie, Bayer, Novartis, and Roche.

TM: Employee of Bayer AG.

HA and XZ: Employees of Bayer Consumer Care AG.

CB: Honoraria: Alimera Sciences, Apellis, Bayer, and Roche; advisory boards: Apellis, Bayer, Boehringer Ingelheim, Janssen, and Roche.

Funding:

SPECTRUM was sponsored by Bayer Consumer Care AG (Basel, Switzerland). The sponsor participated in the study design and conduct, data analysis, and abstract preparation. Medical writing support, under the direction of the authors, was provided by ApotheCom and funded by Bayer Consumer Care AG, Basel, Switzerland, in accordance with Good Publication Practice.

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P161

STRUCTURAL AND FUNCTIONAL ALTERATIONS IN RED BLOOD CELLS ASSOCIATED WITH DIABETIC RETINOPATHY IN TYPE 2 DIABETES

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Introduction: Diabetic retinopathy (DR) is the most common microvascular complication (MVC) of type 2 diabetes mellitus (T2DM) and a leading cause of visual impairment among working-age adults. Chronic hyperglycaemia in T2DM alters the structure and function of red blood cells (RBCs), contributing to various MVCs. However, there is limited evidence of changes in RBC membrane structure, osmotic fragility, and oxygen transport capacity, along with alterations in the activity of antioxidant enzymes, which may lead to tissue hypoxia and accelerate the pathogenesis of DR. Thus, the present study aimed to determine if structural and functional alterations in RBCs are associated with DR in T2DM.

Method: This cross-sectional study included 132 randomly selected T2DM subjects: 62 without DR (DNR) and 70 age- and gender-matched subjects with different stages of DR. Five millilitres of venous blood were collected from each subject. RBCs were isolated from the whole blood by centrifugation and prepared for scanning electron microscopy (SEM) by fixing the sample in buffered glutaraldehyde. The surface roughness of RBCs was assessed from SEM images using 'ImageJ' software, where roughness average (RA), root mean square roughness (RQ), skewness (Rsk), and kurtosis (Rku) of the roughness profile were analysed to evaluate surface characteristics. The osmotic fragility of RBCs was measured using a series of hypotonic sodium chloride (NaCl) solutions, where RBCs were suspended,

centrifuged, and the supernatant was spectrophotometrically analysed to determine the extent of haemolysis. Glycated haemoglobin (HbA1c) and RBC indices were evaluated using the Cobas 6000 and

DxH 800 automated analysers. The activity of antioxidant enzymes in RBCs, such as glucose-6-phosphate dehydrogenase (G6PDH) and glutathione peroxidase (GPX), was measured through enzyme kinetic assays from hemolysates. The 2,3-diphosphoglycerate (2,3-DPG) content in RBCs, a key regulator of oxygen transport to tissues, was also determined from hemolysate by Enzyme-Linked Immunosorbent Assay (ELISA). Plasma samples, separated by blood centrifugation, were analysed for vascular endothelial growth factor A (VEGF-A), typically induced by tissue hypoxia, using the ELISA technique.

Results: The study revealed significantly increased RA, RQ, Rsk, and Rku in DR subjects compared to DNR individuals (all at $p < 0.0001$). RBC samples from DR subjects exhibited both starting ($p < 0.0001$) and complete ($p < 0.0001$) haemolysis at higher NaCl concentrations compared to DNRs. DR subjects had lower RBC counts ($p = 0.031$) and haemoglobin (Hb) levels ($p = 0.014$), but higher HbA1c levels ($p = 0.034$) than DNRs. The enzymatic activities of G6PDH ($p < 0.0001$) and GPX ($p < 0.0001$) were significantly reduced in DR subjects. Additionally,

DR subjects had substantially lower 2,3-DPG levels ($p < 0.0001$) and higher levels of VEGF-A ($p < 0.0001$) compared to DNRs. In DR, RA ($p = 0.031$), and NaCl concentrations at which RBCs started ($p < 0.001$) and completed ($p < 0.001$), haemolysis was positively correlated with VEGF-A, while 2,3-DPG ($p = 0.024$) was negatively correlated. G6PDH and GPX activities and VEGF-A levels were significant (all at $p < 0.001$ level) predictors of DR.

Conclusion: The study identified significant structural and functional changes in RBCs of DR subjects compared to DNRs, including increased surface roughness, osmotic fragility, and reduced antioxidant enzyme activity. These alterations, along with a decrease in 2,3-DPG content in RBCs, reduce the ability of Hb to release oxygen to tissues, potentially contributing to tissue hypoxia and the subsequent secretion of VEGF-A, which may accelerate the progression of DR.

Conflicts of interest: None

Funding: None

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P162

THE GUT MICROBIOTA IS ALTERED IN CRB1-RELATED RETINOPATHIES

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Introduction: The human *CRB1* gene encodes the CRB1 protein, primarily expressed in retinal Muller cells and photoreceptors, where it regulates essential processes such as apical-basal polarity, outer limiting membrane integrity, cellular signalling and adherence junctions. Dysfunction in CRB1 results in a range of retinal phenotypes although systemic implications are still unknown. While *Crb1* expression has been identified in the gastrointestinal (GI) system in rd8 mouse models (*Crb1*^{-/-}), where barrier dysfunction allowed translocation of bacteria into the retina, its role in human GI system remains unclear. Metabolomic

profiling may provide insights into disease pathophysiology and aid the identification of potential systemic biomarkers.

Method: A cross-sectional metabolomic analysis was conducted on blood plasma samples from 25 molecularly confirmed *CRB1*-retinopathy patients from Moorfields Eye Hospital and age- and gender-matched healthy controls.

Results: Of 872 identified compounds, 244 were significantly abnormal in *CRB1* patients (141 elevated, 103 reduced). Key findings included disrupted bile acid metabolism, with elevated primary and secondary bile acids alongside increased gut microbial phenylalanine pathway metabolites, indicative of altered enterohepatic circulation and potential gut microbiome dysbiosis. However, sucrose levels remained unchanged amongst groups, suggesting no gut barrier dysfunction. Significant reductions in key antioxidants and neuroprotective agents, such as α -tocopherol, and bilirubin, were detected. Finally, energy metabolism dysregulation, with elevated glucose, decreased lactate, and impaired fatty acid oxidation.

Conclusion: This metabolomic profile of *CRB1*-retinopathy uncovers systemic metabolic dysregulation. The observed deficits in key antioxidants and metabolic pathways suggest targeted supplementation that could potentially slow *CRB1* disease progression. These findings open avenues for therapeutic strategies aimed at metabolic restoration in *CRB1*-retinopathies.

Conflicts of interest: None

Funding:

Authors are grateful to the Wellcome Trust 205174/Z/16/Z, Fight for Sight, Moorfields Eye Charity and the NIHR Biomedical Research Centre at Moorfields Eye Hospital NHS Trust and UCL Institute of Ophthalmology.

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P163

THE REAL WORLD IMPLEMENTATION OF THE '2020 HYDROXYCHLOROQUINE RETINOPATHY SCREENING PROGRAMME' IN A LARGE TEACHING HOSPITAL - ARE WE GETTING IT RIGHT FOUR YEARS ON?

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Introduction: The Royal College of Ophthalmology introduced the latest Hydroxychloroquine and Chloroquine retinopathy screening guidance in 2020, streamlining testing to spectral-domain optical coherence tomography (SD-OCT) and widefield fundus autofluorescence imaging (FAF) only, and removed the need for baseline testing. Screening need not commence until 5 years of treatment, unless deemed high risk, and the use of virtual clinics or community screening is encouraged.

We looked at how this guidance had been implemented across our trust, and whether the guidance is being adhered to. Just like most trusts across the UK, medical retina has capacity issues, and this latest guidance provides a clear framework to reduce the number of patients needing screening, reduces the burden of tests they require each visit and effectively discourages face to face appointments. The guidance is an evidenced based road map to reducing the burden of HCQ screening, and thus to not stick to it could be viewed as an 'own goal'.

Method: We retrospectively reviewed the medisoft data of the last 30 patients seen within the trust that had hydroxychloroquine listed in their medications. We then went on to excluded those attending for anything but screening, 10 were excluded: 3

cataract, 2 had no referral and did not attend, 2 corneal, 1 ocuoplastics, 1 uveitis and 1 squint.

Of those attending for screening, we looked at whether the correct tests were performed, whether they met criteria to be screened and where the referral originated.

Results: Four patients had been taking HCQ for less than 5 years, and none of these had had risk factors stipulated in the guidance to suggest they should be screened earlier. Therefore 20% we inappropriately seen in clinic for screening.

Of the 20 we screened, 14 had both OCT and FAF, so only 70% had the correct testing. Of the 16 correctly in screening, 13 had both OCT and FAF, meaning 81% of those who meet screening criteria had the correct screening tests.

The referral sources were: 35% rheumatology, 30% GP, 10% trust eye A+E, 10% optometrist, 10% haematologist, and 5% dermatologists.

Just 13% were seen in a diagnostic / virtual clinic, so 87% were seen in a face to face medical retina clinic.

Conclusion: The adherence to the RCOphth 2020 HCQ guidance was suboptimal in our trust, in particular we were screening patients too early, not with the correct tests and nearly 90% in a face-to-face setting.

The issue of seeing patients too early is a combination of incorrect referrals, and incorrect triage. Education to both of these groups have been implemented and will need to be periodically reviewed.

Funnelling all HCQ patients into a diagnostic clinic, allows for a clear testing protocol, and thus full compliance with guidance. This also reduces the time burden on patients of face-to-face appointments and frees capacity in overburdened medical retina clinics.

We are surely not the only trust who could benefit from formally reviewing our implementation of the guidance as a mechanism to optimise patient care, with the secondary benefit of reducing the need for precious face-to-face medical retina clinic appointments.

Conflicts of interest: None

Funding: None

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P164

THE ROLE OF BODY MASS INDEX AND BLOOD PRESSURE IN PAEDIATRIC DIABETIC RETINOPATHY: A STUDY OF RISK AND PROGRESSION

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Introduction: Diabetic Retinopathy (DR) is the most common microvascular complication of diabetes mellitus (DM), leading to visual impairment and blindness. In the UK, annual screening for retinopathy is recommended for patients with type 1 diabetes (T1DM) from age 12. The National Paediatric Diabetes Audit of England and Wales (NPDA) 2021/2022 reported 11.4% of children and young people aged 12–24 years to have an abnormal retinopathy screening result. Mean HbA1c outcomes improved from 2020/2021 to 2021/2022 despite the challenges of Covid-19.

It is crucial to investigate the various risk factors associated with diabetes and their association with diabetic retinopathy, as this could help alleviate the progression of this condition. Recent studies have demonstrated an association between patient weight in the adult population and diabetic retinopathy progression, with increasing abdominal obesity associated with worsening diabetic retinopathy. Hypertension has also been shown to worsen the prognosis of diabetic retinopathy in adult population. However, there is still limited research in studying the association of weight, hypertension and diabetic retinopathy in the paediatric diabetic population.

The objective of this study is to examine the relationship between blood pressure and body mass index (BMI) and their impact on diabetic retinopathy progression within paediatric T1DM patients.

Method: The regional retinopathy surveillance database was reviewed and the retinopathy screening reports of paediatric T1DM patients stored from 2015 to present under the University Hospitals Birmingham Paediatric Service was accessed and analysed.

Patients between the age of 12 to 17 with a diagnosis of T1DM and enrolled in the national diabetic retinopathy screening programme by the 1st April 2024 are included.

Demographic information, result of both initial (post diagnosis) screening and recent screening retinopathy grades with contiguous blood pressure, BMI and HbA1c were collected and analysed using RStudio.

Results: Of 353 patients with T1DM, 138 patients (52.9% male, 47.1% female) fit the inclusion criteria, with a mean age of 16.1 ± 2.2 . Patients with R1 grade diabetic retinopathy in either one or both eyes increased from 0.029% to 0.123% from the initial screening to their most recent screening. Over the duration between the initial and the most recent screening, 2 patients observed an improvement in their diabetic retinopathy grade, 121 patients had no grade changes, 11 had worsening in one eye and 4 had worsening in both eyes.

BMI readings were the highest in the group of patients with worsening grade in one eye

(mean = 30.9 ± 16.8), followed by the patients with worsening grade in both eyes (mean = 29.2 ± 6.7), patients with no changes (mean = 22.4 ± 4.5) and patients with an improvement (mean = 19.4 ± 0.9). Systolic blood pressure readings were found to be the highest in the group of patients with worsening grade in both eyes (mean = 132 ± 16.2), followed by the patients with no changes (mean = 119 ± 12.7), improvement in grade (mean = 119 ± 2.8) and patients with worsening in one eye (mean = 116 ± 7.9).

A generalised linear model revealed BMI to be a significant ($p = 0.00471$) predictor of diabetic retinopathy grade progression, with a higher BMI being associated with the increased odds of worsening of the grade.

Conclusion: Overall, a higher BMI of approximately 29–31 in our study population appears to be associated with worsening diabetic retinopathy. The association between higher BMIs and worsening diabetic retinopathy is well-supported by the literature.

Assessing the impact of blood pressure and BMI revealed that BMI is a relatively more significant predictor of diabetic retinopathy progression in our paediatric population. Since BMI is a modifiable risk factor, it is crucial to develop strategies to maintain an appropriate BMI in this patient group. Future research should explore the impact of various factors, such as diet, physical activity, and lifestyle choices, on changes in diabetic retinopathy progression through alterations in BMI.

Conflicts of interest: None

Funding: None - local departmental funding from UHB

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P165

TOP 12 "WET MACULAR CONDITIONS" THAT DO NOT REQUIRE INTRAVITREAL ANTI VASCULAR ENDOTHELIAL GROWTH FACTOR (ANTI-VEGF) INJECTIONS

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Introduction: Intravitreal injections of anti-VEGF drugs have revolutionised the management of wet age related macular degeneration (ARM), retinal vein occlusion (RVO) and diabetic maculopathy over the last 18 years in the UK.

In the year 2022 over 677,000 intravitreal injections were performed in England alone (4796 per 100,000 population).

In the UK over 75% of intravitreal injections are performed by allied health professionals such as nurses, optometrists and orthoptists.

Although anti-VEGF injections are only NICE approved for 4 main conditions (wet ARM, retinal vein occlusion, diabetic maculopathy and myopic choroidal neovascular membrane(CNV))often clinicians are faced with clinical presentations that mimic the 4 main indications thereby making the clinical decision to treat or not treat a particular patient with anti-VEGF injections difficult.

Method: We have retrospectively reviewed the case records of patients attending rapid access clinics and intravitreal injection clinics over a 4 year period to identify macular and retinal conditions that mimic the four major NICE approved indications for intravitreal anti-VEGF injections.

Results: We identified 12 such clinical conditions:

- 1) Adult vitelliform macular degeneration
- 2) Peripapillary pachychoroid syndromes
- 3) Central serous retinopathy
- 4) Intraretinal fluid overlying the old disciform macular scars
- 5) Retinal pigment epithelial rips involving the fovea
- 6) Old BRVO/ CRVO with extrafoveal macular oedema/ thickening
- 7) Macular oedema/ haemorrhage secondary to a macroaneurysm
- 8) Peripapillary CNV without involvement of the macula
- 9) Macular oedema secondary to epiretinal membrane
- 10) Inactive CNV: slow growing lesion with minimal fluid leakage and stable vision.
- 11) Transient foveal oedema due to microaneurysms close to foveola
- 12) Macular oedema mistaken for vein occlusions and Diabetic maculopathy
 - a) Post operative cystoid macular oedema (CMO)
 - b) CMO in hereditary dystrophies
 - c) Drug (toxic) related macular oedema

Conclusion: We have identified the most common macular / retinal conditions for which anti-VEGF therapies are not necessarily of benefit.

In a large cohort of injection patients it is not unusual to find some patients with other varied pathologies receiving anti-VEGF injections with limited or no benefit.

We believe that it is important to highlight these varied pathologies to all clinicians to avoid inappropriate treatments.

We demonstrate each category of patients with evidence on multimodal imaging to support the diagnosis and also of suboptimal outcomes.

Conflicts of interest: None

Funding: None

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P166

VARIATIONS IN AGE OF ONSET AND LONG-TERM TREATMENT OUTCOMES IN MEN AND WOMEN WITH SORSBY'S DYSTROPHY

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Introduction: To highlight the variations in the age of onset, disease progression and final treatment outcomes of Sorsby dystrophy in men and women.

Method: Ten eyes of 5 patients presented to our centre over a period of 19 years (2005–2024) were included in this study.

All patients were symptomatic and presented with subfoveal choroidal neovascular membranes in at least one eye.

All patients had optical coherence tomography (OCT) scans, Fundus fluorescein angiography and genetic testing to confirm the diagnosis.

All were positive TIMP 3 missense mutation chromosome 22.

The treatment regime included oral steroids, photodynamic therapy(PDT) and intravitreal anti-vascular endothelial growth factor (veg) injections.

Results: Age: Average age at presentation was 34 years for females and 46 years for males.

Sex: 3 females and 2 males from 2 unrelated families.

Follow up period: Ranged from 3 to 19 years with average follow up period of 11.7 years.

Three eyes received Argon laser, 2 received PDT, one oral steroid, 6 eyes had intravitreal anti-Vegf injections and 3 eyes had combination treatments.

Argon laser treatment proved to be least effective with high rate of recurrences of choroidal neovascular membranes with subsequent loss of central vision. Anti-Vegf treatments were universally effective.

Final Visual acuity varied from Hand movements to 6/6.

Both male patients deteriorated rapidly, lost vision in both eyes and were registered blind within just 2 years of presentation. Whereas the three female patients on the contrary fared much better retaining useful vision for decades after their male counterparts lost their vision.

Conclusion: In spite of the common gene defect, the age of onset and final visual outcomes seem vary widely between men and women. These variations were not reported previously.

The disease manifested on average 12 years earlier in females than their male counterparts but the females achieved better visual outcomes. Females also required fewer intravitreal injections.

Males also suffered severe form of maculopathy and lost all useful vision within 2 years of onset of disease. Smokers did much worse, and treatments proved less effective.

These new findings would help to deliver the prognosis more accurately when counselling patients. Male members of the Sorsby family may also benefit from a closer monitoring.

Conflicts of interest: None

Funding: None

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P167

YOUNG PATIENTS WITH CHOROIDAL NEOVASCULAR MEMBRANE: A TERTIARY CENTRE EXPERIENCE OF TREATMENT WITH INTRAVITREAL ANTI-VASCULAR ENDOTHELIAL GROWTH FACTOR INJECTIONS

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Introduction: The treatment of choroidal neovascular membrane (CNVM) in patients under 50 years is challenging due to diverse causes and a lack of consensus on effective therapies. While anti-VEGF intravitreal injections (IVI) are the primary treatment for myopic CNVM, their effectiveness in younger patients with idiopathic or other forms of CNVM is not well established.

Method: We conducted one of the largest retrospective cohort studies to date involving 67 patients aged 18–45 years old who received IVI for CNVM at the Royal Victoria Infirmary in Newcastle-upon-Tyne. The Friedman test was utilized to assess changes in visual acuity (VA) from baseline to one year after treatment.

Results: The study included 67 patients (43 females) aged between 20–45 years old (mean 36, median 39, SD 7.7). A total of 433 injections were administered to 72 eyes, averaging 6 injections per eye (1–34, SD 7.7). Nearly all patients showed an increase in logMAR scores, with myopic CNVM patients demonstrating a significant improvement in VA ($z = -3.077$, $p = 0.002$). Conversely, patients with pseudoxanthoma elasticum, SFD, and Wiskott-Aldridge syndrome exhibited a decline in ETDRS scores, though none lost more than 15 letters.

Conclusion: This study highlights the varied pathology of CNVM in younger individuals, indicating a favourable response to anti-VEGF treatment in myopic cases, while idiopathic cases present greater challenges. Our findings support the safety and efficacy of IVI anti-VEGF therapy for CNVM across different underlying conditions in younger patients. Ongoing research is crucial to enhance tailored treatment approaches and improve visual outcomes in this age group.

Conflicts of interest: None

Funding: None

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MISCELLANEOUS

P168. Abstract not included

P169. Abstract not included

P170

BATTLE OF THE ALGORITHMS: A HEAD-TO-HEAD COMPARISON OF FOUR LEADING AI MODELS IN RETINAL DISEASE DETECTION

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Introduction: The global burden of retinal diseases represents a significant healthcare challenge, with conditions such as diabetic retinopathy, glaucoma, and age-related macular degeneration remaining the leading causes of preventable blindness worldwide. Early detection and intervention are crucial for preventing vision loss, yet traditional screening methods face substantial limitations in terms of accessibility, consistency, and scalability. This has created an urgent need for more efficient and accurate diagnostic tools.

The primary objective of this study was to evaluate and compare the diagnostic accuracy and reliability of four artificial intelligence (AI) language models (ChatGPT-4.0, Grok, Claude, and Microsoft Gemini) in diagnosing various retinal conditions through fundus image analysis across different pathologies and disease severities.

Method: A comprehensive evaluation assessed four AI models' performance across retinal conditions, including diabetic retinopathy ($n = 500$), central retinal artery occlusion ($n = 300$), age-related macular degeneration ($n = 250$), and glaucoma ($n = 500$). Images were sourced from reputable databases and underwent rigorous screening by two independent clinicians. Only high-quality images without artefacts or concurrent pathologies were included in the final dataset.

A standardised prompt was then applied consistently across all AI models, reading, "As an expert ophthalmologist, analyse this fundus photograph systematically following these steps: 1) Confirm image quality is suitable for interpretation; 2) Examine and describe the optic disc (margins, cup-to-disc ratio, colour), retinal vessels (calibre, crossing changes), macula (foveal reflex, lesions), and peripheral retina; 3) Identify any pathological features present; 4) Grade condition severity if applicable; 5) Provide a definitive diagnosis with certainty level. Document all findings in order of clinical significance."

Diagnostic metrics, including sensitivity, specificity, and accuracy, were calculated for each condition and model. Statistical analysis employed chi-square tests for categorical outcomes, t-tests for continuous variables, and McNemar's test for paired nominal data, enabling robust comparison between models. This methodical approach provided a reliable foundation for assessing the comparative efficacy of AI models in retinal disease detection.

Results: ChatGPT-4.0 consistently outperformed other models across all conditions, demonstrating superior diagnostic capabilities. For diabetic retinopathy, ChatGPT-4.0 achieved a sensitivity of 99.0% (95% CI: 0.984–0.993) with a corresponding specificity of 98.5% (95% CI: 0.979–0.989), significantly higher than Grok (95.5%), Claude (94.8%), and Microsoft Gemini (93.2%).

ChatGPT-4.0 maintained its superior performance in AMD detection with 99.0% sensitivity and 97.8% specificity. For macular oedema assessment, ChatGPT-4.0 achieved a sensitivity of 99.5% (95% CI: 0.991–0.997). Disease severity analysis revealed notable variations in performance across all models.

ChatGPT-4.0 accurately detected severe diabetic retinopathy (282/282 cases) and CRAO (300/300 cases). However, all models, including ChatGPT-4.0, showed significantly lower performance for mild diabetic retinopathy (0/102 cases). In glaucoma assessment, ChatGPT-4.0 demonstrated the highest accuracy in identifying advanced cases (467/500 cases, 93.4%), compared to Grok (85.6%), Claude (84.2%), and Microsoft Gemini (82.2%). Statistical analysis revealed significant performance differences between models ($p < 0.05$), with ChatGPT-4.0 consistently demonstrating superior accuracy across all conditions, particularly in moderate to severe cases.

Conclusion: This comparative analysis demonstrates ChatGPT-4.0's superior performance in retinal disease detection compared to other leading AI models. While all models showed promising capabilities, ChatGPT-4.0's consistently higher accuracy across multiple conditions suggests its potential as a preferred tool for retinal disease screening. However, the universal challenge of detecting mild cases across all models emphasises the need for continued technological advancement in early disease detection.

These findings provide valuable insights for the future development and implementation of AI in ophthalmological diagnostics.

Conflicts of interest: None

Funding: None

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P171

CERTIFICATION OF VISION IMPAIRMENT: ANALYSIS OF TRENDS WITHIN EYE CARE CENTRES IN WALES BETWEEN 2019 AND 2024

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Introduction: The Certification of Vision Impairment (CVI) enables people with visual impairments who meet certain criteria to be registered with their local authority social services as sight impaired (SI) or severely sight impaired (SSI). Registration ensures access to benefits and support services that aim to improve quality of life and independence. Close surveillance of certification rates aids identification of delays or lack of certification within ophthalmology services and provides useful epidemiological information to monitor preventable causes of sight loss. The purpose of this study was to analyse trends in new certifications in a single health board in Wales between 2019 and 2024 and to identify socioeconomic factors that impact certification rates.

Method: This was a retrospective analysis of data consisting of all CVIs completed between 1 February 2019 and 30 May 2024 within four hospitals in the Cwm Taf Morgannwg University Health Board, Wales. Anonymised details about each patient's age, gender, postcode of residence, date of completion of CVI form, diagnosis, and status of visual impairment (SI or SSI) were collated by an Eye Clinic Liaison Officer (ECLO) in an Excel sheet. Main causes of certification were identified. The impact of socioeconomic factors on visual loss was measured using the most recent version of the Welsh Index of Multiple Deprivation (WIMD) which assessed various domains including income, employment and health.

Results: A total of 668 CVIs were completed, consisting of 340 SI and 328 SSI certifications. The lowest number of new certifications recorded was in 2020. There was a 11% decline in new certifications in 2023 when compared to 2019. In the SI group, age-related macular degeneration (AMD) (55%) remains the leading cause of certification, followed by glaucoma (12%) and diabetic retinopathy/ maculopathy (5%). In the SSI group, the largest category was AMD (53%), followed by glaucoma (14%) and optic atrophy (6%). The mean age was 76.7 and 35% of patients fell into the 80–89 age range. The most deprived areas within Wales had the most number of new certifications (17%).

Conclusion: Decline in certifications in 2020 most likely reflected the lockdown of eye care services during the COVID-19 pandemic. CVIs are expected to increase due to aging and growing populations, a lack of certifications may reflect undiagnosed eye conditions or refusal of certification. Delays within the certification process, poor understanding of certification protocols and lack of rapid access to the certification process could also play a part. AMD accounted for more than half of certifications which identified an important area for future research and the necessity for fast-track access to services in patients with early AMD to delay progression of the disease. Facilitating and enabling primary care optometrists to certify may be another solution. Continued monitoring of certifications is crucial in streamlining processes and improving support services,

particularly in older patients and patients living in more deprived areas. Rapid access to eye care services, improvement of screening programmes and involvement of everyone involved in eye health at early stages of eye disease are important in reducing preventable causes of sight loss and improving health outcomes.

Conflicts of interest: None

Funding: None

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P172

CLINICAL CHARACTERISTICS OF PATIENTS WITH NANOPHTHALMOS IN GLOUCESTERSHIRE

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Introduction: A nanophthalmic eye is defined as an eye with a short axial length (AL), in this study we defined this as less than or equal to 20 mm. Traditionally these eyes are thought to have a high rate of ocular comorbidity and associated low visual acuity. However, due to nanophthalmos being a rare condition, with an estimated prevalence of around 0.27% in the UK population, there is a relative paucity of literature describing its clinical characteristics. Therefore, we carried out a retrospective review of all patients with nanophthalmos who had undergone cataract surgery at Gloucestershire Hospitals NHS Foundation Trust (GHNHSFT) aiming to describe the demographic, visual and clinical features of these patients.

Method: Cases were found using the medisoft audit tool for cataract surgery outcomes for all cataract surgeries performed at GHNHSFT between 1 Jan 2006 and 31 Dec 2023. Data was collected from electronic patient record (Medisoft®) and paper notes. Patients under 17 years of age and those with inaccurate biometry were excluded. Collected data included age, gender, ethnicity, age at cataract surgery, ocular biometry, refraction, intraocular pressure, number of intraocular pressure lowering medications, details of procedures undertaken to lower intraocular pressure, ocular comorbidities and family history of ocular comorbidities.

Results: Between 1 Jan 2006 and 31 Dec 2023, 62364 cataract operations on 41366 patients above the age of 17 were performed at GHNHSFT. 86 nanophthalmic eyes underwent cataract surgery over this 18 year catchment period. On average, every year our unit completed 5 cases of cataract surgery on nanophthalmic eyes. There was one case of nanophthalmic eye in every 725 cases of cataract surgery. Of the 59 patients included in the study, 14 were male and 45 were female. Around 90% of the patients (53 out of 59) were ethnically white British. The mean age at the time of cataract operation in this cohort was 66.5 years (range: 17 to 86 years). The mean AL was 19.24 mm (range: 14.9 to 20.0 mm); 71 eyes had an AL between 19.00 and 20.00 mm, and 15 eyes had an AL less than 19.00 mm. The mean anterior chamber depth was 2.4 mm (range: 1.87 to 4.05 mm). Pre-operatively, the best-corrected visual acuity (BCVA) was 0.50; 16 eyes had vision worse than 1.0 logMAR and 5 eyes had vision worse than counting fingers. The mean spherical equivalent was + 7.75 (range: -1D to 17.25D). The mean intraocular pressure (IOP) was 15.2 mmHg, and 17 eyes required IOP lowering medication. 9 eyes were diagnosed with angle closure glaucoma. Amblyopia was present in 7% of eyes (n = 6). Family history of glaucoma was reported in one patient, while another had a family history of nanophthalmos. Approximately 25% of eyes (n = 22)

required laser iridotomy. None of the eyes in this cohort required trabeculectomy for IOP control, although three eyes underwent cycloiodide treatment for pressure management.

Conclusion: At Gloucestershire Hospitals NHS Foundation Trust, for every 725 cataract surgeries, there was one case of nanophthalmos. In our study 25% of nanophthalmic eyes required laser iridotomy, 20% of required IOP medication, 10% had angle closure glaucoma and 7% had amblyopia. Traditionally it is taught that nanophthalmic eyes have high rates of ocular comorbidities and lower visual acuity, our study is consistent with this.

Conflicts of interest: None

Funding: None

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P173

CO-OCCURRENCE OF INCOMPLETE ACHROMATOPSIA AND MICROCORNEA SYNDROME IN A SINGLE FAMILY: GENETIC INSIGHTS AND CLINICAL CHARACTERIZATION

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Introduction: Achromatopsia is a rare retinal dystrophy characterized by decreased visual acuity, pendular nystagmus, photophobia, impaired colour discrimination, and central scotoma. Whole-exome sequencing (WES) was employed on a family from Iran, revealing two novel variants in the PDE6H and ADAMTS18 genes. Disease-causing variant in PDE6H and ADAMTS18 have been extremely rarely reported in the literature. This study aimed to describe the clinical and genetic findings in this family and contribute to the understanding of these genetic disorders.

Method: The presence of the identified variants was confirmed using Sanger sequencing, and the zygosity status of other family members was determined. The clinical features of affected individuals were thoroughly assessed, focusing on visual acuity, colour vision, electroretinogram (ERG) results, optical coherence tomography (OCT) imaging, and fundus examination.

Results: Two novel pathogenic variants were identified: NM_006205.3:c.35C>G (p.Ser12Ter) in PDE6H and NM_199355.4:c.3139C>T (p.Arg1047Ter) in ADAMTS18. The proband and two of his sisters were homozygous for the PDE6H variant and heterozygous for the ADAMTS18 variant. Two affected individuals were heterozygous for the PDE6H variant and homozygous for the ADAMTS18 variant. Clinical findings in these individuals included decreased visual acuity, impaired colour discrimination, photophobia, myopia, ellipsoid zone disruption on OCT, and pendular nystagmus in some cases. Two younger affected individuals (the second generation of the family) exhibited microcornea, myopia, and telecanthus, with additional findings of punctal atresia and strabismus. These individuals were homozygous for the ADAMTS18 variant and heterozygous for the PDE6H variant.

Conclusion: The study reports the first known case of concurrent inheritance of incomplete achromatopsia and microcornea, myopic chorioretinal atrophy, and telecanthus (MMCAT) in a single family. The identification of the novel pathogenic variants in PDE6H and ADAMTS18 expands the phenotypic and genotypic spectrum of these disorders. This family represents a unique genetic combination that highlights the diverse range of ophthalmic manifestations associated with these mutations. Our findings underscore the importance of genetic testing and counselling, particularly in regions where rare genetic conditions may be more prevalent. Further studies on the molecular mechanisms underlying these disorders are needed to better understand their pathophysiology and potential therapeutic approaches.

Conflicts of interest: None

Funding: None

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P174

COMPARING AI PERFORMANCE WITH OPHTHALMOLOGY RESIDENT DOCTORS: A STANDARDISED ASSESSMENT OF CHATGPT-4O

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Introduction: Artificial intelligence (AI) is increasingly integrated into medical education, providing personalised learning opportunities and efficient knowledge assessment. ChatGPT-4o, released in May 2024, represents the latest and most advanced iteration of large language models. This study aimed to compare the performance of ChatGPT-4o with that of residents who had recently passed the Part 1 Fellowship Examination of the Royal College of Ophthalmologists (FRCOphth), using a standardised set of multiple-choice questions (MCQs).

Method: A cohort of 15 trainees, ranging from Foundation Year 1 (FY1) to Ophthalmic Specialty Trainee Year 2 (ST2), each of whom had passed the Part 1 FRCOphth examination within the preceding year, were recruited. Participants completed a set of 50 MCQs sourced from the Royal College of Ophthalmologists' sample question bank. Their scores were recorded and subjected to statistical analysis to determine the mean and range. ChatGPT-4o was subsequently tasked with answering the identical MCQ set, and its responses were evaluated against the official answer key. The performance of ChatGPT-4o was then compared directly to that of the resident cohort.

Results: The residents achieved a mean score of 31.6 out of 50 (63.2%), with individual scores ranging from 23 to 41 (46% to 82%). By contrast, ChatGPT-4o attained a perfect score of 50 out of 50 (100%). For additional context, the average pass mark for the Part 1 FRCOphth examination over the preceding year was approximately 62.4%.

Conclusion: ChatGPT-4o's outstanding performance in this controlled assessment underscores its potential as a valuable supplementary resource for ophthalmic education. Its capacity to provide highly accurate answers, accompanied by detailed explanatory content, may enable residents to refine their study strategies and enhance examination preparedness.

Nevertheless, important limitations warrant recognition. ChatGPT-4o, like other AI models, lacks direct clinical experience and cannot engage in nuanced clinical reasoning within ambiguous contexts. Moreover, its knowledge base is derived from pre-existing data, which may not always reflect the most up-to-date clinical guidelines.

or emerging evidence. As AI continues to permeate medical training, it should be employed as a complementary tool rather than a substitute for expert human instruction. Future research should explore the long-term impact of AI-assisted learning on knowledge retention, clinical reasoning skills, and patient outcomes. Further investigation is also required to assess the reliability and utility of AI-driven tools in more complex, scenario-based assessments prior to their broader integration into educational programmes.

ChatGPT-4o, released in May 2024, demonstrated exceptional performance in a standardised ophthalmology MCQ examination, surpassing the scores of residents. These findings highlight the potential of advanced AI models as adjunctive educational tools. Ongoing evaluation and thoughtful integration of systems like ChatGPT-4o into medical education frameworks may enhance traditional training methods, ultimately better preparing future clinicians for the evolving demands of contemporary healthcare.

Conflicts of interest: None

Funding: None

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P175

CONFIDENCE OF ED CLINICIANS IN EXAMINATION AND DIAGNOSIS OF OPHTHALMOLOGY CONDITIONS IN EMERGENCY DEPARTMENTS

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Introduction: Medical students often spend only weeks in Ophthalmology during their training and often have little training exposure thereafter. As a result, there is often reduced confidence and clinical acumen in diagnosing ocular emergencies as clinicians.

Method: We undertook a survey to assess the knowledge and confidence of ED clinicians in performing relevant examinations in ophthalmic emergencies with the aim of identifying key areas of improvement.

Results: 25 ED clinicians were recruited into the survey with 92% response. The majority, 86%, were comfortable with detecting pupillary signs. While 61% of clinicians were very confident performing VA and general external examination, very few felt confident in performing detailed anterior segment examination (26%) and fundoscopy (17%), and only 48% were confident in diagnosing and managing eye-related conditions.

Conclusion: There is need for increased teaching and training in use of specialist eye equipment among non-specialist clinicians and there is need for more teaching in examination techniques for eye related presentations.

Conflicts of interest: None

Funding: None

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P176

DEBUNKING MISCONCEPTIONS ABOUT KOHL: A SYSTEMATIC REVIEW REVEALING THE UNTESTED POTENTIAL OF AUTHENTIC ITHMID FOR OCULAR HEALTH

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Introduction: For decades, kohl has been dismissed as harmful due to lead toxicity, with modern studies labelling it unsafe for ocular use. However, these conclusions are based on adulterated, lead-contaminated products that do not reflect the authentic ithmid kohl historically used in various cultures and traditions. This systematic review challenges the validity of existing data, revealing that authentic ithmid kohl, rich in antimony, has been overlooked in research despite its potential therapeutic benefits for eye health.

The objective of this review is to expose the flaws in previous research that generalized lead-based kohl findings to all kohl products and to reexamine the potential safety and efficacy of authentic ithmid kohl in light of historical and modern evidence.

Method: Using PRISMA guidelines, a systematic review was conducted across PubMed and Google Scholar with search terms such as "ithmid kohl," "antimony-based eyeliner," and "kohl lead toxicity." Studies were included if they examined kohl products, regardless of whether they differentiated authentic ithmid from lead-based adulterated products. Special attention was given to studies that provided data on the composition of commercially available kohl, particularly those highlighting the prevalence of lead contamination in popular brands in Saudi Arabia. Historical texts and traditional medical references were also analysed to contextualize the documented benefits of ithmid. Data were extracted to identify methodological flaws, assumptions, and gaps in the differentiation of kohl types.

Results: A significant proportion of modern studies on kohl fail to differentiate between authentic antimony-based ithmid and lead-contaminated kohl, leading to generalized conclusions about its safety. Historical sources consistently describe ithmid as beneficial for ocular health, yet limited scientific studies exist that directly evaluate its effects. Emerging evidence suggests that authentic ithmid may possess therapeutic properties, such as antimicrobial activity and potential benefits in managing blepharitis. The lack of targeted research on pure ithmid highlights a critical gap in understanding its true safety and efficacy.

Conclusion: This review reveals a critical gap in existing research: authentic ithmid kohl has been largely ignored in favour of studying lead-contaminated products. Historical evidence and limited contemporary studies suggest that ithmid kohl, when pure, offers significant ocular health benefits. This work calls for rigorous, targeted clinical trials to validate its safety and efficacy, challenging the status quo of misconceptions surrounding kohl use in ophthalmology.

Conflicts of interest: None

Funding: None

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P177

DELIVERY OF LEADERSHIP COURSE TO MEET NEW OPHTHALMIC SPECIALIST TRAINING CURRICULUM LEVEL 4 LEADERSHIP

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Introduction: In August 2024, the new Ophthalmic Specialist Training (OST) curriculum was introduced, primarily in response

to the GMC's Excellence by Design Standards for Postgraduate Curricula and the need to incorporate its Generic Professional Capabilities framework. The new curriculum assesses capabilities in seven domains, with each domain divided into four level of competence. We highlighted a gap in opportunities for London trainees to achieve Level 3 and 4 in the Leadership and Team Working domain. The aim of our project was to assess the effectiveness of a new ophthalmology Leadership course and ease the transition for both trainees and trainers to the new curriculum.

Method: A four-step multimodal training package was created to help trainees achieve Level 3 and Level 4 competence in the Leadership and Team Working domain. Pre-course learning was shared via the RCOphth online learning platform, INSPIRE, and consisted of examples of Leadership Projects, outline of a serious incident, a recorded talk on feedback as well as instructions to prepare a presentation on one of their own leadership projects. In-person teaching consisted of a 1-day course. The morning session involved interactive lectures on common leadership themes including feedback and appraisal, Goleman's leadership styles and serious incident analysis. The afternoon session consisted of peer appraisal of attendee presentations of their own leadership projects. Post-course consolidation was the completion of a reflection on how they would adapt their leadership styles had based on the course material. Subsequently each learner completed a feedback form that was used to improve future iterations of the teaching.

Results: 10 London ophthalmic trainees attended the teaching and all completed the feedback. Self-assessed confidence ratings significantly increased for giving feedback ($p < 0.01$), appraisal ($p < 0.01$), quality improvement methodology ($p < 0.01$) Goleman's Leadership framework ($p < 0.0001$) and PSIRF ($p < 0.001$). 90% of trainees reported they would recommend the course to a colleague.

Free-text feedback highlighted particular usefulness of small group breakout sessions, receiving and giving feedback on leadership projects as well as pre-course learning material. All 10 attendees were deemed competent in Level 4 Leadership and Team Working following completion of the course.

Conclusion: Four-step teaching of Leadership and Team Working was effective in developing competence in trainees in Level 4 Leadership and Team Working. This is the first course of this kind to help support the new OST curriculum. The teaching package with the pre-course material has been widely shared via the RCOphth online learning platform, INSPIRE and also improved to facilitate future expansion of the course at a national level.

Conflicts of interest: None

Funding: None

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P178

ENHANCING CAREER INTEREST AND COMPETENCY IN OPHTHALMOLOGY: THE IMPACT OF A STUDENT-FRIENDLY INTERACTIVE CONFERENCE

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Introduction: Ophthalmology has drastically increased in popularity, sharply spiking in competition ratios to 14.41:1 in 2024. Despite this, medical students often have little exposure to

ophthalmology before graduation and even fewer resources to help understand the application process. Furthermore, ophthalmology conferences are often tailored to trainees and above, which may create a barrier of entry to students.

Student conferences are a popular and accessible method to increase interest and awareness of specialties. This study sought to explore the impacts of an interactive ophthalmological student conference and provide recommendations based off analysis and reflections for future potential events.

Method: An interactive, hybrid, national conference, awarded 6.5 CPD points from the Royal College of Ophthalmologists, was held at the Western Eye Hospital on November 30th 2024. The conference featured eight speakers, including a consultant's day in the life and an ST1 application guide. There were also three specialist workshops, including ophthalmology suturing and 'trapplectomy', a published trainee-friendly version of trabeculectomy. Pre- and post-conference surveys were distributed to all attendees via Microsoft Forms and the MedAll platform. Surveys explored attendee perceptions of ophthalmology and conference feedback. These results were analysed with simple statistical measures and thematic analysis.

Results: The conference welcomed 30 in-person and 26 online attendees. There were similar proportions of attendees in each year group, with a slight majority of third- and fourth-year students. Only 25% of students had previously attended an ophthalmological conference, but an average score of 4.3/5 was given for interest in ophthalmology before the conference. Students' incentives to approach ophthalmology were the vast range of subspecialties (28.5%), the blend between medicine and surgery (26.5%) and the advancements in research and technology (10.2%).

On average, students self-reported a low baseline of understanding in ophthalmology (2.7/5), as well as poor competency in ophthalmology examinations (1.8/5) and ophthalmology suturing (1.2/5). This was also complimented by the large proportion of attendees interested in practicing key skills (33.3%) prior to the conference. Following the conference, we observed an increase in average self-reported competence, including their baseline understandings of ophthalmology (4.4/5), competence in ophthalmology examinations (3.7/5) and ophthalmology suturing (4.25/5).

A thematic analysis of pre- and post-conference surveys into the students' perceived barriers of entry into ophthalmology highlighted the lack of exposure and knowledge of ophthalmology (30.6%) and confusion around the application process (12.2%) as key barriers. These reduced to 3.7% and 0% post-conference respectively. Competition ratios were also a key perceived barrier, with 38.8% of attendees highlighting it preconference, increasing to 66.7% post-conference. Furthermore, the financial burden of examinations and training emerged as a new barrier post-conference.

Workshops and the talks were rated highly in post-event feedback regarding relevancy and in engagement. Particularly highly rated sessions were the Day in the Life and ST1 application guide, as well as the ophthalmology suturing and 'trapplectomy'. The event itself was deemed to be pitched at an appropriate level by attendees, and whilst there were some technical barriers in the livestream feature, the conference was highly recommended to future delegates.

Conclusion: Results show that student-friendly ophthalmology conferences are vital to reducing some barriers of entry into ophthalmology, such as a lack of knowledge about the specialty and application criteria. Whilst competition ratios are projected to rise, similar events should aspire to empower medical students with the knowledge and resources to best prepare them for the future. We aim to explore the differences in conferences between other subspecialties in future work.

Conflicts of interest: None

Funding: None

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P179

EVALUATING OCULAR ONCOLOGY REFERRALS FROM THE PRIMARY CARE SERVICE (A SERVICE EVALUATION PROJECT)

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Introduction: The opticians used to send any probable uveal melanomas urgently to the ocular oncology service while atypical nevi are forwarded to the local eye department. Because the eye department in Royal Hallamshire Hospital (RHH) already has an ocular oncology service, all such referrals (from both categories) are dealt with by the ocular oncology team in the department. Some of these referrals are unnecessary and can be managed by the primary community eye service. Decreasing the number of such referrals will help to save more time to evaluate and manage more complex referrals.

Aims of the project:-

- Assessing the optician's referrals related to ocular oncology sent to the eye department in RHH.
- Assessing the outcome of the virtual assessment of such referrals to see how much of these referrals really needed to be referred.
- To plan ways of improving the primary referral process service and make it more efficient so decreasing the number of unnecessary referrals

Method: Analysing referrals coming from opticians in the region to ocular oncology service in Royal Hallamshire Hospital in the period between December 2022 and December 2023. Results

Total 157 referrals. 46 referrals were excluded. 103 referrals were assessed virtually. 87 patients were discharged. 49 referrals were confirmed to be choroidal nevi, while 4 were confirmed to be choroidal melanomas (2 new, 2 previously treated). 22 referrals were found to have CHRPEs, 3 choroidal haemangiomas, 1 choroidal metastasis, 1 vasoproliferative tumour, 1 solitary idiopathic choroiditis. 23 eyes had a variety of other benign diagnoses (like scars, scleral nodules, PED, vortex vein varix, and coloboma). 7 eyes with no fundus pathologies

Conclusion: Primary optician assessment of retinal pigmented lesions has the potential to be improved to make the referral process more efficient and decrease the burden on the already stretched tertiary ocular oncology service.

keys concern about some referrals:-

- Inaccurate description of retinal lesions
- Deficient referrals with incomplete history and examination, no associated images
- Referrals that depend on imaging observations rather than examination finding

Areas for improvement

- Referrals need to be based on clinical examination, including dilation, not only on imaging
- The description of lesions needs to be full and, more importantly, accurate.
- More awareness of the officially agreed referral guidelines
- Differentiating orange pigments from drusen and CHRPEs from nevi is a good skill to acquire

Conflicts of interest: None

Funding: None

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P180

EVALUATING THE EDUCATIONAL UTILITY OF OPHTHALMOLOGY-RELATED HASHTAGS ON INSTAGRAM: A CONTENT ANALYSIS OF EDUCATIONAL, SOCIAL, AND ADVERTISING POSTS

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Introduction: Social media platforms, such as Instagram, are increasingly utilised within medical education due to their visual and interactive nature, engaging both healthcare professionals and the public. In ophthalmology, where imaging and diagnostic precision are key, Instagram provides an ideal platform for disseminating high-quality educational content. With over 38 million users in the UK, Instagram offers a substantial opportunity for educational outreach. However, such content can be overshadowed by social or commercial material. This study aims to evaluate Instagram's utility for ophthalmology education by analysing posts associated with the most commonly used ophthalmology-related hashtags.

Method: An Instagram search was performed using the specified hashtags, identified from <https://besthashtags.com/> as the most popular ophthalmology-related terms. For each hashtag, the first 30 posts were reviewed and categorised as educational (e.g. diagnostic imaging, surgical techniques, case discussions, patient guidance), social (e.g. lifestyle tips, personal experiences), or advertising (e.g. posts promoting products, services, or commercial interests). The proportions of these categories were then calculated and compared across all hashtags.

Results: The classification of the first 30 posts under each hashtag was as follows (educational, social, advertising):

- #ophthalmology: 26 educational, 4 social, 0 advertising
- #ophthalmologist: 22 educational, 6 social, 2 advertising
- #eyecare: 20 educational, 3 social, 7 advertising
- #optometry: 28 educational, 2 social, 0 advertising
- #eyedoctor: 23 educational, 5 social, 2 advertising
- #vision: 2 educational, 28 social, 0 advertising
- #eyehealth: 29 educational, 1 social, 0 advertising
- #eye: 4 educational, 26 social, 0 advertising
- #optometrist: 25 educational, 5 social, 0 advertising
- #eyes: 1 educational, 29 social, 0 advertising

Hashtags such as #ophthalmology, #optometry, #eyehealth, and #optometrist were predominantly educational. In contrast, more generic terms like #vision, #eye, and #eyes yielded mainly social content. Some hashtags, including #ophthalmologist, #eyecare, and #eyedoctor, also contained a noteworthy proportion of advertising posts.

Conclusion: Our findings suggest that Instagram can serve as a valuable platform for ophthalmology education, though the educational yield varies significantly according to the hashtag chosen. Specialty-specific hashtags associated more closely with professional practice (e.g. #ophthalmology, #optometry) hosted a higher proportion of educational posts. In contrast, more general hashtags attracted predominantly social content, while a few others carried a notable advertising presence. This highlights the importance of employing targeted hashtags and promoting

recognised, standardised tags to increase the visibility and credibility of educational material. Establishing professionally endorsed hashtags—potentially introduced by bodies such as the Royal College of Ophthalmologists—could unify the community, encourage quality control, and guide both professionals and learners towards reliable, evidence-based content.

Conflicts of interest: None.

Funding: None.

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P181

EVALUATION OF ORTHOPTIC STROKE SERVICE AT SOUTHPORT AND ORMSKIRK NHS TRUST (MWL)

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Introduction: Research suggests that the prevalence of visual problems post-stroke is approximately 70%.¹

Therefore, it is recommended that patients have either visual screening or formal Orthoptic assessment so that “Crucial information can thus be provided on visual status and its functional significance to the stroke team, patients and carers, enabling early intervention.” The Royal College of Physicians and NICE updated their guidance in 2023 to state patients should either be “screened using a standardised approach” or offered “a specialist orthoptist assessment as soon as possible”.

This evaluation specifically compares the new service with the previous ad-hoc service to establish whether it is more effective.

Method: Since 2021, all stroke patients referred to the Orthoptic team have been documented in an Excel database.

This outlines the date of referral, stroke date, reason for referral, referrer, date seen, etc., which is inputted at the time of the referral and then onwards.

This audit briefly outlines the data from 2021 onwards, focusing specifically on 2022 (the old service) and 2023 (from May onwards—the new service).

The stroke team and Sentinel Stroke National Audit Programme (SSNAP) have provided data regarding the total number of stroke patients coming through the trust during these time periods.

Inhouse Electronic patient records were also reviewed as needed.

Results: Comparing the old and the new services showed that the new service was in line with the BIOS standards on 10 out of 12 points as opposed to the old standards, which only met 3 of the standards.

While the new service significantly improves meeting the standards, there is still a discrepancy between recommended ward time and what is currently allocated.

Despite the increase in the time allocated for seeing patients and the creation of dedicated clinic sessions, waiting times have unfortunately increased overall.

However, when comparing the number of referrals from 2022 to 2023, there has been a 154% increase.

Conclusion:

- The current Orthoptic stroke service meets 10 of the 12 factors BIOS recommends for a “high-quality service”.
- The factors not being met are one element of the staffing and a rota of Orthoptic staff.
- Two out of three staffing recommendations are being met, with the remaining one due to falling short of the recommended WTE specifically for the ward.
- The quantity of referrals to Orthoptics has increased

significantly, starting around Sept 2022 when the HASU opened.

- Although patients are now waiting longer, they are triaged to dedicated clinics. They have a pathway explicitly catering to the most frequently seen clinical conditions, with options for appointments that may be more suited to those with a poorer recovery post-stroke.

Conflicts of interest: None

Funding: None

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P182

EVALUATION OF SINGLE POINT OF ACCESS (SPOA) FOR THE MANAGEMENT AND TRIAGE OF OPHTHALMOLOGY REFERRALS TO MAXIMISE THE USE OF PRIMARY CARE OPTOMETRY SERVICES

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Introduction: Ophthalmology has the highest number of out-patient appointments within the NHS, with ever-increasing demand challenging waiting times and the provision of timely appointments. As such, the avoidance of unnecessary or false positive referrals to secondary care is a priority across many specialities.

SPoA (Single Point of Access) hubs manage all speciality referrals through a single system, ensuring that patients are seen in the right place within the most efficient timescale. Some services further seek to shift appropriate care from hospital settings by deflecting patients to commissioned primary care services which have been shown to be effective in reducing the burden on secondary care.

We evaluate the outcomes of such SPoA ophthalmology referral management and triage services Provided by Primary Eyecare Services in localities across England.

Method: Primary Eyecare Services provides SPoA ophthalmology referral management and triage in five Integrated Care Board (ICB) areas in England (East of England, South East, Midlands, and two in London). Ophthalmology referrals came into the service via primary care optometrists or GPs.

Referrals were reviewed by optometrists specifically trained in triage, with the following outcomes available: urgent or routine deflection to a commissioned primary care optometry service; urgent or routine referral to secondary care or returned to the referring clinician (e.g. requesting further clinical detail to allow for accurate triage or suggesting alternate management including where referral may not be appropriate).

The outcomes of ophthalmology referrals received across the five localities from April 2023 to March 2024 were analysed.

Results: 33931 ophthalmology referrals were received over 12 months.

21.7% (7361) referrals were deflected to a primary care optometry service and 9.4% (3194) were returned to the referring clinician. The remaining 68.9% (23376) of referrals were referred on to an appropriate secondary care clinic with the required urgency (with the triaging optometrist able to change the secondary care clinic type and urgency as appropriate).

Of the 7361 referrals deflected to primary care, 34.4% (2533) were deflected to a community eyecare service, CES (Community Eyecare Service), CUES (Community Urgent Eyecare Service) or MECS (Minor Eyecare Service), 47.2% (3477) to a glaucoma referral

filtering pathway, 16.9% (1246) to an Enhanced Cataract Referral Service and 1.4% (105) to a Children's Post-Vision Screening Service. 90.3% (6649) of primary care deflections were routine and 9.7% (712) urgent.

Of the 23376 secondary care referrals, 83.4% (19499) were routine and 16.6% (3877) urgent.

Conclusion: SPoA services deflected over a fifth of ophthalmology referrals to a primary care service, maximising the use of commissioned community optometry services to shift appropriate care from hospital settings. A further 9% were deflected to the referring clinician. Unnecessary secondary care appointments were avoided both by patients accessing commissioned primary care services close to home, and by ensuring that referrals with insufficient information did not progress to potentially inappropriate clinics, with the resulting impact on patients and secondary care capacity.

SPoA services can be key enablers of NHS strategy, including the movement of appropriate activity from secondary to primary care, bridging the gap between primary and secondary care and maximising out of hospital care and using technology as an enabler. As primary care optometry practices do not have universal access to eRS, SPoA services can support NHSE planning guidance calling for direct referral pathways from community optometrists to ophthalmology services for all urgent and elective eye consultations. SPoA services also have the potential to work effectively with key stakeholders, providing feedback to positively impact future referral quality.

There is substantial potential to further reduce pressure on secondary care systems with the consistent commissioning of both SPoA and extended services in primary care optometry.

Conflicts of interest: None

Funding: None

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P183

IDENTIFYING AND ADDRESSING THE USE OF PERSONAL DEVICES FOR CAPTURING PATIENT IMAGES FOR DISCUSSION, AMONGST RESIDENT DOCTORS

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Introduction: The vast majority of clinicians now carry a smart phone with consistent network coverage, continuous GPS coverage and high-resolution cameras all in one device. Images captured now contain metadata and some devices will automatically sync photos across accounts, unless the user has manually changed settings. This has created an environment where it is easier than ever to take a photo of a patient and share it for the purposes of advice, clinical discussion or simple intrigue. However, it can be easy to overlook the significant fines and potential legal consequences for mishandling patient identifiable information, and what needs to be done to a personal device to ensure this does not happen.

In an acute setting, especially out of hours, the ability to take a photo, show or share it with a colleague for advice, can allow for quicker management plans and hopefully better prognoses. But how often are these image capture moments in keeping with the data protection act and GDPR? This project aimed to identify how many of our Resident Doctors were using their personal devices to take clinical photos, explore how they shared them, what they did with those images after and if they felt GDPR worried them when taking a photo.

Method: An anonymous, voluntary survey was sent to all Resident Doctors within the Ophthalmology Department. The questions identified: if a clinician was using or had ever used their personal device to obtain an image, how they shared that image and if applicable which apps they used, did they delete the image, did they obtain consent, were they aware of any established photo sharing methods the Trust had already initiated and how many clinical photos did they have on their personal device at the time of completing the survey. All responses were anonymous for the purposes of honest answers.

Results: 13 Resident Doctors, all of whom worked the on-call rota, responded to the survey. Responses came from multiple doctors in each training level from Junior Clinical Fellow to ST7. All respondents felt that having a photo available enhanced case discussion and gained verbal consent from their patients. 11 respondents used their personal device to take clinical photos on a weekly or monthly basis. 10 respondents admitted to storing images to their personal devices memory, rather than a dedicated app/server. Only 4 respondents admitted to utilising dedicated sharing apps which bypass personal storage. 10 respondents did not think or were unsure if the Trust had an established encrypted method of photo sharing. 79% of residents admitted to worrying about GDPR when taking a photo of a patient. At the time of survey completion, 7 respondents admitted to having between 1–20 images in their devices personal storage and only 5 respondents admitted to always deleting images from their device.

Conclusion: This short project identified a significant issue. Doctors clearly felt that the ability to use their own devices for image capture helped them on a daily or monthly basis, however the education behind how to manage these images needs further development. Based on the responses, clinicians are putting themselves into possible medicolegal situations simply because they do not know how to avoid it. We propose raising this at the next suitable teaching session and highlight the use of apps such as Pando or MS Teams to share clinical images. We wish to expand this survey regionally and identify if other trusts have a similar issue and what they have done to address any possible problems

Conflicts of interest: None

Funding: None

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P184

IMPOSTER SYNDROME IN OPHTHALMOLOGY TRAINEES – ARE WE OUR OWN WORST ENEMIES?

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Introduction: Imposter syndrome (IS) describes the experience of feeling fraudulent or doubting one's achievements despite evidence to the contrary. IS was first described in 1978 by Clance and Imes who described the characteristics within high achieving women (Bhama et al. 2021). The high stakes and evaluative culture within medicine has been argued to correlate to both physicians and surgeons demonstrating more frequent rates of IS characteristics (Chodoff et al. 2023).

Although explored in other surgical fields, including general surgery, oral and maxillofacial surgery and neurosurgery, the prevalence within ophthalmology residents has not previously been investigated (Bhama et al. 2021; Zaed et al. 2022; Deek et al. 2024). We aim to explore the prevalence and degree of IS within United Kingdom based ophthalmology residents.

Method: An online survey was sent to ophthalmology residents from our School. This included the Clance Imposter Phenomenon Survey, a validated IS measure, plus participant demographics and items querying experience of potential compensatory mechanisms for IS based on another prior IS publication.

The CIPS is a positively worded 20 item question which asks the respondents to rate their answers on a Likert scale, from not at all true to very true. The higher the score, the more frequently and seriously imposterism is present. A score of 40 or less indicates few characteristics, 41–60 indicates moderate characteristics, 61–80 indicates frequent characteristics, and greater than 80 indicates intense imposter feelings.

Results: 12 of 26 ophthalmology residents within the School completed the survey (46% response rate). All participants identified as either female or male gender. 75% (9/12) of all ophthalmic residents demonstrated either moderate or frequent imposter characteristics.

87.5% (6/7) of female residents demonstrated moderate or frequent characteristics of IS, compared to 60% (3/5) of male residents. 57% (4/7) of female residents demonstrated frequent characteristics compared to 40% (2/5) of male residents. No ophthalmic residents demonstrated intense IS characteristics.

Mean scores per CIPS question for both genders were generally higher for female residents. One question – ‘I rarely do a project or task as well as I’d like to do it’ – showed a higher mean score for male residents (mean score 2.8 for males compared to 2.0 for females).

There were no demonstrable additional characteristics linked to expression of IS characteristics, including, hours worked per week, journals read per week, or conferences attended per year. Nor was there an association evident between differing age brackets or stage of training with levels of IS characteristics.

Responses were anonymous. Ethical approval was not required but permission was granted by the Head of School. Data was analysed using Excel.

Conclusion: Our results demonstrate that IS is prevalent to a high degree within ophthalmology residents, correlating with the literature present in other surgical specialties. Similarly to other specialties investigated, IS scores are heightened within the female cohort. However, remains present within the male cohort.

IS can result in the downplaying of accomplishments and underestimation of abilities, which has been suggested to lead to lower levels of recognition and slower career advancements (Vaa Stelling et al. 2023).

Importantly, IS is an independent predictor for burnout and anxiety, both of which have implications linked to patient safety outcomes (Joseph et al. 2023).

There is an unmet requirement for surgical educators to understand IS and its’ implications upon a residents’ clinical and surgical development. Further research is required to ensure appropriate resources and support can be provided to residents in training.

Conflicts of interest: None

Funding: None

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P185

IMPROVING ACCESSIBILITY OF PATIENT INFORMATION LEAFLETS FOR VISUALLY IMPAIRED PATIENTS: A REVIEW OF BEST PRACTICES

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Introduction: Patient information leaflets are integral to effective patient communication and the informed consent process. Designing these materials to be accessible for visually impaired patients presents unique challenges, which as ophthalmologists we should be particularly mindful of. Despite growing emphasis on accessibility, there is a lack of consolidated guidelines easily available to ophthalmologists, and general awareness of this issue remains limited. To address this gap, we sought to identify and compile evidence-based recommendations for improving the visual accessibility of written information for patients with visual impairments.

Method: A review of publicly available resources on visual accessibility was conducted, focusing on recommendations for individuals with visual impairments. These were compared and analysed to identify the most frequently cited and practical guidelines.

Results: Key recommendations were compiled into a practical resource for designing patient information leaflets. These include considerations such as font size and type, contrast levels, colour choices, and layout adaptations. Despite the availability of international guidance on web accessibility—such as those adhered to by the RCOphth website—there are no specific guidelines available on the RCOphth website for members creating written information for their patients.

Conclusion: We have developed a consolidated list of recommendations to improve the accessibility of patient information leaflets for visually impaired patients. By raising awareness and providing practical guidance, we aim to promote inclusivity in patient communication. This project highlights the need for the RCOphth and other professional bodies to make accessible communication guidelines readily available to members. Adoption of these principles has the potential to improve equity in healthcare for visually impaired patients.

Conflicts of interest: None

Funding: None

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P186

IMPROVING THE QUALITY AND APPROPRIATENESS OF INPATIENT OPHTHALMOLOGY REFERRALS: A RETROSPECTIVE ANALYSIS

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Introduction: Efficient and accurate triage of ophthalmology referrals is essential in optimising patient care and with increased waiting lists, this becomes particularly relevant. Recent research has shown a gradual reduction in ophthalmology teaching in medical schools and decline in clinical confidence among junior doctors with eye related presentations. This raises concerns regarding insufficiency or inaccuracy of clinical information in inpatient referrals to the on-call ophthalmology team. This may be further exacerbated with constraints on the ward with the absence of Snellen charts or ophthalmoscopes that prevents clinical examination. This study seeks to assess the quality of inpatient referrals to ophthalmology by assessing whether essential clinical information is provided for triage and identifying inappropriate referrals. These include those not requiring Hospital Eye Service (HES) review (e.g. gradual vision loss) or those better managed through alternative pathways (e.g. orthoptist-led stroke pathways).

Method: A retrospective analysis was conducted on all inpatient ophthalmology referrals submitted via ICE system over a 4 week period. Data extracted included presence of key clinical information (such as laterality and duration of symptoms, examination findings) and appropriateness of the referral based on clinical context. The criteria was based on retrospective judgement. Referrals were classified as “sufficiently detailed” if they included all relevant components. Descriptive statistics were used to report the findings, including the proportion of inappropriate referrals.

Results: A total of 65 referrals were analysed. Only 18 (28%) were deemed sufficiently detailed, containing all critical components. Specifically, key missing information included laterality of symptoms (64.6%), duration of symptoms (60%) and examination findings (43%). Visual acuity, a vital component for triage, was recorded in only 20% of referrals. Additionally, 9.2% of referrals were deemed inappropriate as they included patient who could have been managed through other pathways.

Conclusion: The findings highlight significant gaps in the completeness and appropriateness of ophthalmology referrals. A free-text box format is insufficient for capturing essential clinical information. To address these issues, the structure of referring template needs adjusting. Firstly, prompts are needed to direct referrers to alternative pathways where appropriate, such as orthoptist-led stroke pathways or to opticians post-discharge. Secondly, mandatory fields should be added that include symptoms (laterality, duration, description), past ocular history, relevant medical history and examination findings. Template should also include links to printable Snellen charts or information regarding smartphone-based visual acuity tools, like Peek Acuity, that can enhance referral quality. Finally, a section should be added to provide logistical details such as patient mobility to help assess whether they would be able to attend outpatient clinics. Implementing these changes is likely to enhance referral reliability, streamline triage and improve patient care. Further studies are needed to assess the impact of these recommendations on referral quality and clinical outcomes.

Conflicts of interest: None

Funding: None

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P187

INSIGHTS FROM THREE YEARS OF THE NATIONAL STUDENT OPHTHALMOLOGY CONFERENCE

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Introduction: The shift to virtual medical conferences during the COVID-19 pandemic offered opportunities to engage broader audiences whilst overcoming geographic and logistical barriers. However, studies focused on the value of virtual ophthalmology conferences remain limited.

This study aims to retrospectively analyse the annual National Student Ophthalmology Conference (NSOC) 2022–2024 for attendee satisfaction, overall benefits for attendees and efficacy of a virtual venue. Identifying factors influencing the value of virtual ophthalmology conferences would be exponentially useful for future organisers of similar events.

Method: The NSOCs were aimed at UK medical students but were attended by healthcare professionals internationally. Data from attendee feedback across NSOCs was reviewed. Each of the three conferences were hosted virtually, on Zoom or MedAll, but structured similarly to an in-person event including guest speakers, poster/presentation competition and in 2024 a virtual surgical skills workshop. Post-conference questionnaires collected feedback, with response rates of 46% (n = 58) in 2022, 64% (n = 103) in 2023, and 50% (n = 125) in 2024. Quantitative data from Likert scales underwent statistical analysis, while qualitative data was thematically analysed to identify recurring themes.

Results: Across the three years, all attendees rated the NSOCs positively. The conference content was rated “excellent” across the years, increasing from 60% to 63%. Similarly, the academic standard was perceived as “excellent”, also increasing from 70% to 75% in 2022 to 2024. The majority of attendees agreed that the 2023 conference provided better insights into the specialty (76%) and increased interest in ophthalmology (72%).

Regarding the virtual setting, overall, there was high satisfaction, with 100% “excellent” or “good” rating in 2022 and 2024, and 76% “excellent”, “good” or “neutral” in 2023.

The thematic analysis highlighted three consistent themes: multifaceted benefits for attendees, good organisation and online platform efficacy. In 2023, 55% of attendees strongly agreed the conference enhanced their understanding of ophthalmology.

Conclusion: The NSOCs demonstrated the potential of virtual conferences in delivering accessible, affordable, and high-quality ophthalmology education. It has attracted an increasing number of attendees whilst maintaining high satisfaction ratings and increasing interest in ophthalmology. The findings emphasise the success of a virtual platform, highlight the importance of diverse and relevant talk topics, speaker expertise and opportunities for attendees to present research. Overall, this paper serves as a benchmark for future events, providing valuable feedback for future organisers.

Conflicts of interest:

Salyha Mughal was conference lead of the 2024 conference.

Areen Tawil was communications and relations officer of the 2023 conference.

Kamal El-Badawi was conference lead of the 2023 conference.

Funding: None

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P188

IS THE FUTURE OF OPHTHALMOLOGY TOO BRIGHT FOR ITS OWN GOOD? A LOOK INTO THE VIEWS OF MEDICAL STUDENTS REGARDING OPHTHALMOLOGY AS A CAREER OPTION

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Introduction: The Health Education England (HEE) Report on speciality applications highlights increasing challenges in securing training posts, particularly in ophthalmology, where the ST1 competition ratio has reached 14.41. This rise is attributed to a growing number of applicants without an expansion in training posts. HEE has acknowledged the potential for increasing training numbers but has yet to implement active measures. This study investigates medical students’ perceptions of ophthalmology, their exposure to the specialty, and the influence of competition ratios on their career aspirations.

Method: An anonymous survey consisting of 16 questions was distributed to medical students from five UK universities ($n = 47$). The survey included demographic data, questions specific to ophthalmology using a 5-point Likert scale, and free-text responses. Topics covered included exposure to ophthalmology, perceptions of its lifestyle and prestige, and the relevance of teaching in medical school.

Results: The survey revealed that 80.8% of respondents felt they received inadequate exposure to ophthalmology during medical school, and 72.3% found the teaching clinically irrelevant. Despite this, 73.3% viewed the speciality favourably for its work-life balance and lifestyle, and 91.5% believed ophthalmologists significantly impact patient quality of life. A majority (57.4%) also considered ophthalmology a prestigious speciality among physicians.

However, dissatisfaction with the training pathway was notable, with 93.3% of respondents stating it was not optimal for them. Nonetheless, 64.4% indicated that the increasing competition ratios had not dissuaded them from applying for ophthalmology training. Free text box responses further supported this, as reflected in one respondent's comment: *"The high competition hasn't dissuaded me because I see ophthalmology as a fulfilling and balanced career."*

Conclusion: The findings highlight an intriguing paradox within ophthalmology training: while medical students express significant interest in the speciality due to its lifestyle benefits, work-life balance, and substantial impact on patient quality of life, their experiences in medical school often fail to provide adequate exposure or clinically relevant teaching. This disconnect could hinder early engagement and preparation for a competitive speciality.

Despite these challenges, the survey results suggest that students are not deterred by the high competition ratios. Their resilience likely stems from the perceived benefits of a career in ophthalmology, including lifestyle advantages and the speciality's significant role in improving patients' quality of life. As competition ratios rise and the number of training posts remains static, the potential for a bottleneck in applications is concerning.

This study underscores the positive perception of ophthalmology among medical students, shaped by its lifestyle, work-life balance, and transformative impact on patient outcomes. However, the lack of adequate exposure and clinically relevant teaching during medical school undermines this enthusiasm. HEE must prioritise actionable interventions to ensure the speciality remains attractive and accessible to future trainees while maintaining high standards of training and care. By doing so, the field can capitalise on its growing appeal and continue to meet the increasing need for ophthalmologists in clinical practice.

Conflicts of interest: None

Funding : None

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P189

LEARNING BEST PRACTICE FROM THE ORGANISERS OF OPHTHALMOLOGY POST GRADUATE REGIONAL TEACHING ACROSS THE UK: A QUALITATIVE APPROACH

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Introduction: Ophthalmology post-graduate teaching is organised on a regional (divided by Deanery) and local basis. Much of the learning undertaken by trainees throughout the course of their training is self directed, and based on clinical experience. Therefore, these planned learning events are unique opportunities for trainees be taught in a structured way, supported by

senior clinicians, acting also as educators. Despite the importance and complimentary nature of this style of learning, there is no unified guidance or model for the organisation and delivery of post-graduate Ophthalmology teaching across the UK.

Therefore, this project aims to clarify exactly how different Deaneries organise and deliver Ophthalmology teaching, and what they have found works well. We hope that by canvassing the experiences of educators across the country and understanding what has been of value in different settings, we can identify themes of best practice to help improve the delivery of regional teaching.

Research Questions

1. How is Ophthalmology Regional Teaching delivered by different deaneries across the UK?
2. What aspects have organisers of Regional Teaching found work well?

Method: Data was collected from the Ophthalmology consultants/trainees who lead the organisation of Regional Teaching in each Deanery. There were two elements to data collection: an online questionnaire and semi-structured interviews. The interviews were conducted online on Microsoft Teams, recorded and transcribed by author FT.

Reflexive thematic analysis was used to analyse the interview and survey responses. The data was coded with the research questions in mind, and themes were identified.

Results: We received survey results from 18 out of the 26 regions across the UK, equating to a 69% survey response rate. 12 had teaching on a weekly basis, 3 on a monthly basis, 2 on a 2–3 monthly basis and one deanery had both weekly and monthly teaching. The monthly teaching sessions were all face to face (F2F), whereas the weekly teaching sessions were more often delivered in a hybrid format. Two deaneries provided their teaching entirely virtually. The benefits and limitations of virtual, hybrid and F2F teaching was acknowledged by all, but the delivery was influenced ultimately by many factors: resources, geography, inclusivity and leadership.

Content of the teaching was often determined by the consultant leading the session, as opposed to a curriculum based approach. Varied content, pitched at different levels for trainees was aimed for, with aspects such as interactive sessions, guest speakers, simulation and exam preparation frequently noted.

A recurrent theme was engagement and enthusiasm for the teaching programme. Engaged consultants provided good quality, interesting teaching, whilst engaged trainees kept up attendance and the interactive aspects of the teaching. As well as allowing them to get the most from the educational opportunity through involvement in teaching delivery and content. Leadership appeared to be instrumental to this engagement. The teaching organisers took pride in their role and did not see it as a burden.

Another theme from the interviews is the social aspects of F2F teaching. It is a time to compare notes, network and to develop a shared culture with comradery at its centre.

"It's an opportunity for everyone to get in the same room to discuss issues that are going on. People need to know that they're not alone."

Conclusion: Ophthalmology post-graduate teaching is organised and delivered across the UK in a variety of ways. Different models work well, each having their own benefits and limitations.

Conflicts of interest: None

Funding: None

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P190

LOCAL ANAESTHESIA UNDER SEDATION VERSUS GENERAL ANAESTHESIA FOR BRACHYTHERAPY-TREATED PATIENTS

WITH UVEAL MELANOMA: A PATIENT-REPORTED OUTCOME STUDY

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Introduction: Ruthenium-106 brachytherapy for uveal melanoma has traditionally been routinely performed under general anaesthetic (GA) in the UK, however many ocular surgeries are safely performed under local (LA) or regional (RA) anaesthetic. In the absence of clinical guidelines to assist the clinician, patient experience and preference are important factors to consider when choosing the mode of anaesthesia delivery. LA may confer a number of advantages to patients over GA in certain procedures.

Method: Retrospective patient-reported outcomes study. Patients who had at least one Ruthenium106 plaque-related operation performed under local anaesthetic with sedation (LAS) were included in the study. Recruitment was opportunistic and based entirely on clinician availability and patient choice. All participants were interviewed by telephone and comparison of their experiences between LAS and GA documented.

Results: Fifteen patients were included; of these 9 patients (60%) reported no perioperative side effects, while 3 (20%) experienced drowsiness with GA that was either absent or less following LAS. Fourteen patients (93%) would recommend LAS to friends and family and choose LAS again for future operations.

Conclusion: LAS provides a safe, effective and desirable alternative to GA for Ruthenium-106 brachytherapy. This small study demonstrated a strong patient preference towards LAS due to reduced side effects and quicker recovery. Further work is warranted to explore if these benefits are also seen in other ophthalmic procedures.

Conflicts of interest: None

Funding: None

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P191

MIND THE GAP: WHERE IS THE PSYCHOLOGICAL SUPPORT FOR EARLY-STAGE EYE DISEASE PATIENTS?

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Introduction: The psychological burden of ophthalmic conditions extends beyond vision loss, encompassing anxiety, depression, and reduced quality of life. While clinical psychology services exist for CVI (Certificate of Visual Impairment) patients, those in early disease stages—who maintain functional vision but experience significant emotional distress—are often overlooked. Evidence highlights a bidirectional relationship between psychological stress and conditions like Dry Eye Disease (DED), Central Serous Retinopathy (CSR), and Thyroid Eye Disease (TED). Such distress exacerbates disease progression and hinders treatment adherence and decision-making. To address this gap, we propose a nationwide survey to assess the availability of clinical psychological support for early-stage eye disease patients across UK ophthalmology services, aiming to quantify disparities,

identify barriers, and advocate for integrating psychological care into ophthalmic practice.

Method: A meticulously designed questionnaire will be distributed to eye care professionals across NHS trusts in the UK. The survey encompasses five distinct domains:

1. **Demographics and Clinical Practice:** Capturing professional roles, years of experience, and subspecialty focus to contextualize responses.
2. **Perception of Psychological Impact:** Assessing the extent to which psychological challenges, such as anxiety and social isolation, impact patients' well-being and clinical outcomes.
3. **Availability of Psychological Support:** Quantifying the existence and accessibility of psychological resources for non-CVI patients, including in-house psychologists, peer support groups, and referral pathways.
4. **Impact of Resource Gaps:** Exploring how the absence of psychological services affects consultation efficiency, patient satisfaction, and holistic care delivery.
5. **Strategies for Improvement:** Soliciting feedback on potential interventions to bridge the identified gaps.

The survey is designed to uncover variability in service provision and elucidate barriers such as limited funding, inadequate clinician awareness, and patient reluctance. The findings aim to serve as a catalyst for developing targeted business cases advocating for the introduction of clinical psychological services tailored to the unique needs of ophthalmology patients.

Results: We anticipate the survey will reveal stark inconsistencies in the provision of psychological care across NHS trusts, with significant disparities between services available for CVI registered patients and those in the early disease stages. Preliminary observations suggest that stress-induced exacerbation of ophthalmic conditions—particularly DED, CSR, and TED—is insufficiently addressed, despite its profound implications for both patient outcomes and healthcare efficiency. We also expect to identify key barriers, including long waiting times for referrals, lack of integrated mental health services within ophthalmology departments, and the limited awareness among clinicians regarding available psychological resources.

The results are expected to emphasize the need for proactive psychological intervention strategies. Findings will likely indicate that the absence of psychological support leads to prolonged consultation times, reduced adherence to treatment protocols, and delays in decision-making for critical interventions such as surgery. The survey findings will provide robust data to inform business cases advocating for the integration of clinical psychology services, tailored specifically for ophthalmology. This approach has the potential to transform patient care by addressing both the physical and psychological dimensions of ophthalmic disease.

Conclusion: The psychological needs of early-stage eye disease patients represent a critical and largely unmet dimension of ophthalmic care. This planned survey seeks to uncover systemic inequities in psychological service provision and provide evidence to support the integration of mental health care into routine ophthalmology services. By addressing this pressing gap, we aim to enhance patient outcomes, optimize clinical workflows, and ensure a truly holistic approach to managing eye disease. This initiative not only underscores the urgency of integrating psychological care into ophthalmology but also sets the stage for advancing patient-centred innovation in the field.

Conflicts of interest: None.

Funding: None

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P192

OCULOMICS FOR CARDIOVASCULAR DISEASE: A REVIEW OF ARTIFICIAL INTELLIGENCE APPLICATIONS USING RETINAL IMAGING FOR SCREENING AND RISK STRATIFICATION

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Introduction: Oculomics is a novel field aimed at identifying ocular biomarkers of systemic disease, often with the use of artificial intelligence (AI) models. The aim of this review was to evaluate the current use of AI in retinal fundus photography (RFP) and optical coherence tomography (OCT)/optical coherence tomography angiography (OCT-A) for the prediction and risk stratification of cardiovascular disease (CVD).

Method: A literature search was conducted on PubMed, Web of Science, Scopus, IEEE Xplore and medRxiv for studies published between 1st January 2014 and 31st January 2024. Screening was carried out according to the predefined eligibility criteria.

Results: A total of 15 primary studies were selected for this review, of which 11 reported the use RFP, 2 reported the use of OCT/OCT-A, and 2 reported the use of both modalities in the development of their AI models. Reviewed outcomes include: (1) Prediction of CVD markers (2) Prediction of CVD risk factors and risk scores (3) Prediction and risk stratification for incident CVDs, CVD events, and CVD mortality.

For predicting CVD markers such as retinal vessel-calibre, retina vasculometry, and retinal age gap, the reviewed models showed excellent agreement via intraclass correlation coefficients (ranging 0.823–0.948), Bland-Altman plots, and mean average error (3.55). Author-proposed markers such as RetiCAC, Reti-CVD, and DL-FAS showed moderate to high area under receiving operating characteristic curves (AUROC; ranging 0.729–0.843) in predicting coronary calcium scores and atherosclerosis.

For predicting CVD risk factors such as age, blood pressure, HbA1c etc., both RFP and OCT/OCT-A models demonstrated comparable accuracy (ranging 0.605–0.863) and AUROCs (ranging 0.686–0.970) against traditional risk calculators.

For the prediction and risk stratification of incident CVDs, CVD events and CVD mortality, AUROCs (ranging 0.662–0.720) and C-statistics (ranging 0.543–0.810) were generally moderate to high. Hazard ratios (95% CI; ranging 1.12–8.83) also showed positive associations between model predictions and CVD outcomes.

Whilst RFP models performed consistently well, OCT/OCT-A models yielded inconsistent findings. One study reported poor AUROCs (ranging <0.5–0.59) for prediction of CVD events, whilst another showed good prediction rates (ranging 63.05–96.23%) for CVD risk scores.

Convolutional neural networks were the most prominent neural network employed across the included studies, featuring in 9 out of 12 studies. Other deep learning architectures include masked autoencoder, variational autoencoder, and fully convolutional network.

Conclusion: The models reviewed generally demonstrated good performance and substantial promise, especially for RFP models. However, further studies with larger, more heterogenous datasets and external validation are needed for stronger evidence and generalisability, particularly in OCT/OCT-A models.

Conflicts of interest: None

Funding: Dr Nay Aung and Dr Tao Yang acknowledge the funding support from QMUL Impact Fund. Dr Nay Aung also acknowledges the funding support from the Medical Research Council Clinician Scientist Fellowship (MR/X020924/1).

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P193

OPHTHALMOLOGY AS A SPECIALTY – A GROWING OR WANING INTEREST?: A MULTICENTRE EVALUATION OF TRENDS IN INTEREST AMONG UK MEDICAL STUDENTS IN LONDON

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Introduction: Ophthalmology is arguably an overlooked speciality within UK medical school curricula, despite accounting for the largest proportion of outpatient appointments over the preceding decade across all medical specialities (Veeramani et al., 2024). Whilst ophthalmology remains highly competitive in the UK (NHS England, 2023), there remains a lack of clarity of understanding within medical students to manage ophthalmic conditions. This stems from the current non-mandatory status of ophthalmology in the UK undergraduate medical curriculum, coupled with a paucity of data assessing individual medical student's competency levels in ophthalmology (GMC, 2018). This study aims to evaluate the proficiency and perceptions regarding undergraduate ophthalmology education among UK medical students.

Method: During the academic year of 2023/2024, London-based UK medical students were invited to complete a cross-sectional questionnaire. This was disseminated through official university emailing lists and social media. The questionnaire consisted of 13 questions evaluating various themes including respondent's demographics, ophthalmology exposure, level of ophthalmology knowledge, and career interest. Statistical analysis was performed via Mann-Whitney U-Test using IBM SPSS software.

Results: In total, 126 medical students completed the survey (54.0% females; 4 London medical schools represented). The grade of the medical students ranged from final-year (16.2%), fourth-year (33.3%), third-year (19.2%), pre-clinical years (31.3%). Forty-four per cent of students reported completion of their clinical ophthalmology rotation. Although individuals who completed their rotation demonstrated a significantly higher level of confidence compared to those who had not—particularly in conducting fundoscopy and managing common eye conditions such as acute red eye, acute angle-closure glaucoma (AACG), and diabetic and hypertensive retinopathy (all $p < 0.05$)—33.3% still reported low confidence in managing conditions like AACG. 79.5% of respondents who completed their rotations still perceived their ophthalmology exposure as insufficient. 63.6% of all respondents indicated career interest in ophthalmology.

Conclusion: Our study findings highlight a significant appetite for ophthalmology education and specialty interest, underscoring a gap educators may hope to bridge through spiral revision of ophthalmology and fostering constructive discourse regarding potential curriculum changes within the medical education community. Current teaching modalities have contributed to increased confidence levels towards ophthalmology, however, we

contend that there is significant potential for further enhancement, particularly regarding the amount of time devoted to ophthalmology as a specialty within the medical curriculum.

Conflicts of interest: The authors declare no conflict of interest.

Funding: None

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P194

PATIENT-REPORTED OUTCOME MEASURES (PROMS) AND PATIENT-REPORTED EXPERIENCES MEASURES (PREMS) IN GRANULOMATOSIS WITH POLYANGIITIS (GPA) MANAGED WITH RITUXIMAB

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Introduction: Granulomatosis with polyangiitis (GPA) is a rare autoimmune disease characterised by granulomatous inflammation of small- to medium-sized blood vessels. Ocular symptoms occur in 16–78% of patients and are the initial manifestation in 27% (1). Rituximab, a monoclonal antibody targeting CD20-positive B-cells, is effective for relapsing/refractory disease. Patient-reported outcome measures and experience measures (PROMs/PREMs) tools offer valuable insights into the impact of treatment on quality-of-life (QoL), symptom burden, and satisfaction with care. This data is vital for personalised, effective care, yet published studies are scarce. This survey aims to evaluate PREMs and PROMs of Rituximab therapy for orbital GPA.

Method: Validated PROMs and PREMs questionnaires were adapted for GPA and Rituximab treatment, with 36 questions on symptoms, adverse effects, and QoL. Patients with at least 6 months of Rituximab treatment for orbital GPA were surveyed via standardised telephone interviews conducted by a single author.

Results: 6 patients were identified and completed the questionnaire. The cohort comprised of 3 male and 3 female participants; ages ranged from 52 to 70 years.

Orbital GPA symptoms: Proptosis was observed in 50% (3 patients) prior to starting rituximab therapy. Notably, no participants reported new or worsening of eye bulging/swelling whilst undergoing Rituximab therapy. 2 patients (33.3%) reported diplopia, though its impact on their activities of daily living was reported as minimal. Since starting rituximab 83.3% (5 patients) reported no episodes of discomfort or pain, the same proportion (83.3%) also reported no worsening of vision. One patient demonstrated worsening vision due to progression of cataract. Despite Rituximab therapy, 66.7% (4 patients) suffered from sinus or nasal symptoms secondary to GPA.

Rituximab side effects: Rituximab was generally very well tolerated. Mild infections such as seasonal flu were noted in 50%, whereas 50% reported no infections. Mild cases of fatigue, muscle and joint stiffness and hair-loss were reported by 1 patient. No infusion-related reactions, skin reactions or gastro-intestinal symptoms were reported.

Rituximab compared to previous therapies: Among the participants, 2 patients had initiated Rituximab as the first-line therapy for GPA. Of the remaining 4 patients, 3 had previously been managed with Cyclophosphamide, 2 with Azathioprine, 1 with Methotrexate and 1 with Mycophenolate Mofetil. Participants unanimously reported favourable outcomes with Rituximab. All patients found Rituximab more effective, with fewer side effects. Additionally, all felt it managed the duration of symptoms

much better. All reported a significantly positive impact on daily life and a greater willingness to continue Rituximab treatment compared to the previously trailed treatment(s).

Patient-reported experience of Rituximab treatment: Patients uniformly felt well informed, involved in decision making and aware of potential risks and side effects prior to initiation of Rituximab. All found it easy or very easy to access information regarding Rituximab. All participants felt scheduling infusions was very convenient. During infusions, 100% reported they felt very comfortable and staff communication was exceptional. Transfusions were timely with no reported delays. Participants highlighted Rituximab therapy had a very positive or positive impact on their QoL and all were either very satisfied or satisfied with Rituximab therapy overall. However, 50% (3 patients) of participants reported their emotional needs were only somewhat addressed by the healthcare team. Furthermore, 66.7% (4 patients) were unaware of support services such as counselling, suggesting there are areas of care that can be improved.

Conclusion: Rituximab effectively managed ocular and orbital symptoms of GPA in this cohort, stabilising proptosis but not preventing nasal or sinus symptoms. No significant adverse effects were reported. Patients overwhelmingly praised the treatment experience and healthcare team support during consultations and infusions. Enhancing emotional support and signposting services could further improve patient experience.

Conflicts of interest: None

Funding: None

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P195

REACHING THE UNREACHED: A JOURNEY OF ARUNODAYA CHARITABLE TRUST (ACT)

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Introduction: Arunodaya Charitable Trust (ACT), founded in 1990 by Drs. Reena and Arun Sethi, a couple of ophthalmologists, has embarked on a remarkable journey aimed at providing quality eye care to underserved communities in India. The Trust's idea was to create a community-based eye care project where private practitioners could offer their expertise, time and resources to help those who would otherwise be deprived of proper medical attention.

The couple began by offering free consultations and eye care services under a banyan tree in a village on the outskirts of New Delhi. They believed that providing quality eye care was not just a matter of medical intervention, but a way of improving the lives of entire communities by offering access to services they could not afford.

Method: ACT's mission was aimed specifically at the economically weaker sections of society. In the early stages of the project, the Trust's services were offered entirely on a voluntary basis. However, the couple quickly realized that the vast need for quality eye care couldn't be addressed by one team alone. A scalable, sustainable model was needed—one that could bring in multiple stakeholders and support long-term growth.

The Trust's core objective extended beyond just offering treatment. ACT focused on creating a system that included outreach, education, and awareness. Their goal was to make eye care both accessible and affordable, especially for those in rural areas and

urban slums who had little to no access to medical services. They also recognized the importance of educating communities on eye health, preventative care, and early detection.

ACT's outreach model involved conducting free eye check-up camps in various villages and remote areas. These camps provided basic diagnostics and distributed spectacles and medications where needed. As the initiative gained momentum, ACT expanded its services to cover more advanced treatments, including cataract surgeries and care for conditions like glaucoma, diabetic retinopathy, and other eye disorders. These services were offered either free of cost or at a nominal fee, depending on a patient's financial capacity.

Results: The impact of ACT's work is evident in the numbers. To date, over 1.6 million patients from underprivileged backgrounds have benefited from the Trust. ACT's outreach has expanded across rural communities, urban slums, and low-income neighborhoods in and around New Delhi.

Beyond direct medical services, ACT has made a point of training local community workers and volunteers. These individuals play a vital role in ensuring the continuity of the Trust's work. They assist at eye care camps, help spread awareness about eye health, and support the logistical operations that make ACT's efforts possible. This community involvement has been essential in creating sustainable and lasting change.

ACT has also built partnerships with other organizations, government departments, and corporate entities to strengthen and expand its impact. These collaborations have allowed the Trust to serve more people and extend its reach to new areas.

Conclusion: ACT's journey—from treating patients under a banyan tree to becoming a leading provider of eye care for the economically disadvantaged—is a testament to the power of vision, compassion, and sustained commitment. What started as a personal mission by two dedicated doctors has grown into a significant movement that has set an example for community-based healthcare in India.

The Trust's work shows that grassroots efforts, when combined with professional expertise and a clear purpose, can create long-lasting positive change. ACT's story is one of reaching the unreached, breaking healthcare barriers, and empowering communities. By focusing on equitable access, the Trust continues to prove that the difference between blindness and sight often lies not in wealth or location, but in opportunity.

Conflicts of interest: None

Funding: Self/Institution

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P196

SERVICE EVALUATION OF A BESPOKE VULNERABLE ADULT OPHTHALMOLOGY CLINIC: SUCCESS AND OPPORTUNITIES AFTER 7 YEARS

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Introduction: Adults with special needs are living longer and have unmet needs for their eyecare. Current estimates for the UK population state that there are 944,000 people with dementia (1 in 11 over 65 years old) and 1 million with learning disability, 60% of whom have a refractive error and almost 10% with partial or severe visual impairment. The Vulnerable Adult Ophthalmology Clinic (VAOC) was set up to address the needs of adults with special needs (including purely physical) which cannot be adequately met in a "typical" outpatient clinic setting, where

space, noise and time afforded to these patients and their carers is not in keeping with their needs. This started in 2017 and this is an evaluation of this service 7 years on.

Method: Retrospective review of 73 consecutive patients referred to the VAOC, documenting demographic data, source of referral, time to first VAOC, systemic diagnoses, ophthalmic diagnoses, first visit outcomes, type and timing of any interventions.

Results: Of the 123 patients known to the VAOC, the data for 73 was available for submission. 35 males, 38 females, average age 71.5 years (101-21). 57 were internal referrals from other clinics in the eye department. Systemic diagnoses were either singular or combinations of neurological disorders (dementia, Alzheimers, Parkinson's, post stroke), learning disability, autism, severe psychiatric illness and/or personality disorders, syndromic diagnoses, and severe musculoskeletal disorders. Time to first appointment in the VAOC ranged from 4 days to 100 weeks (average 24 weeks). Ophthalmic diagnoses were singular or a combination of glaucoma, wet AMD, need for diabetic eye screening, cataract and the lack of suitable community refraction venues. Diagnostic imaging and/or biometry was possible on 33 patients and 6 required a combination of examination under anaesthetic (EUA) and/or cataract surgery and/or retinal laser.

Conclusion: The bespoke nature of the VAOC in terms of space, duration of appointment slots and allocated staff has been very well received by patients and carers usually after much less satisfactory experiences in more confined, busier and over running conventional eye clinic settings. As the clinic requires the presence of designated doctor, orthoptist, optometrist, technician and clinic nurse, it has been limited to a once a month slot. This coupled with the delays due to the pandemic has heavily impacted the time from referral to first appointment. When patients need an intervention, a further problem has been identified causing delays due to the lack of a formal pathway for anaesthetic assessment of those patients needing an EUA and/or other surgical or procedural interventions. This delay is prolonged further due to the lack of adequate theatre capacity, especially as patients with special needs needing an EUA, diagnostics and an interventional procedure do take up disproportionately longer theatre time, not conducive to the current metrics of financial viability of a given clinical session. The full data set will be used to inform a specialist commissioning of this service given the increasing demand post pandemic competing with existing back logs and increasing demand on conventional capacity.

Conflicts of interest: None

Funding: None

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P197

SIMULATION TRAINING IN OPHTHALMOLOGY: THE PAN-LONDON EXPERIENCE

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Introduction: Surgical training in Ophthalmology has been negatively impacted during COVID and the post pandemic era. The effect of simulation-based training has been extensively investigated and multiple studies have identified the need for simulation to enhance surgical competence and in managing complications.

In 2023, Health Education England has invested in funding a one-year Ophthalmology Educational Fellowship based at Western Eye Hospital, Imperial College Healthcare NHS trust as part of the training recovery fund. The main role of the Educational fellow was to lead on and deliver surgical simulation courses targeting Pan-London Ophthalmic Deanery specialty trainees.

In this review, we summarise 6 simulation courses delivered to PanLondon ophthalmic specialty trainees and analyse the impact on their confidence levels in performing laser/surgical techniques.

Method: Six simulation courses have been delivered in total during the year 2023/2024 at Western Eye Hospital: two laser simulation courses, two basic surgical courses and two advanced surgical simulation courses. Anonymous Likert scales of confidence have been utilised and distributed before and after delivering each course to assess the improvement of confidence levels of trainees for each laser/surgical skill simulated in the course.

Results: A total of 79 pan-London ophthalmic specialty trainees (OST1-OST7) attended our courses throughout the year. 28 trainees attended the two laser courses on 24th September 2023 and 9th June 2024 showing a 138% and 64% increase respectively in confidence levels performing YAG laser capsulotomy, YAG laser iridotomy and selective laser trabeculoplasty. 27 trainees (OST1-OST4) attended the two basic surgical courses on 12th December 2023 and 23rd February 2024 showing a 50% and 67% increase respectively in confidence levels performing basic surgical simulation skills.

24 trainees (OST5-OST7) attended the two advanced surgical course on 23rd January 2024 and 22nd March 2024 showing a 98% and 57% increase respectively in confidence levels performing advanced surgical simulation skills.

Conclusion: Simulation-based ophthalmic surgical training significantly enhances confidence levels of training in all procedures. The highest confidence rates were for laser procedures for junior residents and for advanced surgical skills for senior residents. Simulation training has a favourable role in skills attainment confidence regardless of procedure or level of training.

Conflicts of interest: None

Funding: None

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P198. Abstract not included

P199

THE ROLE OF A SIMULATION CHAMPION IN FACILITATING CLINICAL SIMULATION SESSIONS, THE KING'S EXPERIENCE

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Introduction: Clinical simulation is a transformative educational strategy in healthcare, providing realistic, risk-free environments for skill acquisition, decision-making, and team collaboration. As simulation programmes gain prominence, the role of a simulation champion has emerged as vital to their success. Simulation champions are healthcare professionals who guide and enhance simulation-based education by fostering engagement, managing logistics, and ensuring alignment with curricular goals. We outline the essential responsibilities and impact of the simulation champion at King's and their contributions to education and patient safety.

Since its inception in August 2023, the Simulation Suite hosts EyeSi cataract surgery simulator, four microscopes, two complete

refraction trial lens sets, surgical instruments, oculoplastics materials for suturing including drapes, and Philips Studios model eyes to practice a host of surgical techniques including corneal suturing, trabeculectomy, intraocular lens insertion, and specially designed Topcon model eyes for retinal laser are available to practise. The room is equipped with a sharps bin to ensure the safe disposal of needles and other sharp instruments. Specialist equipment is available to residents and fellows for advanced surgical skills: corneal graft surgery, tube surgery and Preserflo insertion. The role of the Simulation Champion has been critical in achieving educational and training targets.

Method: The multifaceted roles of simulation champions were examined, drawing from existing literature and practical case studies in ophthalmology simulation. Key functions such as scenario development, facilitation techniques, technological management, and post-simulation debriefing were analysed. Emphasis was placed on the champion's ability to create psychologically safe learning environments, promote reflective practices, and advocate for institutional adoption of simulation. Outcomes were measured through learner feedback, program effectiveness, and alignment with educational objectives. Within the role of the simulation champion, time spent on the EyeSi has been analysed and anonymous Likert scales of confidence questionnaires distributed before and after delivering each simulation event to assess the improvement of confidence levels of Residents in skills attainment.

Results: Over 71 hours of EyeSi operating has been undertaken in the last academic year 2023–2024. 29.5% of this time was undertaken by ST1s. We ran 6 simulation sessions in all subspecialties with feedback.

Simulation champions fulfil several critical functions:

Scenario Development: Champions collaborate with multi-disciplinary teams to design realistic, high-impact clinical scenarios tailored to learners' needs.

Facilitation: Adopting a learner-centred approach, champions guide participants through simulations, encouraging active engagement and critical thinking.

Debriefing Expertise: Effective debriefing, led by champions, facilitates reflection, allowing learners to identify strengths and areas for improvement.

Technological Proficiency: Champions ensure the smooth operation of simulation tools and troubleshoot technical issues, enhancing the learning experience.

Program Advocacy: Simulation champions promote integration of simulation into institutional curricula, demonstrating its benefits in improving clinical skills and patient outcomes.

Learner feedback consistently highlights the role of champions in creating supportive environments and enhancing the authenticity of simulations. Programs led by a skilled champion report higher satisfaction rates, improved learner confidence, and increased retention of clinical competencies as per analysed questionnaires.

Conclusion: Simulation champions are essential facilitators in ophthalmology education, bridging the gap between theoretical knowledge and practical application. Their expertise in scenario design, facilitation, and debriefing fosters an effective and immersive learning environment. By ensuring alignment with educational goals and advocating for simulation's value, champions contribute to improved patient safety and care quality. The findings underscore the need for ongoing training and support for simulation champions, empowering them to adapt to technological advancements and evolving clinical demands. Institutions investing in the development of simulation champions are better equipped to prepare healthcare professionals for the complexities of modern practice. We share our experience and advocate for simulation Champions in ophthalmology units.

Conflicts of interest: None

Funding: None

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P200

THE SAFETY PROFILE OF LONG-TERM INTRAOCULAR OPHTHALMIC VISCOELASTIC DEVICES

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Introduction: Since first being introduced over 50 years ago, ophthalmic viscoelastic devices (OVDs) have revolutionised the fields of cataract, cornea and glaucoma surgery, with extensive published anterior segment safety data available. More recently, these devices have also gained popularity in posterior segment surgery, with particular focus on their potential use as long-lasting, non-toxic, biocompatible, and biodegradable vitreous substitutes. The most commercially available products are based on hydrophilic polymers such as hyaluronic acid (HA), hydroxypropylmethylcellulose (HPMC) and chondroitin sulphate (CS). This remains a novel area of research, with very few studies examining their long-term safety profile.

Method: We conducted a literature review on Medline and Embase via OVID, using key words “viscoelastics” and “intraocular”. Both databases were searched from inception date, with no language limitations.

Results: There is a paucity of high-quality evidence addressing the long-term safety of OVDs in the eye. A total of 38 studies were included for final review. These studies were unsuitable for any direct head-to-head comparisons, therefore a descriptive analysis is presented.

Conclusion: By far, HA-based OVDs have been the most well described, with several long-term intraocular applications. They display highly favourable biophysical properties with minimal adverse reactions in short and long-term in vivo, in vitro and animal studies. There is limited safety data on HPMC and CS-based products, despite their popularity. This represents a gap in the literature and invites the need for high quality evidence to support their continued use, particularly in the realms of vitreoretinal surgery and chronic hypotony.

Conflicts of interest: None

Funding: None

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P201

THE USE OF ARTIFICIAL INTELLIGENCE (AI) IN OPHTHALMOLOGY WHITE SPACE QUESTIONS IN SPECIALTY JOB APPLICATIONS: AN ETHICAL DILEMMA

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Introduction: The advent of artificial intelligence (AI) in recent years has generated excitement and debate regarding its potential to transform industries and influence new technologies. As such the medical field has been no exception to such advancement in this time. In particular AI applications such as ChatGPT, Google Gemini and Meta AI have been developed,

improved and widely used for their ability to produce instant and in-depth responses to any question posed. In this study, we examine the use of AI in answering white-space questions (WSQs) in the specialty application for ophthalmology and the ethical ramifications of using such an accessible tool. In the first instance, AI generated responses risk not accurately reflecting a candidate's specific qualifications or skill set and may otherwise present misleading information.

Method: Data was collected from applications shortlisted for the vacancy of specialty doctor in ophthalmology within the West Hertfordshire Teaching Hospitals NHS Trust in the 2024 application window. From these applications, data was collected from white space questions (WSQs) related to procedures, leadership, teaching, research, team work and supporting information within the application. These answers were analysed and scanned using the GPTZero application Quick AI scan function. WSQ answers that returned with a more than a 10% AI generated score were then re-scanned using the Advanced AI scan function to verify.

Results: Over the 2 application cycles in 2024 to the specialty doctor in ophthalmology job vacancy, 96 individual applications (n = 96) were shortlisted and reviewed for use of AI to generate responses to white space questions (WSQs). Of these 96 applications, 20 were reviewed to have at least 25% of WSQ content to be AI generated. Of those 20 flagged applications 14 returned with at least 50% of WSQ content to be AI generated and 7 individual applications showed 100% of WSQ answers to be AI generated. From this data collection over 7.3% of all shortlisted applications in 2024 that were reviewed demonstrated 100% AI generated WSQ answers.

Conclusion: In summary, the review and discovery of specialty applicants using AI such as ChatGPT to generated answers to WSQs in the application process poses an ethical and potential probity dilemma. At present no stipulation or regulations are in place that explicitly prohibit the use of such AI tools for applicants to generate answers to specialty application WSQs. However, the ethical issue presented is that a false representation of the applicant may be read on paper (as generated by AI) that does not align with the doctor seen at interview in person. In this regard an applicant that did not use AI of any kind could lose their spot at interview to an applicant that has used entirely AI generated WSQ answers. Therefore, clear and robust regulations are necessary across all specialty application processes to resolve such a dilemma at shortlisting. This could include amendments to making WSQs more specific ensuring an adequate depth of understanding in responses, declarations of transparency and formal AI monitoring tools.

In a wider context, attitudes towards AI use especially within education may see change where AI is used as a tool to guide content creation by both students and teachers alike, rather than generate the finished product. With AI use becoming only more prevalent, an adaptation to application process policy is pertinent to address and guide its appropriate use.

Conflicts of interest: None

Funding: None

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P202

TRANS-SCLERAL ESSEN FORCEPS UVEAL BIOPSIES: HOW MANY SAMPLES ARE NEEDED TO ENSURE DIAGNOSTIC SPECIMENS?

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Introduction: The majority of uveal tumours and tumour-mimics are diagnosed on clinical examination and non-invasive imaging techniques alone, without the need for a tissue diagnosis. However, some uveal lesions present a diagnostic challenge. In these cases, intraocular diagnostic biopsy may be necessary.

Recently, the trans-scleral Essen Forceps biopsy (TSEB) technique has been shown to improve diagnostic yield without increasing complications. The technique involves taking a number of tissue samples, at the surgeon's discretion, to ensure adequate lesional tissue is harvested for analysis. Each pass of the biopsy forceps into the uvea causes trauma and risks complications such as inadvertent retinal or vitreous sampling. This study aimed to establish the optimum number of samples required to achieve a tissue diagnosis when performing TSEB, and to see if the surgeon could tell when diagnostically useful samples had been harvested.

Method: A prospective study was performed of all patients undergoing TSEB of uveal tumours at our national ocular oncology centre between January 2023 and November 2024. Data on patient demographics and lesion dimensions were collected. TSEB was performed as per published techniques, with one exception; the sample from each pass of the Essen forceps biopsy was collected and fixed in formalin separately (e.g. first pass, second pass etc.). Intraoperatively, the surgeon stated which sample they believed would be diagnostic. Data regarding diagnostic yield, histopathological findings, and postoperative outcomes were collected.

Results: In total, 10 patients were included in the study: 6 male and 4 female patients with a mean age of 58 years (range: 26–87). Tumours were within the ciliary body in seven cases, and within 10 mm of the limbus in two cases. Only one was posteriorly located >15 mm from the limbus. Tumours had a mean maximal thickness of 4.5 mm (range: 2.3–8.5 mm) and a mean basal diameter of 10.3 mm (range: 4.8–17.5 mm). All patients underwent TSEB for diagnostic purposes, with an average of three tissue bites taken per case (range: 2–5). Transillumination shadows were visible intraoperatively in 6 patients, aiding lesion localisation, two patients had 360-degree ciliary body involvement.

In all cases, the surgeon thought the first pass of the TSEB yielded diagnostic tissue. Histological diagnosis was made on the first pass in 8 of 10 cases. In 90% of cases, histological diagnosis was made on the first or second pass. Only one case was nondiagnostic on all samples: three bites were taken, revealing paucicellular biopsies, suggesting chronic inflammation but with insufficient tissue for a definitive diagnosis. This case had 360-degree ciliochoroidal effusions and no distinct mass or tissue to sample. No cases benefitted from more than two samples, though in two cases where the first sample was diagnostic, the second sample improved cytological clarity.

Final diagnoses included uveal melanoma ($n = 4$), melanocytoma ($n = 3$), inflammation ($n = 2$), and lymphoma ($n = 1$). No intraoperative or postoperative complications were reported across the cohort.

Conclusion: These results demonstrate that the first two tissue samples with TSEB are sufficient for achieving a diagnosis in the majority of cases (90%). Where the lesion is easily identifiable, the first bite is usually diagnostic. This study shows a high degree of correlation between surgeon confidence of tissue yield and histopathological diagnosis. Though there was no evidence to suggest taking multiple tissue bites increased the risk of intra- or postoperative complications, this study reassures surgeons that two good samples from TSEB are sufficient for diagnosis and they can avoid unnecessary further passes into a traumatised surgical site, minimising the risk of complications.

Conflicts of interest: None

Funding: None

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P203

TREATMENT ALLOCATION IN OPHTHALMOLOGICAL RANDOMISED-CONTROL TRIALS (TAO-RCT): A FREQUENTLY UNCONTROLLED AND POORLY REPORTED PARAMETER

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Introduction: Randomised-control trials (RCTs) in ophthalmology are complicated by most patients having two eyes. Investigators can allocate interventions and controls in different ways; participants can be randomised to unilateral or bilateral treatment, the eyes of each participant may be treated differently (providing a convenient paired control), or a single eye from each participant may be included and randomised. Further complication arises from pathologies affecting one or both eyes, and study endpoints involving one or both eyes. Here, a representative sample of ophthalmological RCTs were appraised to explore the relative frequency of 'uncontrolled' treatment allocation: where there is an inconsistent relationship between participant and intervention, such that relationships between individuals' eyes may complicate or compromise analysis.

Method: A systematic review was undertaken to identify all ophthalmological RCTs published between 1 January 2022 and 31 December 2022. The Cochrane Library, Embase, and MEDLINE were searched. Quality assessment was performed using the Joanna Briggs Institute Checklist for Randomised Controlled Trials. We developed a nomenclature system to describe treatment allocation: 'A' (and subsequent letters) denote treatments (including placebo), 'X' denotes no treatment, '+' between letters used between contralateral eyes within an individual, and '&' used to separate individuals. The frequency of different treatment allocation patterns was calculated, in addition to the relative clarity of reporting (either 'explicit', 'implicit', or 'unclear'). p -values for the primary outcome were analysed for variation across allocation patterns to determine whether uncontrolled patterns were more likely to yield statistically significant results. Further analysis was undertaken to explore potential risk factors for uncontrolled treatment allocation, including funding source, country and region of origin, and subspecialty.

Results: Of 3687 studies identified by the search strategy, 359 were ultimately included. The three most frequent treatment allocation patterns were controlled: A + A & B + B; A + X & B + X; and A + B. The most frequent uncontrolled allocation patterns were A + X & B + X & A + A & B + B; and A + A & B + B & A + X & B + X. Uncontrolled treatment allocation was present in 14.5% of RCTs in 2022 and was associated with unclear reporting ($\chi^2 = 44.31$, $p < 0.0001$).

Uncontrolled treatment allocation varied significantly with funding source ($\chi^2 = 17.65$, $p = 0.024$), but not specifically with industry funding ($\chi^2 = 2.93$, $p = 0.087$). Uncontrolled treatment allocation was not associated with world region (Fisher's exact test, $p = 0.127$). Uncontrolled treatment allocation was twice as common in surgical trials relative to medical trials (OR = 2.06, $\chi^2 = 4.74$, $p = 0.029$), but was not associated with subspecialty ($\chi^2 = 11.83$, $p = 0.159$).

Conclusion: Uncontrolled treatment allocation is a frequent issue in ophthalmological RCTs and is more common where

reporting is unclear. While uncontrolled allocation patterns were more common in surgical trials than medical trials, there were no other risk factors for uncontrolled allocation; including region of reporting, subspecialty, and funding source. In trials with uncontrolled treatment allocation, statistical analysis may be compromised by an inconsistent relationship between individuals (whose eyes correlate and interact) and interventions. Therefore, the validity and quality of RCTs may be improved by strategies that support reporting and evaluation of allocation pattern. Our nomenclature system may be used in critical appraisal tools or trial registries to facilitate concise and clear reporting.

Conflicts of interest:

Arun Thirunavukarasu has received funding from HealthSense for unrelated research work.

Funding: None.

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P204

VACCINATION AGAINST SHINGLES AND PREVENTION OF LONG-TERM OCULAR MORBIDITY—A RETROSPECTIVE STUDY OF COST IMPLICATIONS TO THE NHS

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Introduction: Shingles is caused by reactivation of latent Herpes Zoster Virus (VZV) consequent to infection with the virus in childhood in the form of chicken pox. This reactivation can occur many years later, usually as a result of declining immunity because of age or immunosuppression. Herpes Zoster Ophthalmicus (HZO) is defined as shingles that involves the ophthalmic division of the trigeminal nerve and so can affect the scalp, the forehead, the eye and the side of the nose, as well as the sinuses, the tear ducts and the intraorbital structures. As HZO is not a reportable condition in the United Kingdom, its incidence is inferred. The lifetime risk of developing shingles is estimated at 25%; HZO is thought to account for 10–20% of all shingles cases. Long term sequelae of ocular shingles infection include keratoconjunctivitis with corneal scarring, relapsing anterior uveitis, retinal and optic nerve involvement, and ocular motor palsies. All these conditions carry a risk of long-term ocular and/or visual morbidity.

A national public sector shingles vaccination program was introduced in the UK in 2013 offering routine vaccination to those over the age of 70 years; the vaccine used was Zostavax as a single injection. More recently, from 1st September 2023, the age for vaccination has been reduced to 65 years. Whilst Zostavax is no longer being manufactured, patients are still receiving this vaccine until the supplies are depleted. Zostavax has now been superseded by the Shingrix vaccine, given as two separate doses. Shingrix is, however, also available to those aged 50 years and over in the private health sector in the UK.

Method: We undertook a retrospective study determining the number of patients aged 50–69 years who had attended our hospital with HZO from 1st April 2022 to 31st August 2023, and aged 50–65 years from 1 September 2023 to 31st July 2024. We determined how many were affected with long term ocular sequelae of infection with this virus and the subsequent costs to the NHS of multiple attendances to casualty/clinics during the course of their treatment.

The costs tallied further included the outpatient/inpatient attendances and the investigative costs (e.g. OCT imaging), but excluded the pharmaceutical costs.

Results: Our study shows that during this 27-month period, we had a total of 48 cases of HZO, with 9 patients (18.8%, approximately 1 in 5) going on to develop long term ocular complications requiring hospital attendance and hence long-term care. The total cost of treatment to the NHS of these 48 cases at our hospital was £41,671.

Conclusion: We concluded that for the cost of treatment of these 48 patients by hospital eye services, and taking into account the previous and current costs of the vaccine, a total of 340 patients could have been vaccinated against shingles (seven times the number of affected patients). Not only is this number significantly more than the total number of patients afflicted by shingles in the time period of our study, but long-term debilitating ocular comorbidity (1 in 5 of this cohort) could also have been avoided, thereby reducing the burden on hospital eye services. This study has not taken into account the further costs incurred to the patient in terms of time off work and travel costs to and from the hospital, to name a few. Although further study is required, it would seem advisable in terms of cost benefit ratio, to reduce the age of vaccination against shingles down to 50 years.

Conflicts of interest: None

Funding: None

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P205

WHICH UK MEDICAL STUDENTS ARE INTERESTED IN A CAREER IN OPHTHALMOLOGY, AND WHY? RESULTS FROM THE FAST STUDY

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Introduction: Ophthalmology is a specialty to which UK medical students often receive relatively limited exposure during their studies. Despite this, levels of competition for entry to its specialty training pathway are among the highest of any specialty, and rising each year. Understanding levels of interest in the specialty among the current medical student population could help to inform future workforce planning efforts.

This investigation aimed to compare the demographics, extracurricular achievements, and factors influencing career interest among UK medical students interested in ophthalmology with those pursuing other specialties. The study also assessed confidence, certainty, and knowledge of the ophthalmology training pathway, with subgroup analyses by year of study, and gender.

Method: This was a subgroup analysis of the Factors Affecting Specialty Training preference (FAST) study, a national, cross-sectional survey of UK medical students. A novel electronic questionnaire was developed and distributed via a network of more than 200 collaborators representing all 44 UK medical schools accredited by the General Medical Council.

Results: A total of 8395 medical students responded to the survey, of whom 180 (2.1%) identified ophthalmology as their preferred specialty. In the overall sample, 3.1% (n = 79) of male participants selected the specialty, versus 1.7% (n = 98) of female participants. 2.5% (n = 33) of final-year students were interested in pursuing ophthalmology, rising from 1.7% (n = 21) of first-year students. 33.3% (n = 80) of ophthalmology aspirants have a parent or sibling in medicine, markedly higher than the national average of 20.5%. Students favouring ophthalmology were more likely than average to have co-authored research publications

(10.0% vs. 6.52%) and hold national or international medical prizes (7.2% vs. 4.2%).

Across all year groups, those expressing a preference for ophthalmology were less certain about their career choice than the national average (35.6% vs. 38.3%) and markedly less confident about their chances of gaining entry to the specialty (15.0% vs. 23.1%). However, they were more likely to self-report feeling well-informed about the training pathway (51.2% vs. 44.8%). Among final-year students interested in a career in ophthalmology, several factors emerged as influential in their decision-making regarding which specialty to pursue. These included lifestyle factors such as work-life balance (93.3% vs. 79% nationally), compatibility with family life (81.8% vs. 76.7%), financial remuneration (81.8% vs. 66.1%), as well as intellectual challenge (81.8% vs. 70.6%) and the future outlook of the specialty (81.8% vs. 74.9%).

Conclusion: Among medical students interested in pursuing a career in ophthalmology, financial and lifestyle factors were identified as being very influential when selecting their preferred specialty. These factors have been highlighted in other similar investigations conducted in the UK and overseas. Higher levels of interest among final-year students could be the result of increased exposure to the specialty through teaching or clinical placements. Though potentially skewed by their seniority, our findings suggest that ophthalmology-inclined medical students may be more involved in extracurricular pursuits than many of their peers. Low self-reported levels of career certainty (35.6%) and confidence in gaining entry to ophthalmology specialty training (15.0%) may have been informed by their familiarity with the training pathway and high competition ratios – which were roughly 14:1 in the most recent application cycle.

Conflicts of interest: None

Funding: None

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NEURO OPHTHALMOLOGY

P206

EVALUATING RETINAL GANGLION CELLS IN INHERITED OPTIC NEUROPATHIES USING PERCEPTUAL TASKS AND FLAVOPROTEIN FLUORESCENCE IMAGING

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Introduction: Inherited optic neuropathies (IONs) are characterised by degeneration of retinal ganglion cells (RGCs). Different types of RGCs serve different visual functions, which can be preferentially driven by perceptual tasks. Midget RGCs subserve high acuity vision and red-green chromatic sensitivity. Parasol cells are associable with fast temporal and coarse spatial frequencies, while small bistratified cells are responsible for blue-yellow sensitivity. The purpose of this study was to investigate patterns of RGC loss in Leber hereditary optic

neuropathy (LHON) and dominant optic atrophy (DOA) using psychophysical regimes, and to explore structural and functional imaging correlates.

Method: Twenty-four participants with a molecular diagnosis of LHON or DOA were recruited into a prospective natural history study. Cross-sectional data from baseline visits available to date is presented. Participants underwent colour assessment using the low vision Cambridge colour test. Temporal and spatial contrast sensitivity functions were assessed using two alternative forced choice tasks. Stimuli were displayed using the Metropsis (Cambridge Research Systems). Additionally, participants were assessed for best-corrected visual acuity (BCVA), ocular coherence tomography (OCT) evaluation of retinal layer thickness (Spectralis, Heidelberg Engineering), and retinal flavoprotein fluorescence (FPF) intensity (Ocumet Beacon, Ocuscience). OCT and FPF assessments were sectorised using Garway-Heath grids. Colour discrimination was decomposed into two directions (red-green and blue-yellow), and grouped using k-means clustering. Contrast sensitivity was analysed using linear mixed models, and the correlation between measures was assessed with linear regression.

Results: Participants had a mean age of 33.5 years (M:F = 17:7). Nineteen had LHON, while the remaining five had *OPA1* DOA. Colour deficits could be clustered into four groups: relative preservation of both red-green and blue-yellow discrimination axes, deficits primarily in only one axis, or deficits in both axes. Red-green deficits were correlated with BCVA ($R^2=0.2937$, $p=0.0009$); blue-yellow were not. Blue-yellow deficits were correlated with age in DOA ($R^2=0.5935$, $p=0.0253$). Participants overall had poorer sensitivity to slow temporal ($p=0.0004$) and fine spatial ($p=0.0063$) patterns. Contrast sensitivity deficits were correlated with BCVA across all spatial frequencies, but only slower (≤ 5 Hz) temporal frequencies. No significant correlations emerged between any psychophysical functional measures and OCT structural measures. FPF assessment was elevated in all optic nerve head sectors compared with normative mean values. No significant relationships emerged between FPF and either OCT or BCVA, but all optic nerve head sectors, except nasal and temporal inferior, were correlated with deficits in sensitivity to coarser (≤ 4 cpd) spatial frequencies.

Conclusion: Clustering of colour deficits suggests that IONs are heterogenous in their manifestation of type-specific RGC deficits. The correlation of red-green colour sensitivity with visual acuity is expected given both assessments reveal primarily midget cell status. The correlation of blue-yellow deficits with age in DOA suggests a relationship between disease progression and small bistratified cell dysfunction. A relative preservation of sensitivity to coarse spatial and fast temporal patterns in both LHON and DOA may indicate the relative preservation of parasol cells. Psychophysical outcomes were not correlated with OCT measurements, perhaps because OCT cannot resolve type-specific structural deficits, and exhibits a floor effect. FPF is a marker of mitochondrial status (via increased fluorescence in states of oxidation) and was expectedly increased: mitochondrial dysfunction being the key pathogenic mechanism in IONs. While FPF intensity was not cross-sectionally correlated with OCT or BCVA, it may have a predictive role that is better revealed longitudinally. These findings together support the further investigation of type-specific RGC deficits in IONs as a possible biomarker of disease status and subgroup. Highlighting surviving RGC types may also help to nominate novel therapeutic targets.

Conflicts of interest:

Patrick Yu-Wai-Man receives consultancy fees from GenSight Biologics, Stoke Therapeutics, PYC Therapeutics.

Funding: NIHR

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P207

TELEMEDICINE EVALUATION OF PRIMARY CARE REFERRALS FOR SUSPECT PAPHILOEDEMA: A 12-MONTH STUDY OF 224 CONSECUTIVE REFERRALSSimon Hardman Lea¹, Lyn Price¹, Claudia Quijano¹¹Evolutio Care Innovations, Reading, UK

Introduction: Since 2016, referrals from primary care to hospital neuro-ophthalmology clinics have increased by 600%, placing a considerable strain on specialist services. One of the diagnostic groups sent for urgent assessment is suspect pathological disc swelling, but previous hospital-based studies have indicated that up to 60% of possible papilloedema patients seen in clinic (from all referral sources) are false positives. This study was undertaken to evaluate how a system of comprehensive data collection in the community with asynchronous ophthalmologist telemedicine assessment could help to address this problem.

Method: The records of all patients referred from primary care as suspect papilloedema/swollen discs in the twelve months December 2023/November 2024 were evaluated. All cases had their initial community assessment performed by trained optometrists, collecting a standard set of observations including visual acuity, colour vision and pupil reactions, threshold field analysis, disc photographs and OCT images, before being scrutinised by asynchronous telemedicine by ophthalmologists. Telemedicine decisions about possible papilloedema -and the need for onward referral for a specialist neuro-ophthalmology opinion -was made on the basis of symptoms and disc appearance (on 2-D photographs) including haemorrhages, cotton wool spots, vascular obscuration by the retinal nerve fibre layer and peripapillary wrinkles, plus nasal retinal nerve fibre layer thickness and disc OCT features.

Results: In the 12 months, 224 patients were referred from primary care with possible swollen discs - 63 were sent in from General Practice, 161 from their own optometrist. 145 had been referred as urgent cases, 79 routine. Ages ranged from 2 to 81 with 29 cases under the age of 16. 78% were female (175/224).

Symptoms of headache were recorded in less than half the group (104/224). No patient had noted transient visual obscurations or diplopia.

After assessment, 35 /224 patients (15.6%) were referred on with possible pathological disc swelling. 10 had been sent in by their GP, 25 from their optometrist. 80% (28/35) were female and 2 were under 16 years old. 18/35 had symptoms of headache. Visual fields were either normal or inconsistent in the 33 patients where performed. Only one case had disc swelling of Frisen grade 2 or worse.

Conclusion: After community assessment with telemedicine, 84% of primary care referrals for papilloedema /disc swelling were false positives and did not require onward hospital referral. There was no difference between GP and optometry referrals in need for referral. Headaches were commoner in the group of patients needing referral on to hospital, but not enough to be a sole discriminator of the need to refer. In this group, visual field abnormalities were not a useful indicator of possible papilloedema. Combinations of signs including nasal retinal nerve fibre layer thickness, obscuration of vessels on the disc and peripapillary wrinkles were more indicative of the need to refer.

Combining community-based data collection with telemedicine assessment, focusing on specific papilloedema features, effectively reduced unnecessary hospital referrals. This could help to address the surge in neuro-ophthalmology referrals seen since 2016, therefore optimising capacity for genuine urgent cases in need of hospital neuro-ophthalmology assessment.

Conflicts of interest: None**Funding:** None**Corresponding Author:** Simon Hardman Lea, simon.hardman-lea@evolutio-uk.com**ORBIT AND OCULOPLASTICS**

P208. Abstract not included

P209

BICK'S ENTROPION SURGERY PERFORMED BY AN ADVANCED CLINICAL PRACTITIONER, AN AUDIT OF RESULTSAlice Dyer¹, Owuraku Titi-Lartey¹, Paul Rainsbury², Sarah Levey²¹University Hospital Plymouth NHS Trust, Plymouth, UK

Introduction: NHS ophthalmology services are under significant strain, with widespread consultant shortages and extensive waiting lists, in particular ¾ of ophthalmology departments have insufficient consultant numbers to meet current demands.

As of September 2024, over 600,000 patients awaited their first ophthalmology appointment, and the speciality faces the largest backlog of patients requiring follow-up care.

This pressure is exemplified by the 6,471 patients awaiting treatment for ophthalmic surgical procedures including entropion surgery, within our department, increasing the burden on oculoplastic services.

To address this challenge, we explored the potential of Advanced Clinical Practitioners (ACPs) to contribute to surgical care. This abstract presents the outcomes of a pilot program training an ACP in the surgical management of entropion under consultant supervision

Method: A retrospective audit of the first consecutive 20 patients who had entropion surgery with a Bick's procedure (lateral wedge excision) and everting sutures between September 2021 and November 2024.

We included patients who were operated on by an ACP with consultant supervision. Cases were selected at the consultant's discretion. Cases requiring redo surgery, or alternative procedures were not performed by the ACP. Data was collected by reviewing our electrical patient records on MediSIGHT. We recorded patient demographics, the date of surgery, the duration of time the patient was on the waiting list, the outcome of the surgery and associated complications.

Results: In total 20 entropion repair procedures were carried out by the ACP over a three-year period, all procedures were under the supervision of a consultant oculoplastic surgeon.

Patients were seen in the ophthalmic clinic first before being listed for an entropion surgery. Patients were listed either by the consultant, or from the ACP clinic and reviewed by the consultant on the day of listing.

The average age of the patient cohort at the time of the surgery was 78 years old. The mean duration of time on the waiting list was 10.5 months from listing to surgery with the longest being 18 months and the shortest being 1 month due to risk of corneal damage.

At the post operative review 19 patients had a satisfactory lower lid position, one patient required a repeat entropion procedure due to wound dehiscence. The success rate for the procedure was therefore 95%.

Conclusion: Our audit shows that ACP led entropion surgery using the Bick's procedure and everting sutures can be safe and

effective with a success rate of 95%. To the authors knowledge, this is the first instance in the United Kingdom of an ACP performing Bick's entropion surgery under consultant supervision. As a result of our findings, we now run independent ACP led Bick's entropion surgery lists, with consultant support in an adjacent theatre. We would recommend this model of care to other units to help reduce waiting times for this procedure.

Conflicts of interest: None

Funding: None

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P210

BONE GRAFTS VS. TITANIUM MESH IMPLANTS IN ORBITAL FRACTURE MANAGEMENT: A SYSTEMATIC REVIEW AND META-ANALYSIS

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Introduction: Orbital fractures constitute a significant portion of maxillofacial trauma and pose considerable challenges due to the subsequent functional and cosmetic impairments such as diplopia, enophthalmos, and disruptions to facial aesthetics and eye movements. Surgical reconstruction aims to restore orbital volume, improve structural integrity of the orbit, and optimise visual and cosmetic outcomes. Bone grafts and titanium mesh implants are both widely used materials in the repair of orbital fractures, however, their comparative effectiveness remains contested. This systematic review and meta-analysis compares the effectiveness of these materials in managing orbital fractures, focusing on diplopia, enophthalmos, postoperative complications, and aesthetic outcomes.

Method: A systematic search of MEDLINE, EMBASE, CINAHL, and CENTRAL databases was conducted per PRISMA guidelines. Studies directly comparing bone grafts and titanium mesh in orbital fracture management were included. Outcomes assessed included diplopia, enophthalmos, complications (e.g. infection, implant extrusion, donor site morbidity), and aesthetics. Data extraction followed the Cochrane format, and meta-analyses were performed using fixed and random-effects models. Risk of bias was evaluated using the ROBINS-I tool for non-randomised studies.

Results: Eight studies involving 291 patients met inclusion criteria. No significant difference in diplopia incidence was observed between bone grafts and titanium mesh (OR: 1.87; 95% CI: 0.52–6.71; $p = 0.34$). Titanium mesh was associated with a significantly lower risk of enophthalmos (OR: 2.83; 95% CI: 1.19–6.75; $p = 0.02$). Complication rates did not significantly differ between groups (OR: 2.39; 95% CI: 0.55–10.35; $p = 0.25$). Aesthetic outcomes were insufficiently reported for meta-analysis but suggested favourable results with both materials.

Conclusion: Titanium mesh demonstrates advantages in preventing enophthalmos and restoring orbital volume, making it preferable for high-risk cases. Both materials show comparable effectiveness in minimising diplopia and have similar safety profiles. Material selection should consider individual patient factors, defect characteristics, surgeon expertise as well as

material availability. Future research should explore tailored approaches to material use based on fracture type and clinical context.

Conflicts of interest: None

Funding: No financial or proprietary interests were received by any author for this work. Additionally, no sponsors were involved in the study design, data collection, analysis, interpretation, writing of the report, or decision to submit the article for publication.

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P211

CHARM NEEDLES EXTRACTION ON PATIENT'S FACE: AN UNIQUE CASE REPORT

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Introduction: Charm needles, or susuk, are needle-shaped metallic objects inserted subcutaneously in different parts of the body. The practice of inserting susuk is a cultural and superstitious traditional belief common in the south-east Asian region, especially to the people of Malaysia, Thailand, Singapore, Indonesia and Brunei. Most needle-wearers wish to keep the existence of the needles as a secret.

Method: To present the background and management of Charm needles user in ophthalmology field.

Results: A 41-years-old woman came with the chief complaint of pain on her right eye for one month before admission. Her complained was accompanied by migraine, watery eyes, and blurry vision on her right eye. Patient had history of using charm needles when she was young fifteen years ago on her both superior palpebra, mandibula, and metacarpal dextra. Ophthalmological examination of the eye showed 0.63 with pin hole 0.8f1 for visual acuity, and palpable charm needle on her both superior palpebra, other anterior segment examination within normal limit. Three position Schedel x-ray examination showed multiple charm needle on patient's superior palpebra and mandibularis. Based on the examination, patient was diagnosed with multiple corpus alienum on her both superior palpebra, mandibula, and metacarpal dextra. Charm needles extraction was performed based on patient's concern.

Conclusion: Charm needles are unique foreign bodies that are inserted subcutaneously into various parts of the body with potential complications of migration to critical anatomical structures and organs. This cultural practice is largely confined to South-East Asia. Better understanding of their imaging appearance and potential complications is required. Surgical therapy is indicated based on patient concern or complications happened.

Conflicts of interest: None

Funding: None

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P212

HYALURONIDASE-INDUCED ORBITAL COMPARTMENT SYNDROME FOLLOWING SUB-TENON'S ANAESTHESIA: REVIEW OF LITERATURE AND A CASE REPORT

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Introduction: Hyaluronidase is widely used as an adjunct in ophthalmic local anaesthesia to improve the distribution and effectiveness of local anaesthetic agents. Hypersensitivity to hyaluronidase is a rare complication, resulting in orbital inflammation and, in severe cases, orbital compartment syndrome. This study investigates the clinical features and management strategies for hyaluronidase-induced orbital inflammation and presents a rare case of orbital compartment syndrome caused by a severe reaction to hyaluronidase.

Method: We conducted a comprehensive literature review using PubMed, Embase, and Cochrane databases from inception to December 2024, focusing on studies reporting hyaluronidase-induced orbital inflammation following ophthalmic local anaesthesia. Additionally, we present a case report of a 57-year-old female patient who developed orbital compartment syndrome after sub-Tenon's anaesthesia with hyaluronidase, necessitating urgent canthotomy and cantholysis.

Results: Orbital compartment syndrome resulting from severe orbital inflammation is a rare complication of hyaluronidase. The incidence of hypersensitivity to hyaluronidase is reported to be approximately 1 in every 2,000 cases. The risk is increased in individuals with previous exposure to periocular local anaesthesia containing hyaluronidase. Other risk factors include a history of allergic reactions to bee or wasp venom. The onset and severity of symptoms are variable, typically manifesting from a few minutes to several weeks post-injection, and may include pain, periorbital oedema, proptosis and restricted eye movements. Both type I (immediate) and type IV (delayed) hypersensitivity reactions are thought to be implicated in the pathogenesis.

Diagnosis of hyaluronidase hypersensitivity relies on a combination of clinical presentation, imaging studies, and allergy testing if the diagnosis is in doubt. Treatment strategies include systemic corticosteroids, antihistamines, and, in severe cases, emergency orbital decompression.

Our case report details a patient who developed severe left periorbital swelling, chemosis, proptosis, restricted eye movements, and elevated intraocular pressure (50 mmHg) on day 1 post uncomplicated cataract surgery. She exhibited no intraocular inflammation, with quiescent anterior and vitreous chambers. Orbital compartment syndrome was diagnosed, and she underwent an emergency canthotomy-cantholysis. She was initially treated with intravenous antibiotics for presumed orbital cellulitis. CT scan findings showed significant left periorbital soft tissue oedema with proptosis and orbital fat streaking, with no collections or sinus pathology visualised. The patient recovered well on topical steroids. Notably, she had a similar but milder episode of postoperative day 1 periorbital inflammation in 2018, which was treated as a presumed pre-septal cellulitis and was likely a sensitising event.

Conclusion: We conclude that hyaluronidase hypersensitivity is the most probable cause of orbital compartment syndrome in this unusual case. Although hypersensitivity to another agent cannot be completely excluded, hyaluronidase and lignocaine were the only agents administered into the sub-Tenon's space, and an

infectious aetiology is highly unlikely given negative cultures and the absence of risk factors. This study highlights the importance of considering hyaluronidase hypersensitivity as a rare but potentially sight-threatening complication of ophthalmic local anaesthesia. Ophthalmologists should include hyaluronidase hypersensitivity in the differential diagnosis of acute post-operative orbital inflammation, especially in patients with a history of prior exposure to the drug.

Conflicts of interest: None

Funding: None

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P213

PERIOCCULAR TUMOUR EXCISION AND RECONSTRUCTION: IMPACT ON QUALITY OF LIFE

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Introduction: Eyelid tumours can significantly impact a patient's quality of life, both physically and psychologically. Two-stage tumour resection followed by eyelid reconstruction is the primary management, but patient satisfaction with aesthetic and psychological outcomes is an important factor to consider.

Aim: To assess the impact of periocular tumour excision followed by delayed reconstruction on quality of life measures.

Method: A retrospective review of 147 patients who underwent periocular tumour resection followed by delayed eyelid reconstruction was performed. Impact on quality of life was assessed using a questionnaire evaluating pre-operative information, surgical experience, postoperative recovery, aesthetic outcomes, and psychological impact.

Results: Of the 104 patients who responded to the questionnaire, 98% (102 patients) were satisfied with the information and preoperative care provided. 96% (100 patients) rated their surgical experience positively, and 94% (98 patients) were satisfied with their outcomes.

However, 17 patients felt the surgery impacted their appearance, with 6 of them feeling very self-conscious about their appearance. Six patients reported an impact on self-confidence, among which 4 patients reported a negative psychological impact. 37 (36%) patients reported visible scarring, and 7 of these felt it significantly affected their appearance.

It is also noted that among patients who experienced negative psychological effects, a majority of them (63%) underwent a type of skin graft surgery. This suggests a potential correlation between graft size and colour mismatch and adverse psychological outcomes.

Overall, it was felt that cosmetic satisfaction was influenced by scarring, graft concerns, surgical pain and experience, and surgeon preference.

Conclusion: Whilst most patients were satisfied with their post-operative outcomes, a minority (11%) experienced negative psychological effects, including concerns about appearance and diminished self-confidence. This underlines the importance of highlighting the likely impact on cosmesis and addressing any concerns during pre-operative counselling. Comprehensive pre-operative counselling related to factors like postoperative scarring, lash loss, tissue colour match, etc., can help with improving patient satisfaction and overall quality of life in this cohort of patients.

Conflicts of interest: None

Funding: None

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P214. Abstract not included

P215

THE IMPACT OF WARMED LOCAL ANAESTHESIA ON PATIENTS WITH VARYING PAIN THRESHOLDS DURING OCULOPLASTIC SURGERY

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Introduction: Injecting local anaesthetics (LA) is a routine procedure in oculoplastic surgery, but it can often cause considerable burning or stinging pain. Various strategies are used to alleviate the pain during LA administration. In this study, we aimed to assess the impact of using warmed (38 °C) versus non-warmed (room temperature) LA on pain reduction in patients undergoing oculoplastic surgery.

Method: Patients scheduled for bilateral oculoplastic surgery were included in the study after obtaining informed consent. Equal amounts of 2% lignocaine with adrenaline (1:80,000) and 0.5% bupivacaine were used as the LA. The warmed LA was heated in a warmer for 15 minutes prior to administration. Each patient was randomly assigned to receive either warmed LA or non-warmed LA in the right eye, with the opposite type of LA in the second eye. Injection was given by the same surgeon for both eyes. Pain was assessed immediately after the LA injection using a linear analogue pain scale ranging from 0 to 10. During each injection, the lowest oxygen saturation, highest heart rate, and highest mean blood pressure were recorded.

Results: Twenty patients were enrolled in the study. The average reported linear pain score was 5.3 with warmed LA and 6.3 with non-warmed LA. Eighty percent of the patients reported lower pain score with the warm LA, 10% reported no difference and 10% reported lower pain with non-warmed LA. Subgroup analysis showed that patients with low pain threshold, i.e. pain score ≥ 7 , showed statistically significant lower pain score with warm LA. Warm LA was also associated with lower pulse rate with compared to non-warmed LA but the difference was not statistically difference. There was no difference in mean arterial pressure or oxygen saturation. Patients with high pain threshold, i.e. pain score < 7 , showed no difference in pain score (average score was 3.5 in each group). There was no difference in pulse rate, mean arterial pressure or oxygen saturation.

Conclusion: Patients with a low pain threshold (high pain score) may benefit more from warmed LA, as indicated by the statistically significant different pain score between warmed and nonwarmed LA in this group. Warming LA is a simple and effective method to reduce pain and improve patients' experience during oculoplastic surgery.

Conflicts of interest: None

Funding: None

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P216

VELIGROTUG (VRDN-001), A FULL ANTAGONIST MONOCLONAL ANTIBODY TO IGF-1R, IN ACTIVE THYROID EYE

DISEASE (TED): THRIVE PHASE 3 EFFICACY AND SAFETY TOPLINE RESULTS

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Introduction: Veligrotug (VRDN-001), a full antagonist humanized monoclonal antibody to the IGF-1 receptor (IGF-1R), is an investigational treatment for thyroid eye disease (TED). Clinical and preclinical evidence indicate a central role for IGF-1R antagonism in reducing the inflammation and proptosis that occur in TED. Topline efficacy and safety results at 15 weeks were assessed from an ongoing phase 3 randomized double-masked placebo-controlled trial of veligrotug vs placebo in patients with active TED (NCT05176639).

Method: Adults with moderate-to-severe active TED (onset ≤ 15 months, proptosis ≥ 3 mm, and clinical activity score [CAS] ≥ 3) were randomized to receive 5 IV infusions 3 weeks apart of either 10 mg/kg veligrotug or placebo. The proptosis responder rate (PRR), defined as ≥ 2 mm reduction vs baseline by Hertel exophthalmometry, PRR by MRI/CT, complete resolution of diplopia, and mean changes from baseline in proptosis and CAS were assessed. Treatment-emergent adverse events (AEs) were assessed through 15 weeks, with follow-up ongoing through 52 weeks.

Results: A total of 113 patients were randomized to veligrotug ($n = 75$) or placebo ($n = 38$) and included in the intention-to-treat population. At baseline, mean proptosis was 23.2 mm in each group; CAS was 4.5 vs 4.8 and diplopia was present in 67% vs 68% of patients for veligrotug vs placebo. At 15 weeks, PRR by Hertel was 70% vs 5% ($p < 0.0001$) for veligrotug vs placebo, with a mean proptosis reduction of 2.9 mm vs 0.5 mm ($p < 0.0001$). PRR by MRI/CT was 69% vs 9% ($p < 0.0001$) for veligrotug vs placebo, with a mean proptosis reduction of 2.9 mm vs 0.6 mm ($p < 0.0001$). Mean CAS decreased by 3.4 vs 1.7 ($p < 0.0001$) for veligrotug vs placebo. In patients with diplopia, complete resolution of diplopia occurred in 54% (27/50) vs 12% (3/26) ($p < 0.0001$) for veligrotug vs placebo. AEs occurred for 66 (88%) veligrotug vs 24 (63%) placebo patients, and most were mild; 4 patients (veligrotug) had serious AEs (all unrelated to treatment). Hearing impairment AEs were reported for 12 (16%) veligrotug vs 4 (11%) placebo patients.

Conclusion: Topline results from the THRIVE phase 3 trial show 5 IV infusions of 10 mg/kg veligrotug were well tolerated and led to significant and clinically meaningful improvements in proptosis, CAS, and diplopia at 15 weeks. Additional follow-up through 52 weeks is ongoing.

Conflicts of interest:

Vickie Lee, Antonio Manuel Garrido Hermosilla, Peerooz Saeed, Michael Schittkowski, Julian Bermúdez Pío-Rendón, Eckart Bertelmann, Edwina Eade, Marta Pérez-López, Marco Sales Sanz, and

Patrice Rodien are clinical research investigators for Viridian Therapeutics, Inc.

Cathy Michalsky was an employee of Viridian Therapeutics, Inc. at the time the work was completed; Abhijit Narvekar and Thomas Ciulla are employees of Viridian Therapeutics, Inc.

Funding: This study was sponsored by Viridian Therapeutics, Inc.

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PAEDIATRIC OPHTHALMOLOGY

P217

A MARKED INCREASE IN “SMARTPHONE SQUINTS” IN CHILDREN DURING LOCKDOWN: EFFECTIVELY TREATED WITH BOTOX

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Introduction: Purpose: Firstly, to summarise a surprising increase in incidence of Acute Accommodative Concomitant Esotropia (AACE or “smartphone squint”) in children and young people coinciding with the Covid-19 pandemic and lockdown measures. Secondly, to report on the beneficial effect of a botulinum toxin injection to the medial rectus muscle/s in successfully treating this condition.

Method: Cases presenting to paediatric ophthalmology in NHS Highland with AACE over a 7 year period were identified retrospectively. Demographic and clinical details were obtained, including age, degree of esotropia, response to treatment and complications. Patients were offered a procedure under GA of 2.5 IU botulinum toxin A (Botox, Allergan, Irvine, CA, USA) into the medial rectus muscle/s.

Results: 19 initial presentations of AACE were identified between 2018 and 2024. Prior to this period very few cases of AACE were encountered. There was a marked jump in the incidence of AACE in 2020, corresponding to national Covid-19 lockdowns. The following year (2021), treatment with Botox injection to the medial rectus muscle/s was performed when GA access restrictions were relaxed. The incidence of AACE reduced considerably following this period, up to 2024.

There were 10 male and 9 female patients. They ranged in age from 8 to 17 years old, with a mean age of 12. Three cases were myopic, 2 were hypermetropic and wearing their full cycloplegic correction, and the remaining cases were emmetropic with no latent hypermetropia on cycloplegic refraction.

17 patients were treated with Botox: 13 injected into both medial recti and 4 cases unilaterally. Nine cases experienced an initial temporary exotropia and 5 resulted in a temporary ptosis.

All those treated with Botox achieved straight eyes with no diplopia within 6 weeks. Normal motor and sensory fusion were re-established in all cases. One case was managed surgically with a recess/resect procedure with an excellent outcome. A further case was managed with strict abstinence of near screen use, which resulted in resolution of the squint.

The standard protocol for follow up was for orthoptic review up until 1 year post procedure.

Two patients returned with a further presentation of AACE, 3 years after their initial treatment. For both these patients, a further treatment with Botox did not result in long term benefit; they subsequently underwent conventional strabismus surgery.

Conclusion: We recommend Botox injection into the medial rectus as a safe, quick and effective treatment of AACE as an alternative to surgical management. Public health information on the risks of prolonged near screen use may encourage prevention.

Our data show a significant and temporary increase in the incidence of AACE in the paediatric population during the Covid-19 lockdown. This coincides with the time period when children had to increasingly rely on screens for education, leisure, and social activities. We suggest that the increase in screen use over this period caused a corresponding increase in AACE.

For the majority of patients, a single treatment with Botox, to one or both medial recti depending on the presentation, gives an excellent clinical outcome. In patients who have a further episode of AACE (several years after initial treatment in this series), direct progression onto squint surgery may be the best option.

We would encourage consideration of this treatment for teens and tweens presenting with AACE due to its favourable safety and efficacy profile.

Conflicts of interest: None

Funding: None

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P218

ASSESSMENT AND OUTCOMES OF PAEDIATRIC OPTIC NEURITIS IN A TERTIARY CHILDREN’S HOSPITAL: A 10-YEAR RETROSPECTIVE REVIEW

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Introduction: Paediatric optic neuritis (PON) is a rare condition, with an incidence of 0.15–0.57 per 100,000 children, marked by distinct characteristics compared to adult optic neuritis (ON). While adults typically present with unilateral visual loss and pain on eye movement, PON more often manifests bilaterally, with severe visual impairment and less frequent pain. Despite significant visual loss at presentation, most children achieve normal visual acuity (VA) within a year. The aetiology of PON includes associations with demyelinating conditions such as multiple sclerosis (MS), neuromyelitis optica spectrum disorder (NMOSD), and myelin-oligodendrocyte glycoprotein-associated disease (MOGAD).

However, long-term data on clinical outcomes and management practices remain limited. This study aims to evaluate clinical presentations, long-term visual outcomes, and treatment strategies for PON in a tertiary paediatric hospital.

Method: A retrospective review was conducted on children aged 0–16 years diagnosed with ON at Alder Hey Children’s Hospital from May 2014 to April 2024. Data on demographics, clinical presentation, diagnostic tests, and treatments were collected. Visual acuity was recorded in logMAR, and optic nerve function was evaluated using pupil response, colour vision, and electrodiagnostic tests. Imaging, including MRI, and serological investigations for aquaporin-4 and MOG antibodies, were performed. Outcomes were assessed at multiple follow-up intervals, with a minimum follow-up of two weeks. Statistical analyses, including McNemar’s test and paired t-tests, were conducted to evaluate changes in visual outcomes over time.

Results: Sixteen patients (19 eyes) were included, with a mean age of 11 years and a female predominance. Most cases were unilateral (81%), and MOGAD was the most frequent diagnosis

(56%). At presentation, 78% of affected eyes had VA worse than 0.3 logMAR, significantly improving by three months (86% achieving VA $\leq$ 0.3 logMAR, $p = 0.002$). Optic disc swelling was present in 46% of unilateral cases at diagnosis, with resolution in 83% by three months. Bilateral cases exhibited greater severity, with one requiring plasma exchange for steroid-refractory symptoms, resulting in substantial visual improvement. MRI findings revealed intra-orbital and papillitis as the most common sites of inflammation. Treatment primarily involved intravenous methylprednisolone, followed by oral corticosteroid tapering, with outcomes comparable across different diagnoses.

Conclusion: This study highlights the clinical features, management, and outcomes of PON, reinforcing trends observed in prior research. Despite severe visual impairment at onset, most children experience significant recovery within the first 3 months, however, this may continue up to 1 year post-presentation. The predominance of MOGAD over MS as an underlying diagnosis underscores the importance of comprehensive diagnostic workup, including antibody testing, in paediatric cases. Our study found no significant difference in LogMAR visual acuity testing between key diagnoses of optic neuritis. Variation in treatment approaches reflects the absence of standardized protocols, with intravenous corticosteroids as the preferred initial therapy. These findings emphasize the need for individualized management and long-term follow-up to optimize visual outcomes in children with PON.

Conflicts of interest: None

Funding: None

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P219

CLINICAL OUTCOMES OF RETROPUPILLARY IRIS CLAW IOL FIXATION IN PAEDIATRIC AGE GROUP

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Introduction: To study the clinical outcomes of retro-pupillary iris claw IOL fixation in the paediatric age group.

Method: We conducted an ambispective study where we took 50 eyes of children of the ages (5–15 years) with subluxated lens or aphakia with insufficient capsular support who underwent retro-pupillary iris claw IOL fixation from Jan 2019 till August 2020 (retrospective arm) and from Aug 2020 till Aug 2021 (prospective arm) at our centre and recorded their clinical outcomes in terms of visual acuity, pupil size, central macular thickness, effective lens position on UBM and complications, if any.

Results: With a mean follow-up period of 15 months (6 months to 30 months) with a mean age of 8.32 (± 1.52). The BCVA on last follow up was 0.30 (± 0.24) which improved from 0.82 (± 0.45). The mean post-up spherical equivalent was +1.12 (± 1.91) Dioptre sphere at our last follow-up. The endothelial cell count dropped from 3239 (± 242.34) mm² to 3150.58 (± 253.89) mm² making it a 2.8% loss over a mean follow-up of 15 months. The maximum pupil dilation wasn't affected by retrofixated iris claw IOLs. About 44% eyes had early horizontal pupil ovalisation, 14% had corneal oedema on post-operative day one, five had IOL disencapsulation out of which two were traumatic and three were spontaneous, all five were successfully re-encapsulated. About 2% had wound leak and subsequently Choroidal detachment on post-op day one which required resuturing. One eye developed retinal detachment on 18 months follow-up. None of the eyes developed uveitis, glaucoma or macular oedema.

Conclusion: From our study we conclude that retro-pupillary iris claw IOL fixation is a safe method of correcting paediatric aphakia with insufficient capsular support which yields good clinical outcomes.

Conflicts of interest: None

Funding: None

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P220

CURRENT MANAGEMENT IF RETINOPATHY OF PREMATURITY IN THE UNITED KINGDOM

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Introduction: Retinopathy of prematurity (ROP) is a sight-threatening condition, but prompt intervention can prevent blindness. With BEAT-ROP and RAINBOW trials, anti VEGF (vascular endothelial growth factor) has transformed the management of ROP. Despite The Royal College of Ophthalmologists guidelines (2022), there remains variations in management.

Method: We surveyed the UK ROP treatment group to obtain a snapshot of current practice.

Results: Seventeen treating units responded. When treating ROP, 16/17 use intravitreal anti-VEGF injection first line for zone 1 disease and AP-ROP, and 13/17 for posterior zone 2 disease.

The choice of anti-VEGF was Bevacizumab 10/17, Ranibizumab 4/17, while 3/17 use either. For treatment of anterior zone 2 disease, laser is used by 16/17.

After reactivation following anti-VEGF, 7/17 units proceed to laser, 10/17 would consider re-injection.

For management of persistent avascular retina (PAR), 10/17 adopt a conservative approach, 7/17 would consider laser.

Conclusion: This survey demonstrates a consistent approach for treatment of posterior ROP. The main variations found were the choice of anti-VEGF, how to manage reactivation post injection and the management of PAR. There were different stated rationales for this variation in management.

This survey provides a snapshot of current management of ROP in the UK and demonstrates some of the variation in approaches. There is a need for further studies and consensus on the management of reactivation of ROP post intravitreal injection and PAR.

Conflicts of interest: None

Funding: None

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P221

INTERESTING PAEDIATRIC OPHTHALMOLOGY CASE OF GIANT HOGWEED PERIORBITAL BURN

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Introduction: The patient is a 5 year old Caucasian boy who was referred to Ophthalmology by Paediatrics and he presented with a 1 day history of unilateral vesicular painful periorbital rash around his left eye and on his left forehead.

The characteristics and pattern of the rash suggested it could be left-sided Herpes Zoster Ophthalmicus.

But after further detailed history taking, it was ascertained that the most likely cause of the rash was Giant Hogweed Burns due to the patient's face coming into contact with the toxic plant sap.

Method: His visual acuity, anterior segment and posterior segment examinations were normal.

Initial swabs taken from the ruptured vesicles were negative for virus and bacteria.

Subsequently, more in-depth history was obtained from the patient's father that the patient had been playing and rolling around in a grass field before the facial lesions started.

The case was discussed with Consultant Paediatric Ophthalmologist and then referred for onward discussion with the Paediatrics for further dermatological opinion. The patient was commenced on steroid cream.

Results: The patient was scheduled a follow-up in Paediatric Ophthalmology Clinic to monitor his progress.

Phone consultation with his parents 1 month later revealed that the facial rash resolved with steroid cream treatment.

Conclusion: This is a very fascinating case report as facial and periocular lesions caused by Giant Hogweed Burns are not commonly encountered in an ophthalmology setting.

Although clinical suspicion points towards a herpes zoster ophthalmicus or herpes simplex rash, this case reminds us that a unilateral vesicular facial and periocular rash can also be caused by contact dermatitis from skin exposure to an irritant.

A key learning point is that Giant Hogweeds are relatively common in the British outdoors and their toxic sap can cause severe skin rashes, burning blisters and long-lasting scars.

The treatment is to wash it off immediately with soap and water upon contact and sunlight should be avoided due to photosensitivity induced by the Giant Hogweed Burn. If the sap has entered the eye, please contact an ophthalmologist immediately. For serious skin inflammation, steroid creams are effective and for superficial burns, lotion or aloe vera can be applied.

Conflicts of interest: None

Funding: None

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P222

LEVEL OF EXPERIENCE OF OPHTHALMOLOGISTS AND OPHTHALMIC HEALTH PROFESSIONALS IMPACTS IDENTIFICATION OF FOVEAL HYPOPLASIA

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Introduction: Diagnosis and prognosis of arrested foveal development or foveal hypoplasia (FH) can be ascertained using the Leicester grading system for FH using optical coherence tomography (OCT). In clinical practice, ophthalmologists and

ophthalmic health professionals with varying experience consult patients with FH, however, to date, the FH grading system has only been validated amongst experts. We compare inter-grader and intra-grade agreement of healthcare professionals against expert consensus across all grades of FH.

Method: Hand-held and table-mounted OCT images (n = 341) obtained retrospectively were graded independently at a single centre by experts (n = 3) with over six years of experience, and "novice" medical and allied health professionals (n = 5) with between less than one to three years of experience. The images showed no FH (n = 133), grade 1 FH (n = 106), grade 2 FH (n = 43), grade 3 FH (n = 42), grade 4 FH (n = 11), and atypical FH (n = 6). Sensitivity, specificity and Cohen's kappa scores were calculated for each grader, and expert vs novice performance was compared.

Results: All graders showed high sensitivity (median 97% (IQR:94–99)) and specificity (median 94% (IQR:90–95)) for identifying the presence or absence of FH. No significant difference was seen for specificity between expert and novice graders, but experts had significantly greater diagnostic sensitivity (median difference = 5.3%, H = 5.00, p = 0.025). Expert graders had the highest agreement with ground truth and inter-grader agreement. Novice graders with one year or less experience in grading had the lowest agreement with ground truth (kappa 0.7–0.9) and showed high variability in inter-grader agreement (pairwise kappa 0.7–0.54) particularly for atypical FH. Review of misclassified images with atypical FH suggest lack of awareness that changes at the inner segment ellipsoid may exist despite observation of features of normal foveal structure, such as the presence of a foveal pit. Our results also suggested difficulty identifying the fovea in handheld OCT images in children where the macula is decentred due to lack of fixation while acquiring the images.

Conclusion: Health professionals of a variety of backgrounds and levels of experience can accurately identify FH using handheld and table-mounted OCT images. Considering the diagnostic and prognostic value of accurate grading of FH, grading in clinical practice by less experienced ophthalmologists can be improved by including handheld OCT and FH grading in the Royal College of Ophthalmologists training curriculum. Creating education resources for ophthalmologists and ophthalmic health professionals can upskill all professional groups to ultimately improve patient outcomes.

Conflicts of interest:

No conflicting relationship exists for any authors.

AA is currently an employee of F. Hoffmann-La Roche Ltd.

Funding:

Medical Research Council (MRC), London, UK (grant number: MR/J004189/1, MRC/N004566/1 and MC_PC_17171)

Ulverscroft Foundation and Fight for Sight (grant ref: 24NN181) RS and SRR are supported by National Institute for Health and Care Research (NIHR).

MGT acknowledges prior support from NIHR during study period (CL-2017-11-003).

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P223

MYELINATED RETINAL NERVE FIBRES - CONGENITAL OR ACQUIRED?

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Introduction: Myelinated retinal nerve fibres (MRNFs) are described as congenital developmental anomalies, however there is increasing evidence of acquired MRNFs; with or without ocular or systemic associations. The purpose of this study is to report a documented case series of acquired MRNFs.

Method: Three cases of new onset acquired myelinated retinal nerve fibres of the optic disc were referred to our tertiary centre. The history, the ocular features, the systemic findings, and retinal imaging are reported for each case.

Results: Case one is an 18 year old male who was diagnosed with juvenile idiopathic arthritis with right anterior uveitis at the age of 7 years. He subsequently developed secondary glaucoma requiring a trabeculectomy. Right-sided MRNFs developed eight years after a brief post-operative hypotony and disc swelling, which has remained static over two years follow-up.

Case two and three, 13 and 15 year old females respectively, were referred by their optometrist for new onset optic disc changes. Neither had any past ocular, past medical, nor family history of systemic disease. Case two had unilateral (RE) isolated MRNFs with no progression, and she remained asymptomatic over two years follow-up. Case three also had isolated unilateral (RE) MRNFs.

Conclusion: MRNFs are rare structural anomalies. We report one case of acquired MRNFs associated with ocular disease, and two cases of isolated unilateral acquired non-progressive MRNFs. The presence of MRNFs in this newly described group of patients challenges traditional teaching of the congenital onset of all MRNFs.

Conflicts of interest: Not applicable

Funding: Not applicable

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P224

OCULAR INVOLVEMENT IN THE MUCOPOLYSACCHARIDOSIS: AN 8-YEAR LONGITUDINAL STUDY

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Introduction: Mucopolysaccharidoses (MPS) are inherited disorders characterized by impaired lysosomal degradation of glycosaminoglycans. Ophthalmic manifestations, including corneal clouding, optic neuropathy, elevated intraocular pressure, and retinopathy in MPS types I, II, III, IV, and VI, can lead to significant visual impairment. This study presents clinical observations and corneal changes in MPS patients.

Method: A prospective study assessed visual acuity, corneal clouding, funduscopy, OPTOS imaging, optical coherence tomography (OCT), and electroretinography (ERG) in our MPS patients.

Results: 80 MPS patients, aged 3 to 58 years, were evaluated. Various degrees of corneal clouding were present in 71 patients. The cohort included 30 MPS I Hurler, 14 MPS I Hurler-Scheie, 4 MPS I Scheie, 2 MPS II, 13 MPS IV, and 8 MPS VI. Two patients underwent corneal transplants, one had penetrating keratoplasty and the other bilateral deep anterior lamellar keratoplasty (DALK). Median LogMAR visual acuity was 0.62 in patients with MPS I Hurler, 0.18 in MPS I Hurler/Scheie, 0.08 in MPS I Scheie and MPS II, 0.54 in MPS IV, and 0.32 in MPS VI. Retinopathy was found in 32

patients: 25 MPS I, 4 MPS II, 1 MPS IVA, and 2 MPS VI. All patients received enzyme replacement therapy (ERT), hematopoietic stem cell transplantation (HSCT), or both except (2 MPS II & 3 MPS IV).

Conclusion: Corneal clouding was prevalent across all MPS subgroups, mostly mild to moderate changes, and 32 patients had retinopathy. While it was challenging to assess progression due to additional ocular conditions, visual acuity remained relatively stable.

Conflicts of interest:

O.Mehana, M. Noor, O. McGrath and K. Sornalingham have no financial disclosures.

T. Aslam has received funding and educational grants from Bayer, Novartis, Roche, Allergan, Laboratoires Thea, Oraya, Bausch and Lomb, Topcon, Heidelberg, and Canon.

J. Ashworth has received funding and educational grants from Santen, BioMarin, and Inventiva and Orchard therapeutics.

Funding:

This study was supported by a research grant from BioMarin Pharmaceutical Inc. (Grant number: Ashworth/Grant G00423). BioMarin had no role in the design or conduct of the study.

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P225

OPTOMETRIST REFERRALS TO HOSPITAL EYE SERVICES FOR CHILDREN WITH SUSPECTED OPTIC DISC SWELLING: A DISTRICT GENERAL HOSPITAL EXPERIENCE

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Introduction: Following a high profile medicolegal case of negligence in 2017, in which a UK optometrist failed to refer for papilloedema, there has been a surge in referrals for suspected optic disc swelling in children. This has placed further stress upon the Paediatric Ophthalmology service. This study evaluates the appropriateness and quality of such referrals, as well as presenting their management outcomes.

Method: A single centre retrospective review was conducted of all consecutive referrals for suspected optic disc swelling to a district general hospital. Exclusion criteria included patients age 18 or over, and those for whom complete data on the Electronic Patient Record was unavailable. A total of 333 cases were included between January 2018 and August 2024.

Results: The majority of referrals were of female sex (59.9%). DNA rate for the initial consultation at the hospital eye clinic was 12% (40 out of 333 referrals). Headaches were reported in 16% of patients. In 127 cases (38%) the clinician documented normal optic disc examination with no pathological signs found. Other common diagnoses following ophthalmology review included tilted disc (11%), optic disc drusen (22%) and crowded disc (1%). A total of 25 patients underwent MRI imaging while 3 patients had CT of the brain. A total of 4 patients underwent lumbar puncture. 4 patients in this study received a diagnosis of Idiopathic Intracranial hypertension. Other neurological diagnoses included congenital hydrocephalus (1 case), meningioma (1 case) and craniosynostosis (1 case).

Conclusion: Inappropriate referrals to secondary care prolong waiting lists and may cause undue anxiety to patients and their families. Creating a standardised referral proforma documenting symptoms, OCT images if available and exact disc findings such as haemorrhages or vessel obscuration, may facilitate more streamlined triaging and assessment of these patients.

True optic disc swelling in children is a paediatric emergency necessitating urgent review in the hospital setting to exclude life-threatening aetiologies. Our study suggests the vast majority of optometrist referrals do not comprise true disc swelling.

Conflicts of interest: None

Funding: None

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P226

OUTCOMES OF INFERIOR RECTUS MUSCLE RECESSION SURGERY USING ABSORBABLE VERSUS NON-ABSORBABLE SUTURES

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Introduction: Inferior rectus (IR) recession surgery is used in the management of vertical strabismus.

However, overcorrection remains a significant risk, reported to range from 21% to 50%. Several variables contribute to this risk, such as the impact of the lower lid retractors, dissolution of absorbable sutures and muscle tightness in thyroid eye disease (TED) patients.

Non-absorbable sutures have shown to reduce risk of overcorrection; a study has shown 6% fewer overcorrections when compared to absorbable sutures. Studies involving the medial rectus in animals and humans have shown non-absorbable sutures reduce incidence of stretched scar formation, diminishing the risk of overcorrection.

However, non-absorbable sutures have their own drawbacks. These sutures can erode through overlying tissue over time, increasing the risk of debris and mucous accumulation, and therefore infection along suture tracks. Long term irritation, suture exposure and patient discomfort can be a problem.

This study aims to contribute novel insights by comparing clinical outcomes of IR recession using absorbable and non-absorbable sutures. With focus on evaluating post operative deviation control and overcorrection incidence, these findings aim to provide valuable contributions to clinical practice, addressing the limited literature surrounding this topic.

Method: A retrospective study was conducted at a tertiary referral centre on patients undergoing IR recession with absorbable or non-absorbable sutures between 1st December 2020 to 31st January 2023. Patients were categorised based on the type of suture used: absorbable or non-absorbable. The following patient demographics were collected: age at operation, sex, date of operation, surgical indication and the presence of TED. All patients were anonymised with case numbers in compliance with patient confidentiality regulations. Details of the operation included adjustable vs non-adjustable suture technique, surgeon training level, number of muscles recessed, IR distance recessed, and time from surgery to final follow-up.

Results: 32 IR recession procedures were evaluated, 26 of whom underwent procedure with absorbable suture and the other 6 with non-absorbable, with mean follow-up across both groups was 3.6 months (range 1 month–25 months). Success was defined as ≤ 5 prism dioptres deviation.

Chi square analysis showed no significant difference between success in absorbable vs non absorbable sutures in vertical deviation at near or distance fixation. Success rates for vertical deviation at near fixation were 57.7% and 33.3% ($p = 0.48$, $p > 0.05$) for the absorbable and non-absorbable suture group respectively. At distance fixation, success rates were 65.4% for

absorbable sutures and 50% for non-absorbable suture procedures ($p = 0.73$, $p > 0.05$).

Mean vertical deviation in primary position at near fixation at final follow-up was 6.96 PD and 6.3 PD for the absorbable and non-absorbable suture group respectively [$U = 76$, $p = 0.47$]. The mean vertical deviation in primary position at distance fixation at final follow up was 5.54 PD in the absorbable, compared to 5.5 PD in the non-absorbable group [$U = 76.5$, $p = 0.48$].

Overcorrection was found in 6 patients (23.1%) in the absorbable group and 1 patient (16.7%) in the non-absorbable group. There was no significant difference in the incidence of overcorrection between the groups ($p = 0.61$, $p > 0.05$).

Logistic regression analysis, accounting for absorbability of suture as a covariate, assessed impact on success in at least one of near or distance fixation. It showed that adjustable suture use (coefficient = -1.51 , $p = 0.214$), number of muscles involved (coefficient = -1.81 , $p = 0.120$), TED (coefficient = 2.0731 , $p = 0.090$) and consultant involvement (coefficient = 0.30 , $p = 0.762$) had no significant impact.

Conclusion: This study demonstrates that IR recession surgeries using absorbable sutures yield outcomes comparable to those using non-absorbable sutures, with no significant difference in success rates and overcorrection rates consistent with existing literature. These findings suggest that absorbable sutures are equally viable in use for IR recession surgeries.

Conflicts of interest: None

Funding: None

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P227

PER-OPERATIVE EPIPOLE OPTIC DISC IMAGES ENHANCE CRANIOFACIAL CARE

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Introduction: Craniofacial disorders can be associated with raised intracranial pressure due to early fusion of the cranial sutures and altered CSF dynamics. This may result in papilloedema, which may be evident before cranial vault surgery, appear at a later stage in care or never be seen at all. Optic nerves in craniofacial disorders can remain elevated, even after normalization of ICP – thus presentation of papilloedema in children with craniosynostosis sheds little light on the timing of raised ICP unless serial recordings are made and interpreted in light of what has gone before. This can result in challenges in treatment planning if no baseline or previous recordings have been made.

Craniofacial surgery is generally undertaken prior to the age of 18 months, meaning preoperative disc imaging may not be possible. The Epipole camera is a handheld device which can be used by the craniofacial surgical team to acquire optic disc images at the time of surgery, providing a permanent record of optic disc health before any intervention has taken place and as a baseline. The field of view is adequate for evaluation of the optic disc and first order vessels.

Method: Our institute is one of four nationally commissioned craniofacial centres in England. Between September 2022 and January 2024, all children undergoing a craniofacial procedure had Epipole imaging whilst in the anaesthetic room prior to surgery. Tropicamide 1% eye drop was instilled and images acquired by the craniofacial surgeons. All images were reviewed and graded for papilloedema by the ophthalmology team. The

optic disc findings were collated with pre-operative examination findings and correlated with craniofacial phenotype.

Results: The majority of Epipole images were of adequate quality to be able to be graded for papilledema. 150 children were imaged, of which 2% had papilledema at time of surgery. This resolved post procedure in all cases. These baseline images are invaluable in management of the small percentage of cases who go on to develop raised ICP despite an initial transcranial procedure.

Conclusion: We conclude that the Epipole camera, is an easy, reliable and effective method to record peroperative disc appearance. It is a practical and easy-to-use device that non-ophthalmologists may use to acquire disc images for analysis by an ophthalmologist. We plan to extend the use of the device to the wider paediatric neurosurgical team.

Conflicts of interest: None

Funding: None

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P228

PREDICTION OF LONG-TERM VISUAL OUTCOMES IN INFANTILE NYSTAGMUS USING HANDHELD OCT: A 12-YEAR LONGITUDINAL STUDY

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Introduction: Infantile nystagmus is often associated with foveal hypoplasia. It is currently unknown whether foveal hypoplasia grading can predict long-term visual outcomes. We evaluated the clinical utility of foveal hypoplasia grading using handheld optical coherence tomography (OCT) as a predictor of long-term visual outcomes in infantile nystagmus.

Method: We conducted a single-centre, prospective, longitudinal study from 2012 to 2024. Handheld OCT was performed in preverbal children with nystagmus aged under 33 months. Foveal tomograms were graded using the Leicester Grading System for Foveal Hypoplasia. Participants were followed up as pre-adolescent children aged 9–12 years. The main outcome measure was distance visual acuity (VA), measured in logMAR (Thomson Test Chart, Thomson Software Solutions). Secondary outcome measures included near VA (Sloan ETDRS, Precision Vision), colour vision (Ishihara's Tests, Kanehara Trading Inc.), contrast sensitivity (Thomson Test Chart) and stereopsis (Frisby Near Stereotest, Frisby Tests). Data were analysed using a linear mixed regression model.

Results: Nineteen children with infantile nystagmus (idiopathic infantile nystagmus [IIN]: 5; albinism: 12; achromatopsia: 2) were recruited. Median age at first examination was 22 months (range: 1–33; IQR: 13–30). Median follow-up was 140 months (range: 110–147; IQR: 135–143). Foveal hypoplasia grade at first examination was significantly associated with future distance VA ($R = 0.86$, $F = 98$, $P < 0.0001$), near VA ($R = 0.82$, $F = 74$, $P < 0.0001$), colour vision ($R = 0.60$, $F = 20$, $p < 0.0001$), contrast sensitivity ($R = 0.82$, $F = 76$, $P < 0.0001$) and stereopsis ($R = 0.70$, $F = 16$, $P = 0.001$).

Conclusion: In the world's first study of its kind, we find handheld OCT performed during early childhood can predict long-term future VA, near vision, colour vision, contrast sensitivity and stereopsis in children with nystagmus. This can help guide diagnosis, prognosis, clinical management and counselling.

Conflicts of interest: None

Funding: Supported by the Medical Research Council, UK (MR/N004566/1 and MR/J004189/1); Fight for Sight, UK (24NN181 and 5009/5010); Ulverscroft Foundation, UK; Nystagmus Network, UK. Sohaib Rufai is funded by the National Institute for Health Research (NIHR), UK (Academic Clinical Lectureship: CL-2022-11-001).

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P229

SAFETY PROFILE AND VISUAL OUTCOMES OF FLUOCINOLONE ACETONIDE IMPLANTS IN PAEDIATRIC NON-INFECTIOUS UVEITIS

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Introduction: This audit reports the outcomes of 0.18 mg fluocinolone acetonide (FAC) implants in paediatric patients with non-infectious uveitis (NIU) and chronic cystoid macular oedema (CMO).

Method: This retrospective, single-centre audit was conducted at Moorfields Eye Hospital, London, UK. It included all patients under 16 years old who received FAC implants for NIU with chronic CMO. Data on patient demographics, uveitis diagnosis, treatment history, lens status, intraocular pressure (IOP), best-corrected visual acuity (BCVA), and central subfield thickness (CST) were collected. Follow-up assessments were conducted three months postFAC implantation or lens extraction.

Results: Five patients (7 eyes) were included, with a mean age of 12.9 years. Pre-existing ocular hypertension (OHT) was noted in 5 eyes. No new cases of OHT were observed post-FAC, and no patients required additional IOP-lowering treatments. 4 out of 7 eyes were pseudophakic at time of FAC implantation. At three months, 5 out of 7 eyes showed improvement in BCVA after either FAC implantation or lens extraction. CMO resolved in 3 out of 7 eyes, with all patients except one (Immune Recovery Uveitis) showing improvements in CST. There were no severe adverse events such as permanent vision loss.

Conclusion: The 0.18 mg FAC implant exhibited a favourable safety profile in a paediatric population, with no significant adverse events and stable IOP outcomes. The sustained-release nature of the FAC implant offers a promising long-term treatment option for paediatric uveitis, though further long-term and randomised controlled trials are needed to further assess sustained efficacy and safety.

Conflicts of interest: None

Funding: None

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P230

SCLERAL FIXATED INTRAOCULAR LENS IN CHILDREN: META-ANALYSIS AND SYSTEMATIC REVIEW

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Introduction:

Purpose: To compare outcomes of scleral fixated Intraocular lenses in the paediatric population. We included sutureless scleral fixated intraocular lenses (SLSFIOL), glued scleral fixated intraocular lenses (GSFIOL), and sutured scleral fixated intraocular lenses (SSFIOL) using polypropylene, and Gore-Tex sutures.

Method: A systematic review and meta-analysis were performed in line with the Preferred Reporting Items for Systematic Reviews and meta-analysis guidelines. An electronic search was performed to include studies from 1999 to 2023. We included studies of 1–18 years of age. The analysis was based on fixed and random effect models. Primary outcomes included visual acuity, refraction, and intraocular pressure (IOP). Secondary outcomes were lens tilt and extrusion of suture material.

Results: Thirty-three studies (1531 eyes) were included (GSFIOL $n = 75$, SLSFIOL $n = 162$, SSFIOL $n = 1294$ (Gore-tex $n = 30$, and polypropylene $n = 1264$)). Vision improved from preoperative (mean 0.75 ± 0.50 SD) to postoperative (mean 0.39 ± 0.19 SD) logMAR. The mean spherical equivalent was 0.59 ± 0.91 . Transient high IOP post-OP was highest among polypropylene SSFIOL, 8.1%. Polypropylene SSFIOL had glaucoma in 0.31% and corneal decompensation in 0.15%. A suture break or erosion was reported in 1.19%. None of the other groups develop glaucoma or corneal decompensation. Suture erosion was not reported in Gore-tex SSFIOL.

GSFIOL had a 5.3% reoperation rate, mostly for IOL repositioning.

Conclusion: All types of scleral lens fixation showed improvement in initial outcomes. SLSFIOL and Gore-Tex SSFIOL have low complication rates.

Conflicts of interest: None

Funding: None

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P231

STRABISMUS: AN ASSESSMENT OF EDUCATIONAL CONTENT ON SOCIAL MEDIA

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Introduction: Patients with strabismus use social media to access health information related to their condition. Previous studies have assessed ophthalmology content on TikTok, but there has been no in – depth analysis of strabismus related content on this and other social medial platforms. The aim of this study was to assess the quality of educational content for patients with strabismus on social media.

Method: TikTok and Instagram were searched for 3 hashtags: #strabismus, #squint, and #lazyeye, using freshly created accounts. The top 50 videos under each hashtag which met inclusion criteria were analysed for metrics including views, likes, comments, type of creator and type of content. Educational videos aimed at patients were analysed by 3 separate investigators using the Global Quality Scale, mDISCERN and PEMAT-AV tools (understandability and actionability). Statistical analysis was

conducted using Pearson's Correlation Coefficient to assess association between the views and likes of educational videos against their quality scores using the tools.

Results: The hashtag #squint was excluded from the study as it generated significant irrelevant video content. From the top 50 videos for #strabismus and #lazyeye on each platform ($n = 200$), the included educational videos aimed at patients ($n = 54$) had an average mDISCERN of 2.8 (/5), Global Quality Scale (GQS) of 3.3 (/5), PEMAT-AV understandability score of 78.5%, and PEMAT-AV actionability score of 42.3%.

Pearson's correlation analysis revealed significant associations between video quality scores and engagement metrics. Specifically, video views were negatively correlated with both the mDISCERN score ($r = -0.3199$, $p = 0.01833$) and the GQS score ($r = -0.3399$, $p = 0.011888$). However, views showed no significant correlation with the PEMAT-AV understandability score ($r = -0.2472$, $p = 0.071764$). Interestingly, views did show a positive correlation with the PEMAT-AV actionability score ($r = 0.292$, $p = 0.032155$). Likes were also correlated with quality metrics. A significant negative correlation was found between likes and the mDISCERN score ($r = -0.3661$, $p = 0.006494$) as well as the GQS score ($r = 0.3669$, $p = 0.006338$). This revealed that lower quality videos had a greater number of views and likes in comparison to high quality educational videos. Additionally, there was a negative correlation with PEMAT-AV understandability, although the association was weaker ($r = -0.2520$, $p = 0.066022$). Similar to views, likes were also positively correlated with the PEMAT-AV actionability score ($r = 0.3364$, $p = 0.012878$).

With regards to creator type, 66.7% (36/54) of educational videos were created by a healthcare professional, with 35.2% ($n = 19$) created by Ophthalmologists, and 27.7% ($n = 15$) by Optometrists. There was no content found in the search on either platform which had been created by a UK based ophthalmologist or orthoptist.

Conclusion: This study highlights the significant role of social media in disseminating health information on strabismus. While TikTok and Instagram provide platforms for patient education, the quality of the educational content varies widely. The findings demonstrate that lower-quality videos tend to garner more engagement in terms of views and likes, which could lead to patients following misleading or incomplete medical advice. The positive correlation between actionability and lower-quality videos is particularly concerning, as these videos, while actionable, may encourage patients to take steps based on incorrect or poorly explained information. To improve the effectiveness of social media as an educational tool, there is a clear need for more content created by qualified healthcare professionals, especially UK – based ophthalmologists and orthoptists, to ensure that patients receive accurate, comprehensive, and safe guidance.

Conflicts of interest: None

Funding: None

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P232

THE ROLE OF NON-LEUKOCORIA RED REFLEX REFERRALS IN DETECTING OCULAR PATHOLOGY IN AN ETHNICALLY DIVERSE POPULATION: A MULTI-CENTRE STUDY

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Introduction: The 'red' reflex test is a key screening tool used to identify serious ocular pathologies, such as cataracts and retinoblastoma, in children and infants. However, recent studies suggest variability in its positive predictive value (PPV). Unnecessary urgent referrals create a burden for already-stretched paediatric ophthalmology services in addition to parental distress.

It is well established that leukocoria (a white reflex) is an important indicator of urgent sight- or life-threatening conditions. Non-leukocoria referrals (absent or abnormal 'red' reflexes) may indicate non-urgent conditions, such as strabismus or amblyopia based upon Brückner's theory, in addition to urgent conditions. Although referrers are advised to compare the child's reflex to their parents' reflexes in the Newborn and Infant Physical Examination protocol, non-leukocoria is often misinterpreted in non-white children and infants due to ethnic differences in reflex appearance. A recent study on decolonizing the term 'red' reflex raises the possible impact of this terminology. Our study aimed to evaluate the outcomes of non-leukocoria referrals at two large centres in London.

Method: We retrospectively reviewed 'red' reflex referrals at Moorfields Eye Hospital (January to December 2022) and Royal London Hospital (January 2021 to February 2023). Referrals were stratified into absent or abnormal reflexes (non-leukocoria) and white reflex (leukocoria) referrals. We excluded leukocoria referrals. Examination outcomes, patient demographics, referral source, reason for referral, and lag time between referral and appointment were assessed. Urgent conditions included sight-threatening (e.g. cataract) and/or life-threatening ocular pathology (e.g. retinoblastoma) requiring intervention. Nonurgent conditions included strabismus and amblyopia.

Results: The median age was 3 months (range 1–84) for 43 children. 20.5% were white. The main source of referrals was general practitioners (86.1%), followed by neonatologists and paediatricians (11.6%). 76.7% of referrals were seen within 14 days as part of the two-week wait cancer referral pathway. The median waiting time was 6 days (range 0–71). 92.8% were normal on examination, with no cases of urgent conditions. Two non-urgent strabismus and one high refractive error were found, resulting in a PPV of 6.98 % for non-urgent conditions. No referral mentioned an examination of parents.

Conclusion: Our study confirmed a low diagnostic value of absent or abnormal 'red' reflex testing in an ethnically diverse population of children and infants for non-urgent conditions. This increases the burden on paediatric ophthalmology services as these children need to be seen urgently. The low proportion of white babies suggests that there is misinterpretation of the 'red' reflex in an ethnically diverse population. We recommend using a racially inclusive term ('glow') reflex or anatomically based ('fundus') reflex to make best use of the referral pathway, while maintaining an emphasis on leukocoria screening. In addition, we are keen to emphasise the importance of assessing the parents before referral.

Conflicts of interest: None

Funding: None

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P233

UK OPHTHALMOLOGISTS' EXPERIENCE WITH THYROID EYE DISEASE IN CHILDREN (A NATIONAL SURVEY)

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Introduction: The risk of developing Thyroid eye Disease (TED) in children and adolescents is similar to adults. However, TED is rare in paediatric ophthalmology clinics and orbital clinics. The aim of this study was to survey the experience of UK ophthalmologists involved in the management of TED in children in the last 10 years with regards to signs, severity, treatments and outcomes.

Method: An online survey was circulated to ophthalmologists with a subspecialty interest in paediatric ophthalmology and strabismus. A questionnaire was designed to capture the number of cases seen and their demographics in the last 10 years. Details were collected regarding examination findings of dysthyroid optic neuropathy, proptosis, ocular motility dysfunction, eyelid signs, exposure keratopathy, their management plan and outcome. Data regarding thyroid hormonal profile (Hyper, Hypo or Euthyroid) and smoking status was also collected.

Results: There were 61 respondents from UK & Ireland and 40 had experience with TED in children. The total reported cases were 92 over the last 10 years. 86% of cases were female and the most common affected age group was 11–15 years 46/92 (50%). 86/92 (93%) cases presented with hyperthyroidism and the remaining 6/92 (7%) were euthyroid. 2/92 (2%) were known smokers. The most common presentations were lid retraction 76/92 (83%), proptosis 74/92 (80%), and lid lag 70/92 (76%). Orbital decompression was undertaken in 7 cases (8%). There were 3 cases (3.2%) with permanent visual loss.

Conclusion: Our survey demonstrates that paediatric TED can result in permanent visual loss. Close collaboration with the paediatric endocrinologists is important as the control of the thyroid status can improve TED and also help in identifying vision threatening cases.

Conflicts of interest: None

Funding: None

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SUSTAINABILITY (INCLUDING ABSTRACTS THAT AIM TO DECARBONISE HEALTHCARE, OR CONTRIBUTE TO SOCIAL AND ENVIRONMENTAL WELLBEING)

P234

AN INVESTIGATION INTO THE AUTOMATION OF 3D MODELLED BESPOKE EYEWEAR USING INDIVIDUAL-SPECIFIC PARAMETRIC DATA AND ADDITIVE MANUFACTURING

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Introduction: Eyewear frames are mass-produced, reducing choice. There is no room or incentive to accommodate individual-specific facial features.

With the new era of mass customisation provided by the advent of 3D printing, the possibility of manufacturing bespoke eyewear frames on a production line of one can become a reality.

Eyewear is an essential part of life for most people, irrespective of age and social status. Its use is evident in prescription eyewear: myopic, hypermetropia, presbyopia, squints, the Leisure/fashion industry, protective wear in manufacturing, the sports sector: skiing, swimming, and for Health workers: loupes and protection.

Using 3D printing will drive the development of mini-factories located close to the end user. Styles can be chosen, altered, and reproduced in less than a day. Production will be on demand, with inventory and storage being discarded.

Presently, the bulk manufacturing of Eyewear is possibly harming the environment. Eyewear is mainly manufactured by centralized manufacturing systems (CMS). A few disadvantages are transportation costs, which result in CO₂ emissions, and the cost of disposing of excess stock.

Distributing manufacturing systems (DMS) support regional manufacturing, allow the decentralisation of production, and provide personalised products to customers in the surrounding region. The advantages are cost reduction, short delivery time, and reduced CO₂ emissions.

The 3D Printed eyewear with DMS will enable on-demand manufacturing whilst reducing transport and emissions.

3D printing will streamline the multiple steps and personnel involved in the manufacture of single eyewear.

Method:

- This was an analytical, non-comparative, non-interventional, observational study.
- Data collected and stored on a password-controlled personal computer
- Participants sent in digital photographs taken with a measuring ruler in specific positions to enable accurate measurement reference points.
- Calibrated Measurements are taken from photographs.
- Eyewear frames were created using Fusion 360 (Autodesk) parametric digital workflows
- Demographics: 20 participants, 80% male, All adults
- The image was also rendered in CAD
- Data extracted from measurements were as follows:
- Interpupillary distance
- Right and Left Nasopupillary distance
- Ear to ear – Frontal
- Forehead to ear – Straight
- Forehead to ear – Slant

This data was used in the parametric table, within the computer-assisted design (CAD) for sketching, and in the additive manufacturing of personalised eyewear frames tailored to individuals rather than the mass market.

Results: These measurements were used to create digital workflows for 3D printing personalised eyewear frames tailored to individuals rather than the mass market.

There was evidence of facial asymmetry in 95% of the participants.

There was a wide variation in all the measurements. Also, asymmetry in comparing the Right vs Left Nasopupillary distance (NPD) showing Nasopupillary distance was observed.

Opticians usually use the Interpupillary distance (IPD) to fit prescription eyewear. This places the midpoint at the nasal bridge.

IPD measurement is a significant part of eyewear prescription. It measures the distance between the left and right pupils.

At the opticians, individuals with facial asymmetry will benefit from NPD instead of IPD when purchasing prescription eyewear.

The use of 3D printing for the manufacturing of eyewear is not new. However, it is used in rapid prototyping and mass customisation.

Conclusion: This technology is a proof of concept that will enable the Rapid prototyping of eyewear frames for any industry or individual.

In addition, rapid prototyping could facilitate the introduction of a clinic that makes frames for people with facial deformities, especially children with craniofacial abnormalities.

The above findings show that NPD's sensitivity to identifying asymmetry will ensure that prescription eyewear is accurately fitted and reduce errors.

Similarly, for facial asymmetry and, consequently, nasopupillary asymmetry (NPA), a prismatic effect in prescription eyewear can arise, as can negative effects of off-optical-axis viewing if used in prescription eyewear.

3D printing will enable the easy, quick, and cost-effective manufacture of eyewear and the customization of style and facial features.

Should this technology become an actualisation, there will be no need to hold stock of eyewear frames for sustainability. This will reduce the carbon footprint associated with transporting and manufacturing eyewear.

Conflicts of interest: none

Funding: None

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P235

COMPARATIVE CARBON FOOTPRINTING STUDY OF REUSABLE VS. DISPOSABLE INSTRUMENTS IN CATARACT SURGERY

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Introduction: Theatre activity is a major source of hospital waste, accounting for 21–30% of total output, and is 3–6 times more energy-intensive than any other hospital department.

Cataract remains the leading cause of blindness worldwide, and cataract surgery is the most commonly performed surgical procedure worldwide, but to date has received little analysis in this field.

The aim of this study was to evaluate the carbon impact of streamlining single-use packs, and transitioning from single-use items to reusable equivalents in cataract surgery.

Method: A focus group and prospective audit was carried out to identify single-use pack items for streamlining. A bottom-up, process-based carbon footprint analysis was conducted of single-use items and reusable alternatives using activity data provided by the suppliers and greenhouse gas (GHG) conversion factors from the UK Government GHG Conversion Factors for Company Reporting database.

Results: Total carbon savings from streamlining was 935 kgCO₂e/ year, and switching, was 309 kgCO₂e/year. Single-use items had a 27 times greater carbon footprint than reusable equivalents (322 kgCO₂ vs. 12 kgCO₂e respectively). A net carbon saving was achieved from 19 or more re-uses of the reusable items compared to single-use equivalents.

Conclusion: This study provides evidence of carbon emissions savings both from streamlining disposable instruments in cataract surgical packs, and from switching from single-use to reusable instruments. Mitigation strategies targeting hotspots in resource intensive areas will be an important aspect of reducing the climate burden of surgery. Change in practice and policy is required in all surgical specialities to realize net zero carbon healthcare.

Conflicts of interest: None

Funding: None

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P236

EVALUATING SUSTAINABILITY IN OPHTHALMIC THEATRES: THE SINGLE-USE SHARPS PERSPECTIVEJohnny Ma Kwok Kit¹, Jack Lisle², Jennifer HY Tan²¹School of Medicine, University of Sheffield, Sheffield, UK; ²Sheffield Teaching Hospitals NHS Foundation Trust, Sheffield, UK

Introduction: The NHS produces 586,000 tonnes of waste annually, generating a carbon footprint of 30,000 tonnes of CO₂. Healthcare waste consists of various streams, each with a different disposal method and associated carbon footprint. High-temperature incineration, used for disposable surgical sharps in the ubiquitous yellow bins, has one of the highest carbon footprints at 1074 kg CO₂ per tonne. Reusable sharps, while more sustainable, can compromise sharpness and precision when used repeatedly.

In ophthalmology, particularly in oculoplastics, demand for high-performing instruments often necessitates the use of disposable sharps. Recycling these instruments presents a more sustainable option, with hospital waste emissions estimated between 21–65 kg CO₂ per tonne. Grey sharps bins, designed for recycling metallic waste, offer an environmentally friendly alternative. However, many healthcare practitioners are not familiar with or accustomed to using these bins. This project aimed to enhance sustainability by promoting the use of grey sharps bins for recyclable materials within the ophthalmology operating department at the Royal Hallamshire Hospital in Sheffield.

Method: Over a 10-week period, the utilisation of recycling sharps bins was tracked across 3 ophthalmology operating theatres and one Minor Procedures Suite (MOPS). Six bins were included, with their weights recorded on a weekly basis. Initial measurements established the baseline usage for two weeks. After two weeks of initial measurements were collected to establish baseline use, utilisation was promoted by raising awareness through face-to-face presentations with each theatre team, and posters above each bin. The remaining 8 weeks were recorded to determine any change in disposal patterns. Carbon emission estimates were derived using the Inventory of Carbon and Energy (ICE) database, a leading repository of estimated carbon emissions for materials.

Results: Over the 10-week period, a total of 79.5 kg of metallic waste was collected in the sharps recycling bins. During the initial two-week baseline period, an average of 0.683 kg of disposable instruments were collected in each theatre each week (range: 0.1–1.4 kg; SD: 0.534). In the eight weeks following interventions to promote their use, this increased to 1.639 kg per theatre per week (SD: 1.333). Although this represents a 2.4-fold increase, a short baseline period and inherent variability within the nature and number of operations each week meant it fell short of statistical significance ($p = 0.0906$).

Conclusion: This study shows that relatively simple interventions can have positive impacts on waste disposal practices. 79.5 kg of predominantly surgical stainless-steel waste was collected.

Using ICE database emission factors of 4.18 kg CO₂/kg for stainless steel, the estimated carbon emissions offset was 332.31 kg CO₂, averaging 33.23 kg CO₂ per week. Financial benefits are harder to calculate, but NHS data show that the financial cost of waste disposal mirrors that of the carbon footprint. Overall, this study highlights the significant environmental impact of metallic waste in surgical settings and the potential benefits of improved waste segregation and recycling. Further studies with longer baseline periods and enhanced tracking systems could refine these findings and help embed more sustainable practices within healthcare.

Conflicts of interest: None

Funding: None

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P237

GOING THE EXTRA MILE: CERTIFYING VISUAL IMPAIRMENT IN GLAUCOMAElfride Sanjana¹, Emily Ball¹¹Shrewsbury, Shrewsbury and Telford NHS Hospital Trust

Introduction: Certification of visual impairment (CVI) provides an indication of the incidence of sight loss. In England, between 2022/2023, there were 20,330 new CVI registrations, a three percent decrease compared to 2019/20. CVI involves individuals being registered as sight impaired (SI) or severely sight impaired (SSI) and is determined by the patient's visual acuity and visual field defect severity. This study aimed to provide an indicator of the number of patients seen in the primary care setting who were referred to the Hospital Eye Service (HES) between 01 July 2023 and 31 July 2024, without receiving CVI.

Method: A retrospective study that analysed 680 new patients, assessed between July 2023 and July 2024, was retrieved via the Trust's electronic clinical noting software, Medisight®, to identify the number of patients who were referred to ECLO for CVI registration at their first appointment with the Doctor. All patients referred via community ophthalmic services or community optometrists with advanced glaucomatous visual field loss were included within this study and their demographics, including age and sex, family history, referral pathway, visual acuity, intraocular pressure (IOP) measurements, cup-disc ratios (CDR), mean deviation values for visual field defects with corresponding severity grading, glaucoma diagnosis and CVI severity (SI or SSI) were analysed. Each patient's visual field defect severity grading was determined by utilising the Hodapp-Parrish-Anderson (HPA) Visual Field Criteria of Glaucoma Severity.

Three patients were excluded from this study due to their pre-existing comorbidities rendering them unsuitable for inclusion, such as AMD and retinal vascular diseases. One patient suffered from advanced dementia, so visual field testing was not possible. However, this patient's presenting IOP and disc appearance was sufficient to stage the severity of their glaucoma.

Results: Among the 680 new patients assessed within HES, between 01 July 2023 and 31 July 2024, 3% ($n = 20$) were referred to ECLO services for CVI. The average patient age was 77 years old and 55% ($n = 11$) of the 20 patients who received CVI were male. Furthermore, among the 20 patients analysed within this study, 35% ($n = 7$) were referred without receiving CVI by Community Health Eye Care (CHEC), 55% ($n = 10$) were referred by their community optometrist and 10% ($n = 3$) were referred internally by an ophthalmologist.

All patients included within this study presented with a CDR of 0.8 or worse in at least one eye. Consequently, only 15% of patients were recorded as having moderate VF damage in their right eye, compared to 70% of patients having severe VF damage in their right eye. Comparatively, only 5% of patients suffered moderate VF damage in their left eye, opposed to 85% suffering from severe VF damage. 6 patients were certified as SI and all remaining were severely sight impaired.

Eighty five percent of patients were diagnosed to have POAG ($n = 17$). Ten percent were diagnosed to have NTG ($n = 2$) and one patient was diagnosed with aqueous misdirection.

Conclusion: CVI is a means of identifying patients in need of support and social care and it is an important health indicator to the local and national level in the United Kingdom.

Health policies prioritise prevention and CVI data provides information on preventable causes of blindness. The incidence of SI or SSI is an indicator of the effectiveness of policies and the need for reforms. Visual impairment from glaucoma is asymptomatic, unlike other causes, such as Age-Related Macular Degeneration and Diabetic Retinopathy.

Improved patient education, robust screening programmes within the community setting and strengthening the workforce and health infrastructure at different tiers of healthcare is the need of the hour. The role of timely referral to the HES, awareness amongst health professionals on the benefits of CVI registration and a continuous dialogue between the community and HES cannot be over-emphasised.

Conflicts of interest: None

Funding: None

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P238

REDUCING THE CARBON FOOTPRINT OF INTRAVITREAL INJECTIONS

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Introduction: Intravitreal injections are quickly becoming one of the most common procedures in Ophthalmology with a significant increase in their numbers over the years fuelled by the advancement of healthcare and rise in aging population. In England alone, an estimated 677,000 injections were performed in the year 2022. However, the growing volume of these procedures raises concerns regarding their environmental impact, particularly the carbon footprint associated with materials, logistics, patient and staff travel, and clinical processes. It's high time that we as healthcare workers respond to the call for balancing clinical efficacy with sustainability.

Method: A hybrid methodology of Life cycle analysis and Input-output model was used to calculate the carbon footprint of an intravitreal injection.

A travel questionnaire was drafted to collect data on patient and staff travel. Data were collected regarding procurement of materials, cost of intravitreal packs, drugs and building energy use. Greenhouse gas emissions from waste disposal from the injection was estimated using most recent UK Government GHG conversion publication.

Additionally, in our study, we are looking at two centres within a single Trust, but with different population densities and demographics, and we expect to shed light on how significant this difference is for the IVTI carbon footprint. Previous publications have looked at very different healthcare systems and demographics. Ours will bring something new by focusing on a UK NHS urban and semi-urban population.

Results: Although an accurately reproducible estimate is difficult, carbon emissions from anti-VEGF agents were the single highest contributor. Patient travel is the next largest contributor to the carbon footprint of an intravitreal injection, accounting for up to three quarters of emissions. The Injection pack and sourcing accounted for around 20 percent of emissions. The least contributing factor was waste disposal followed by building energy use.

It's worth noting that previous studies have understandably excluded the "cradle-to-grave carbon footprint" of the drug, and as a result may have underestimated the overall IVTI carbon footprint, and in the process overweighted the contribution of staff and patient transport. We will show the differential results when we estimate the carbon footprint both with and without inclusion of the drug.

Discussion: We conducted a review of current practices and literature to identify strategies for reducing the carbon footprint of intravitreal injections.

1. **Supply Chain:** Examining the use of eco-friendly materials, reducing single-use items, and adopting pre-filled syringes, reducing unused items from the injection pack.
2. **Clinical Practices:** Evaluating batching of appointments, reusable instruments, bilateral injections and vial-sharing protocols where feasible.
3. **Waste Management:** Exploring safe disposal methods and minimising excess packaging.
4. **Travel Reduction:** Considering telemedicine and optimised scheduling to the nearest centre to reduce patient and staff travel.

Conclusion: Reducing the carbon footprint of Intravitreal injections presents a valuable opportunity for ophthalmology to align with broader sustainability goals while maintaining high standards of patient care. On a large scale, these strategies could have a significant impact.

Implementing these changes across clinical settings requires collaboration between healthcare providers and pharmaceutical companies. Ultimately, these practices go a long way in leaving behind a healthy planet for our future generations.

Conflicts of interest: NA

Funding: N/A

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P239

SHINING A LIGHT ON DOMESTIC ABUSE IN OPHTHALMOLOGY: IMPROVING CLINICIAN CONFIDENCE IN TRIAGING AND MANAGING ACUTE PATIENTS WHO DISCLOSE DA AT IMPERIAL COLLEGE LONDON

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Introduction: It became apparent when two cases of Domestic Abuse (DA) presented in close succession to our Eye Casualty at Imperial College that there was a gap in knowledge in how to discuss and handle these cases. Doctors and nurses alike did not feel confident in the safeguarding guideline and necessary steps to protect these vulnerable patients, nor how to fully ascertain level of risk the individual is in. On further investigation it became clear that we as a Trust, and more widely Ophthalmology as a speciality, has lacking data on how DA presents in the acute Ophthalmic setting.

Method: We addressed this issue by gathering anonymous feedback from clinical staff to understand baseline knowledge. We included doctors, nurses and healthcare assistants in our data. This feedback demonstrated our concerns were accurate and only 7.5% of our staff body were routinely asking patients about their safety at home, and over half were unaware of the escalation plan if patients were to disclose DA. With this information, we also

investigated the Imperial DA guideline for our Ophthalmic unit and it became evident that in reality there was no clear framework in place which would need to be addressed.

It was evident that clinicians were being put off asking questions given they felt unequipped to handle the answer, which was a patient safety concern. We aimed to improve this by firstly, creating a robust DA framework with the help of our Trust Safeguarding team which was relevant for the Ophthalmic unit, and secondly, to provide teaching sessions to our clinicians. These sessions included routine prompts when triaging patients, assessing dependents or risk in the house, and familiarising clinicians with the new guideline.

Results: After our teaching sessions we gathered further data from those that had attended and found that 91.7% of people felt more confident discussing this topic and 87.5% agreed that they would want to change their practice to routinely ask patients about DA. 100% were aware of the local guideline.

Some of the challenges that arose during this project was a lack of traction from staff. It was unclear whether this is due to the emotion of this topic, or that it is just easier to ignore. If you don't ask patients about DA they will rarely offer the information which means it feels simpler to continue to sweep it under the carpet. This has meant data is difficult to gather when it comes to how big an issue we have. This project has also shone a light on lacking data in a wider sense within Ophthalmology. There is minimal British data on how DA presents in the acute Ophthalmic setting compared to that of other safeguarding or non-accidental injury.

Conclusion: The main aim of this project was to shine a light on this really important issue. This is truly one of the instances that an Ophthalmologist can save a life and we are pleased that within our Trust we have improved our framework and clinician confidence on how DA should be handled for patients that disclose within the acute setting. We hope to continue our work by raising regional and national awareness with further research and data collection on what types of pathology present to Eye Casualty in a DA setting over the next year.

Conflicts of interest: None

Funding: None

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P240

SUSTAINABILITY ASSESSMENT OF CATARACT SURGERIES USING THE EYEFFICIENCY APPLICATION

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Introduction: Cataract surgery is one of the most frequently performed procedures worldwide, and its environmental impact, particularly carbon emissions and waste, has become a growing concern. The Eyeefficiency application, developed by the RCOphth Sustainability Working Group and funded by a World Bank research grant, offers a tool to assess sustainability in cataract surgeries. The app collects key data on productivity, patient complexity, staffing, and environmental factors, allowing comparisons between global units. The study aimed to assess the sustainability of cataract surgery using Eyeefficiency.

Method: Data was collected at a district general hospital in the southeast of England. Information on service details, appointment patterns, staffing, building and energy use, theatre operations, waste disposal, consumables, reusable instruments and linen was gathered and input in the Eyeefficiency application. Permissions to use the application were obtained prior to data collection. A time and motion

study was conducted between 29/11/24 and 6/12/2024 to assess the time and surgical efficiency of a theatre session and calculate the waste generated per session. This data was incorporated into the Eyeefficiency toolkit to generate a sustainability report.

Results: The sustainability assessment revealed that each cataract surgery at generated 695.47 kg CO₂e which was significantly higher than the global average of 96.69 kg CO₂e. The breakdown of CO₂ emissions included 641.46 kg from the disposal of supplies, 30.46 kg from the production of disposable/consumable supplies (excluding pharmaceuticals) and 14.38 kg from pharmaceutical supplies. Additionally, the hospital generated 3.29 kg of waste per cataract surgery, compared to the global average of 0.82 kg. The total cost per surgery at our setting was GBP 652.23, while the global average cost was GBP 235.91.

Conclusion: The sustainability assessment of cataract surgeries highlighted significant disparities in carbon emissions, waste generation and cost compared to the global averages. These findings provide insight into areas where changes could be made to reduce the carbon footprint, including optimizing consumable use, reduction in single use instruments, improving waste management and reducing energy consumption. This toolkit offers valuable data to help eye units, governments and NGOs benchmark their performance and work toward more sustainable practices in cataract surgery.

Conflicts of interest: None

Funding: None

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P241

USING A THREE-STAGE DELPHI PROCESS TO REDUCE THE CARBON FOOTPRINT OF NHS INTRA-VITREAL INJECTION SERVICES

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Introduction: The National Health Service (NHS) has set the ambitious goal of becoming the world's first net zero health service by 2040. Intravitreal injections (IVIs) are one of the most common eye procedures performed in the United Kingdom (UK), with an estimated 800,000 IVIs performed in 2022 alone, likely proving a major contributor to the carbon footprint of NHS ophthalmic services.

As the UK population ages, and the demand for IVIs increases, the need to design environmentally sustainable NHS IVI services is of increasing importance. Previous studies have found that the Delphi process is a useful tool for identifying strategies to reduce the carbon footprint of NHS cataract services. Our study aimed to evaluate the role of the Delphi process for improving the environmental sustainability of NHS IVI services.

Method: The Delphi process has previously improved sustainability of local cataract services. Using it, we conducted the following:

1. Collected ideas from all staff members involved with IVI services at an NHS teaching hospital on how to reduce the carbon footprint of IVI services locally.
2. Suggestions were de-duplicated, anonymised, and ideas were grouped into themes.
3. Staff were invited to discuss ideas and discard those that were unrealistic.

4. Refined ideas were put into an online poll and ranked by the same staff members.
5. Implemented the highest ranked ideas.
6. Estimated the financial savings, carbon, and waste savings before and after implementation.

Waste generated per IVI was weighed before and after the Delphi process and waste disposal costings were calculated using data provided by local hospital management.

Results: The three-stage Delphi process generated 23 suggestions from a wide range of staff cadres, refined to 17 suggestions following de-duplication and grouping. Of these suggestions, eleven focused on reducing the use of single-use products, five targeted strategies for streaming logistics within the IVI service, and one explored the potential to introduce greener waste disposal practices. We sought to implement the top five ranked suggestions.

One suggestion looked to improve waste disposal practices, including recycling. Staff were educated on recycling, and recycling bins were purchased, with an additional 108.9 g of waste per injection being recycled.

Our waste reduction plan involved ceasing to use plastic aprons as they were deemed unnecessary for IVI by infection prevention, as well as removal of three eye spears, plastic forceps, and a cardboard tray from the IVI pack. These ideas reduced waste by up to 39.3 g and saved 42 pence per IVI.

The first and fifth ranked suggestions were to ask manufacturers to reduce packaging, and to shift prescriptions and consent forms online but efforts were limited by manufacturer and management restrictions respectively.

As around 12,000 IVIs are performed locally per annum, our Delphi process is projected to increase waste recycled by 1,306.8 Kg, reduce overall waste by 471.6 Kg, and save £5,040, per year.

Conclusion: Our study suggests that the Delphi process is an effective tool for reducing the carbon footprint of NHS IVI services and improving engagement with the sustainability agenda amongst staff members. Its implementation within our NHS trust is proving promising, with a significant reduction in plastic consumption, and an increase in waste being recycled, coupled with associated cost reduction. With the already-high numbers of IVIs increasing as the population ages, the Delphi process proves a useful tool in reducing the carbon impact of IVIs.

Conflicts of interest: None

Funding: None

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P242

UTILISATION OF THE SUSTAINABILITY INDEX FOR DISPOSABLES IN CATARACT SURGERY (SIDICS) TO MEASURE THE SUSTAINABILITY AND CO₂E EMISSIONS OF A STANDARD NHS CATARACT PACK

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Introduction: Cataract surgery, one of the most commonly performed procedures globally, with around 30 million surgeries annually, contributes significantly to the healthcare sector's carbon footprint. This environmental impact is exacerbated by the extensive use of disposable consumables. However, this is an area where small changes could have a large impact on the overall carbon footprint, especially considering the high volume

of cataract surgeries performed each year. The Sustainability Index for Disposables in Cataract Surgery (SIDICS), developed by the European Society of Cataract and Refractive Surgeons (ESCRS), serves as a tool to evaluate the sustainability of cataract packs and calculate the carbon footprint measured in carbon dioxide equivalents (CO₂e), of consumables. This study aimed to assess the sustainability of cataract packs and calculate the CO₂e emissions associated with these consumables using SIDICS.

Method: Data was prospectively collected at a district general hospital in the South East of England. The SIDICS tool requires information on the annual number of cataract surgeries performed and the corresponding number of cataract packs used at the trust. The details of the 19 single-use components of the standard NHS cataract pack were also entered into SIDICS. Additionally, a proforma was used to assess the usage of these pack components during 22 cataract surgeries conducted between 19/08/24 and 30/08/24.

Results: The hospital used 6345 cataract packs, producing a total CO₂ footprint of 40,336 kg CO₂e. When compared with the ESCRS benchmark, a potential CO₂ saving of 15,543 kg CO₂e was identified. During the 22 cataract surgeries in this study, certain consumables—two cannulae and an 18gauge blunt fill needle—were unused and discarded. Recommendations for reducing the environmental impact included using smaller facial drapes (with a potential saving of 9,666 kg CO₂e), replacing single-use gallipots and kidney trays with reusable alternatives (potential saving of 2,818 kg CO₂e) and eliminating postoperative shields. These modifications could substantially lower the carbon footprint of cataract surgeries.

Conclusion: Considerable variation in the usage of single-use items in cataract packs presents opportunities to reduce CO₂ emissions. Optimizing the pack contents and transitioning to reusable or more sustainable alternatives will result in a significant reduction in the carbon footprint of cataract surgery procedures.

Conflicts of interest: None

Funding: None

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UVEITIS

P243

ANTI-ADALIMUMAB ANTIBODY DEVELOPMENT IN ADULTS TREATED FOR NON-INFECTIOUS UVEITIS WITH ADALIMUMAB BIOLOGIC THERAPY

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Introduction: Adalimumab is a fully human therapeutic monoclonal antibody directed against tumour necrosis factor- α used in the treatment of non-infectious uveitis. Anti-adalimumab antibody (AAA) levels are used in combination with clinical features to determine adalimumab treatment failure, where levels greater than 10 AU/L are considered positive. Currently, there are no recommendations regarding AAA testing and no clinical guidelines for the management of patients who develop AAAs. Management decisions are made on a case-by-case basis.

Our objectives were to investigate the rate of AAA development in patients with non-infectious uveitis treated with

adalimumab and to review clinical management decisions for patients with detectable AAAs.

Method: This was a retrospective cohort analysis of all adults treated with adalimumab for non-infectious uveitis, managed within the uveitis service at a single tertiary centre, between September 2008 and September 2024. Data was collected from electronic and paper patient records. AAA levels, adjuvant immunosuppressive treatment and clinical management decisions for patients with detectable AAAs were recorded and analysed.

Results: 149 patients received adalimumab for non-infectious uveitis. 20/149 (13.5%) patients were on adalimumab alone, 16/149 (10.7%) patients were on a sub-therapeutic adjuvant immunosuppressant and 113/149 (75.8%) were on full therapeutic adjuvant immunosuppressant.

68/149 (45.6%) were tested for AAAs. Of those tested, 21/68 had active uveitis and 47/68 had no active uveitis but underwent random testing. 81/149 (54.4%) patients had no active uveitis and were not tested. The rate of positive AAAs (>10 AU/L) in our cohort was 23/149 (15.4%). 14/21 (66.7%) who were tested due to a flare of their uveitis were positive for AAAs. 9/47 (19.1%) who had no active uveitis who were randomly tested, were positive.

Of the 23 patients positive for AAAs, 5/23 patients had detectable levels between 10–20 AU/L, 4 of these patients continued with the same adalimumab regime and 1 patient ceased adalimumab therapy. Subsequent AAA levels in this group became undetectable or stayed the same. 8/23 patients had detectable AAAs between 20–150 AU/L. 1 patient ceased adalimumab therapy, 1 patient continued on the same adalimumab regime and 6 patients had adalimumab increased from fortnightly to weekly. Of the 6 patients where adalimumab frequency was increased to weekly, subsequent AAA levels in 4 became undetectable and 2 reduced but were still detectable. 10/23 patients had detectable AAAs >150 AU/L. All 10 patients ceased adalimumab therapy.

Conclusion: Adalimumab is an effective treatment in our cohort of adults with non-infectious uveitis. We report a low incidence of AAAs. AAAs are more likely to be raised in the presence of active uveitis. However, our study shows that AAAs can be elevated in patients on adalimumab with quiet eyes, therefore we recommend checking AAAs in stable patients as their treatment may require alteration. Of those with detectable AAAs, a level greater than 150 AU/L was considered definite treatment failure and adalimumab was discontinued. For the majority of patients with AAAs between 20–150 AU/L, increasing adalimumab from fortnightly to weekly led to undetectable AAA levels. Our data indicates that uveitis in patients with detectable AAAs of less than 20 AU/L remain controlled on their current adalimumab regime and did not require treatment adjustment. We infer that the threshold of positive adalimumab antibodies that necessitates a change in treatment can be raised from 10 AU/L to 20 AU/L.

Conflicts of interest: None

Funding: None

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P244

ANTI-ADALIMUMAB ANTIBODY DEVELOPMENT IN CHILDREN AND YOUNG PEOPLE TREATED FOR NON-INFECTIOUS UVEITIS AND RHEUMATIC DISEASES WITH ADALIMUMAB BIOLOGIC THERAPY

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Introduction: Adalimumab is a monoclonal antibody that inhibits the action of Tumour Necrosis Factor alpha and downregulates inflammation. It is an effective, evidence-based, biologic therapy approved by National Health Service England (NHS England) in the UK to treat Children and Young People (CYP) with non-infectious uveitis, regardless if the CYP has juvenile idiopathic arthritis (JIA) or not. However, development of anti-Adalimumab antibodies (AAA) is an emerging recognised phenomenon and has previously been linked to treatment failure and disease relapse. Currently, there is limited evidence and guidance on the use of AAA testing to inform treatment management, and the factors influencing AAA development; prompting our observational cohort study at a single tertiary referral centre, in a national children's hospital in the UK.

Method: All CYP treated with Adalimumab within a specialist paediatric rheumatology service, between 2015–2023, were included in this retrospective observational study. Potential factors influencing AAA development and its implications for CYP were explored. Data was collected from hospital systems including patient databases, clinical records and blood test result software; clinical diagnosis, length of treatment, concurrent use and dosage of disease modifying anti-rheumatic drugs (DMARDs), reason for testing, AAA levels, and outcomes following results.

Results: 390 CYP were initiated on Adalimumab treatment between 2015–2023. 112/390 (28.7%) CYP were tested for AAA, with 156 individual tests performed. 135/156 (86.5%) tests were performed due to flare of disease. Other reasons for testing included suspected noncompliance with medication, clinical interest, and monitoring.

51/112 (45.5%) CYP had AAA detected (defined as >10 mg/L) after an average of 2.4 years (range: 0.3–5.8yrs) of being treated with Adalimumab. There was no direct correlation between AAA level and time from treatment onset; some showed high antibody levels within the first 4 months of starting treatment, while others did not have significant antibody development after 6 years of treatment.

21/156 (13.5%) CYP were not taking concurrent DMARDs alongside Adalimumab. 52.3% (11/21) CYP without concurrent DMARD versus 44.4% (24/54) CYP on sub-therapeutic dose (<15 mg/m2) methotrexate had detectable AAA, compared to 21.1% (12/57) on full therapeutic dose (15–20 mg/m2) methotrexate.

7 CYP demonstrated fluctuations in AAA levels (range: 13–522 mg/L) despite no changes to regular medications (excluding use of corticosteroids); 1 CYP had a level of 677 mg/L which decreased to 155 mg/L one week later, despite no change to treatment.

In 53.1% (34/64 individual tests) of positive AAA tests, Adalimumab was subsequently discontinued in 34 CYP, compared to in 15.2% (14/92 individual tests) of negative AAA tests, where Adalimumab was subsequently discontinued in 14 CYP.

Conclusion: There is notable AAA incidence amongst CYP receiving Adalimumab, with no significant difference found between non-infectious uveitis and JIA cohorts. 45.5% of those tested had detectable AAA with large variability in antibody levels. Concurrent DMARD use, at full therapeutic dosage, alongside Adalimumab can reduce the risk of developing AAA. Significant fluctuations of AAA levels were found at sequential testing, despite no changes to long term Adalimumab treatment in a small subgroup. This questions the clinical relevance of a positive antibody test. However, those with positive AAA were at least 3x more likely to discontinue Adalimumab treatment compared to those with a negative result. This suggests that AAA are associated with a loss of clinical effectiveness in treating uveitis in CYP.

Decisions regarding Adalimumab discontinuation should be based on comprehensive clinical review rather than AAA levels alone, given the fluctuations of AAA in some patients despite no change in treatment. Collaborative multi-centre study looking into the factors affecting antibody formation and ways to suppress AAA

development would be useful given the pivotal role of Adalimumab in controlling inflammatory eye and joint disease.

Conflicts of interest: None

Funding: None

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P245

AUDITING THE USE OF SYSTEMIC CORTICOSTEROID THERAPY FOR NON-INFECTIOUS UVEITIS AND SCLERITIS

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Introduction: The ocular inflammation steroid toxicity risk (OSTRICH) study analysed the systemic use of corticosteroids in UK uveitis practice, comparing it to the international consensus of ≤ 7.5 mg/day of prednisolone for maintenance systemic corticosteroid prescribing in uveitis.

A maintenance dose of >7.5 mg/day is associated with a significant side effect profile, and hence patients who require these higher doses are best moved onto biologic therapies. We audited the use of corticosteroids in uveitis practice within our own Trust and compared it to the international consensus.

Method: A 3-month period of all uveitis clinics at Queen Mary's Hospital, Sidcup (King's College Hospital NHS Foundation Trust) between November 2022 and January 2023 was chosen.

Inclusion criteria

1. Patients with non-infectious uveitis diagnosed at least 12 months prior
2. Patients who have been under ophthalmic care for at least 12 months
3. Adults only (16 years or older)
4. Any uveitis (chronic anterior uveitis, intermediate uveitis, posterior uveitis, panuveitis).

Exclusion criteria

1. Patients who were prescribed steroids for systemic disease rather than ocular inflammation
2. Children (<16 years)
3. Newly diagnosed patients (within the last 12 months)
4. Patients who have been under ophthalmic care for <12 months
5. Patients who have been in any perioperative period in the last 12 months.

Normal/ideal maintenance steroid use

1. ≤ 7.5 mg/day of prednisolone, or if this is exceeded it is for no longer than 3 months.

Excessive maintenance steroid use

1. Maintenance >7.5 mg prednisolone for >3 consecutive months
2. >1 course ≥ 40 mg oral corticosteroids in the past 12 months
3. ≥ 500 mg intravenous (IV) methylprednisolone in the past 12 months.

Excessive maintenance steroid dose – acceptable reasons

1. Required to control an acute uveitis flare-up
2. Immunosuppressant increase not tolerated
3. Immunosuppressant increase not effective
4. Immunosuppressant increase declined by patient

Excessive maintenance steroid dose – unacceptable reasons

1. No consideration for steroid-sparing agents
2. Prescribing error

Results: 109 patients satisfied the inclusion criteria. Of these, there were:

- 47 anterior uveitis
- 24 intermediate uveitis
- 8 posterior uveitis
- 15 scleritis

Analysing over-use of steroids

- 13 patients had a maintenance dose of >7.5 mg prednisolone for >3 consecutive months
- 9 patients had 1 course of ≥ 40 mg oral corticosteroids in the past 12 months
- None of the patients sampled had ≥ 500 mg intravenous (IV) methylprednisolone in the preceding 12 months
- Some patients had *both* a high maintenance dose and a high-dose course of tapering steroids over the preceding 12 months.

The vast majority of reasons for using these high dose corticosteroids were acceptable:

- 13 patients required them due a flare up in the uveitis or scleritis
- 5 patients required them because the immunosuppressant increase was not effective.

There was only one case where it was an unacceptable reason (prescriber error and lost to follow up).

Conclusion: The vast majority of uveitis patients at King's College Hospital NHS Trust are referred early to the appropriate specialists for initiation of immunomodulatory therapy in cases where the uveitis is chronic. The excessive use of steroids observed in the sampled patients were mostly for acceptable reasons, with only a single patient being the outlier.

Conflicts of interest: None

Funding: None

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P246

EVALUATING A DECADE OF JUVENILE IDIOPATHIC ARTHRITIS-ASSOCIATED UVEITIS SCREENING: THE SHEFFIELD EXPERIENCE

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Introduction: Juvenile Idiopathic Arthritis (JIA)-associated uveitis is the most common extraarticular complication of JIA and the leading cause of uveitis in children and young people (CYP). This

sight threatening condition often presents as silent chronic anterior uveitis, frequently remaining asymptomatic until vision is significantly affected. This study aims to describe our experience in setting up a regional JIA-uveitis screening service in a tertiary centre and evaluates the compliance with recommended national guidelines, consecutive cohort characteristics and demographic influences.

Method: Annual audits were conducted based on the 2006 guidelines on screening for uveitis in JIA published by British Society for Paediatric and Adolescent Rheumatology (BSPAR) and The Royal College of Ophthalmologists (RCOphth). Data from 466 CYP enrolled in the JIA-uveitis screening service between 2009 and 2022 were analysed.

Results: The mean age at referral was 9.08 ± 4.17 years, and 55% were female. JIA subtypes were oligoarticular (42%), polyarticular (29%), enthesitis-related (9%), psoriatic (6%), systemic onset (2%) and non-specific JIA (12%). Before 2009, Sheffield had no formal JIA uveitis screening service, and CYP were screened inconsistently by different clinicians. In 2009/10, the multidisciplinary team (MDT) service was established. Early audit showed that 115/183 (63%) CYP were screened within 6 weeks of referral, 47/183 (26%) were screened bimonthly in the first 6 months, and 114/183 (62%) screened every 3–4 months after the first 6 months until age 11 as recommended by the national guidelines. By 2016, compliance had improved to 40/51 (78%) CYP screened within 6 weeks of referral, 29/49 (59%) screened bimonthly in the first 6 months, and 26/43 (60%) screened every 3–4 months until age 11. These improvements were driven by introduction of screening proforma, trained paediatric optometrist screeners and developments within the MDT uveitis clinic.

From 2017 to 2022, appointments offered by the hospital were also audited based on the guidelines and demographics were analysed. With a 100% uptake, audits showed 191/232 (82%) CYP were offered screening appointment within 6 weeks of referral and 178/232 (77%) screened within timeframe, with a mean offered duration of 5.0 ± 3.9 weeks. CYP from the middle deprived tertiles were significantly less likely to receive timely appointments as compared to least deprived tertiles within 6 weeks of referral (odds ratio [OR] 0.38, 95%CI 0.16–0.91, $p < 0.05$). Compliance with bimonthly screening appointment offered for the first 6 months were improved to 96/107 (89%), but a decline was observed during the COVID-19 period (2020–2022), with 66/89 (74%) offered (OR 0.33, 95%CI 0.15–0.72, $p < 0.05$) and 56/89 (63%) attended. CYP aged 10 to 13 years were less likely to receive timely appointments compared to those under 10 years (OR 0.41, 95%CI 0.180.92, $p < 0.05$). For screening every 3–4 months after first 6 months until age 11, 160/182 (88%) of CYP were offered timely appointments, and 134/182 (74%) attended. CYP from the most deprived tertiles were significantly less likely to continue attending longitudinal appointments (OR 0.34, 95%CI 0.12–0.80, $p < 0.05$). Among symptomatic CYPs, 26/39 (67%) were seen within 1 week, as recommended by the guidelines. No statistically significant difference was found in gender and types of JIA in appointment being offered and attended for all criteria. Additionally, travel distance (median: 12.45 miles, range: 0.4–82.4 miles) and travel time (median: 37 minutes, range: 3–115 minutes) did not significantly affect attendance rates.

Conclusion: In this study, we described how repeated audit cycles can drive sequential improvement over a 10-year period. Those from the most deprived tertile were less likely to engage well with uveitis screening requirements over time. This needs to be focused for targeting of future efforts. This study offers a framework for developing equitable screening strategies to prevent vision loss in CYP with JIA-associated uveitis.

Conflicts of interest: None

Funding: None

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P247

EVALUATING SLIT LAMP PERFORMANCE IN DIAGNOSING ANTERIOR UVEITIS: A SIMULATED MODEL APPROACH – ARE ALL SLIT-LAMPS THE SAME? AND, DOES IT MATTER?

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Introduction: Uveitis is a leading cause of preventable vision loss in children and young people (CYP). Diagnosing uveitis in CYP can be challenging as it often is painless and completely asymptomatic, yet without interventions, it can lead to permanent ocular complications.

Anterior uveitis, the most common form of intraocular inflammation, is often diagnosed by detecting inflammatory cells in the anterior chamber using a slit lamp. Slit lamps are critical for accurate diagnosis, but variability in light intensity and emission quality can affect diagnostic accuracy and subsequent clinical management.

This study investigates the light-emission of slit lamps in a tertiary referral eye centre in the United Kingdom, assesses their influence on diagnosing anterior uveitis and highlights the clinical implications of performance variability.

Method: 43 slit lamps were evaluated, with light emissions recorded on each device, reference numbers and service due dates. Lux measurements were recorded with a photometer under six standardised settings: full brightness head-on, at a 45° angle, 1 × 1 mm aperture at a 45° angle, all taken with and without room lighting.

A range of simulated cell chambers was designed, mimicking the anterior chamber, with floating particles representing inflammatory cells. To minimise intra- and inter-observer differences, multiple observers independently counted the number of cells visible in the simulated chamber using each slit lamp.

Observers adhered to the Standardisation of Uveitis Nomenclature (SUN) classification, grading cells from 0 to 4+ using a 1 × 1 mm slit beam at maximal brightness and standardised magnification. Results were then analysed to assess the accuracy and consistency of slit lamps with different light emissions in visualizing anterior chamber cells.

Results: A total of 21 tungsten and 22 Light emitting diode (LED) slit lamps were evaluated. Preliminary findings indicate significant variability in the ability of different slit lamps to accurately detect inflammatory cells. Notably, slit lamps with (LED) light bulbs consistently produced brighter illumination, resulting in more accurate disease grading and enhanced visibility of cells in the anterior chamber. At a 45° angle with a 1 × 1 mm aperture, tungsten BQ900 lamps emitted an average of 1011 lux with the light on (793 lux off), while LED BQ900 lamps emitted 1413 lux on (1275 lux off). Analysis of variance (ANOVA) showed significantly higher variance in tungsten lamp emissions ($F[1, 39] = 4.50$, $p = 0.04$), indicating inconsistencies in ambient lighting conditions.

Lux emissions from the tungsten Bq900 ranged from 24500 lux to 116900 lux, while the LED Bq900 had a lux emission range of 78100 lux to 197000 lux. The standard deviation (SD) for the tungsten Bq900 was 23970lux, while for the LED BQ900, it was 35689lux. The Coefficient of Variation (CV) revealed that the LED BQ900 (24.99%) was lower compared to the tungsten BQ900 (43.5%), indicating greater relative consistency in light emissions for the LED model despite its higher average intensity.

Simulated anterior chamber testing showed LED slit lamps, with brighter and more consistent light, were more accurate in detecting and grading low-grade inflammation according to the SUN

classification. Tungsten BQ900 models exhibited greater variability in observer-reported measurements, with discrepancies especially evident at lower cell densities. The consistent, brighter light from LED lamps reduced variability and improved the accuracy of inflammation detection, particularly for subtle inflammatory changes.

Conclusion: The results underscore that not all slit lamps are the same. Performance significantly varies depending on the bulb type and age, which can adversely impact light emission and affect clinical grading. This study highlights the limitations of slit lamp technologies despite periodic maintenance and calibration to ensure optimal performance.

An objective, repeatable, qualitative and quantifiable recording method to detect intraocular inflammation accurately would be the logical way forward to bypass mechanic and human factor-related errors.

Conflicts of interest: None

Funding: None

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P248

FROM BLADDER TO EYE: A RARE CASE OF DISSEMINATED BCG INFECTION FOLLOWING INTRAVESICAL THERAPY

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Introduction: Bacillus Calmette-Guérin (BCG) therapy is commonly used in bladder cancer treatment, yet it can, in rare cases, lead to disseminated infections, posing significant diagnostic and therapeutic challenges. This case study aims to explore the clinical presentation, diagnostic approach, and therapeutic response in a patient with disseminated BCG infection, providing insights that may guide future management of this rare complication.

Method: This observational case study described the management and clinical progression of an 80-year-old patient receiving maintenance intravesical Bacillus Calmette-Guérin (BCG) therapy for bladder cancer (transitional cell carcinoma) since 2019. The patient presented 3 years after that with pan-uveitis in the right eye, initially suspected to result from trauma. However, a comprehensive diagnostic workup, including vitreous biopsy and screenings for infectious and inflammatory uveitis (syphilis, HIV, Quantiferon, CMV, Hepatitis B, and C), yielded negative results.

Further investigation with a PET scan and identification of a conjunctival mass raised suspicion of tuberculosis (TB). This suspicion was supported by a multidisciplinary team (MDT) review, leading to a diagnosis of probable disseminated BCG infection. Anti-TB therapy was promptly initiated, and intravesical BCG instillation was discontinued. The patient's progression was closely monitored following the start of anti-TB treatment.

Results: Two months after starting anti-TB therapy, the ill-defined conjunctival mass had become well defined and reduced in size, with improvement in ocular comfort. By five months, the mass had completely resolved. Although the affected eye became phthisis, mortality and subsequent systemic morbidities were prevented.

Conclusion: This case study emphasises the importance of recognising that disseminated BCG infection can manifest in various organs, including sites like the eye, in patients with a history of BCG therapy. Notably, standard tests for *Mycobacterium tuberculosis*, including Quantiferon, may yield negative results, as

BCG contains attenuated *Mycobacterium bovis*. In such cases, a high clinical suspicion of disseminated BCG infection based on patient history should justify the initiation of anti-tuberculosis therapy. The patient's rapid response to empirical TB treatment highlights its effectiveness in managing BCG-related systemic infections and supports the importance of early intervention in similar cases.

Disseminated BCG infection (ocular) is a rare, sight threatening and also a critical consideration in patients receiving intravesical BCG therapy. This case demonstrates that early diagnosis and coordinated management can prevent subsequent morbidity and mortality.

Conflicts of interest: None

Funding: None

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P249

INTERMEDIATE OUTCOMES OF GATT IN UVEITIC GLAUCOMA

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Introduction: To report the 5-year clinical outcomes of gonioscopy-assisted transluminal trabeculotomy (GATT) in a Uveitic glaucoma cohort.

Method: This was a single-centre, multi-surgeon, retrospective consecutive case series of all patients with uveitic glaucoma who underwent GATT, with a minimum follow-up of 3 years at the Birmingham Midland Eye Centre. The primary outcome was surgical success as defined by WGA recommendations. Secondary outcome measures included intraocular pressure (IOP), number of medications, visual field parameters, retinal nerve fibre layer (RNFL) changes, further procedures and complications.

Results: A total of 29 eyes from 28 patients were included. The cohort was heterogeneous, with anterior uveitis (6 eyes, 20.69%) and juvenile idiopathic arthritis (JIA)-related uveitis (5 eyes, 17.24%) being the most common aetiologies. Fourteen eyes (48.28%) were phakic at the time of the procedure. The median follow-up was 5.43 years (range: 1.33–6.43 years). Complete surgical success was achieved in 34.48% (10 eyes) and qualified success in 55.17% (16 eyes), while 44.83% (13 eyes) experienced surgical failure, with a mean time to failure of 1.89 ± 1.69 years. Among the failures, 6 eyes (20.69%) did not meet IOP criteria, and the remaining cases failed due to glaucoma re-operation criteria. All failures proceeded to further glaucoma surgery, with a mean time to subsequent surgery of 2.35 ± 1.83 years. One eye was no light perception (NPL) at final follow-up.

Mean preoperative IOP was 31.83 ± 10.27 mmHg on 4.28 ± 0.92 medications, which significantly improved to 17.45 ± 9.01 mmHg on 2.14 ± 2.00 medications at final follow-up ($p < 0.0001$). Oral acetazolamide use reduced significantly, from 20 eyes (68.97%) preoperatively to 8 eyes (27.59%) postoperatively. Visual acuity (VA) showed no significant change, from 0.65 ± 0.66 preoperatively to 0.66 ± 0.82 postoperatively ($p = 0.8$). VF parameters showed a non-significant worsening in mean deviation (MD) but a nonsignificant improvement in visual field index (VFI). Mean global RNFL thickness significantly worsened at final follow-up ($p = 0.0080$).

Conclusion: GATT is a safe and effective procedure for uveitic glaucoma, with significant reductions in IOP and medication use.

However, progressive RNFL loss underscores the need for close long-term monitoring in this challenging cohort.

Conflicts of interest: None

Funding: None

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P250. Abstract not included

P251

LASER CYCLOPHOTOCOAGULATION FOR TREATMENT OF REFRACTORY GLAUCOMA SECONDARY TO UVEITIC GLAUCOMA: LONG-TERM FOLLOW-UP

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Introduction: Cyclophotocoagulation for intraocular pressure (IOP) control in refractory uveitic glaucoma has fallen out of favour due to concerns regarding long-term complications, particularly hypotony. Our aim was to describe the long-term outcomes of pressure control and complications such as hypotony and phthisis

Method: This is a retrospective single tertiary centre study involving 25 eyes of 27 patients (16 males, 11 females) in a diverse cohort with an average (SD) follow-up of 5.97 (3.4) years. The cohort mean age was 49.7 (20.9) years with a range from 14–83 years

Results: At baseline, the mean IOP (SD) was 35.9 (15.7) mmHg and 12.4 (8.0) mmHg with a best corrected visual acuity (BCVA) of 1.6 (1.12) on 3.88 (1.21) medications. Thirteen eyes (48.1%) required oral acetazolamide due to suboptimal visual field (MD = −16.8, PSD = 4.49). Eight eyes (29.6%) required repeat cyclophotocoagulation, and nine eyes (33.3%) required further glaucoma-lowering procedures. At final follow-up, the mean IOP significantly improved ($p < 0.001$) to 12.4 (8.0) mmHg; however, BCVA had worsened to 1.98 (1.67) on 1.70 (1.66) medications, with only one eye (3.7%) requiring oral acetazolamide. At final follow-up, 6 eyes (22.2%) developed hypotony(OR), 6 eyes (22.2%) developed band keratopathy, and 3 eyes (11.1%) became significantly phthisical (OR 0.016).

Conclusion: This research provides evidence that cycloablation should only be used as a final resort for uveitic glaucoma.

Conflicts of interest: None

Funding: None

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P252. Abstract not included

P253

NOVEL THERAPEUTIC STRATEGIES TO RESTORE VISION IN OCULAR HYPOTONY (STRETCH): RESULTS FROM A PROSPECTIVE PILOT SERIES

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Introduction: To report visual and anatomical outcomes following intravitreal hydroxypropylmethylcellulose (HPMC) for the treatment of chronic structural hypotony.

Method: This is a prospective interventional case series. Patients with chronic structural hypotony were evaluated by a dedicated multidisciplinary team. All patients had evidence of chronic structural hypotony for more than three months with an intraocular pressure less than 6.5 mmHg. Evidence of visual potential was primarily assessed using flash visual evoked potentials and full field electroretinograms. Prior to treatment, reversible causes were assessed and managed. Detailed ocular phenotyping was carried out before and after treatment. Eligible patients were managed as per protocol with multiple injections of HPMC varying in volume between 70 µl–1400 µl. Injections were carried out under topical anaesthesia in an injection suite, and repeated every two to four weeks until the pre-morbid axial length was reached (or, if not available, an alternative pre-specified endpoint). Patients were then followed in a hypotony clinic and further managed as appropriate.

Results: A total of eight eyes were included in the study. Primary outcome measures were best corrected visual acuity 12 months after initiating treatment. Secondary outcome measures were intraocular pressure and axial length 12 months after the first injection. Best corrected visual acuity improved in seven of eight eyes (87.5%) and remained stable in one. Intraocular pressure increased in seven eyes (87.5%), whereas axial length increased in six eyes (75%).

Conclusion: This is the first case series to use a structured protocol for delivery of intravitreal HPMC for the management of chronic hypotony. This approach not only prevents complications associated with chronic hypotony, but also improves vision and restores ocular anatomy. Treatment with HPMC shows promise in managing structural hypotony in selected eyes, with potential improvements in both functional and structural outcomes.

Conflicts of interest: none

Funding: None

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P254

OCULAR INFLAMMATORY SIDE EFFECTS OF IMMUNE CHECKPOINT INHIBITORS IN A UK TERTIARY REFERRAL CENTRE: OPTIMISING PATIENT CARE

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Introduction: Immune-checkpoint inhibitor (ICI) therapy has revolutionised cancer medicine. Increasingly, ICI are used earlier in disease and in combination with each-other. Although these agents are effective, they have been shown to cause various adverse effects due to dysregulation of the immune system leading to unintended autoimmune phenomena. We aim to share the UK's largest series of patients with ophthalmic complications associated with ICI therapy and create a referral algorithm to optimise future patient care.

Method: A single institution retrospective case series of 20 patients identified from an in-house electronic patient record system and referral screening by the authors at a tertiary hospital.

Results: Underlying malignancies included melanoma (67%) and lung (17%) with renal, breast and oesophageal cancer equally sharing 17%. 39% of patients were receiving pembrolizumab monotherapy. 33% were on a combination of ipilimumab-

nivolumab, with 22% managed with nivolumab and 6% treated with ipilimumab only. Phenotypes included uveitis (anterior and posterior), optic neuritis, myasthenia gravis as well as corneal pathology and reactive dacryoadenitis. Corticosteroid treatment was given in most cases. Some required additional immunomodulatory treatment.

Conclusion: Ophthalmologists need to be aware of cancer patients on ICI immunotherapy presenting with ocular symptoms as a significant number of patients are presenting with vision-affecting uveitis as a side effect. It is preferable to maintain ICI's as life-saving treatment, and often patients can be managed with the addition of systemic or local steroids. Early multidisciplinary intervention and good communication with the oncology team optimises care.

Conflicts of interest: None.

Funding: None.

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P255

OUTCOMES OF ILUVIEN (FLUOCINOLONE ACETONIDE INTRAVITREAL IMPLANT 0.19 MG) FOR NON-INFECTIOUS UVEITIS (NIU): RESULTS OF A RETROSPECTIVE CASE SERIES

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Introduction: The treatment of NIU remains challenging. In recent years, methods have been devised to ensure delivery of the optimal therapeutic drug dosage to the posterior segment to help reduce the inflammatory burden of the disease. As per the 2019 National Institute for Health and Care Excellence (NICE) guidelines, a fluocinolone implant is recommended for NIU, although the evidence backing this remains uncertain. We aim to assess the outcomes of a Fluocinolone implant (ILUVIEN) at 12 months including effects on visual acuity, posterior segment inflammation and central retinal thickness (CRT), including complications of this therapy.

Method: We analysed electronic patient records of all patients treated for NIU with ILUVIEN implant between 2020 and 2022 at Worcestershire Acute Hospitals NHS Trust, UK. The pre-operative and 12-month post-operative follow-up data were collected used for analysis. The change in best corrected visual acuity (BCVA) as measured using the LogMAR scale was noted. Patients identified to have cystoid macular oedema (CMO) were assessed for reduction in CRT at 12 months using a t-test. The severity of posterior segment inflammation was noted and compared pre-operatively and at 12-month follow-up. Complications specifically analysed, included development of glaucomatous change and cataracts. The final outcomes were classified as a success or failure, with success being defined as either reduced disease burden or control till 12-month follow-up. Failure was defined as development of severe complication or no improvement in disease burden.

Results: A total of 78 eyes in 63 patients were included in the analysis. The mean age of the patients was 54.9 years (22–86 years). Final 12-month follow up data was available for 69 of 78 eyes (88.4%). Mean BCVA on the LogMAR scale changed from 0.29 to 0.24 over 12 months. Mean CRT in patients with CME was 404.83 microns pre-operatively which reduced to 336.71 microns ($p = 0.02$). At 12 month follow up, 65 of 69 eyes (94.2%) showed a maintained reduction in vitreous inflammation. Complications encountered included 10 eyes (14.4%) developing cataract with 4

requiring surgery. 7 eyes (10.1%) requiring IOP lowering procedures and 22 eyes (31.8%) remained on anti glaucoma medications. Overall, 60 eyes (86.9%) met the criteria for success and 9 eyes classified as failure.

Conclusion: This study demonstrates that at a follow up period of 1 year, ILUVIEN is an effective therapy for patients with NIU in terms of maintenance of vision and reduction in posterior segment inflammation. It results in significant reduction in CRT in patients who have CMO. Complications such as development of increased IOP and cataract should be closely monitored throughout. Studies with long term follow ups are required to elicit longer term effects of the implant.

Conflicts of interest: None

Funding: None

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P256

OUTCOMES OF PATIENTS WITH UVEITIS IN REMISSION UNDERGOING CATARACT SURGERY

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Introduction: Uveitis is an inflammatory eye disease that can be associated with systemic conditions but often occurs idiopathically. Many patients experience only one or a few episodes of uveitis in their lifetime and may not require ongoing follow-up by a uveitis specialist. When these patients undergo cataract surgery, standard protocols are often followed without considering their previous uveitis history. Consequently, pre-operative work-up may be inadequate. There is a lack of evidence regarding the risks of cataract surgery in patients with a history of uveitis and the potential for post-operative uveitis recurrence. It is well-established that active uveitis complicates cataract surgery, requiring aggressive pre-operative and post-operative corticosteroid therapy to prevent complications such as cystoid macular oedema, elevated intraocular pressure, and post-operative uveitis. This study aims to evaluate the visual outcomes and complication rates of cataract surgery in patients with a history of uveitis who are currently in remission.

Method: This retrospective cohort study evaluated cataract surgery outcomes in patients with a history of uveitis now in remission who were not currently under the care of a uveitis specialist. Data was extracted from electronic medical records (EMR) for patients who underwent cataract surgery between January 1, 2019, and March 1, 2024. Clinical data, including the location of uveitis and any associated systemic diseases, was collected. Pre-operative and post-operative medication regimens were also recorded. Primary outcomes included best-corrected visual acuity (BCVA) at one week, four to six weeks, and eight to sixteen weeks post-operatively. Secondary outcomes included intra-operative and post-operative complications within five months of surgery. Patients were excluded if they had less than three months of follow-up data, isolated post-operative uveitis, or keratouveitis managed by the cornea service.

Results: Twenty-three eyes presented with a history of uveitis in remission underwent cataract surgery during the study period. The mean age of the patients was 65.4 years. Most cases presented with anterior uveitis (79.2%), followed by intermediate uveitis (7.9%), panuveitis (7.9%), and posterior uveitis (4.2%). The majority of cases were idiopathic in origin (70.8%). No pre-operative medication regimen was documented for any patient.

Postoperatively, patients received a different topical medications regimen, including: topical steroids and NSAIDs (37.5%), topical steroids and antibiotics (29.2%), topical steroids, antibiotics, and NSAIDs (8.4%), topical steroids alone (12.5%), or no documented regimen (12.5%). Following surgery, the mean best-corrected visual acuity (BCVA) at one week, four to six weeks, and eight to sixteen weeks was 0.223, 0.354, and 0.268 LogMAR, respectively. No intraoperative complications were reported. Post-operatively, complications included cystoid macular oedema in five eyes (20.8%), postoperative uveitis in three eyes (12.5%), and elevated intraocular pressure in two eyes (8.4%).

Conclusion: This study highlights the significant risks associated with cataract surgery in patients with a history of uveitis, even when the disease is in remission for several years. Compared to national data on standard cataract surgery, the incidence of cystoid macular oedema was seven-fold higher in this group of patients. This suggests that inadequate pre-operative and post-operative management, particularly in patients not followed-up by a uveitis specialist, can lead to increased risk of developing post-operative complications. To mitigate these risks, it is imperative to establish evidence-based protocols for the management of cataract surgery in patients with a history of uveitis in remission. These protocols should include referral to a uveitis specialist for pre-operative evaluation and the implementation of appropriate preoperative and post-operative treatment regimens.

Conflicts of interest: None

Funding: None

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P257

SCREENING FOR AXIAL SPA IN HLA-B27 ASSOCIATED UVEITIS: AN AUDIT OF PATIENTS PRESENTING TO THE OPHTHALMOLOGY CLINIC

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Introduction: Axial spondyloarthritis (axial SpA) is a debilitating chronic inflammation of the axial skeleton presenting as back pain, which has been linked with Human Leukocyte Antigen B27 (HLA-B27).^{1,2} HLA-B27 has been implicated in other inflammatory conditions such as anterior uveitis.³ Anterior uveitis is the inflammation of the iris and/or the ciliary bodies of the eye and it represents the most common form of uveitis seen in ophthalmology.⁴ In half of patients diagnosed with anterior uveitis, Human Leukocyte Antigen B27 (HLA-B27) positive status is observed.

In the UK, patients received a delayed diagnosis of axial SpA at 8.5 years from the initial onset of symptoms,⁵ compared to the international mean of 6.7 years.⁶ Diagnostic delays cost the UK economy £18.7 billion annually⁶ as well as result in functional impairment and worsening quality of life for patients.⁷ Despite this, studies into the prevalence of screening for axial SpA in patients presenting to the ophthalmology clinic are sparse.

Objective: We present a novel cross-disciplinary audit assessing the awareness and incidence of screening for axial SpA in patients who present to the ophthalmology clinic with HLA-B27 associated anterior uveitis. We aim to reduce the diagnostic delay in axial SpA through increasing the prevalence of, as well as by identifying any significant demographic correlations.

Method: An audit was carried out to identify if patients presenting from July 2023 to January 2024 with a diagnosis of HLA-B27 associated anterior uveitis were screened for axial SpA

symptoms according to the Assessment of Spondyloarthritis international Society (ASAS) criteria. Patient data was anonymised and patient demographics as well as referral pathway were recorded for identification of any significant correlations.

Results: Out of 23 patients with documented positive HLA-B27 status, 47.8% were screened for at least 1 parameter of the ASAS criteria and the mode was 1. None of the patients were screened for 4 or more parameters. The most screened for parameter was for pain in the lower back which was screened for in 47.6% of patients. 39.1% of patients had a previously diagnosed autoimmune condition while 43.5% of patients had a history of back pain or arthritis. 38.1% of patients were also asked if they had a previous diagnosis of any axial SpA while 42.9% of patients were screened for early onset joint or back pain that necessitated a presentation to primary or secondary care.

Conclusion: The audit shows that the importance of screening for axial SpA has been recognised by clinicians. However, there could be benefit from further education and utilisation of the National Axial Spondyloarthritis Society (NASS) screening tool by clinicians as only 47.8% of patients were screened for, and the sensitivity of screening may not be ideal due to the low number of parameters screened for. Patient's HLA-B27 status could be made accessible in the future to facilitate consultations. Further audits including a larger sample size and multiple facilities would help to provide a more comprehensive picture of the prevalence of screening for axial SpA. We have strategized and carried out interventions targeting clinicians from both regional and national standing to maximise our outreach with a plan for a further re-audit in the future to assess the success of intervention. Plans have also been drawn up to assess from existing data for the utility of the Dublin Uveitis Evaluation Tool (DUET)⁸ among clinicians.

The retrospective nature of the study, as well as the documentational challenges faced in ascertaining patient's HLA-B27 status and screening for axial SpA present potential confounding factors not adjusted for under the current study design.

Conflicts of interest: No conflict of interest to declare

Funding: No funding received

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P258

VALIDATING DIFFERENTIAL GENE EXPRESSION IN SUBTYPES OF CD4+ T CELLS IN UVEITIS

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Introduction: Non-infectious uveitis comprises a heterogeneous group of CD4+ T cell mediated diseases that are characterised by intraocular inflammation. Effector subtypes of CD4+ T cells are known to drive the pathogenesis of non-infectious uveitis.

Method: Here, we performed RNA-seq analysis on pathogenic CD4+ T cells from a murine model of uveitis called experimental autoimmune uveoretinitis (EAU). We selected three significantly upregulated genes for validation at the protein level on the cell surface of activated CD4+ T cells.

Results: Using flow cytometric analysis, we found that the proteins encoded by these genes (CD226, CD314 and TNFSF14) were expressed on the cell surface of CD4+ T cells under Th1, Th17 and ThP polarising conditions in vitro. We also showed that

a proportion of CD4+ cells from the site of active disease in vivo in B10 mice with EAU express CD226, CD314 and TNFSF14. We have therefore shown that CD226, CD314 and TNFSF14 can be considered markers of pathogenic CD4+ T cell effector cells. To understand whether the method of EAU induction influenced marker expression, further in vivo experiments were carried out using active immunisation and adoptive transfer with different immunising peptides (RBP-31-20 or RBP-3629-643) to induce EAU. We found that, in general, higher proportions of CD4+ T cells from retinas of actively immunised mice expressed proteins of interest, irrespective of the immunising peptide used.

Conclusion: Overall, we have identified protein markers that are expressed by pathogenic CD4+ T cell subtypes in EAU. These findings provide further insights into the pathogenesis of non-infectious uveitis.

Conflicts of interest : None

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VITREORETINAL DISEASES AND SURGERY

P259

ENHANCING ROBOTIC OPHTHALMIC SURGERY: A MACHINE LEARNING PERSPECTIVE ON TOOL TRACKING IN VITREORETINAL SURGERY

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Introduction: The retina is the innermost, light-sensitive neural tissue of the eye and is subject to surgical interventions that account for a significant proportion of ophthalmic surgeries globally. Retinal microsurgery (RM) presents unique technical challenges compared to other types of microsurgery. In particular, it requires exceptional precision when operating on micro-scale targets within the confined space of the retina; damage to this delicate tissue can have irreversible consequences given its limited ability to regenerate and surgical outcomes are influenced by the surgeon's experience alongside iatrogenic factors. Robotic systems offer promise in overcoming human limitations, through dampening of tremor and augmentation of surgical video. However, surgical robotic systems must be manually positioned in the region of interest, increasing the complexity of device handling during the surgical workflow. In addition, the efficacy of their imaging modalities relies heavily on maintaining a clear surgical field, which is often hindered by uncontrollable factors such as tool occlusion, haemorrhages, or shadowing artefacts from the tool. Computer vision emerges as a valuable solution in addressing these challenges. Performing efficient and safe retinal microsurgery requires real-time, constant knowledge of the location of the surgical tool and computer vision can help provide this information by understanding the scene context through segmentation powered by deep learning. This capability allows for robust extraction of data about the tool's position even when visual cues are obscured or degraded, thereby enhancing the precision of retinal microsurgery.

Method: We propose a machine learning (ML) algorithm to segment the position of surgical instrumentation within the retina, using data acquired from an intra-operative microscope in real time. Surgical videos were collected in vivo using an intra-operative microscope, and frames were extracted from each

video. Each frame was manually annotated to generate a binary segmentation mask of the surgical tooltip and background pixels. These ground truth masks then underwent a series of data augmentation steps. Multiple machine learning algorithms were built using a U-Net architecture, consisting of a contracting path to capture context and a symmetric expanding path to enable precise localisation. They were then trained, validated, and tested on 80% of the dataset to optimise their hyperparameters using a grid search. Finally, the best-performing model was evaluated on the heldout validation dataset.

Results: The final tuned U-Net model achieved an IOU score of 84% and an AUROC of 0.81, demonstrating its effectiveness in accurate tool segmentation across the surgical field. It performed well on augmented data, suggesting the model is generalisable to noisy surgical landscapes and varying orientations of the tool. In addition, the sensitivity was 85%, indicating the model was able to capture most of the true positive instances of the tool. Furthermore, the specificity of 75% illustrates the model's relatively high degree of confidence in excluding pixels not belonging to the background class.

Conclusion: With our ageing population, retinal microsurgery is anticipated to see a rise in prevalence. The integration of robotics into this field holds great promise in surmounting human limitations but is challenged by the impaired intraoperative visualisation and the need for manual repositioning in the area of interest. In response, we propose an efficient ML approach for supervised segmentation of the tooltip in RM, achieving an IOU of 84%. Tooltip tracking has the potential to provide real-time surgical guidance, facilitating the incorporation of robotics into RM and thus enhancing the safety of retinal surgeries. Moving forward, our focus will be on refining the model and conducting further training on a more diverse dataset.

Conflicts of interest: None

Funding: None

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P260

EVALUATION OF VISUAL OUTCOMES, DISTANCE-TO-DISC, AND MORTALITY RATES IN CHOROIDAL MELANOMA PATIENTS UNDERGOING PROTON BEAM THERAPY : A SERVICE REVIEW AND AUDIT

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Introduction: Choroidal melanoma is the most common primary intraocular malignancy in adults and is associated with significant impacts on both visual outcomes and overall survival. Proton beam therapy (PBT) offers a targeted treatment modality that minimises damage to surrounding healthy tissues. This study evaluates the effectiveness of PBT in treating choroidal melanoma by analysing visual outcomes, distance-to-disc metrics, and mortality rates in a cohort of patients.

Method: A retrospective analysis was conducted on 1,563 patients diagnosed with choroidal melanoma who received proton beam therapy at Liverpool Clatterbridge Cancer Centre. Data including tumour size, distance from the optic disc, pre- and post-treatment visual acuity (logMAR), and survival status were analysed. Statistical methods such as ANOVA, linear regression, logistic regression, and Kaplan-Meier survival analysis were employed to assess the relationship between tumour characteristics and clinical outcomes.

Results: Tumours located within 2 mm of the optic disc exhibited the greatest deterioration in visual acuity, with a mean increase in logMAR of 0.32. Larger tumours were associated with higher mortality rates, with 38.41% mortality for large tumours compared to 12.79% for small tumours. Survival analysis indicated that larger tumours had a higher recurrence probability over time. The study also identified critical thresholds for tumour size and distance to the optic disc that significantly impact both visual outcomes and survival rates.

Conclusion: This audit provides comprehensive evidence supporting the role of proton beam therapy in managing choroidal melanoma. Tumour proximity to the optic disc and tumour size are critical factors influencing visual and survival outcomes. Early detection and adherence to clinical guidelines are essential for optimising patient care and improving treatment outcomes. This work informs clinical practice by identifying thresholds that guide treatment decisions and follow-up care, with potential to enhance patient outcomes across similar healthcare settings.

Conflicts of interest: None

Funding: None

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P261

LONG-TERM OUTCOMES OF SCLERAL BUCKLE RETINAL DETACHMENT REPAIRS DONE USING THE MODIFIED MCHUGH METHOD

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Introduction: Rhegmatogenous retinal detachment (RRD) is primarily treated either by external approach with scleral buckling, an internal approach with pars plana vitrectomy or both. The scleral buckling versus primary vitrectomy in RRD study showed that buckling had superior visual outcomes in phakic eyes [1]. Buckling has been falling out of favour over the last two decades and is quickly becoming a lost art of vitreoretinal surgery. We aim to look at the anatomical and functional outcomes using the modified McHugh method (MMM) of scleral buckling compared to the conventional method. This method is simpler perform and teach residents. It involves 4 muscle slings, identifying breaks, cryotherapy to breaks, placing encircling 276 silicone rubber explant and a 5 mm radial plomb under the 276 if the break is posterior.

Method: We looked at all cases of scleral buckle performed using the modified McHugh method done at a single centre from 1st January 2010 till 31st March 2024. All surgeries were performed by either the senior fellow or consultant, with the consultant teaching the modified McHugh method present. Medical records were analysed for presentation, time to treatment and post-operative follow-up times and outcomes.

Results: A total of 118 eyes from 113 patients who have had scleral buckle surgery since 2010. Of these patients, 32 eyes from 31 patients were performed using the modified McHugh method. The median age of patients was 32 years [IQR 20.5–48] and 16 (51.6%) were males. The right eye was affected in 19 (59.4%) of the eyes. Most of the retinal detachments were rhegmatogenous (n = 27, 84.4%), two (6.25%) were associated with retinal dialysis and one was a schisis detachment (3.2%). The median number of breaks is 2 [IQR 1–3], with the inferior location being the commonest n = 18, 56.3%). The macula was off in 13 cases (40.6%) and off in 7 cases (21.9%). Only one (3.2%) patient was pseudophakic at time of diagnosis and four (12.5%) patients had proliferative vitreoretinopathy at the time of diagnosis. Median best corrected visual acuity

at time of diagnosis was 0.3 [IQR 0.1–0.80]. Median time till surgery (from date of diagnosis) was 4 days [IQR 3–8 days]. Median follow-up time is 18.5 months [IQR 9–48 months]. The best corrected visual acuity achieved post-operatively was 0.3 [IQR 0.1–0.7] (p = 0.36). Anatomy was achieved in all at immediate post-operative FU and at 3-month interval.

Conclusion: The national audit of the outcome of primary surgery for rhegmatogenous retinal detachment found that the overall reattachment rate with a single procedure was 77% (95% CI 73.9–80.2) [2]. They also found that over one third of patients had at least one complication noted at some point in the 3-month follow-up period and 6.35% retinas were not attached at 3 months postoperatively. The outcome of RRD repair by buckling from a tertiary centre in Scotland found a re-detachment rate of 9.23% [3]. Therefore, the modified McHugh method has anatomical and functional outcomes comparable with conventional outcomes with low re-detachment and complication rates. It is a cheap, simple to teach and widely applicable method to a range of RRD presentations.

Conflicts of interest: None

Funding: None

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P262. Abstract not included

P263.

STERILE HYPOPYON FOLLOWING PARS PLANA VITRECTOMY AND SILICONE OIL INJECTION

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Introduction: Intraocular inflammation following ophthalmic surgery is common and is usually well controlled with topical steroids. Less frequently, post-operative inflammation can present with more severe signs including anterior chamber fibrin and hypopyon. In such cases, the differential diagnosis includes infectious endophthalmitis, sterile anterior chamber inflammation, toxic anterior segment syndrome (TASS) and lens-induced uveitis. This is a case series reporting five cases of post-operative sterile endophthalmitis in patients who underwent pars plana vitrectomy with silicone oil injection for retinal detachment.

Method: This is a retrospective case series of 5 patients who underwent surgery for the management of retinal detachment at Southend University Hospital NHS Foundation Trust, and the Royal Eye Infirmary, Derriford Hospital, University Plymouth Hospitals NHS Trust. Medical records were retrospectively reviewed and informed consent for publication was obtained from all participants

Results: All five patients presented with signs of anterior chamber inflammation exhibiting flare, fibrin and hypopyon in the early or late post-operative period following vitrectomy with silicone oil injection. Treatment with topical antibiotics and steroids was commenced, with rapid resolution of the anterior chamber inflammatory reaction and no further intervention required.

Conclusion: In this case series we reported the development of severe anterior chamber inflammation and sterile hypopyon in five patients that underwent pars plana vitrectomy with silicone oil injection. The use of silicone oil as a tamponade agent for

retinal detachment can rarely induce sterile anterior chamber inflammation with signs resembling infectious endophthalmitis. Onset of symptoms, absence of pain and good response to topical steroids can help differentiate between the two entities.

Conflicts of interest: None

Funding: None

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P264

STRUCTURE AND FUNCTION PRESERVED FIVE YEARS AFTER HESC-RPE PATCH TRANSPLANTATION WITH LOCAL IMMUNOSUPPRESSION IN SUBMACULAR HAEMORRHAGE SECONDARY TO NEOVASCULAR AMD

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Introduction: There are currently limited study outcomes for long term survival and function of stem cell derived RPE patches for treatment of advanced neovascular AMD (nAMD). We present 5-year safety, structural and functional outcomes following sub-retinal transplantation of a human embryonic stem cell-derived retinal pigment epithelium (hESC-RPE) monolayer delivered on a coated, synthetic membrane, as a therapeutic strategy for advanced nAMD.

Method: Two subjects with acute vision loss due sub-macular haemorrhage in advanced nAMD received the hESC-RPE patch. Systemic immunosuppression was used peri-operatively followed by local, depot immunosuppression. The subjects were monitored for five years with a series of structural and functional assessments, including multimodal imaging, reading tests and microperimetry.

Results: No serious hESC-related adverse events, no uncontrolled proliferation and no tumourigenicity were observed. Sustained visual acuity improvement of > 15 letters was recorded for both patients up to 2 years and one subject maintained this for 5 years. Fundus photography showed preservation of the patch location and some pigmentation for 5 years. Optical coherence tomography showed RPE characteristics on the patch, preservation of some retinal ultrastructure with signs of remodelling, fibrosis, and thinning. Adaptive optics imaging showed survival of cone photoreceptors in the transplanted area. Both demonstrated significant improvement in contrast sensitivity, reading speed, fixation and microperimetry, comparing to baseline.

Conclusion: HESC-RPE patches did not show any major safety concerns following transplantation. Good structural, safety and functional stability in advanced nAMD, was shown for at least two years and in up to 5 years. Our primary outcomes show that hESC-RPE transplantation shows potential as a therapeutic strategy in severe late neovascular macular degeneration.

Conflicts of interest:

Taha Soomro: Commercial Relationship: Code N (No Commercial Relationship) | Odysseas Georgiadis: Commercial Relationship: Code N (No Commercial Relationship) | Fred Chen: Commercial Relationship: Code N (No Commercial Relationship) | Britta Nommiste: Commercial Relationship(s); Code E (Employment): Tenpoint therapeutics | Amanda-Jayne Carr: Commercial Relationship: Code N (No Commercial Relationship) | Mandeep Sagoo: Commercial Relationship: Code N (No Commercial Relationship) | Peter Coffey: Commercial Relationship(s); Code P (Patent): Patent Application No. PCT/GB2009 /000917; Code P (Patent): International Patent Application No. PCT/GB2011 /051262; Code O (Owner): Tenpoint therapeutics | Lyndon DaCruz: Commercial Relationship(s); Code P (Patent): Patent Application No. PCT/GB2009/000917; Code P (Patent): International Patent Application No. PCT/GB2011/051262; Code O (Owner): Tenpoint therapeutics

Funding: Sir Michael Uren Fund for Moorfields Eye Charity

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